DEVELOPMENT AND IMPLEMENTATION OF ONLINE HEALTH SERVICES SYSTEM

An Adaptive Barangay Information System

A Capstone Project Presented to the Faculty of the Department of Information Science and Technology School of Sciences and Engineering,
University of Asia and the Pacific
Pearl Drive, Ortigas Center, Pasig City, Metro Manila

In Partial Fulfillment of the Requirements for the Degree of Master of Information Technology (MIT)

By

Sarah Andrea B. Viana

February 2022

Table of Contents

Abstract 7

Purpose and Description 10

Statement of the Problem (Business Opportunity) 11

The Existing System 13

Conceptual Model of the Study 14

Figure 1: Process flow of the Existing System 15

Proposed System 15

Figure 2: Process flow of the Proposed System 16

General Objective: 16

Specific Objectives: 16

Letter of Instruction No. 949, s. 1979 19

Primary Health Care (PHC) in the Philippines 19

Community Health Systems Catalog Country Profile: Philippines 20

Figure 3. Philippine Health System Structure (Source: caro.doh.gov.ph) 21

Philippines eHealth Strategic Framework and Plan 2014-2020 22

The eHealth Strategic Guiding Principles. 22

Figure 4. eHealth's Vision (Source: ehealth.doh.gov.ph) 23

Health System Goal or Challenge for Health Consumers : Safer and Quality Health Care. 23

eHealth Key Result Areas 25

Health Consumers: Safer and quality health care 25

Barangay Health Center 27

Development and Implementation of Health Information Systems 27

Figure 5: Components of a Health Information System by WHO as referred from the Development and Implementation of Health Information Systems (Source: researchgate.net/publication/242520887) 30

Figure 6. Components of a Health Information System by WHO as referred from the Development and Implementation of Health Information Systems (Source: researchgate.net/publication/242520887) 31

Related Studies 31

System Development Life Cycle 31

Figure 7: Agile Software Development Life Cycle Model (Source: Grefiel (et.al) 2018) 32

Figure 8: Agile-Waterfall Systems Development Life Cycle Model (Source: Grefiel (et.al) 2018) 34

Large-Scale Health Information System Project Planning, Implementation and Evaluation 34

A literature review covering all aspects of developing and implementation of Health Information systems discussed that implementation is defined as the process of planning, testing, adaptong, and integrating HIS so that the technology becomes routinely used in the organization. The researchers also defined HIS as a software that assists healthcare organizations to gather, process and disseminate information within the organization and their environment. It was also mentioned that HIS incorporates a range of different types of systems, which include patient information systems, administrative systems, radiology and pharmacy information systems, telemedicine and hospital information systems, such as computerized physician entry systems. 34

Initial Investigation and Analysis 35

Barangay Management Information System (BMIS) for Cities and Municipalities in the Philippines 35
Figure 9: SDLC Comparison for Barangay Management Information System (BMIS) for Cities and Municipalities in the Philippines (Source: (Imus, 2018, 2)) 36
Pampanga's Barangay Health Information System (PBHIS):A Decision Support & Health Information System for Rural Health Unit 1 36
Figure 10: Agile Waterfall Hybrid Methodology for Pampanga's Barangay Health Information System (PBHIS): A Decision Support & Health Information System for Rural Health Unit 1 (Source: (Estinar et al., 2018, 2) 36
Data Instrumentation 36
Quantitative Data Collection 36
Figure 11: Five Methods Used for Quantitative Data Collection Method (Source: questionpro.com) 37
Statistical Treatment 37
Sampling Method 38
Sample Size Determination 38
Figure 12: How to calculate the number of completions. (Source: checkmarket.com) 39
Likert Scale 39
Theory Acceptance Model 39
Figure 13: Conceptual Model for Theory Acceptance (Source: Laderer A., 2000) 40
U-S Adaptability 40
Figure 12: Example of Using Access Efficiency (AE) to evaluate User-System Adaptability in Adaptation Process 41
User IT Adaptation Behaviors 42
Figure 13: Ways to discuss User IT Adaptation Behaviors 42
Source: core.ac.uk 42
Electronic Health Record Adaptability and Interoperability 43
Figure 14 Knowledge Management Environment 44
Source: www.kmworld.com 44
Boehm's Software Quality Model 44
Figure 15: Boehm's Software Quality Diagram 45
Source: geeksforgeeks.org 45
Risk Management in Healthcare 45
Figure 15: The Eight Domains of Enterprise Risk Management 46
Source: catalyst.nejm.org 46
Cognitive Fit Theory 46
Figure 16: Extended Cognitive Fit model 46
Source: theorizeit.org 46
PHP Hypertext Processor 46
Gantt Chart 47
Health Privacy Code 47
Role-Based Access Control (RBAC) 48
Figure 17: Illustration of the Role-Based Access Control 49
Source: https://www.techtarget.com/searchsecurity/definition/role-based-access-control-RBAC 49
SSL and TLS 49
Signature Hash Algorithm 49

Scope and Delimitations 50
Scope 50
Delimitation 53
Quantitative Research Methodology 55
Success Indicators 55
Likert Scale 57
Table 1: Illustration of the Likert Scale and Success Indicators 58
Dependencies 59
Data Collection 59
Sample Size Determination 59
Interview 60
Figure 19: Proposed System presentation with the Barangay Captain and Council of Brgy. Sta. Ana 61
Figure 20: Proposed System presentation with the Barangay Health Workers of Barangay Sta. Ana 61
Documents and Records 61
Figure 21: Sample Patient Charts Figure 22: Health Board Indicators 62
Figure 22: Patient File Storage for Barangay Guitnang Bayan 1 62
Figure 23: DOH and Municipal Mandated Forms both for Barangay Guitnang Bayan 1 and Barangay Sta. Ana 63
Surveys 63
Table 2: Alternatives 63
Systems Development Life Cycle 64
Implementation Plan 65
Table3: Implementation Plan for the Health Services Information System 65
Project Cost 70
Table 4: Implementation Plan for the Health Services Information System 70
Key Areas of System Flow 71
Software Acquisition 71
Figure 24: Parallel Testing 73
Table 5. Implementation Plan for the Health Services Social System 73
Use Case Diagrams of the Existing System 75
Diagram 1: Barangay Health Worker 75
Diagram 2: Barangay Midwife 76
Diagram 3: Barangay Nurse 77
Diagram 4: Patient 78
Use Case Diagram of the Proposed System 79
Diagram 5: Patient Registration and Consultation Schedule 79
Diagram 6: Patient Consultation 80
Risks and Mitigation Plan 81
Table 6: Risks and Mitigation Plan for the proposed system 81
Class Diagrams 83
Figure 25: Starting & Stopping 83
Figure 26: Swim Lane 1 84

Test Policy 84
Table 7: Initial tests 84
Figure 27: Test Process 86
Health Privacy Code 86
Figure 28: SMRBHIS Role-based Access Control Setting 89
Figure 29: Only System Administrators can update the System User Details 90
Figure 30: A comparative view of Nurse's System Maintenance Function vs. a System Administrator 90
Figure 31: Password Protected User Access for Patients 91
Figure 32: Password Protected User Access for System Administrators 92
Figure 33: SMRBHIS Security Report Figure 34: SMRBHIS Certificate Information 93
Figure 35: SMRBHIS SSL Report 94
Barangay Guitnang Bayan 1 : Barangay Health Workers 96
Table 8: Barangay Guitnang Bayan 1's response on Reliability 96
Chart 1: Barangay Guitnang Bayan 1's response on Reliability 97
Table 9: Barangay Guitnang Bayan 1's response on Usability 97
Chart 2: Barangay Guitnang Bayan 1's response on Usability 98
Chart 3: Barangay Guitnang Bayan 1's response on Understandability 99
Table 10: Barangay Guitnang Bayan 1's response on Security 100
Chart 4: Barangay Guitnang Bayan 1's response on Security 101
Table 11: Barangay Guitnang Bayan 1's response on Support 101
Chart 5: Barangay Guitnang Bayan 1's response on Support 102
Table 12: Barangay Guitnang Bayan 1's Mean Average 102
Barangay Sta. Ana : Barangay Health Workers 103
Table 13: Barangay Sta. Ana's response on Support 103
Chart 6: Barangay Sta. Ana's response on Reliability 103
Table 14: Barangay Sta. Ana's response on Usability 104
Chart 7: Barangay Sta. Ana's response on Usability 105
Table 15: Barangay Sta. Ana's response on Understandability 106
Chart 8: Barangay Sta. Ana's response on Understandability 106
Table 16: Barangay Sta. Ana's response on Security 107
Chart 9: Barangay Sta. Ana's response on Security 107
Table 17: Barangay Sta. Ana's response on Support 108
Chart 10: Barangay Sta. Ana's response on Support 109
Table 18: Barangay Sta. Ana's Mean Average 109
Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response 110
Table 19: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Reliability 110
Chart 11: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Reliability 110
Table 20: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Usability 111
Chart 12: Barangay Guitnang Bayan 1 and Barangay Sta. Ana's Response to the proposed system's Usability 111

Table 21: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Understandability 112
Chart 13: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Understandability 112
Table 22: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Security 113
Chart 14: Barangay Guitnang Bayan 1 and Barangay Sta. Ana's Response to the proposed system's Security 114
Table 23: Barangay Guitnang Bayan 1 and Barangay Sta. Ana's Response to the proposed system's Support 114
Chart 15: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Security 114
Table 24: Barangay Guitnang Bayan 1 and Barangay Sta. Ana's Combined Mean Average 115
Citizens of Barangay Guitnang Bayan 1 115
Table 25: Citizens of Barangay Guitnang Bayan 1's response on Reliability 115
Chart 16: Citizens of Barangay Guitnang Bayan 1's response on Reliability 116
Table 26: Citizens of Barangay Guitnang Bayan 1's response on Usability 116
Chart 17: Citizens of Barangay Guitnang Bayan 1's response on Usability 117
Chart 18: Citizens of Barangay Guitnang Bayan 1's response on Understandability 118
Table 28: Citizens of Barangay Guitnang Bayan 1's response on Security 118
Chart 19: Citizens of Barangay Guitnang Bayan 1's response on Security 119
Table 29: Citizens of Barangay Guitnang Bayan 1's Average Mean 119
Citizens of Barangay Sta. Ana 120
Table 30: Citizens of Barangay Sta. Ana's response on Reliability 120
Chart 20: Citizens of Barangay Barangay Sta. Ana's response on Reliability 120
Table 31: Citizens of Barangay Barangay Sta. Ana's response on Usability 120
Chart 21: Citizens of Barangay Barangay Sta. Ana's response on Usability 121
Table 32: Citizens of Barangay Barangay Sta. Ana's response on Understandability 122
Chart 22: Citizens of Barangay Barangay Sta. Ana's response on Understandability 122
Table 33: Citizens of Barangay Barangay Sta. Ana's response on Security 123
Chart 23: Citizens of Barangay Sta. Ana's response on Security 123
Table 34: Citizens of Barangay Barangay Sta. Ana's Average Mean 124
Citizens of Barangay Guitnang Bayan 1 and Barangay Sta. Ana 124
Table 35: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Reliability 124
Chart 24: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Reliability 124
Table 36: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Usability 125
Chart 25: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Usability 126
Table 37: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Understandability 126
Chart 26: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Understandability 127

Table 38: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Security 127
Chart 27: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Security 128
User-System Adaptability 128
Figure 36: Use of Access Efficiency Model to evaluate the U-S Adaptability in Adaptation Process for user login and consultation for patients 128
Figure 37: Use of Access Efficiency Model to evaluate the U-S Adaptability in Adaptation Process for access to medical records and profile update for patients 129
Figure 38: Use of Access Efficiency Model to evaluate the U-S Adaptability in Adaptation Process for access to individual records for Barangay Health Workers 130
Figure 39: Use of Access Efficiency Model to evaluate the U-S Adaptability in Adaptation Process for access to appointment records for Barangay Health Workers 131
Figure 40: Use of Access Efficiency Model to evaluate the U-S Adaptability in Adaptation Process for access to all records for Barangay Health Workers 131
Figure 41: Use of Access Efficiency Model to evaluate the U-S Adaptability in Adaptation Process for access to all records for Barangay Health Workers 132
Conclusion 133
Recommendations 134
Figure 42: Gantt Chart for the Implementation of the Proposed System 137
References: 137

Appendix A **143**
Field Service Health Information System (Sample) 144
Entity Relationship Diagrams 149
Gantt Charts for the Proposed System 158

Appendix B **159**
Supporting Documents for the Proposed System 161
Designation of Data Protection Officers Pursuant to Republic Act No. 10173, Titled, Data Privacy Act of 2012 166

Appendix C **166**
Consent Form 169
Privacy Impact Assessment 171

Appendix D **178**
Survey Form for Patients 179

Appendix E **183**
Product Information 184
System How to's 184

Appendix F **192**
Certificate of System Acceptance from Barangay Sta. Ana 193
Certificate of System Acceptance from Guitnang Bayan 1 194
Certificate of Presentation at the 7th International Researach Conference (ITRC7) 195

Certificate of Presentation at the 1st Asian Training Summit and Research Congress on Information Technology (ATSRCIT 2021) **196**

Appendix G **196**

Photos 196

Appendix H **202**

Documents Used for System Design 203

Initial System Page Designs 204

Appendix I **221**

The Proponent 221

Appendix J **222**

Other Documents 222

Abstract

Barangay Health Services has been providing assistance throughout its premise ever since the release of the Letter of Instruction No. 949,s. 1979 by President Ferdinand Marcos mandating the minister of health to develop and implement programs focusing on health development at the community level. This has caused barangay officials to collaborate and create a team of barangay health workers with nurses and midwives that will be able to provide such services for each individuals and families. Thus, in the ever-changing world where technology penetrates in all aspects of human interaction, it is essential to have an automated information system for barangay health services as well. Though the grounds for making such progress were not limited to what one community is using, it is ideal for one barangay health services clinic to work together with the others and be united at its core. This study proposes the development and implementation of an online health services system that will be able to serve its function for the barangays of San Mateo Rizal. It is meant to be user-system adaptive considering the process flows and reports mandated by the Municipal Health office and the Department of Health.

CHAPTER 1

BACKGROUND OF THE STUDY

As one of the settlements in the province of Rizal, San Mateo has grown to become one of the fast growing municipalities (DILG, 2021). It is composed of fifteen (15) barangays with a population of two hundred eighty-eight thousand seven hundred fifty nine (288,759) and a land area of four thousand eight hundred twenty five (4,825 has.) (Wikipedia)

Its barangay halls, which serve as the center of progress planning and implementation, provide different services to its people as mandated by law. It has the issuance of (1) barangay certification and clearance, (2) issuance of Barangay Clearance as a requirement for the Municipal Business Permit, (3) community tax certification, (4) management of barangay records, data and similar documents, (5) provision for the use of barangay facilities and properties, (6) health services and *katarungan* pambarangay or the Barangay Justice System, (7) environment services. Amongst its services, the barangay's health center is at par in need of further improvement.

The barangay health center is located within the premises of the barangay halls. Each health center houses six to ten personnel including the barangay health center nurse and midwife. They cater to twenty-five to thirty families in a day to administer the necessary medical attention per barangay. They are also in-charge of vaccination, case monitoring, and case analysis reports. The center also provides reports to the Municipality Hall where it then submits the data to the Department of Health.

With this information at hand, each barangay administrator believed that attention should be turned on to their Health Services Center most especially given the current pandemic setup. Thus, the development of the proposed Health Information System.

The proposed Barangay Health Services Information System is ideally meant to be adaptive for other barangays within the premise of San Mateo, Rizal as well given that the majority of the procedures are consistent and paralleled with one another.

This project is also supported by different government programs for health services. As per the different administrative projects of the Philippine government, an information system within a

local government unit is an indication of progress. This has been covered in the report for Philippines eHealth Strategic Framework and Plan for 2014-2020 whose strategic guiding principles revolves around (1) client-focus or person centered information, (2) collaboration between stakeholders, (3) user's involvement, (4) strategic approach in phases to achieve the eHealth vision so as to gain more focus, judiciously and efficiently use resources, (5) harmonization and independence and so on.

The proposed system also features the improvement of file management and information safe keeping for the patients. These ideals were anchored in Barangay Guitnang Bayan 1's mission (DOH, 2021) to be able to actively carry out the mandates and ensure transparency, honesty and efficiency in the delivery of services in barangay.

Furthermore, the implementation process of the proposed system was divided into two (2) plans. The first one is the implementation plan for the Health Services Information System which focuses on the technical edges of development. Second, the implementation plan for the Health Services Social System. This second plan will provide a room for user acceptance once the system has been deployed.

Purpose and Description

In 1999, the Department of Health underwent an organizational reform and launched Interlocal Health Zones (ILHZ) enjoining municipalities in optimizing resources and maximizing benefits from local health services. This motion is supported by the Republic Act No. 7160 of 1991 which states that local government units shall ensure and support, among other things, the preservation and enrichment of culture, promote health and safety, enhance the right of the people to a balanced ecology, encourage and support the development of appropriate and self-reliant scientific and technological capabilities and the like. In which, embarking LGUs to handle the general welfare of its jurisdiction to preserve the comfort and convenience. Thus, developing an infrastructure or spearheading the improvement of their existing processes is upon their purview.

Republic Act 7160 (GOVPH, 2021) then relies deeply on the local government unit's ability to undergo the necessary actions as its contribution to the national paradigm. For a municipality, paragraph 3 of section 17 covered health services which include the

implementation of programs and projects on primary health care, maternal and child care, and communicable and noncommunicable disease control services, access to secondary and tertiary health services, purchase of medicines, medical supplies, and equipment needed. For a Barangay, one of its basic services and facilities should include, but not limited to, health and social welfare services which consist of maintenance of the barangay health center and day-care center.

To further support the improvement of health services under local government units, each Barangay should be able to provide the needed data and documentation as required by the Department of Health through its Municipality. With this, the development and implementation of an Online Health Services System for Barangays will help improve the processes and transactions involving health services more specifically for San Mateo Rizal.

Statement of the Problem (Business Opportunity)

The statement of the problem is seen with a business opportunity as stated below to develop and implement the Health Services Information System of San Mateo, Rizal:

- Patient Health Records Accessibility and Security

  Citizens of San Mateo Rizal have their health records kept in a physical cabinet stored within the Barangay Health office's premise compiled per family that may be at risk of theft and effects of natural calamities such as typhoons and earthquakes in consideration to its geographical location. This provides an opportunity for online automation of patient health records providing ease of access as well as secured and safe keeping of records. This may also include health information security in terms of role-based access control, encrypted database features, multi-factor admin user authentication, patient consent, and adherence to privacy and security policies of the government.

- Streamlining Documentation and Reports Generation

  The existing system records its daily transactions to a physical logbook per family with an instance of multiple transaction codes for the same citizen. Automating the records and reports consolidation is an opportunity to avoid rework on filling up forms and generate reports aligned to the Department of Health's FSHIS (Field Service Health Information System *See Appendix A*).

- Efficiency in Health Care Delivery and Administration

  Health care delivery in the Barangay Health Office caters to mostly patients from birth to death.

- Prompt Data Availability for Decision Making in the Barangay Health Services Level

  Barangay Health Services reports to the Municipal Health Office and to the Department of Health for proper reports collection. These reports were transcribed for data analytics and stands as support for future actions and projects decision making. Having a process that can generate these documents in a faster period of time can provide an adequate response for the transition of information and delivery to the corresponding department.

- Provision of an adaptable Barangay Health Information System for the citizens of San Mateo Rizal

The Barangay Health Services procedures and forms could be unique per barangay as per the discretions, procedures and instructions of the Barangay Council. There were some forms that had labels and designs appropriate for its client's transactions such as child vaccination. On the other hand, the health office administrators submit reports in a form provided by the Department of Health known as the Philippine Field Health Services Information System (FHSIS, 2021). This form, once automated and is designed in consideration of the user's familiarity to FHSIS, may provide an opportunity to unify the process amongst Barangays despite the minor differences.

- Sufficiency and Efficiency of Barangay Health Workers

  According to the House Bill 01339 introduced by Hon. Luis Raymund Villafuerte, (Villafuerte, 2016) there should be one (1) nurse for every barangay with a population not exceeding 5000, one (1) midwife for every barangay with a population not exceeding to 2500 and one (1) barangay health worker for every barangay with a population not exceeding 500. This house bill appeared to be at par from the reality since the population of different barangays has gone beyond the given numbers but still, the quantity of nurses, midwives and barangay health workers are the same.

- Increase IT Literacy of Barangay Health Workers

  The Barangay Health Services team is composed of the resident nurse, the midwife and support that facilitates medicine inventory, patient assistance and document controllers. All of these health service providers may have an opportunity to be more efficient in the use of technology for automated processes involving their line of duties and responsibilities to serve the citizens of their barangay within the municipality of San Mateo, province of Rizal.

The Existing System

Barangay Health Services in San Mateo Rizal were coordinated through the Municipal Health Office. The assigned barangay health workers were trained to assist throughout daily transactions as per the guidance of nurse/s and midwives administering proper medical services to the citizens of each barangay. As per the Field Health Services Information System (FHSIS, 2021) of the Department of Health, a barangay health center should be able to provide family planning, maternity checkups, child vaccination, dental checkup (if logistically possible), infectious disease prevention and control services, non-communicable disease prevention and control services, and environmental health and sanitation services for every member of its barangay whether it is categorized as the poorest of the poor, middle or upper class.

The operation of barangay health services also includes monthly visitation to localities and residences covered by its territory to survey citizens with disease, disabilities as well as to identify living members of the community.

The barangay health officers identify a citizen's record based on the name of the head of each family, log them in a bounded logbook and proceed with the needed services. A different officer takes charge of medical inventory including medicines for tuberculosis, hypertension, non-communicable diseases and the like. Also, the assigned days for vaccination/immunization for children and pregnant women takes one day in a week's service. Afterwards, a report required by the Department of Health will be sent to the Municipal Health Office for data collection, documentation and decision making.

Each process offered by the barangay health services corresponds to a form where each of its design and format differs from each barangay. Thus, they adhere to only one final report. A unified form required by the Department of Health which also corresponds to its policies and guidelines.

Conceptual Model of the Study

Presented below is the conceptual model of the existing system for barangay health services in San Mateo, Rizal.

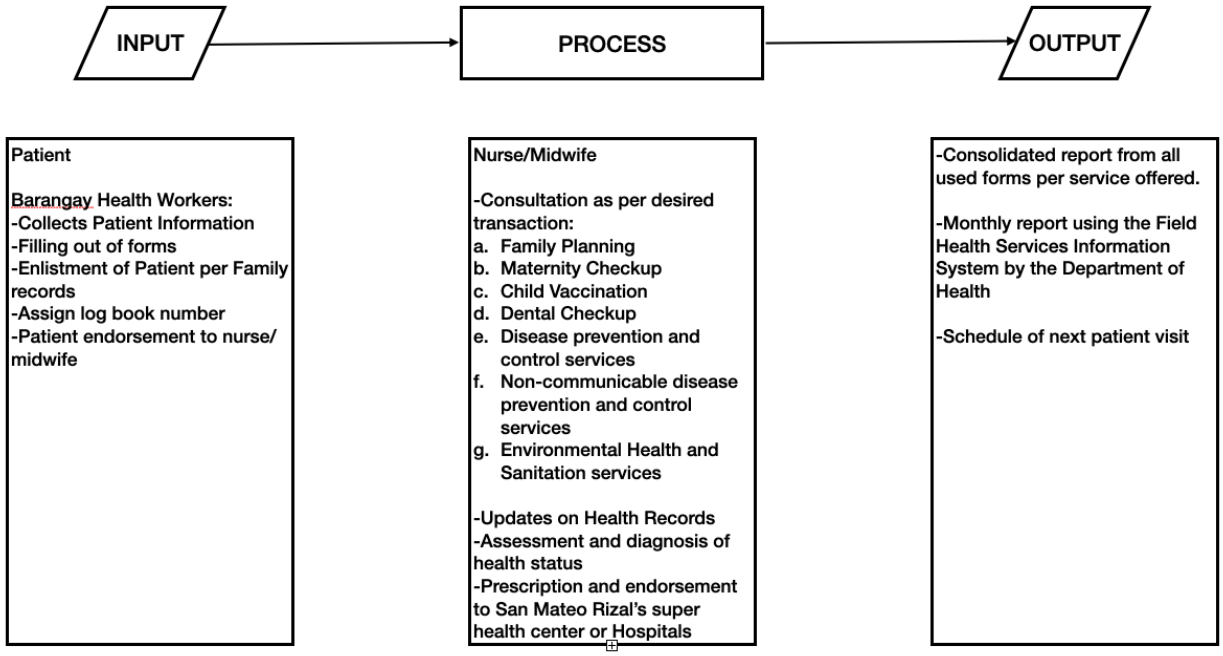


Figure 1: Process flow of the Existing System

Figure 1 represents the flow of community-level data. Where at first, the barangay health worker collects new patient's information through barangay-specific designed forms with basic information such as name, birthday, age and complete address. A file number will be assigned for the patient based on the numerical order on the health center's logbook. Each document will be filed in an envelope tagged based on the head of the family's name (i.e. Father's name). Second, the patient will choose amongst the services offered by the health office and proceed with the process either for the nurse or midwife.

Documents will then be consolidated once every month such as the field health services information system's report as per the Department of Health [*See Appendix A*].

Proposed System

Presented below is the conceptual model of the study. The model outlines the process flow of the proposed system:

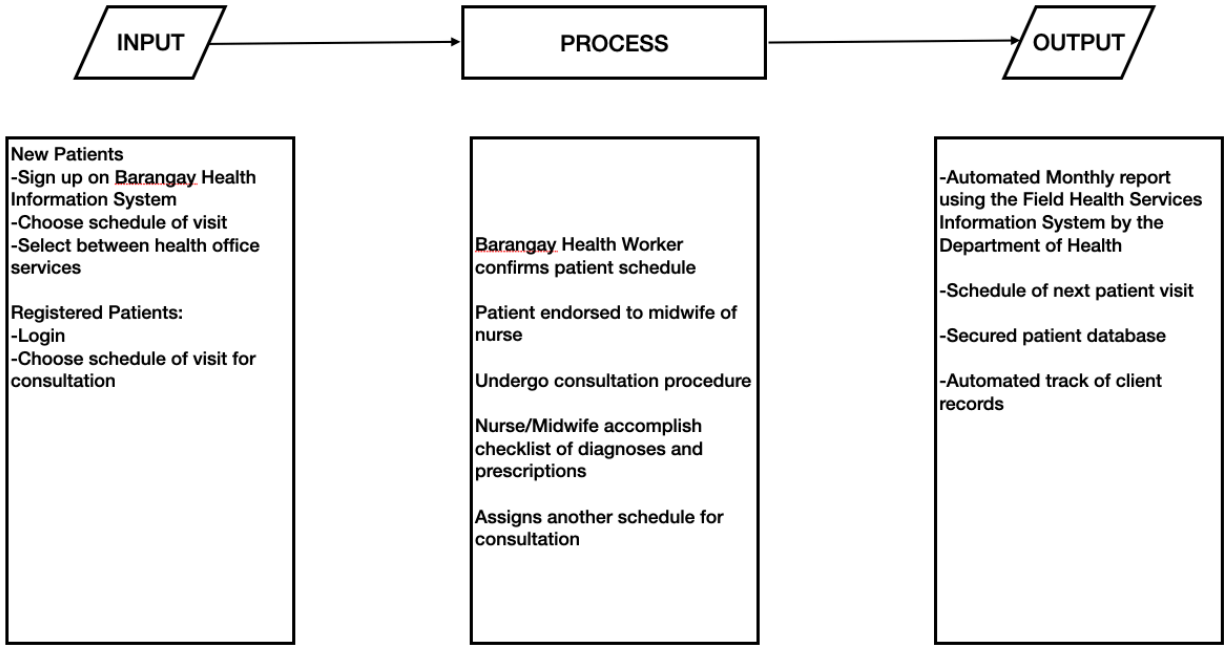


Figure 2: Process flow of the Proposed System

General Objective:

To be able to Implement and Develop an Adaptable On-line Health Information System for the Barangays of San Mateo Rizal.

Specific Objectives:

To develop an Adaptable On-line Health Services System for the Barangays of San Mateo Rizal to improve patient monitoring and profiling for maternal, neonatal, TB treatment compliance, and other health services provided by the center. As well as collecting data and disseminating health information to policy makers, providers and citizens in an organized and secured manner. The detailed objective of the system are as follows:

- To provide an adaptable health information system to the Barangay Health Centers of San Mateo Rizal that will help provide uniformed documentations, report generation and patient records that correspond to the requirement of the Department of Health in subordination by the Municipal Health Office.

- To lessen time consumed and rework in the process of patient profiling, enlistment and patient record updates. *Patient Profiling and enlistment* involves capturing of the patient's basic information such as name, contact details, address, occupation, birthdate, monthly income, number of living children (if applicable), name of spouse and spouse details.

- To have safe, secure and easy access to patient records and information. Allowing barangay health officers to access information through an online system with less risk on possible effects of calamities such as flood and earthquake in consideration to the territories covered by the Municipality of San Mateo, Rizal.

- To generate an online comparative report of diseases within the preferred time as well as to store operational data, communication data submitted to the Municipal Health Office, and documents.

- To allow patient consultation on-line scheduling. Citizens of San Mateo Rizal do often visit the health centers with concerns directed to maternal and child check ups/vaccination, tuberculosis, hypertension, infectious disease and other non-communicable diseases. These visits were all based on the dosage required for patients for a vaccine or for monitoring. On-line scheduling will provide

patients an overview of the barangay health office's schedule and may be catered to based on their preferred date and time.

- To provide an online communication line between the patients and the barangay health officer through instant messaging for any related inquiries.

- To have role-based access for system users. Providing different privileges with stronger security features for the health information system. Patients will be provided with login and password credentials in reference to the existing list of barangay and barangay citizens. Nurses, midwives and other barangay health officers will also have different levels of access for the system.

CHAPTER 2

REVIEW OF RELATED LITERATURE

The review of related literature for this study discusses the guidelines and policies followed in the implementation of Primary Health Care administered in Barangay Health Services office in a Philippine context and is narrowed down to Local Government Units. Also, the literature supporting this study's implementation such as the research methodology.

Letter of Instruction No. 949, s. 1979

In 1979, the late President Ferdinand E. Marcos mandated the then Minister of Health, Hon. Enrique M. Garcia and its officials to design, develop and implement programs that will focus on health development at the community level particular to rural areas; effectively utilize these systems in order to control or eradicate the immediate and specific health problems confronting Filipino communities; and redirect and re-orient the organization of all its agencies at the central level so as to provide efficient and adequate administrative, planning and other support services for health development by (1) strengthening staff functions and capabilities; (2) streamlining programs of activities (DILG, 2021).

This mandate also covers the aspect of strengthening the primary health care system in coordination/cooperation with all the ministries and agencies engaged in social, political and economic development.

Primary Health Care (PHC) in the Philippines

According to the article published by Vera (2013), primary health care is defined as an essential care made universally acceptable to individuals and families in the community by means acceptable to them through their full participation and at a cost that the community and country can afford at every stage of development. In a Philippine setup, there are two types of

primary health care workers, the Barangay Health Worker or Village Health Worker and the Intermediate level Primary Worker.

A primary health care program is also composed of eight (8) elements such as: (1) education for health, (2) locally endemic disease control (3) expanded program on immunization (4) maternal and child health and family planning (5) environmental sanitation and promotion of safe water supply and (6) nutrition and promotion of adequate food supply (7) treatment of communicable diseases and common illness and (8) supply of essential drugs.

Also, Primary Health Care should be run by the following principles. First, the 4A's (accessibility, availability, affordability and acceptability), it focuses on the available resources within the community, health promotion and prevention of illness. Second, community participation should be considered as the heart and soul of primary health care. Third, people are the center, object and subject of development. This principle takes on three concepts such that any undertaking must also be based on the people's needs and problems, partnership between the community and the agencies found in the community and social mobilization and decentralization, and that health work should start from where the people are and building on what they have where challenges like *Scheduling* of Barangay Health Workers in the health center. It is also discussed that there are some barriers considered in community involvement for primary health care provision such as lack of motivation, attitude, resistance to change, dependence on the part of community people and lack of managerial skills.

Self reliance is also one of primary health care's principles. It discusses that through community participation and cohesiveness of people's organization, they can generate support for health care.

Community Health Systems Catalog Country Profile: Philippines

As discussed in this literature, the Philippines has a multitude of strategies, guidelines, manuals and governmental decrees that cover a wide array of health areas through both vertical and comprehensive programming even far back to 1970. Though the Department of Health was able to issue policies and guidelines, mainly for LGU - or in provinces, municipalities and cities - and in barangays, or villages, they may adapt and implement policies and guidelines very differently, (Devlin, 2016).

The community health policies were last updated throughout the *Aquino Health Agenda: Achieving Universal Health Care for All Filipinos* dated 2010 where there are 4 main cadres for the number of community health providers such as Barangay Health Workers (BHWs), Barangay Nutrition Scholars (BNS), Barangay Service Point Officers (BSPOs) and Community Health Team (CHT) members.

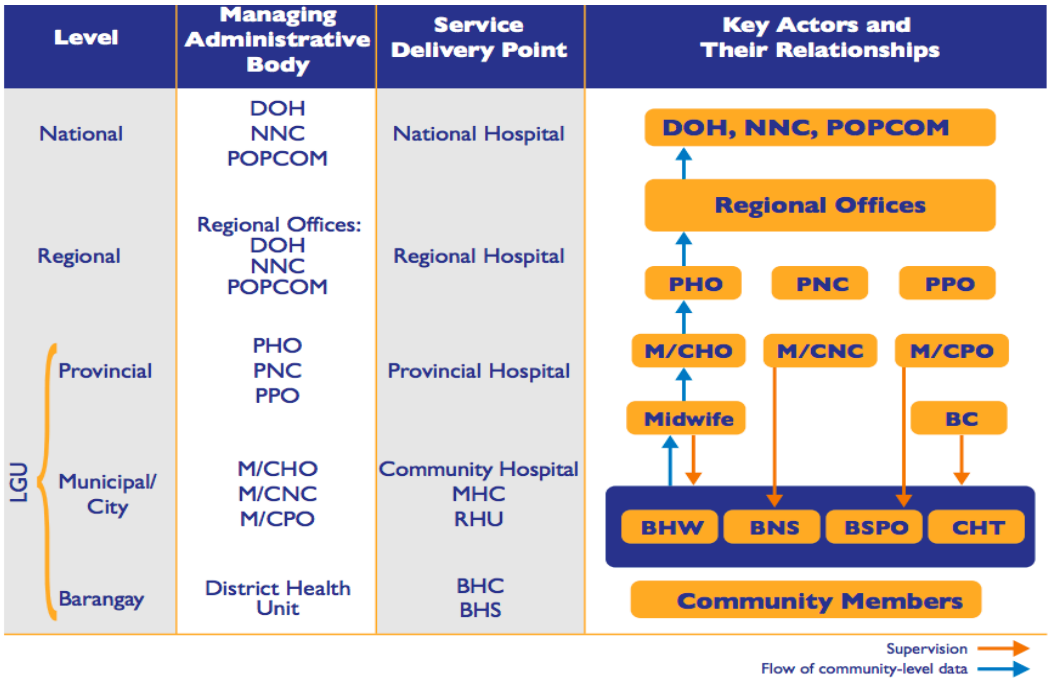


Figure 3. Philippine Health System Structure (Source: caro.doh.gov.ph)

For training, the national health policies have not mentioned any specifications for this aspect. The authors of this reference stated that this could have been left somewhat broad to allow LGUs to adapt these policies to local context, given varying priorities and available resources.

For leadership and governance, under the Barangay level, it is managed and coordinated by the barangay captain. Whereas, the elected leader of the barangay, conducts administrative oversight of the community health providers and supports them alongside a barangay council comprising 6-8 councilors who are the community's key decision makers. It is also stated that the barangay council may provide guidance and support for community health providers, such as honoraria for their work.

Philippines eHealth Strategic Framework and Plan 2014-2020

As reported last April 15, 2014, the Department of Health (DOH) has recognized the use of ICTs in support and development of the functions of the department. It is reported that information systems and data sources are being integrated or harmonized to eventually address other challenges like establishment of the DOH data warehouse, quality database and establishment of a more responsive information system and access to and sharing of knowledge products.

It has been mentioned as well that pilot implementation of telemedicine to underserved areas and population through the National Telehealth Center, National Institute for Health and the University of the Philippines, Manila.

The following opportunities were also identified through this endeavor such as (a) government and multi-sectoral engagement, policy and regulation environment, strategic planning/financing, Human Capital Development, Systems and Data Interoperability, Infrastructure, and Information Use.

The eHealth Strategic Guiding Principles.

The Philippines eHealth development guiding principles were laid down in nine items such as (1) client-focus or person centered information (2) collaboration and / or partnerships with different stakeholders (3) user's involvement (4) strategic approach in phases to achieve the eHealth vision so as to gain more focus, judiciously and efficiently use resources (5) harmonization and interdependence to guide alignment of eHealth activities at the national level without controlling health care providers to implement local eHealth solutions (6) recognize the presence of entities that have already started eHealth so as not to constraint their continuing advancement and gain their support (7) availability of human resource to implement the eHealth agenda in the country and promote transparency and public accountability (8) compliance to laws and regulations (9) optimize use of resources so as not to duplicate time, effort and investments

Also the eHealth's Vision is that by 2020 (eHealth, 2021), eHealth will enable widespread access to health care services, health information, and securely share and exchange patient's information in support to a safer, quality health care, more equitable and responsive health system for all the Filipino people by transforming the way information is used to plan, manage, deliver and monitor health services with the illustration below:

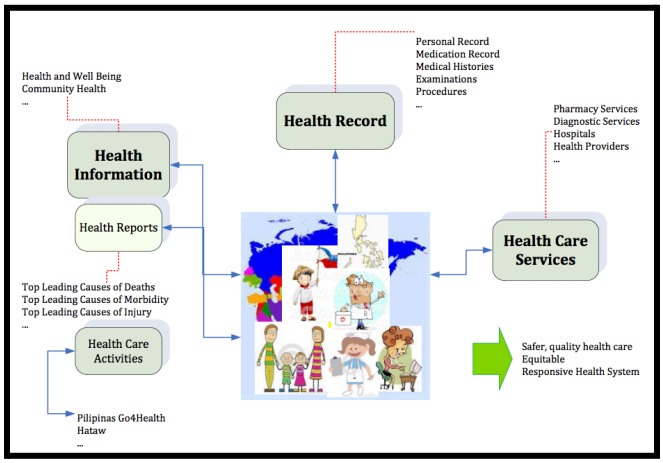


Figure 4. eHealth's Vision (Source: ehealth.doh.gov.ph)

The Desired eHealth Outcomes. This section in the published article for Philippines eHealth Strategic Framework and Plan for 2014-2020 also discussed the desired outcomes should the proposal become operational. The main context focuses on the both the Health System Goal or Challenge and the Desired eHealth Outcomes whereas:

Health System Goal or Challenge for Health Consumers : Safer and Quality Health Care.

This discusses the increased ability to access, control and share health information, minimization of time and effort in providing the same health information to different health care providers. Also, the minimization of health inequalities of those living in remote or rural areas due to poor access to health care and to address shortage of health human resources affecting those in remote or rural areas.

Desired eHealth Outcomes. The desired eHealth outcomes for the aforementioned goals or challenges were the improved access to health information and maintenance of the personal health record, with controlled access to personal health information, and improved management of health care plans. Then, improved access to primary care services for rural and remote locations that reduces travel time and access; improved access to knowledge, services and resources to assist in managing one's health with support for early detection and treatment of diseases, and better management of health conditions and adherence to medication and treatment regimes. (Kashefi, 2015)

The second part as discussed on the Desired eHealth Outcomes is for Health Care Providers: Make more informed decisions where the goals and challenges are (a) improve ability to make informed decisions at the point of care (b) minimize time and effort in performing same or duplicate treatment services or performing unnecessary ones (c) improve the ability to issue orders, prescribe medications and refer individuals to other health care providers and (d)

improve ability to monitor the effectiveness of health care services rendered like treatments and diagnosis, and outcomes.

> It is mentioned that the expected outcomes for these goals or challenges are the improved access to an integrated/single view of the patient's health information at the point of care including an improved sharing and exchanging of health information across different geographical locations and all parts of the health sector. Also, the improved access to systems and health information like clinical decision support tools, medications, clinical knowledge, skills development and others, improved access to doctor's ordering system, medicine prescription and referrals and improved collaboration and coordination among health care providers, and interaction with health consumers.

The third part is for Healthcare Managers, Policy Makers and Researchers: Effective program/research development, implementation and monitoring where the goal and challenges are (a) to maximize or eliminate increasing incidence or spread of diseases in the country like HIV/AIDS, chronic diseases, and others (b) meet increasing demands or expectations brought about by new/emerging technologies (c) improve ability to make informed decisions like developing and implementing program interventions to address health issues and concerns, conducting surveillance activities, improving clinical practices and treatments, and monitoring and (d) minimize time and effort in collecting, consolidating and integrating information from different sources.

To measure the success of the Philippines National eHealth strategy, the following items are identified for each category initially discussed.

eHealth Key Result Areas

Health Consumers: Safer and quality health care

- There is a national health identified that is unique for each person.
- There is access to quality and reliable health information.
- Health consumers have access to individual electronic health records.
- Health consumers can access and manage their personal health records.
- Health consumers can interact with their health care providers.
- Health consumers' health information can be shared and exchanged across the health sector in compliance to established standards, rules and protocols.
- Health consumers in remote areas can use electronic consultations or Telehealth capabilities.
- Health consumers can experience high speed broadband connectivity.
- There are online health reporting systems to generate quality information as support to decision making and health care delivery.

Health Care Providers: Make more informed decisions

- There is an identification that is unique for each health care provider.
- There is access to quality and reliable health information.
- Health providers have access to an integrated/single view of the patient's health information at the point of care.
- Health providers have access to health information like clinical decision support tools, medications, clinical knowledge, skills development and other health care providers, and interact with health consumers.
- Health providers can securely share and exchange health information across different geographical locations and all arts of the health sector, as governed by information sharing and exchange protocols.

- Health providers can provide electronic consultations or Telehealth services.
- There are online test/examination ordering systems and results reporting, referrals and prescriptions.

Health Care Managers, Policy Makers and Researchers: Effective program/research development, implementation and monitoring

- There is access to quality and reliable health information for improved analysis, utilization, decision making and research purposes.
- Health care managers have available tools to support the monitoring of health system activities and outcomes.
- There are defined health data standards and processes for eHealth solutions' development and compliance.

Barangay Health Center

According to the House Bill No. 1339 introduced by Hon. Luis Ray F. Villafuerte, a Barangay Health Center (BHC) shall be one Barangay Health Center in every barangay unit, which functions to (1) deliver basic and primary care services for the community to include, but not limited to, immunization, prenatal check up of well and ill infants, (2) coordinate efforts and actions of the different members of health team, (3) keeps records of health indices, community profile, and other data, (4) serves as an area of congregation for meetings of the health team, (5) serves as training of future and present members of the health team and (6) assists and coordinates with the respective LGUS's and appropriate offices on first aid and procedures in disaster risk reduction.

For barangay health team placement, the barangay health center shall be composed of a nurse, a midwife, and a barangay health worker. Each BHC shall be manned by a health team proportionate with the existing population in the barangay where, (a) there should at least be one (1) nurse for every barangay with a population not exceeding five thousand (5000) (b) one midwife for every barangay with a population not exceeding two thousand five hundred (2500) (c) at least one barangay health worker for every barangay with a population not exceeding five hundred.

Development and Implementation of Health Information Systems

According to Lippeveld et. al (2000), in their book Development and Implementation of Health Information Systems, under the subtitle of 'What is wrong with the current health information systems?', information systems tend to be data-driven rather than action-driven which then resulted in inadequate provision of the needed management support. It is where most healthcare providers in developing countries equate information systems with filling endless registers with names and addresses of patients, compiling information on diseases on certain periods of time without adequate feedback. As expressed as well, the data received are often not congruent with the ideals of decision making due to incomplete, inaccurate, untimely, obsolete and unrelated to priority tasks and functions of the local health personnel. Which then leads to an assumption that health information systems are management obstacles rather than tools.

The referenced publication also itemized the reasons for the aforementioned statements into five points such as (1) irrelevance of the information gathered; (2) poor quality of data, this is without considering the technical skills of the health workers collecting the data; (3) duplication and waste among parallel health information systems, in reference to existing ones; (4) lack of timely reporting and feedback and (5) poor use of information.

On the subtopic, *Developing a 'systems approach' for health information systems,* the author mentioned that each country has to develop or restructure its own specific system, tailored to the prevailing socioeconomic, political and administrative context. Furthermore, it has also been highlighted that despite the differences between health information systems, it is essential to have a minimum criteria such as information generating process whereby data are transformed into information; and a more or less organized structure where persons interact with resources, such as data collection instruments, or with machines such as computers.

Lippeveld et.al itemized the structure of a health information system as composed of two sets of interrelated components grouped under two entities as the information process and the health information system management structure. As seen on figure 4, raw data are transformed into information in a usable form for management decision making. The breakdown of the process starts with data collection, then data transmission, data processing, data analysis and presentation of information for use in planning and managing the health services. This also covers the health information system resources and the set of organizational rules.

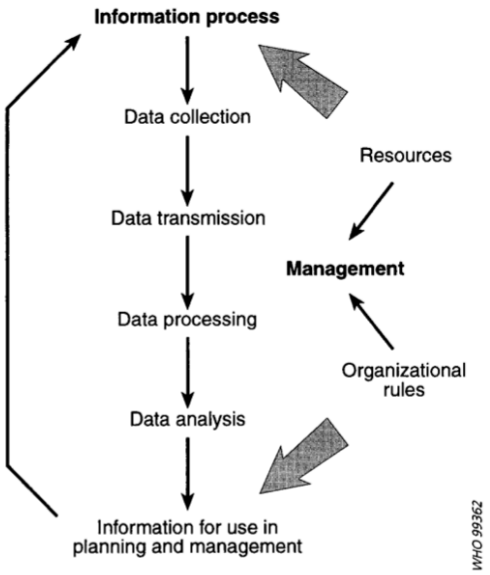


Figure 5: Components of a Health Information System by WHO as referred from the Development and Implementation of Health Information Systems (Source: researchgate.net/publication/242520887)

As discussed in the reference, services and resources should be as available as possible to the periphery. In this context, periphery refers to the different management functions, health services provision and resource availability. Though, limitations should still be applied. As per the decentralization related to the provision of technical competence (technical limit) or to the efficient use of equipment (economic limit) or to the distribution of power (administrative limit). To further illustrate the model of health services, Lippeveld et al (2000) referred to the World Health Organization's representation of this key structure as seen on Figure 5 and further explained the concentration levels such as primary, secondary and tertiary level.

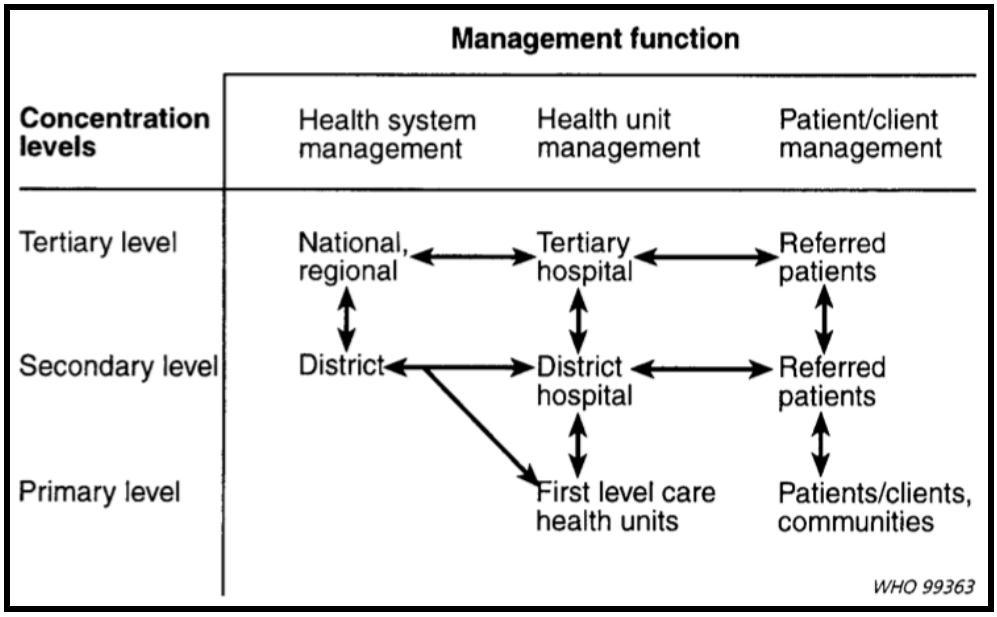


Figure 6. Components of a Health Information System by WHO as referred from the Development and Implementation of Health Information Systems (Source: researchgate.net/publication/242520887)

As referred from Figure 6. The concentration level for barangays is falling on primary level. The *Patient/client management functions,* in the context of user information, is the care provider - the doctor, the health auxiliary, the midwife as well as the community health worker or the traditional birth attendant. According to Lippeveld, et.al (2000), a well-designed health information system can be a major tool in improving the quality of care delivered by care providers, by generating the information they need to make appropriate decisions such as: date, findings, and treatment prescribed during the last visit that will help the care provider to make better decisions for tuberculosis patient visiting a rural health center and the like (continuity of care).

For the *Health Unit Management Functions,* the objective of the health unit was defined as to provide health care to a defined population in the assigned area. It can also be classified to the first level of concentration and referral-level care units. Also, it is noted that health centers are

supposed to provide treatment to tuberculosis patients and provide prenatal care to all pregnant women in the scope of their area.

Related Studies

System Development Life Cycle

The process of developing the proposed Health Services Information System for the Barangays of San Mateo Rizal called for simultaneous consultation, stakeholder visits and feedback from the main users of the system. Thus, the study entitled Pampanga's Barangay Health Information System (PBHIS): A Decision Support & Health Information System for Rural Health Unit 1 (Estinar, et.al. 2018) was cited. This study discusses the different challenges related to health services in relation to timeliness and clinical knowledge. The researchers used the Agile-Waterfall Hybrid methodology.

On the other hand, the use of the Agile method for system development allows an iterative process of development and delivery. This also covers sprint planning where at the start of each development cycle or sprint, tasks are scheduled to be implemented and delivered. Then, a sprint review, where clients review the result and provide feedback. Agile is also defined as a systems development approach that includes values, principles, and practices useful for flexible, interactive and participative approach (Kendall, 2010).

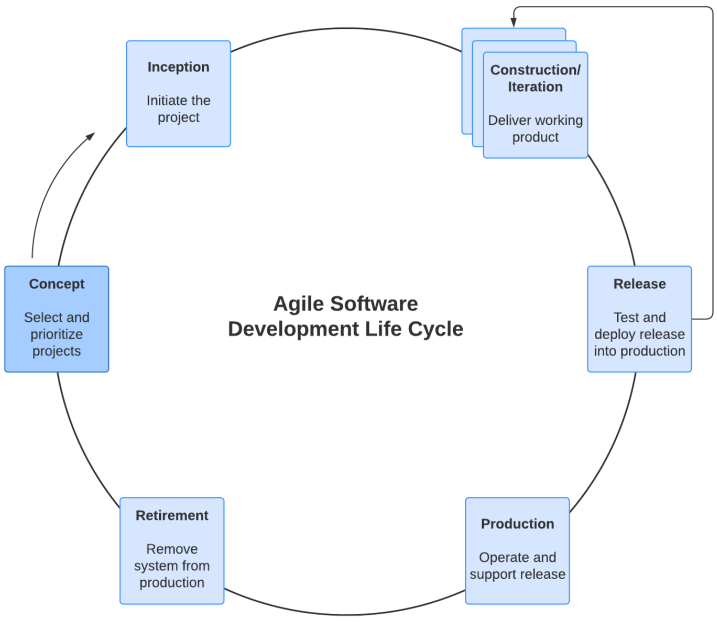


Figure 7: Agile Software Development Life Cycle Model (Source: Grefiel (et.al) 2018)

Since the development of the proposed system started during the period of the COVID-19 pandemic, a flexible approach should be established in consideration as well with the variables encompassing the Health Services of the barangays in San Mateo Rizal. The Agile-Waterfall model helped in the establishment of agreement between parties from Barangay Guitnang Bayan 1, Barangay Sta. Ana and the proponent when it comes to the overview of the system and how it will be implemented together with constant consultation for the transactions or process should there be clarifications and feedback needed as the system develops. This systems development life cycle method also answers to the context of the proposed system as it touches the considerations for the equal importance of both hardware and software components and the expected output of this project which is a software with both backend and frontend (Moksony, 2021).

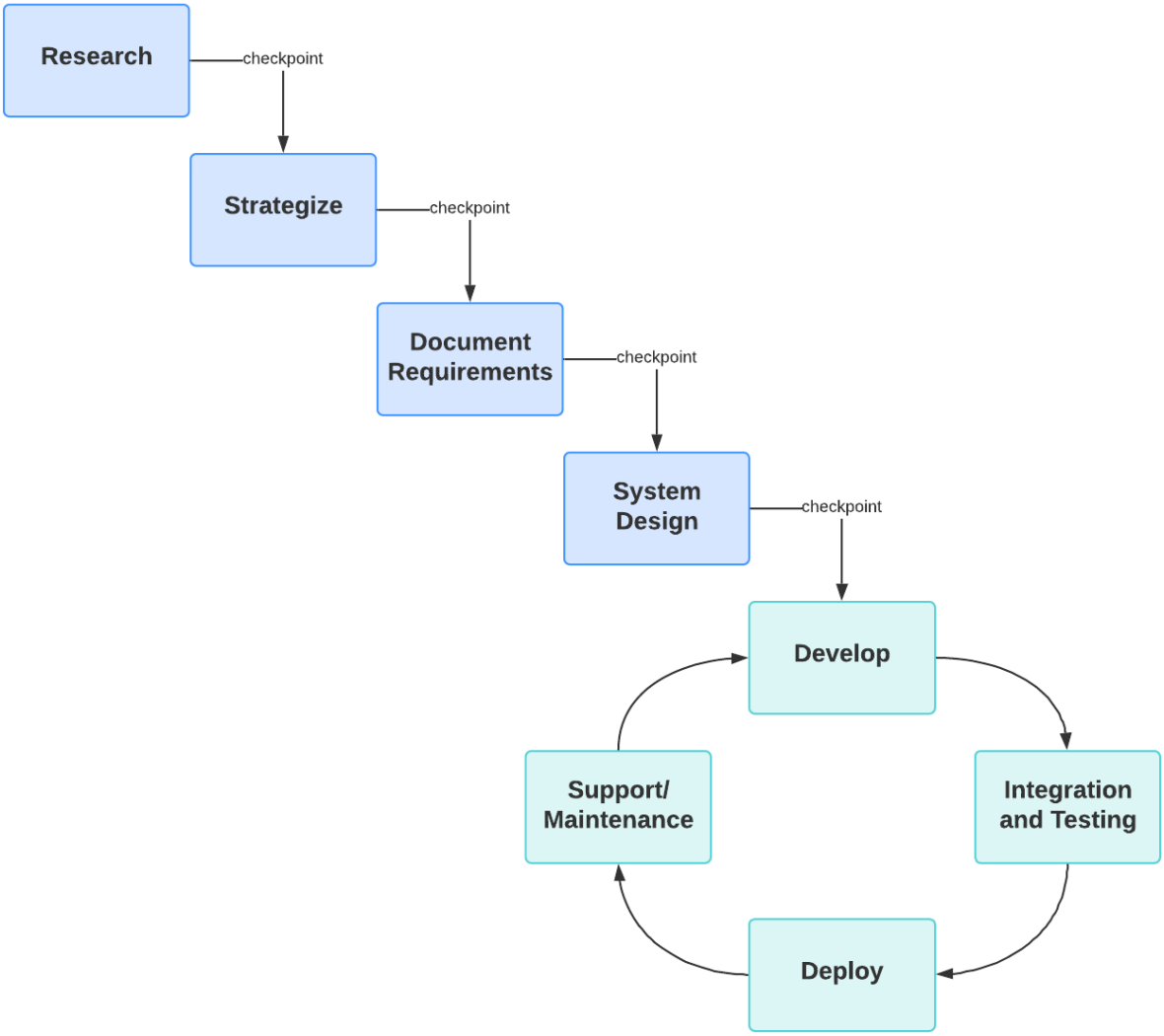


Figure 8: Agile-Waterfall Systems Development Life Cycle Model (Source: Grefiel (et.al) 2018)

According to the referenced presentation of Erick Bergmann and Andy Hamilton (Bergmann,et.al. 2013) by Lucidchart blogs (Lucidchart, 2021), the Agile-Waterfall best fits a system development for reasons such as: to encourage collaboration, the organization is transitioning from waterfall to agile, and the project has set budget, and delivery date but would benefit from Agile's fast design, analysis and planning.

Large-Scale Health Information System Project Planning, Implementation and Evaluation

A literature review covering all aspects of developing and implementation of Health Information systems discussed that implementation is defined as the process of planning, testing, adaptong, and integrating HIS so that the technology becomes routinely used in the organization. The researchers also defined HIS as a software that assists healthcare organizations to gather, process and disseminate information within the organization and their environment. It was also mentioned that HIS incorporates a range of different types of systems, which include patient information systems, administrative systems, radiology and pharmacy information systems, telemedicine and hospital information systems, such as computerized physician entry systems.

The rationale of HIS as discussed in this literature are (1) healthcare organizations are increasingly required to address the growing costs of delivering healthcare to aging populations without compromising quality of care, access, or equity and are required to integrate new scientific evidence into practice. This will also benefit patients in greater engagement with their own care, monitoring of chronic illness, disease prevention and increased efficiency in hospital settings. HIS makes home-based care for patients in remote areas or with chronic conditions a possibility.

As per costs, policy makers and planners need to be aware that investment in telehealthcare will not inevitably yield clinical or economic benefits.

Initial Investigation and Analysis

Barangay Management Information System (BMIS) for Cities and Municipalities in the Philippines

According to the proponents of this study (Imus, 2019), Barangay Management Information System plays a big part to make work efficient. As such, part of the development is the consideration of policies being practiced in the barangay and city hall for execution. This

also uses the modified waterfall model which is, based on the study, uses the same phases as the pure waterfall model.

The study's implementation and testing phase were done after the complete understanding of the system requirements and specifications. While, the deployment phase is planned to have the product available to clients as after beta testing.

This study also compares different system development life cycles such as presented on the table below:

Criteria	Modified Waterfall Model	Iterative Model	Spiral Model	V-Model	Rapid Application Development (RAD) Model
Speeding access to the applications and service	1	3	2	2	3
Visibility of the stakeholders	3	3	1	2	3

Figure 9: SDLC Comparison for Barangay Management Information System (BMIS) for Cities and Municipalities in the Philippines (Source: (Imus, 2018, 2)*)*

Pampanga's Barangay Health Information System (PBHIS):A Decision Support & Health Information System for Rural Health Unit 1

The proponents of this study presented another functionality of a health information system in the context of barangays. It is for decision support. This also represented another system development life cycle which is the Agile-Waterfall hybrid methodology. As presented, (Estinar et al., 2018, 2) the hybrid methodology was chosen as a best fit for the study considering the municipalities dynamic collaboration with different stakeholders and government agencies.

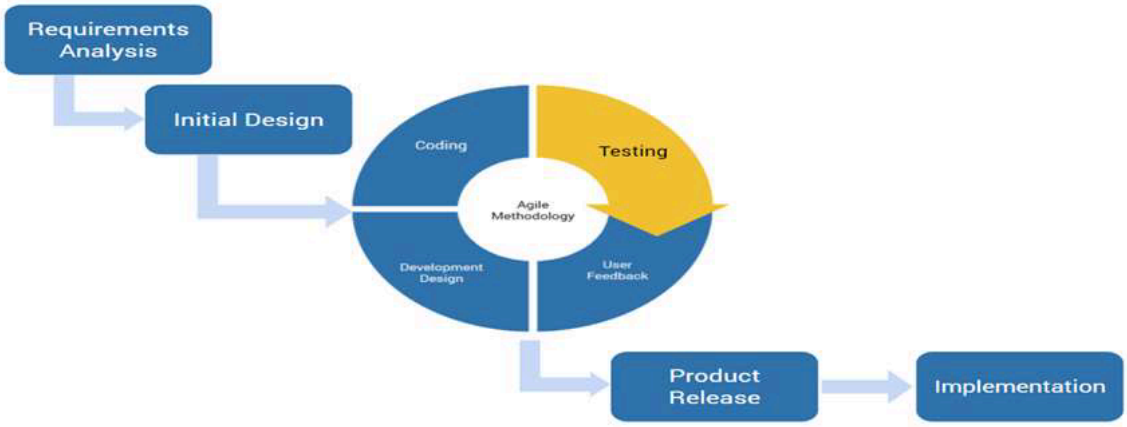


Figure 10: Agile Waterfall Hybrid Methodology for Pampanga's Barangay Health Information System (PBHIS): A Decision Support & Health Information System for Rural Health Unit 1 (Source: (Estinar et al., 2018, 2)

Data Instrumentation

Quantitative Data Collection

To represent both definitive and objective data, researchers use the quantitative method for data collection (QuestionPro, 2021). This method requires computational, statistical, and mathematical tools to derive results from the collected quantitative data.

Figure 11: Five Methods Used for Quantitative Data Collection Method (Source: questionpro.com)

Probability sampling utilizes forms of random selection getting data from the representatives of a population. Second, interviews. Interviews under the quantitative method require researchers to ask standard questions. Third, surveys allow researchers to quantify the responses through checklists or rating scale type questionnaires. Fourth, observation requires researchers to have naturalistic keen observation skills and senses for getting numerical data. Fifth, document review that is used in the existing system. This process makes an effective and efficient way of using records to validate data.

According to the Barangay Management Information System for Cities and Municipalities in the Philippines (Imus, et.al. 2018), different data gathering methods effective on the premise of the study are *Observation, Interview, Internet Research, Library Research* and *Survey Questionnaires.* This has relatively supported the documentation and provision for related aspects needed to evaluate the paper's purpose and usability to its clients.

Statistical Treatment

Applications of statistical methods to a data set to transform it from a group of meaningless numbers into meaningful output. Also, statistical treatment allows researchers to organize and process data.

There are different statistical treatments considered for different research structures. In this context, the following are as follows:

Sampling Method

According to statisticshowto.com (2021), there are two main types of sampling: Probability Sampling and non-probability sampling. The first one uses randomization to select sample members and the second one uses non-random techniques such as judgment of the researcher.

Probability sampling has various methods such as simple random sampling, stratified random sampling, systematic sampling, cluster random sampling. On the other hand, non-probability (Aapor, 2021) sampling (Baker R. 2013) has various methods as well such as convenience sampling, haphazard sampling, purposive sampling, expert sampling, heterogeneity, modal instance sampling, quota sampling, and snowball sampling.

Sample Size Determination

Market Research states that there are different considerations in getting the sample size of population for a survey. It initially defines the sample as a selection of respondents chosen in such a way that they represent the total population as best as possible (Dessel, 2013). Thus considering the two measures for accurate data. First, margin of error (confidence interval) and second, confidence level. Margin of error is the positive and negative deviation allowed for survey reports while confidence level represents how often the percentage of population actually lies within the boundaries of the margin of error.

	Confidence level = 95%			**Confidence level = 99%**		
	Margin of error			Margin of error		
Population size	5%	2,5%	1%	5%	2,5%	1%
100	80	94	99	87	96	99
500	217	377	475	285	421	485
1.000	278	606	906	399	727	943
10.000	370	1.332	4.899	622	2.098	6.239
100.000	383	1.513	8.762	659	2.585	14.227
500.000	384	1.532	9.423	663	2.640	16.055
1.000.000	384	1.534	9.512	663	2.647	16.317

Figure 12: How to calculate the number of completions. (Source: checkmarket.com)

Response rate for surveys also provides important information as it represents the ratio of respondents that fill in the questionnaire. Conventionally, for online surveys, a response rate of 20% is already considered good.

Likert Scale

According to Dr. Saul McLeod (McLeod, 2019) there are various kinds of rating scales developed to measure attitudes directly though the most used is Likert Scale. To define, Likert scale is a five (or seven) point scale which is used to allow the individual to express how much they agree or disagree with a particular statement.

A Likert scale assumes that the strength/intensity of an attitude is linear, for example, on a continuum from strongly agree to strongly disagree, and makes the assumption that attitudes can be measured.

Theory Acceptance Model

TAM (Theory Acceptance Model) as proposed by Fred Davis for his doctoral thesis in 1985 (Laderer A., 2000), proposed that 'system use is a response that can be explained or predicted by user motivation, which in turn, is directly influenced by an external stimulus consisting of the actual system's features and capabilities (TAM, 2021).

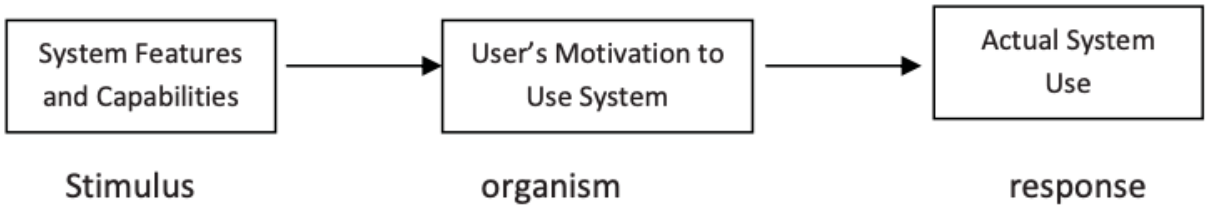


Figure 13: Conceptual Model for Theory Acceptance (Source: Laderer A., 2000)

Davis also considered three factors such as Perceived Ease of Use, Perceived Usefulness, and Attitude Toward Using the system for user's motivation. Thus, identifying these

considerations as a dominant variable for users to determine whether the technology is to be used or not.

To define, perceived usefulness is the degree to which an individual believes that using a particular system would enhance his or her job performance. While perceived ease of use is the degree to which an individual believes that using a particular system would be free of physical and mental effort.

U-S Adaptability

In the year 2016, Bingcong Zeng and Benjamin P. -C. Yen (Zeng, 2016) evaluated user-system adaptability in the use of web based information systems through the model access efficiency. The proponents cited three common approaches for adaptation techniques such as: page splitting, content summary, and transcoding. *Page Splitting*'s main idea is the division of original web pages into several sub-pages in order to fit the user's specific requirements. *Content Summary* on the other hand allowed users to browse the summary of web pages before deciding to visit any page with full details. Lastly, *Transcoding* which is known as the approach in tailoring web content in accordance with the capabilities of the client device and the network connection.

The Access Efficiency model considered two factors: user and system side. For the system, it looked into the ease of being accessed for web content and structure also known as "accessibility" which is in connection with the ability of a user to navigate through the system/website easily and comfortably through inlinks. Thus, incorporating the attractiveness of the inlink and the level of its source page. During the process of adaptation, activities were evaluated and web structure as well as user interfaces are being changed in order to improve user experience.

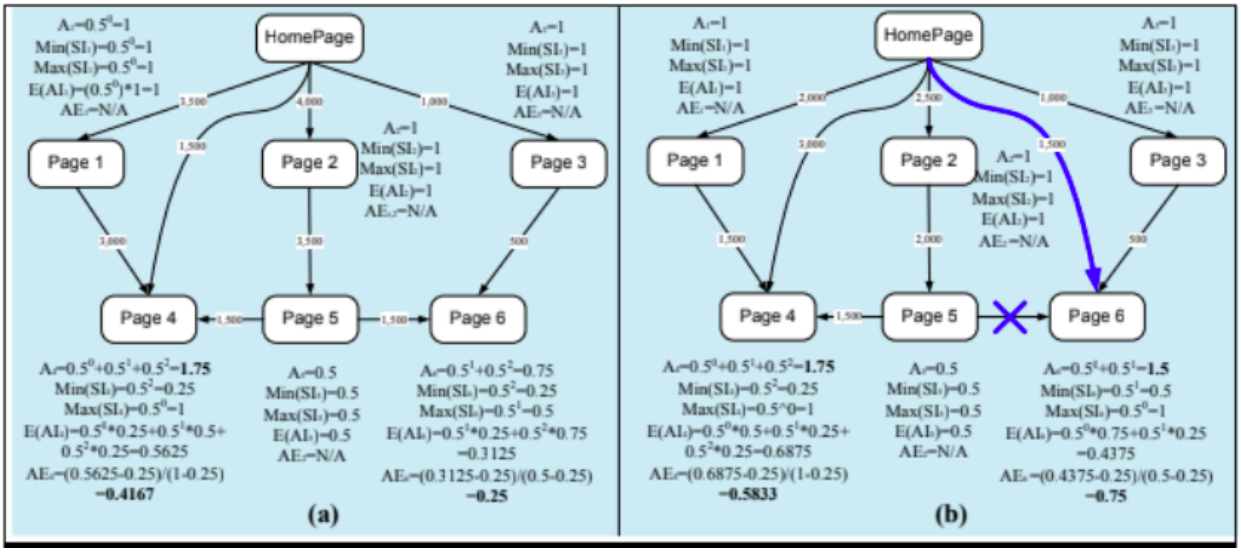


Figure 12: Example of Using Access Efficiency (AE) to evaluate User-System Adaptability in Adaptation Process

The *initial technical adaptation phase*, users browse to try out the functions when they first use the system. At this stage, users are more inclined to accommodate themselves to the technical aspect of the e-environment.

The *content-structure adaptation phase* constitutes the major part of the adaptation process which starts generally after or sometimes during the initial technical adaptation phase. Here, users browse the website with the aim to acquire information they desire.

User IT Adaptation Behaviors

Another study that cited adaptability is the User IT Adaptation Behaviors by Armin Kashefi, Pamela Abbott and Azerikatoa Ayoung (Kashefi, 2015). This study evaluates different researchers in relation to user's adaptation to new technology.

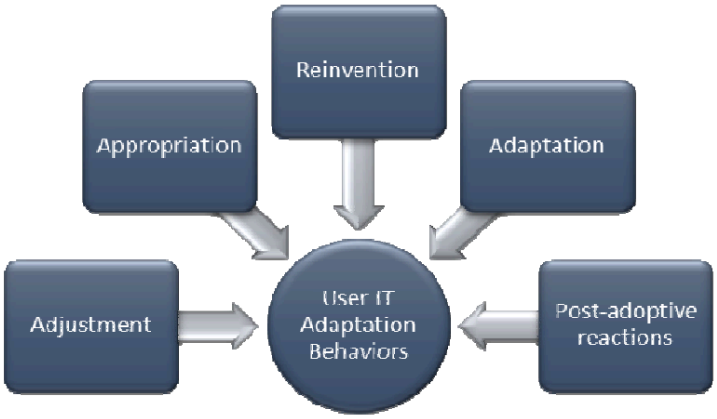


Figure 13: Ways to discuss User IT Adaptation Behaviors

Source: core.ac.uk

Ways to discuss user IT adaptation behaviors manifested that users may engage in different adaptation behaviors directed at each of the dimensions. Users may adapt the work task by modifying work routines and procedures or users may employ adaptation behaviors directed at themselves, such as developing IT skills by attending training sessions or seeking additional knowledge through social support.

Electronic Health Record Adaptability and Interoperability

Knowledge management as cited by Michael Koenig (Koenig, 2018) in his article is the 'process of capturing, distributing, and effectively using knowledge'. In this context, the Conceptual Healthcare Knowledge Management Model for Adaptability and Interoperability of Electronic Health records pointed out that knowledge management is important in guiding health care providers in working on new technology and in juicing out its benefits. The study also defined adaptability (Razzaque, 2021) in various ways such as (1) patient's and clinician's acceptability (2) usability and user friendliness of the interface and (3) security and confidentiality of EHR. Also, user acceptance is salient for the development of an EHR. According to the study, patients will accept data if they are educated and aware of the use of EHR, know that the data is secured with smart interfaces to assist their queries and, if they

have access to readily available computers or a network connection infrastructure. On the other hand, clinicians will accept the EHR if they realize its advantages, they are computer literate, find the user interface friendly, assured that data is secured and know that they are involved in the development process of EHR.

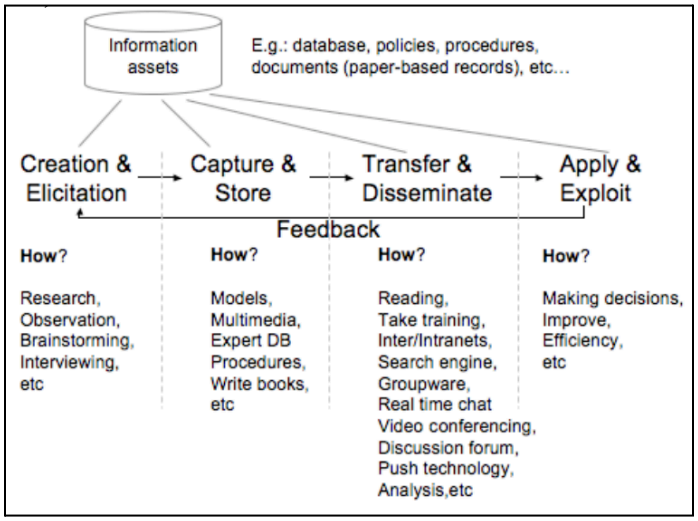


Figure 14 Knowledge Management Environment

Source: www.kmworld.com

As illustrated, knowledge management utilizes tacit and explicit knowledge. Where knowledge is professional intellect-know-how. Know-why etc. Also, it is pointed out in the study that KM needs technologies like infrastructure, internet, intranet and extranet as a pre-requisite for the knowledge management environment.

Boehm's Software Quality Model

The software quality model represents a hierarchical quality model which focuses on a wider range of characteristics introduced by B.W Boehm (Boehm, 1978). This model consists

of *portability, reliability, efficiency, usability (human engineering), testability, understability and modifiability.*

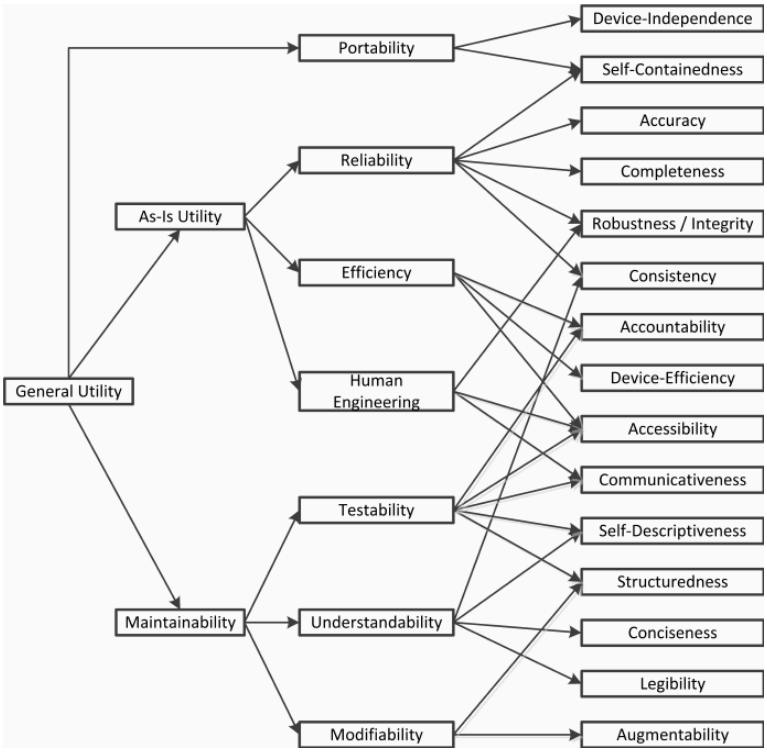


Figure 15: Boehm's Software Quality Diagram
Source: geeksforgeeks.org

The diagram represents how Boehm's software quality diagram focuses on measuring properties and characteristics in such a way that complex and non-technical stakeholders that are involved in the life cycle of a software.

Risk Management in Healthcare

The Healthcare Providers Service Organization (Catalyst, 2021) article on Risk management Plan for Healthcare Businesses and Practices defined risk management involves identifying and evaluating risks, in addition to implementing the most advantageous methods in reducing or eliminating them.

The Evolution of Healthcare Enterprise Risk Management involved eight (8) risk domains such as (a) operational, (b) clinical & patient safety, (c) strategic (d) financial (e) human capital (f) legal and regulatory (g) technological and (e) environmental - and infrastructure - based hazards.

Figure 15: The Eight Domains of Enterprise Risk Management
Source: catalyst.nejm.org

Key components of performing risk management in healthcare include the following procedures which also involve the tasks of a risk manager. First, *identifying risks* involving managing uncertainty and new risks. This process uses data, institutional and industry knowledge, and engagement of patients, employees, administrators and payers, health managers can uncover risks. Second, *quantifying and prioritizing risk.*

Cognitive Fit Theory

As developed by Iris Vessey (1991), this theory proposes that the correspondence between task and information presentation format leads to superior task performance for individual users.

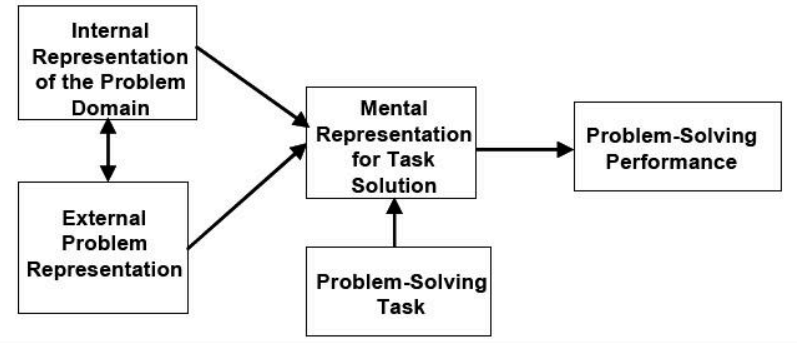


Figure 16: Extended Cognitive Fit model

Source: theorizeit.org

PHP Hypertext Processor

The proponent leaned on using PHP for the proposed system. PHP is a general purpose, server side scripting language that runs a web server that's designed to make dynamic pages and applications. PHP as a web development option is secure, fast and reliable and it offers lots of more advantages to make it accessible to people. (PHP, 2021)

Gantt Chart

The proponent used Gantt Chart to show activities (tasks or events) displayed against time. On the left of the chart is a list of the activities performed to complete the proposed system and along the top is a suitable time scale. Each activity is represented by a bar; the proposition and length of the bar reflects the start date, duration and end date of the activity (What is a Gantt Chart, 2019).

The proponent was able to see a glance of:

- What the various activities are
- When each activity begins and ends
- How long each activity is scheduled to last
- Where activities overlap with other activities, and by how much
- The start and end date of the whole project

The proponent created the Gantt chart for the proposed system.

Health Privacy Code

According to the Health Privacy Code Implementing the Joint Administrative Order No. 2016-2022 (Health Privacy Code, 2021) "Privacy Guidelines for the Implementation of the Philippine Health Information Exchange", the Health Information Privacy should be implemented considering the following rules:

- *Rule #1:* Collection and Processing of Health Information
- *Rule #2:* Access of Health Information
- *Rule #3:* Use and Disclosure Information
- *Rule #4:* Organizational Security Measures
- *Rule #5:* Physical Security
- *Rule #6:* Technical Safeguards
- *Rule #7:* Cloud Services
- *Rule #8:* Use of Social Media
- *Rule #9:* Human Resources
- *Rule #10:* Health Research
- *Rule #11:* Patient Registries
- *Rule #12:* Publication and Public Communication
- *Rule #13:* Health Privacy Board
- *Rule #14:* The Privacy Team of a Health Facility
- *Rule #15:* Compliance, Incident Reporting, Response

- *Rule #16:* Procedures in the Investigation of Complaints Filed Before the Health

  Privacy Board

- *Rule #17:* Penalty Clause

Role-Based Access Control (RBAC)

RBAC is also used in the proposed system provided to nurses, wives, admin and barangay health workers. According to Rosenbrance (Rosencrance, 2021), role-based access control is a method of restricting network access based on the roles of individual users within an enterprise. This process also ensures employee-access-only information they need to do their jobs and prevents them from accessing information that doesn't pertain to them.

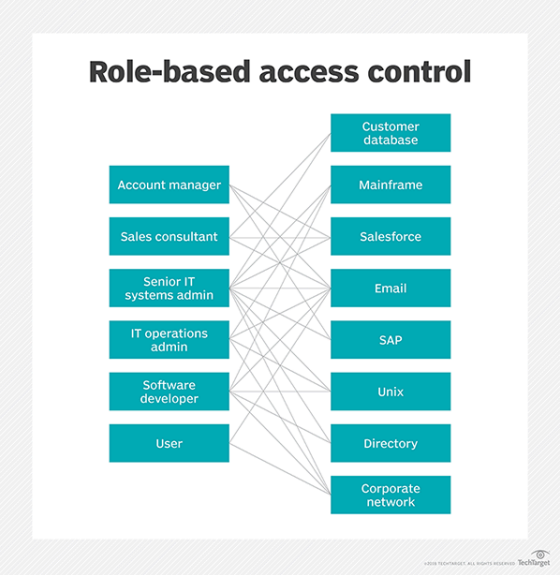


Figure 17: Illustration of the Role-Based Access Control
Source: https://www.techtarget.com/searchsecurity/definition/role-based-access-control-RBAC

SSL and TLS

According to digicert.com (Websecurity, 2021), SSL stands for Secure Sockets Layer and a standard technology for keeping an internet connection secure, safeguarding any sensitive data that is being sent between two systems, preventing criminals from reading and modifying any information transferred, including potential personal details. On the other hand, TLS (transport layer security) is just an updated, more secure, version of the SSL. This also covers the HTTPS (HyperText Transfer Protocol secure) which appears in the URL when a website is secured by an SSL certificate.

Signature Hash Algorithm

According to ivanti.com, the hashing algorithm is a global option which means that only one can be set per configuration. This minimizes the amount of hashing that takes place on an endpoint running the configuration. If the hashing algorithm changes, or a file is updated, a rescan is required to generate a new hash code. SHA 256 is a more complex, but slower hashing algorithm. While, as discussed by pagefreezer.com (Callaghan, 2021) the benefits of SHA256 are (1) it is a secure and trusted industry standard, (2) collisions are incredibly unlikely, (3) the avalanche effect where unlike some older hashing algorithms, even a very minor change to the original information completely changes the hash value - what is known as an avalanche effect.

CHAPTER 3

RESEARCH METHODOLOGY

This chapter provides research methodology and concepts included in the consideration of the Development and Implementation of an Adaptive Barangay Health Information System including the business process flow and success indicators and statistical treatment.

Scope and Delimitations

Scope

The proposed Development and Implementation of an Online Health Services as an Adaptive Information System for San Mateo, Rizal includes processing of health programs and medical services involving the participation of Barangay Health Workers with different user-access privileges depending on the line of work. Also, this includes patient profiling and online registration to accommodate online access to personal medical information and virtual schedule of visit for consultation of medicine administration with the assistance of the barangay's registered nurse or midwife.

The proposed system provides reports in accordance to the Department of Health's Field Health Service Information System (FHSIS) such as family planning, maternity checkup, child vaccination, dental checkup (if applicable), infectious disease prevention and control services, environmental health and sanitation services, and non-communicable disease prevention and control services. This also serves as a reference for the collection of information needed from the citizens patronizing the services of their barangay's health center such as name, age, gender,

date of birth, parents/spouse details and the like including the disclosure agreement presented on the privacy notice.

The proposed system also covers feedback and implementation for two barangays in San Mateo Rizal. The first barangay is the Barangay Guitnang Bayan 1 headed by Barangay Captain Jejomar Cruz while the second barangay is the Barangay Sta. Ana headed by Barangay Captain Henry Cruz. Both parties gave the proponent permission to gather information, administer interviews and develop and implement the proposed system within the premise of health services.

The system will also cover the provision for reports such as charts and documents to be submitted to the municipal office of San Mateo, Rizal and to the Department of Health and user accounts management. Also, that the system is only limited to barangay health services functions and not to other barangay features.

For each procedures, the following will be covered for the proposed system:

Enlistment. The initial step in the processing of Barangay Health Services proceeds by collecting vital information from the patient such as name, address, age, marital status, and the like. Collection of these data will be performed online. After the enlistment process, the Barangay Health Worker will proceed with collecting information for the mandated DOH form on Family Planning. Majority of the information will be extracted from the personal profile adding just the details of the patient's family members.

Revisit. After establishing the patient's information on its first visit, the second visit will only require the patient's barangay health system ID number as a reference for the profile. This will provide an updated patient status and medical administration.

Accounts. Users of the system will create their personal accounts with passwords. This will include the admin, the barangay health workers and the patients. The patient's account is also accessible online should he/she wish to check his profile and medical records for any purpose it may serve him/her. This will also enable the patient to schedule a visit to the barangay health station.

Reports. Barangay officials use reports from the satellite office to further improve and enhance their services. The proposed system will generate reports as per the preferred time frame of the user or barangay health officer.

Scheduling. Involves online scheduling for patients who would like to visit the barangay health services office. The proposed system will inform the patient if the health services center is still accepting patients on that day or not.

Data. Data to be covered in the transition from old system to new will only include the period from when the system has started testing. Only taking into accounts the new patient transactions from the time of the proposed system's approval until implementation. The old files will be migrated gradually by the person-in-charge once the system is already working at the defined premise.

User Adaptability. The proposed system is also designed to be adaptive and is open for use to other Barangays only within the premise of San Mateo. This is to also help the other Barangays automate their Health Services Information System and to promote ease of transaction not only for the barangay health workers, but most importantly, to the people.

Barangays. The proposed system is presented to two (2) barangays in San Mateo Rizal such as Guitnang Bayan 1 and Sta. Ana. Both barangays have their own Health Service Offices who agreed to implement the aforementioned system.

Adding of Barangays. The proposed system is initially tested in barangays Guitnang Bayan 1 and Sta. Ana, though as another feature, the system administrator can also add another barangay. Adding another barangay will also access database input and updates such as new users, its credentials, health records and consultation and barangay health worker features.

Delimitation

The proposed Development and Implementation of Online Health Services as an Adaptive Information System for San Mateo, Rizal limits its coverage to the following:

Location of Implementation. The proposed system is referenced to the guiding principles of the Department of Health for local government unit's health services but the data gathering, interviews, and consultation were performed only within the premise of two barangays to compare and contrast their services. Thus, making the implementation focused on the health centers of San Mateo Rizal with fifteen(15) barangays.

User Adaptability. The proposed system is only limited to the degree of match between users and system in terms of access paths chosen where the proposed system is able to direct users to a relatively high efficiency in information access. This is limited to scheduling, consultations, health record creation, updates and retention within the allowable period of time and report generation. The system doesn't measure adaptability at a back-end level. The barangay health services are following the same set of tasks as implemented by the Department of Health (Kashefi, 2015).

Services. The proposed system does not cover inventory of medicines as well as logistics requirements by the barangay health services as per agreed upon with the barangay health workers due to the versatility and inconsistency of supplies and human resource.

Role-Based Access. The system users - barangay health workers and patients are limited only to the premise of their role description as identified by the administrators of Barangay Health center.

Security. The security of the proposed system were referred to the Health Privacy Code of the Philippines as published by the Department of Health, Department of Science and Technology and the Philippine Health Insurance Corporation. The system only covers electronic premise such as:

a) Collection and processing of health information
b) Access of health information and Use and Disclosure of Health Information
c) Physical security
d) Technical safeguards

Reports. The generation of reports only focuses on the transactions held within the territory of each barangay's health services office. Transactions held outside the functions of the office will not be covered. Also, the reports to be submitted to the municipal hall will follow the FSHIS format as discussed with the representatives of Health Centers (FHSIS, 2021). The FSHIS was based on the shared documents from Barangay Health workers. These are the forms that they workk on with to provide reports to municipal offices. These documents were not directly extracted by the proponent from the Department of Health. Though, based on research and references, this is also the same form mandated by the government to be used for Health related reports from LGUs (Department of Health, 2019)

System limitations. The proposed system is initially running in a single web hosting plan with the following specifications:

- 1 website
- 30GB SSD Storage
- 1 Email Account

- 100GB Bandwidth
- 2 Subdomains
- 1 FTP Account
- 2 CronJobs

Any further changes that the client would prefer to expand the system capability and cater to more than two(2) barangays will require web host plan adjustment given the limitations of the proposed system.

Quantitative Research Methodology

This methodology focuses on measurement and testing using numerical data. Quantitative Research Methodology will be used to process the information gathered from the participants of the study. As part of the objectives, it is necessary to identify the end user's acceptance of the proposed system based on specific success indicators and data analysis through Descriptive Statistics (mean, median, mode).

Success Indicators

The success indicators in this study are based on ePhilippines Key Result Areas, Boehm's Software Quality Model, EHR Adaptability and Interoperability, Technology Acceptance Model and User-System Adaptability by Bingcong Zeng et.al. Though, the scales were adjusted to fit in the context of a barangay.

The following success indicators are as follows:

Health Consumers: Safer and quality health care

- Unique Health Identification for each Patient
- Easy but Secured access to Health Information
- Individual Patient Records

- Interaction to Health Care Providers is faster
- Health consumers' health information can be shared and exchanged across the health sector in compliance to established standards, rules and protocols.
- Online health reporting to generate quality information as support to decision making and health care delivery.

Health Care Providers: Make more informed decisions

- Online ordering system and results reporting, referrals and prescriptions.

Health Care Managers, Policy Makers and Researchers: Effective program/research development, implementation and monitoring

- Health Information produced by the system for Barangay Guitnang Bayan and Barangay Sta. Ana Health Services Center provides charts and reliable reports for analysis.
- The system was able to provide support in monitoring health system activities and outcomes.
- The system was able to adhere to data standards and processes of the Barangay.

Descriptive Statistics. This was used in the study to describe the features of the data. A statistic that provides simple summaries about the sample and the measures together with simple graphics analysis, it forms the basis of virtually even quantitative analysis of data.

Univariate Analysis. As per the article by Prof. William Trochim, *Research Methods Knowledge*

Base Descriptive Statistics involve Univariate Analysis which describes the examination across cases of one variable at a time. This has also provided three major characteristics of a single variable as the distribution, the central tendency and the dispersion. Where in the distribution, a summary of the frequency of individual values or ranges of values for a variable is presented. The central tendency of a distribution is an estimate of the "center" of a distribution of values. This includes, mean, median and mode. Then, for dispersion, this refers to the spread of values around the central tendency.

Standard Deviation. Leads to a more accurate and detailed estimate of dispersion because an outlier can greatly exaggerate the range. The formula for standard deviation is:

$$\sqrt{\frac{\sum(X - \bar{X})^2}{n - 1}}$$

where:

- X is each score,
- $\bar{X}$ is the mean (or average),
- n is the number of values,
- Σ means we sum across the values.

Likert Scale

In its final form, the Likert scale is a five (or seven) point scale which is used to allow the individual to express how much they agree or disagree with a particular statement (McLeod, 2021).

This measuring tool was used in the study to determine the agreement of stakeholders to the content of the barangay health services information system parallel with the success indicators.

Table 1: Illustration of the Likert Scale and Success Indicators

CATEGORIES	Excellent 5	Very Good 4	Good 3	Fair 2	Poor 1
Portability					
Reliability					
Efficiency					
Usability					
Testability					
Understandability					
Modifiability					
Security					

Technology Acceptance Model

The technology acceptance model is also used as part of the methodology to quantify user acceptance of the proposed system in the context of *perceived usefulness, perceived ease of use and trust.*

Dependencies

This section describes the different aspects in consideration for the development of the proposed system.

Budget. The proposed system would require monetary considerations to be able to procure necessary logistics and cost of living for manpower during the duration of system development.

Stakeholder and Administrative Cooperation. The stakeholders are expected to practice their determinative expectations during the process of development and implementation of the proposed system.

Manpower and Client Acceptance. In the context of system development, the clients will be required to undergo a series of training and familiarity for the system's adaptability and transition process.

Data Collection

Sample Size Determination

Barangay Guitnang Bayan 1 and Barangay Sta. Ana has a total population of 40,758 as of 2015 based on PhilAtlas (PhilAtlas, 2021). Given this information, the determinant of survey respondents were based on two measures such as the margin of error (or confidence interval) and confidence level corresponding to a questionnaire as per the use of the proposed system.

The targeted confidence level to the proposed system was set to 95% as a standard in quantitative research as per the check market reference written by Gert Van Dessel (Dessel, 2013) and a 5 % margin of error. Thus, the completion of sample size required for the proposed system's adaptability and acceptance survey is 381 respondents while the estimated response rate is at 20%.

The same procedures were performed in determining the number of respondents for barangay health workers (BHWs). Each barangay has a total of 20 BHWs. Considering the 5% margin of error and the 95% confidence, the result is still 20 respondents. Thus, getting the 100% of barangay health workers population for barangay Guitnang Bayan 1 and Sta. Ana.

Interview

To further understand the Health Services procedures of Barangay Guitnang Bayan 1, an interview was administered with the Barangay Captain. Jojo Cruz and the health services *kagawad* and nurse in-charge. In an overall point of view, the nurse-in-charge provides the necessary information for the decision making within the premise of the health services. The barangay health workers who were trained to do their tasks provided assistance in filling the documents, arranging the reports and assisting patients.

Figure 18: Interview with the Barangay Captain and Nurse-in-Charge of Barangay Guitnang Bayan 1

Figure 19: Proposed System presentation with the Barangay Captain and Council of Brgy. Sta. Ana

Figure 20: Proposed System presentation with the Barangay Health Workers of Barangay Sta. Ana

Documents and Records

The nurse-in-charge provided sample data, sample forms and discussed the processes of handling patient's for a transaction in the barangay health center. The emphasis on the use of family planning form was mentioned since it serves as their main form in identifying family members, assignment of file number and so on.

Figure 21: Sample Patient Charts

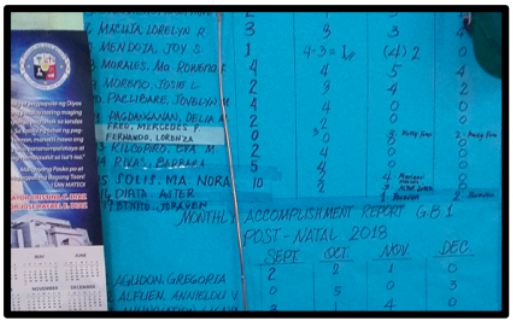
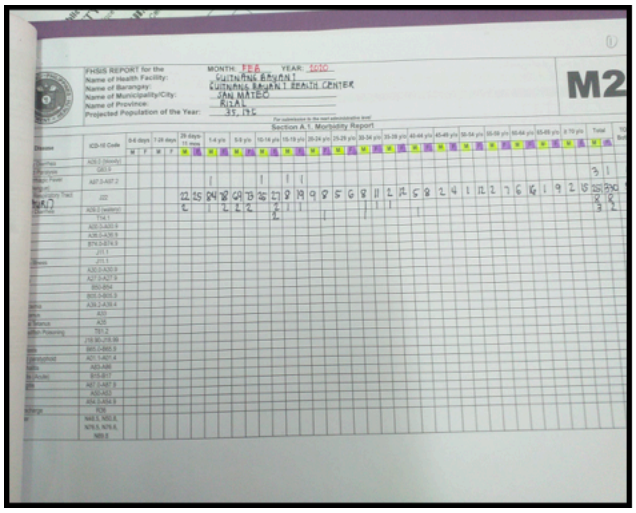


Figure 22: Health Board Indicators

Figure 22: Patient File Storage for Barangay Guitnang Bayan 1



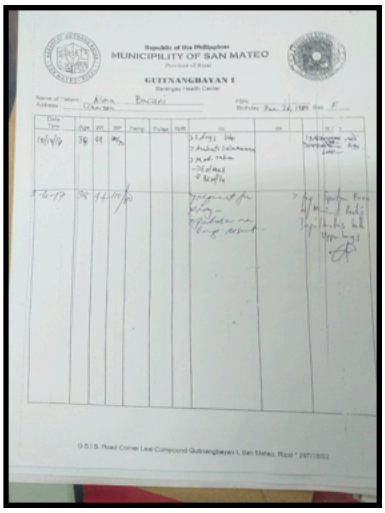


Figure 23: DOH and Municipal Mandated Forms both for Barangay Guitnang Bayan 1 and Barangay Sta. Ana

Surveys

To get the pulse of the barangay health worker's point of view on the possibility of the proposed system, they were then gathered and asked to vote given the alternatives below:

Table 2: Alternatives

Alternatives	Benefits	Requirements	Votes
Retain the Existing System	• Retain same number of employees	• Rework • Expenses will not be reduced • Wait time for Residents with Concerns • Bulk of Ques	1
Remove the use of logbooks and change forms.	• Lessen needed logistics	• Training for new process • Uncharted Paper Trail and Residential inquiries	6
Change the process	• Lessen the need for a number of Information Officer • Ease of doing business (main objective)	• Employee trainings • Process slowdown due to transition period • Rework issues	0
	•	•	9

Based on the presented alternatives and the number of votes gathered from the barangay health workers, nine (9) out of ten (10) voted for the Development and Implementation of the Barangay Health Services Information System.

Systems Development Life Cycle

Since the context of the proposed system is on Health Services, the development life cycle considered for the proposed system is Agile Method.

In this context, Agile is used due to the specification, design, implementation and testing are inter-leaved and the outputs from the development process are decided through negotiation during the software development process.

This process is also used to reduce overheads in the software process and to be able to respond quickly to changing requirements without excessive rework.

Implementation Plan

Below are the tables used to represent the implementation plan. This section is divided into two parts. Table 2 represents the Online Health Services Information System while Table 3 represents the Health Services Social System. The process of dividing the implementation plan into two (2) parts was based on the study of Zia and Bukhari (2011) for the Plan for Implementation of Hospital Information System in Developing Country: Recommendation from socio-technical perspective.

Table3: Implementation Plan for the Health Services Information System

CATEGORY	DESCRIPTION	TIMEFRAME	RESOURCES NEEDED
Key Areas of System Flow	In this stage, key concepts of the existing process should be identified.	September 2020	Inputs from Barangay Health Workers
Software Acquisition	In this stage, decisions regarding the software acquisition are taken as well as the operating system and database and related software.	September 2020 week 4	IT Team
Hardware Acquisition	In this stage, hardware acquisition is done. It includes peripheral devices.	October 2020	IT Team and Barangay Health Workers and Admin
Agreement and Presentation to 2nd Barangay	In this stage, the proposed system was presented to the second barangay, Barangay Sta. Ana in reference to adaptability.	January 2021	IT Team and Barangay Admin (Barangay Capt. and Council)
Data Preparation and Conversion	In this stage, data is prepared for transfer.	October to January 2021	IT Team and Barangay Health Workers and Admin
Installation	In this stage, installation of both hardware and software and entering of relevant data in the system is done.	March 2021 Week 2	IT Team and Barangay Health Workers and Admin
Testing and Feedback (Iterative)	In this stage, testing is done to ensure, either the system works effectively or not after the complete configuration.	March 2021 Week 4	Barangay Health Workers and Admin
System Adjustment	In this stage, feedback was considered in system updates and presented to clients.	April 2021	IT Team and Barangay Health Workers and Admin
System Adaptation	In this stage, features and function of the system were shown once again to clients for adaptation and function matching in connection to the actual tasks for parallel implementation.	May 2021 - October 2021	IT Team and Barangay Health Workers and Admin

System Adaptation (phase 2)	In this stage, the system is being finalized including documentation and training sessions.	November 2021	IT Team and Barangay Health Workers and Admin

Data Protection Officer

The barangay health services is bounded to communicate thoroughly with the Data Privacy Officer from the municipality of San Mateo where there will be Barangay Health Services Data Protection Committee for the following:

Functions of the Data Protection Committee:

a. Rule Making - the committee shall develop, promulgate, review or amend rules, policies and regulations for the effective implementation of the Act.
b. Advisory - the committee shall be the advisory body on matters affecting protection of personal data.
c. Public Information - the Committee shall undertake necessary or appropriate efforts to inform and educate the public of data privacy, data protection, and fair information rights and responsibilities.
d. Compliance and Monitoring - the Committee shall perform compliance and monitoring functions to ensure effective implementation of the Act, these Rules, and other issuances.
e. Complaints and Investigations - the Committee shall adjudicate on complaints and investigations on matters affecting personal data.
f. Enforcements - the Commission shall perform all acts as may be necessary to effectively implement the Act, these Riles, and its other issuances, and to enforce its Orders, Resolutions or Decisions, including the imposition of administrative sanctions, or penalties.

g. Other functions - the Committee shall exercise such other functions as may be necessary to fulfill its mandate under this Act.

Data Protection Officer (DPO)

a. Under the supervision of the Municipality of San Mateo, the DPO, is responsible for the adherence of barangay health centers to the Act, its implementing rules and regulations and related laws, and shall oversee all activities related to the development and implementation of policies and procedures on data privacy and protection of personal information.
b. In particular, the DPO, together with the members of the committee shall perform the following duties and responsibilities.
c. Monitor barangay health services' compliance with the Act, its IRR, issuances by the National Privacy Commission (NPC) and other applicable laws and policies. For this purpose, the DPO may:
d. Collect information to identify the processing operations, activities, measures, projects, programs, or systems of the PIC or PIP, and maintain a record thereof
e. Analyze and check the compliance of processing activities, including the issuance of security clearances to and compliance by third-party service providers.
f. Inform, advise, and issue recommendations to the PIC or PIP.
g. Ascertain renewal of accreditations or certifications necessary to maintain the required standards in personal data processing.
h. Advice the PIC or PIP as regards the necessity of executing a Data Sharing Agreement with third parties, and ensure its compliance with the law;
i. Develop, review and implement policies, guidelines, projects and / or programs of the PIC or PIP relating to privacy and data protection, by adopting a privacy by design approach;

j. Conduct Privacy Impact Assessments relative to activities, measures, projects, programs, or systems of the PIC or PIP

k. Advice the PIC or PIP regarding complaints and / or the exercise by data subjects of their rights (e.g., requests for information, clarifications, rectification, or deletion of personal data)

l. Ensure proper management of data breach and security incident by the PIC or PIP, including the latter's preparation and submission to the NPC of reports and other documentation concerning security incidents or data breaches within the prescribed period;

m. Inform and cultivate awareness on privacy and data protection within the organization of the PIC or PIP, including all relevant laws, rules and regulations and issuances of the NPC;

n. Serve as the contact person of the PIC or PIP, the NPC and other authorities in all matters concerning data privacy or security issues or concerns and the PIC or PIP

o. Cooperate, coordinate and seeks advice of the NPC regarding matters concerning data privacy and security

p. Perform other duties and tasks that may be assigned by the Municipality of San Mateo Rizal/ Mayor's office that will further the interest of data privacy and security and uphold the rights of the data subjects.

Training and Capacity Building

a. All newly-hired personnel shall be oriented on data protection and security. The orientation shall be facilitated by the Data Privacy Committee or the office

responsible for conducting employee and patient orientation. The orientation will include the data privacy structure in San Mateo Rizal, data privacy laws, this manual, and other related policies and protocols.

b. The data privacy committee shall undergo periodic training on:
    i. Applicable privacy laws and policies
    ii. Data protection and security measures
    iii. Addressing breach of privacy and complaints, including proper reporting procedures
    iv. The consequences for violating privacy laws and policies

Project Cost

The development and implementation of Barangay Health Information System covers the project cost for transportation both to the barangay of Guitnang Bayan 1 and Sta. Ana, food allowance, meetings expenses, the fee for actual system development and documentation.

Table 4: Implementation Plan for the Health Services Information System

Item #	**Activities**	**Budget**	**Actual Expenses**
#1	Transportation (Gutinang Bayan 1 Barangay Hall)	Php 3,000.00	Php 2,800.00
#2	Transportation (Sta. Ana Barangay Hall)	Php 3,000.00	Php 2,900.00
#3	Food allowance	Php 5,000.00	Php 5,000.00
#4	Meetings	Php 3,000.00	Php 2,000.00
#5	System Development	Php 50,000.00	Php 70,000.00
#6	Documentation	Php 3,000.00	Php 3,000.00
#7	Web Hosting Plan >Activated SSL >Domain Name >Cloudflare CDN >Daily Backups	Php 3,000.00	Php 2,820.00

	>200 GB SSD Storage >Unlimited Bandwidth >Unlimited MySQL databases >Free domain registration >100 Email Accounts >99.9% Uptime Guarantee >24/7/365 Support >Powerful Control Panel >Webmail Access >DNS Management >Subdomains per Account >Parked Domains (100) >Unlimited FTP Accounts >FTP Over SSL >3GB Database Size >Entry Processes (30) >Active Processes (60) >I/O - 1024 >IOPS - 128		
#8	Hardware >Computer unit and peripherals >Internet Provider Services (est. Php 1,500 per month)	Php 45,000.00	Php 45,000.00
Total Expenses			***Php 132,820.00***

Key Areas of System Flow

To collect the areas related to system flow, the data collection structure started with semi-structured interviews such as notes taking and audio recording. Included in this process is the collection of documents such as the Family Planning Form 1, Barangay Health Center Patient form, CVD/NCDRisk Assessment Form for AB/Diabetic. Preparation of Raw data, validation and verification and analysis follows after the documents to certify the validity of test data.

Software Acquisition

The proposed information system for Guitnang Bayan 1 and Sta. Ana Health centers were established through PHP and MySQL as well as the website made accessible for both the barangay health workers and the client.

Hardware Acquisition. This phase is under the commitment of Brgy. Guitnang Bayan 1 and Barangay Sta. Ana administration who opted to include procurement of hardware/logistic support for the system. Nevertheless, for the testing phase, the proponent will provide the necessary hardware support.

Data Preparation and Conversion. In this phase, the transition from the existing system to the proposed system may take a while to finish since the documents are entirely in manual format. Hence, the proposed system will only include new patient transactions from the time of its approval to implementation. As mentioned on the scope and limitations of this paper.

Installation. This stage will be performed ideally, in the 2nd week of March 2021 where the process of conversion has been accomplished.

Testing. This phase of implementation will take place with support from the proponent to guide the users on how they will perform the workarounds of the system while getting familiar with its interface. Though, the system aimed to have lesser assistance for the user interface is meant to be intuitive.

To further provide an overview of how the implementation of the proposed system will be, the following details were presented including the use of parallel run testing.

Parallel Testing. These tests could be performed to make sure the new version of the application performed correctly, to make sure the consistencies are same between new and old version, to check if the data format between two versions has changed, to check the integrity of the proposed system for Brgy. Guitnang Bayan 1 San Mateo, Rizal.

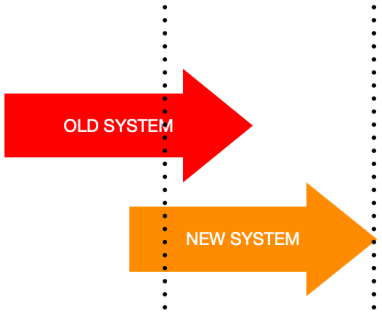


Figure 24: Parallel Testing

Delivery. Once all aspects of the system have been tested and approved by both the health workers and barangay captains, it will be delivered to the center and perform its tasks to provide ease of work and fast transactions where patient data is safe and secured.

Table 5. Implementation Plan for the Health Services Social System

CATEGORY	DESCRIPTION	TIMEFRAME	RESOURCES NEEDED
User Group Formation	This user group is formed with appropriate users who will use the system.	January 2021 2nd week	Barangay Health Workers and Admin
User and Operator Training	This stage includes the training of users. This section also administers to the needs of users/operators of the system	January 2021 3rd week	Barangay Health Workers and Admin

	should there be changes in the procedural aspects in relation to the mandates of the Department of Health.		
User Acceptance	After proper training of the user groups, the system is accepted by the user group. Part of this will be some acceptable testing.	January 2021 4th week	Barangay Health Workers and Admin

Table 4 focuses on the social system implementation plan. This covers user group formation, user and operator training and user acceptance.

User Group Formation. User groups in the case of Brgy. Guitnang Bayan 1's health services center includes all of it's workers such as the health officers and one (1) nurse. Modules for training will also be provided both in soft and hard copy to further support the training in all possible cases.

User and Operator Training. Since the technical skills of the users for the proposed system may not fit in the technical needs of the system, the proponent will save as the operator for the transition period and will be substituted by another representative from the barangay once proper training was administered.

User Acceptance. The success of this phase will be measured through the Technology Acceptance Model by Davis (1989).

Use Case Diagrams of the Existing System

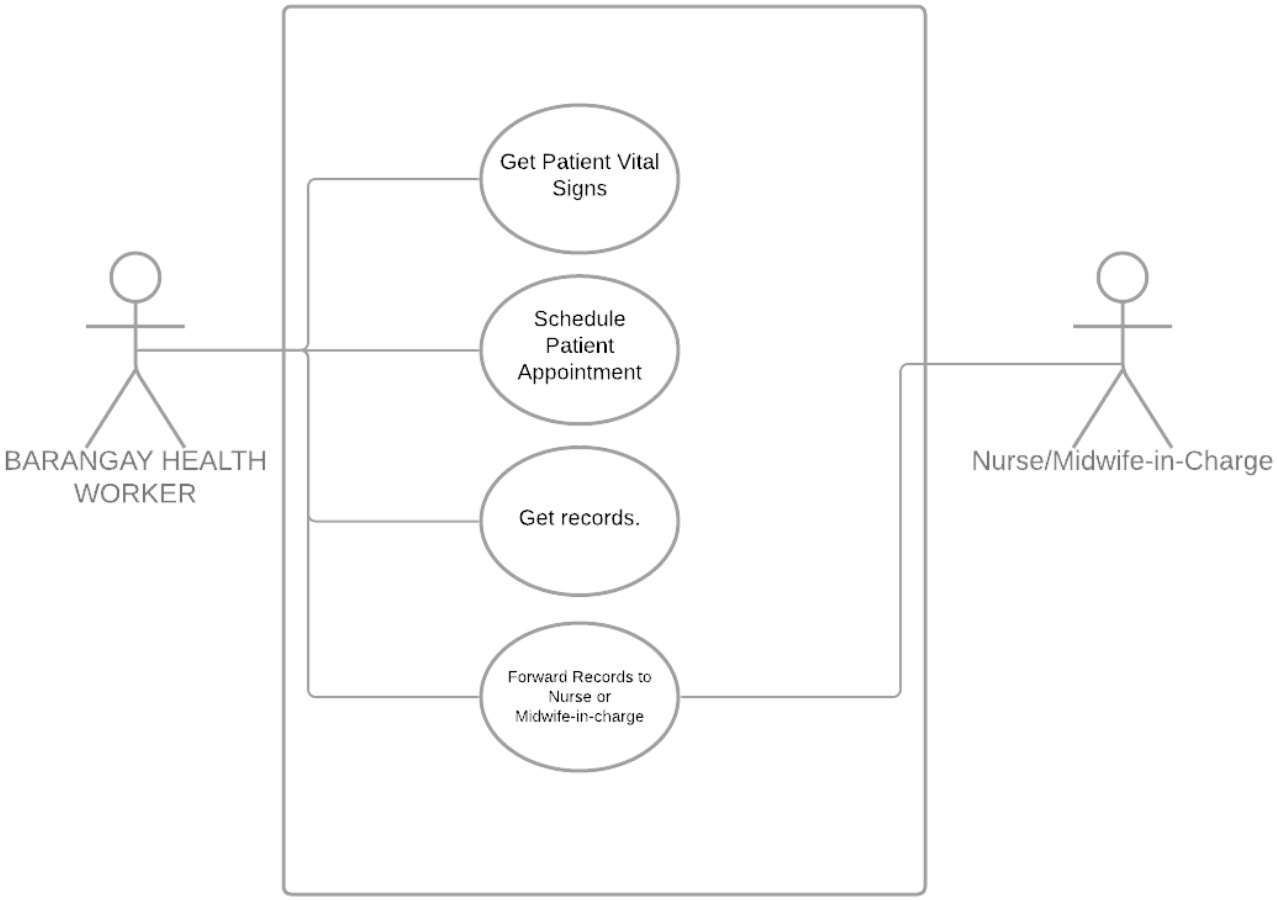


Diagram 1: Barangay Health Worker

The Barangay Health Worker (BHW) monitors height and weight, defaulters in cases of immunization, reports to the barangay health center, family planning, tuberculosis compliance to treatment and others, incidence of disease or any health cases in the community, communicates and coordinates with the community to participate in BHC activities, gathers data about the overall status of health of the community, administers first aid, conduct health visits, conduct health visits, conduct health education and strengthens referral systems.

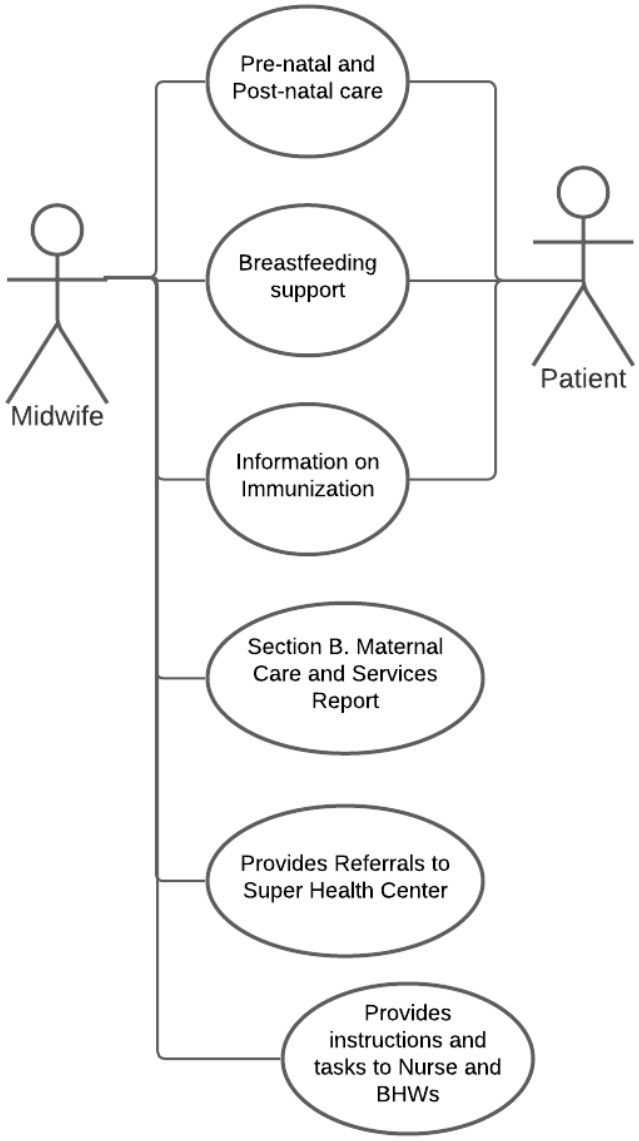


Diagram 2: Barangay Midwife

Midwives to barangays are mandated to ensure that mothers and infants, in their respective jurisdictions, are given quality and essential health care services, including, but not limited to pre-natal and post-natal care, breastfeeding support, provision of information on the importance of immunization and assurance that infants are given the proper vaccines as

provided in the DOH expanded program on immunization, and proper care and nutrition of both the mothers and the infants.

Midwives are also in-charge of providing tasks and service monitoring for barangay health workers and nurses for being mainstays in the barangay health centers and directly reports to the Super Health Center.

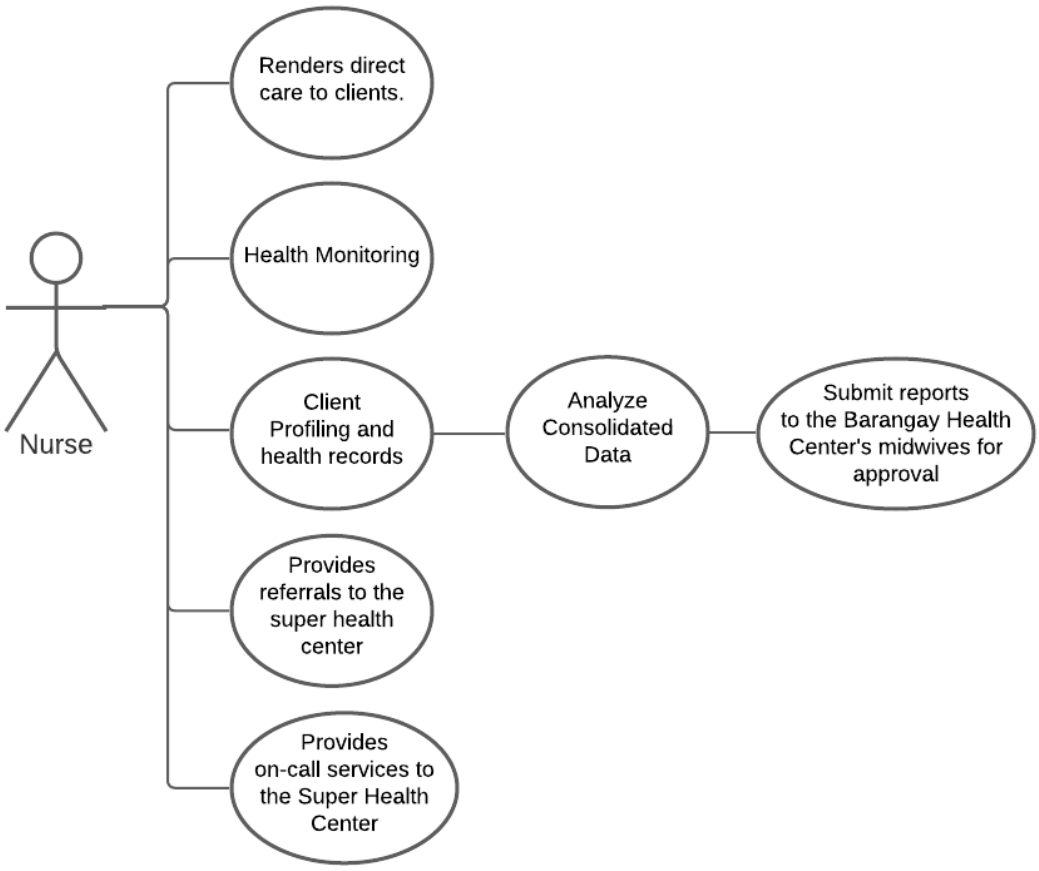


Diagram 3: Barangay Nurse

The barangay nurse helps improve the delivery of public health care services, provides relevant and timely education on wellness, hygiene, sanitation and other health safety measures to the poorest communities in rural areas, provides reports based on the Field Health Services

Information System and referrals to the Super Health Center. They are also assigned to generate patient reports to the midwives for confirmations.

Barangay nurses are also responsible for identifying the needs and concerns of individuals, groups, families and the community, formulates health plans, interprets and implements nursing plans and programs, assists other health team members in implementing health programs, conducts/facilitates necessary training or educational orientation to other health team members, renders direct care to various clients with different needs, monitors and detects the presence of health concerns in community and records and reports health status and presence of health problems.

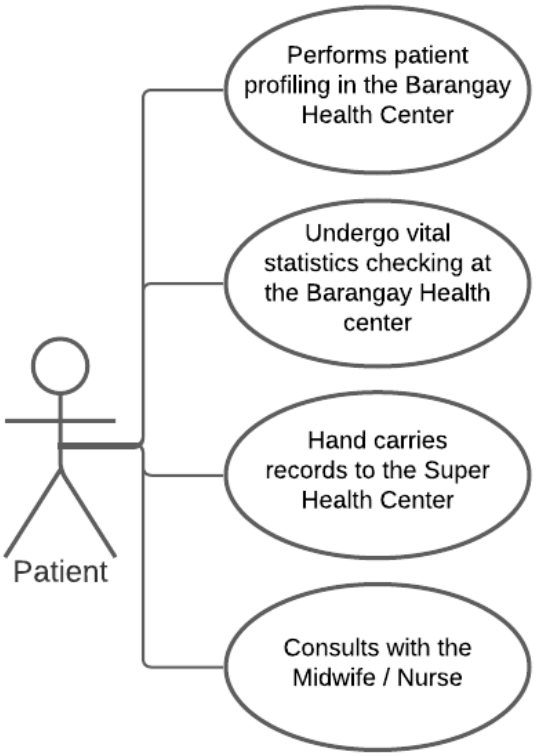


Diagram 4: Patient

The Barangay Health Centers provides services to the citizens of the barangay referred to as patients. Each patient was given forms for profiling and monitoring. The different transaction forms were filled out by each patient depending on their health claim or required assistance.

Needs of pregnant women and child vaccination were mainly addressed in the barangay health center as well as tuberculosis monitoring, hypertension, diabetes and health education for the members of the community and evaluated by either the nurse or midwife with the assistance of the barangay health worker. All patient profiles were kept in the Barangay Health Centers for proper documentation. Patients were also given copies of the document as per need.

Use Case Diagram of the Proposed System

FILLED ITERATION: Development and Implementation of an Online Health Information Services System

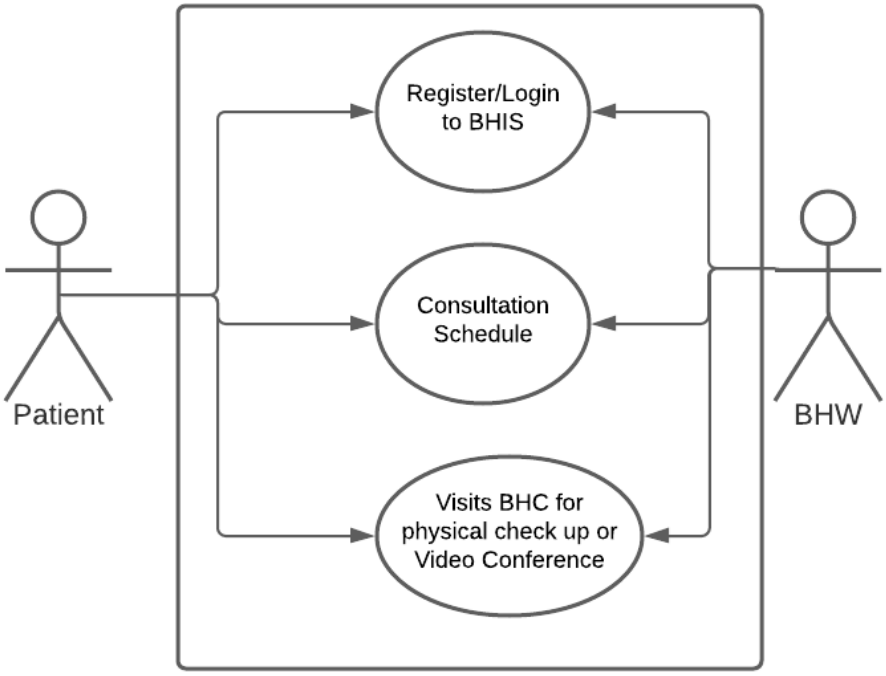


Diagram 5: Patient Registration and Consultation Schedule

Use Case	Enrollment Process
Description	Patients register to the system with pertinent information as required by the Department of Health; Other details will be completed upon

	consultation with the midwife, nurse or barangay health worker as per consultation type.
Triggers	Patients are required to comply with the details for enrollment procedure. Details collected will be for official patient enlistment and data storing. Barangay Health workers will access the registered information for profiling.
Basic Course of Events	1. Patients register online and set a request for a consultation schedule. 2. The assigned barangay Health worker will confirm the schedule and send a consultation confirmation message via text / email.

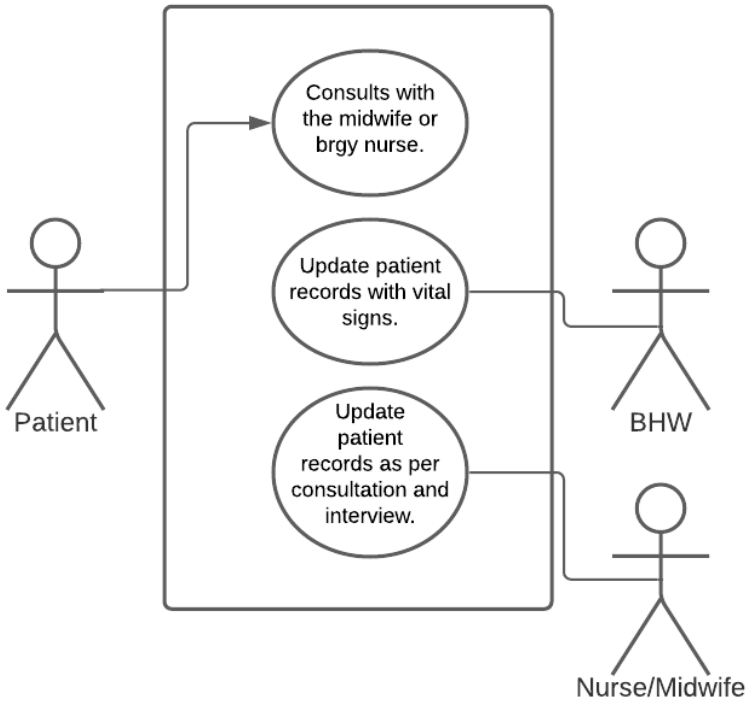


Diagram 6: Patient Consultation

Use Case	Enrollment Process
Description	Patients consults with the barangay Nurse or Midwife, assisted by

	the Barangay Health Worker
Triggers	Records will be pulled out from the barangay health information database as per patient's online registration records.
Basic Course of Events	1. Nurses/Barangay Health Workers/Midwife will access the barangay health information system for patient records and update as per: *Family Planning *Maternity Checkup *Child Vaccination *Dental Checkup *Infectious Disease and Prevention Control Services *Non-communicable Disease Prevention and Control Services *Environmental Health and Sanitation Services

The Use Case Diagram for the proposed system illustrates the procedures implemented both for the Barangay Gitnang Bayan 1 and Barangay Sta. Ana in reference to the instructions and mandate of the Department of Health.

Risks and Mitigation Plan

Table 6: Risks and Mitigation Plan for the proposed system

Risks	Mitigation
Maintenance of New System Infrastructure	Request additional or train existing manpower
Budget	Invoke Budget to next fiscal year through the Local Government budget for Basic Services and Facilities Program
Susceptibility to Cyber Attack	Include Network Security/Firewall features in the improvement stage

Change of Local Government officials	Barangay Health workers are independent of the transition in local government.

Maintenance of New System Infrastructure

The proposed system's implementation should be well-supported by training and learning materials that will encompass the functional coverage of the system to improve user familiarity and ease of use to the new system.

Budget

According to the Department of Budget and Management, the Internal Revenue Allotment (IRA) for each local government unit may allot for basic services and facilities programs where health and nutrition is included which aimed the improvement of health services. It is necessary to note, nevertheless, that the budget should be prior to the identified Fiscal Year. In this case, the proposed systems' implementation and foreseen maintenance expenses should be included.

Susceptibility to Cyber Attack

The inclusion of network security/firewall features in the improvement stage is vital and should be emphasized as another step of progress for the proposed system as per the agreed standards of the Barangay Health Center the Municipality of San Mateo, Rizal.

Change of Local Government Officials

The transition of Local Government Officials under the Local Government code of the Philippines Book III Local Government Units for the role and creation of barangay includes Chief Officials and Offices such as (a) A Punong Barangay, (b) seven Sangguniang Barangay members, (c) Sanguniang Kabataan Chairman, (d) Barangay Secretary, (e) Barangay Treasurer

in accordance as well with Omnibus Election Code - Article VI. The appointment of new officers does not affect the Barangay Health Workers' employment since according to Senate Bill No. 1380 by then, Senator Joseph Victor Ejercito:

"Section 395-A. Appointment and qualifications of Barangay Health Workers - The Barangay Health Worker as accredited and recommended by the Municipal or City Health Board in accordance with the provisions of Republic Act No. 7883, otherwise known as the barangay Health Workers and Benefits Act of 1995, shall be appointed by the Municipal or City Mayor of the area where such barangay health worker will be appointed is located. Barangay Health workers shall be entitled to allowances and such other benefits to which other appointed barangay officials may be entitled to no person duly appointed as a barangay health worker shall be removed except for valid cause as provided under existing civil service rules and regulations and only after due notice and hearing."

Class Diagrams

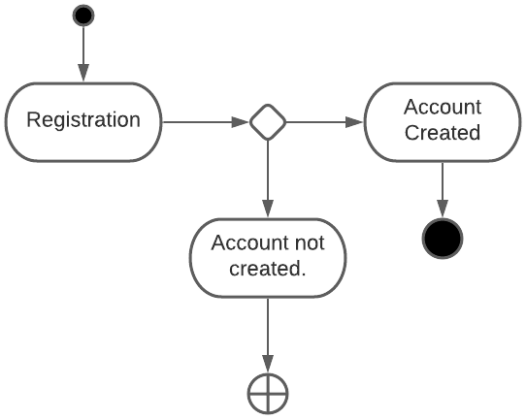


Figure 25: Starting & Stopping

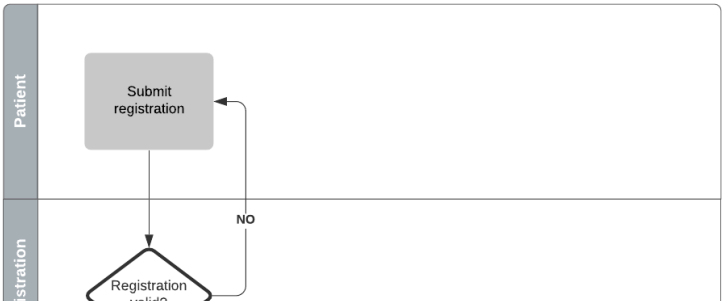


Figure 26: Swim Lane 1

Test Policy

The adaptability of the proposed system aimed to surface through another overarching aspect where stakeholders make use of the system from different Barangays in San Mateo Rizal, focusing mainly on Barangay Guitnang Bayan 1 and Barangay Sta. Ana.

Mission of Testing

To effectively and efficiently provide timely, accurate and useful and adaptive barangay health information systems to the Baranggays of San Mateo, Rizal.

Sequential Test Levels Performed by Participants

The initial set of tests were implemented to the Barangay Health Workers of Barangay Guitnang Bayan 1 and Barangay Sta. Ana.

Table 7: Initial tests

Level	Owner	Objectives	Key Areas of Testing
Unit	Development team	• Detects defective code in different modules. • Reduce risk of unit failure in Production.	Functionality and resource utilization.
Integration	Development team	• Detect defects in user interfaces • Reduce risk of dataflow and workflow failures	Functionality, data quality, performance, reliability, usability, resource utilization, maintainability, installability, portability and interoperability
System	Risk Mitigation and Quality Assurance	• Detect defects in use cases and end-to-end scenarios • Assist in mitigating risk of unmet business requirements in production	Functionality

Acceptance	Barangay Guitnang Bayan 1 and Barangay Sta. Ana	• Demonstrate readiness for deployment • Detect defects in user workflows	

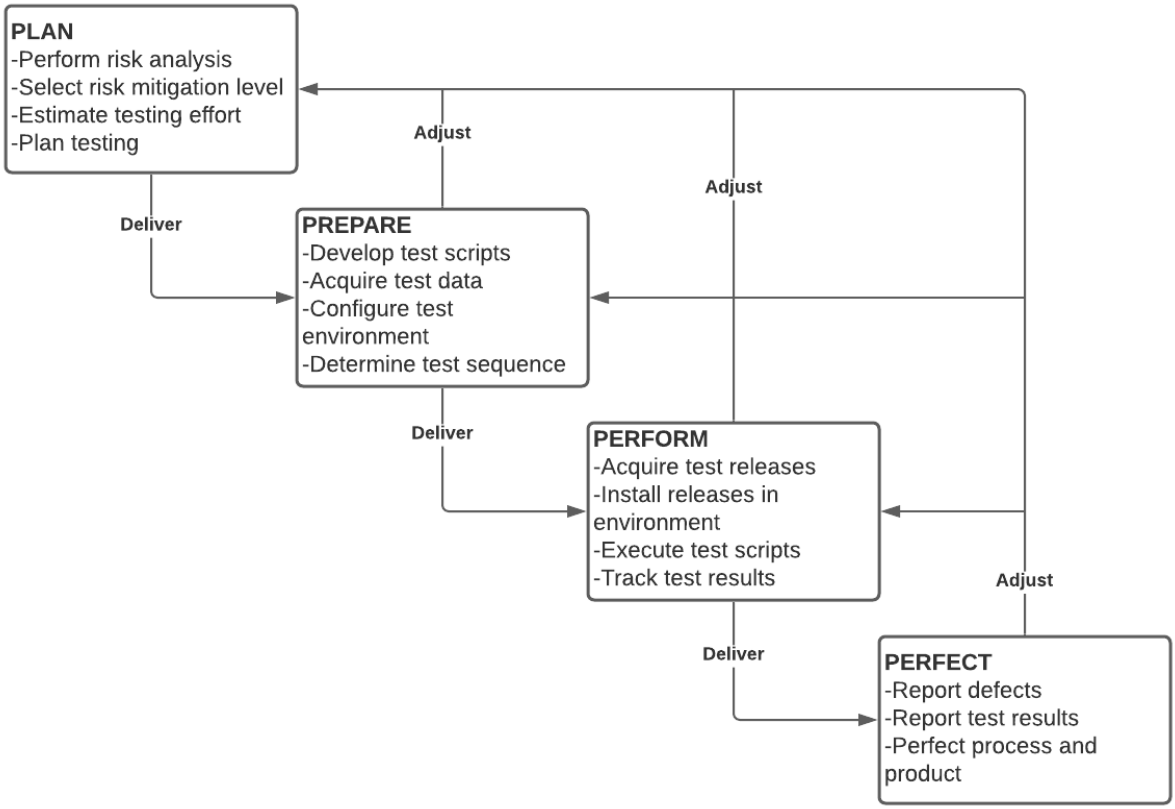


Figure 27: Test Process

In the illustrated test process, both Barangays were given the test timeline to accurately provide needed modules in the context of an adaptive barangay health information system. Most procedures required adjustments and consistent simulation due to the detailed process flow of healthcare providers.

Pursuant to the constitution, the Department of Health, Department of Science and Technology and Philippine Health Insurance Corporation curated the Health Privacy Code implementing joint administrative order no. 2016-0002 entitled "Privacy Guidelines for the Implementation of the Philippine Health Information Exchange.". Whereas, the proponent of the San Mateo Barangay Health Information System adhered to.

Collection and Processing of Health Information

The collection of data should be indicated in the consent form including competence, amount and accuracy of information, patient understanding and voluntariness.

To address this rule, the proponent, together with the barangay health workers and Health *Kagawad* restructured a consent form for patients that should be accomplished at the Barangay Health Center to help patients understand and be well-oriented on the context of the system and its services. (See appendix C for sample Consent form.)

Access of Health Information and Use and Disclosure of Health Information

This section under the health privacy code, discusses access of healthcare providers, accessible information for health care providers, approval of access, access of patient or client to medical records. This documentation was signed by the barangay health workers prior to their employment.

Physical Security

According the Rule #5 of the health privacy code, a maintenance and update on inventory will be the responsibility of all IT personnel assigned to the Barangay Health Center. This includes the following:

- Server Access
  Where, the proposed system for the barangay health centers of San Mateo Rizal will run in a paid web host housing its server. This also provides decent uptimes and fast loading times.

- Computer Access
  Where, since the system is running online, it can be remotely accessed whenever or wherever the client/health care providers are.

- Computer Loss and Bringing of devices outside the health facility
  Where, computer loss will not completely paralyze the operation of the system data it is being stored online. The health care providers and clients can also make use of their mobile devices to provide an input and record patient data.

- Bring your own device
  Where, in this context, barangay health officials are open to the idea of letting health care providers bring their own devices at work. Though, the area for work should be rebuilt for a more conducive working setup considering the latest development in their process.

Technical Safeguards

Rule number 6 of the Health Privacy Code includes access controls whereas technical policies and procedures for electronic information systems that maintain electronic protected health information are present to allow access only to persons or software programs that have been granted access rights. In this regard, the proponent included the following functions to the proposed system:

- Access Authorization (Addressable).

This function was defined by the Health Privacy Code as policies and procedures for granting access to electronic health information, such as those that relate to workstation access, transaction, program, process, and/or other mechanisms shall be implemented by PHCP.

Role-based Access Control

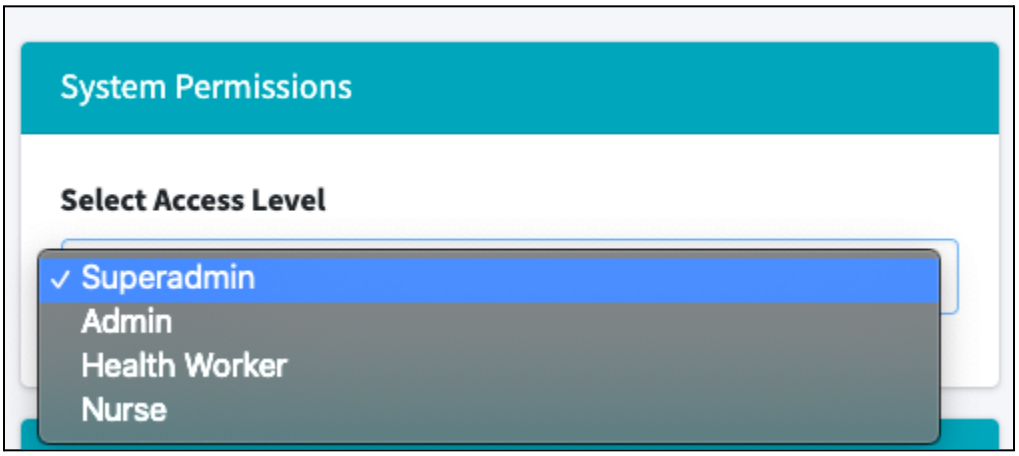


Figure 28: SMRBHIS Role-based Access Control Setting

This feature allows users of the barangay health information system to access functions specific to their tasks in the healthcare center.

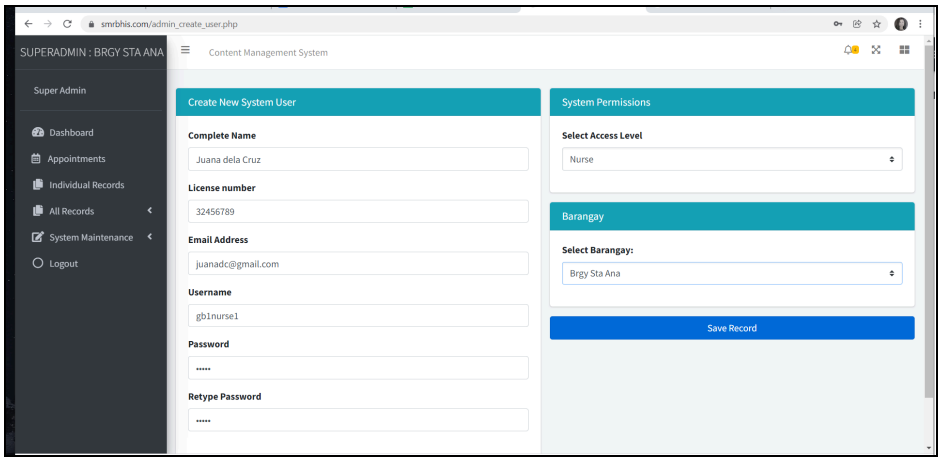


Figure 29: Only System Administrators can update the System User Details

In this function, the system administrator is the only one allowed to provide updates on different information for system users to empower restrictions and close monitoring.

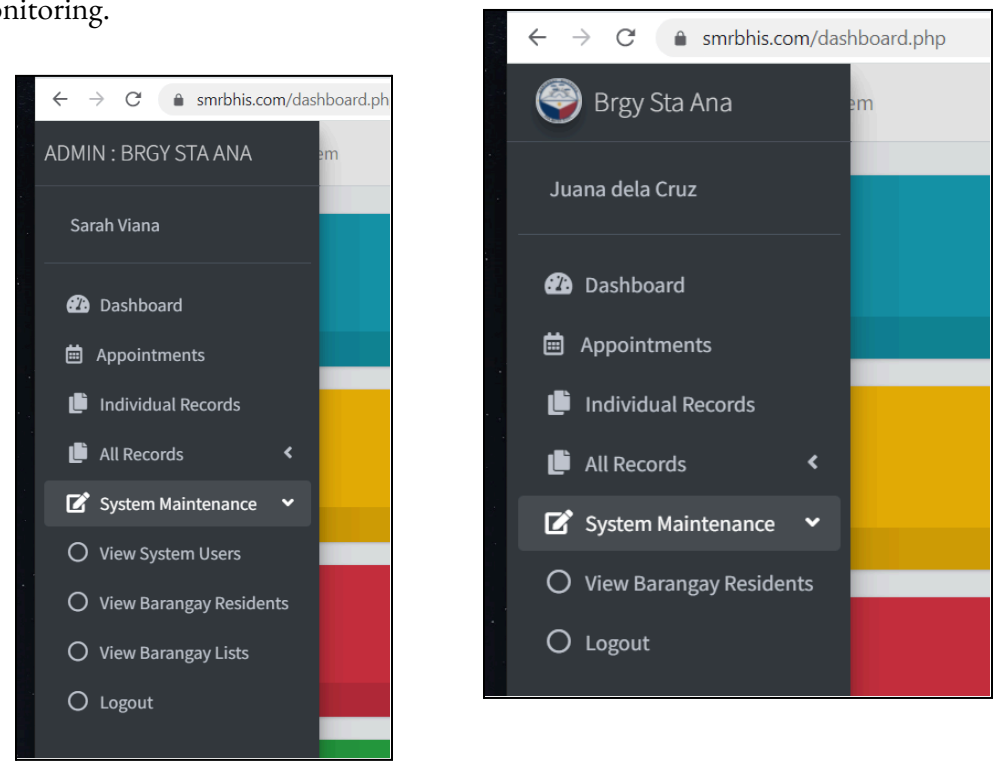


Figure 30: A comparative view of Nurse's System Maintenance Function vs. a System Administrator

- User Identification (Required)

The Health Privacy Code also required the user identification feature on electronic health information which should be made within the policy and procedure of the PHCP.

Below are the following guidelines provided by the aforementioned rule:

A. There shall be a user name and / or number for identifying user identity throughout all levels of the organization.
B. User identity shall not be shared, delegated or assigned to a group or individual.
C. User identity that was previously used shall not be reused for new and/or existing users.

The proponent added the following functionalities to adhere to this rule. Below are the different user identity features provided in the San Mateo Rizal Barangay Health Information System.

Password-protected user access that complies with facility password policies of the health facilities.

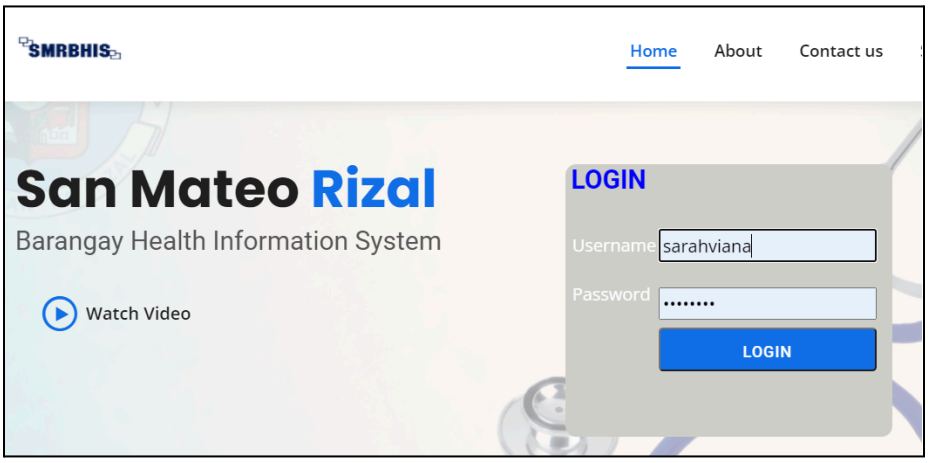


Figure 31: Password Protected User Access for Patients

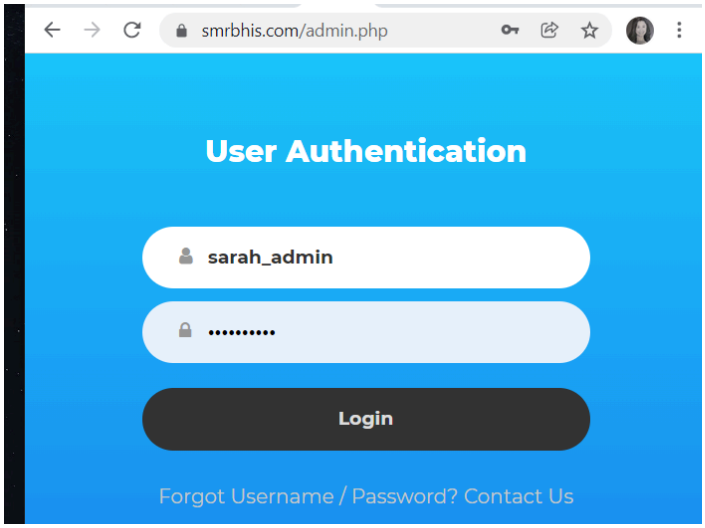


Figure 32: Password Protected User Access for System Administrators

- Encryption and Decryption (Addressable)

  According to the Philippine Health Policy Code, this feature is the method of converting an original message of regular text into encoded text using an algorithm.

  A. For encryption in transit, the standard security technology shall be SSL (Secure Sockets Layer).
  B. Minimum standard requirement for encryption shall be AES (Advanced Encryption Standard) 128.
  C. For encryption in storage, the standard shall be TKE (Trusted Key Entry).

To address the following security requirements, the proponent included the following:

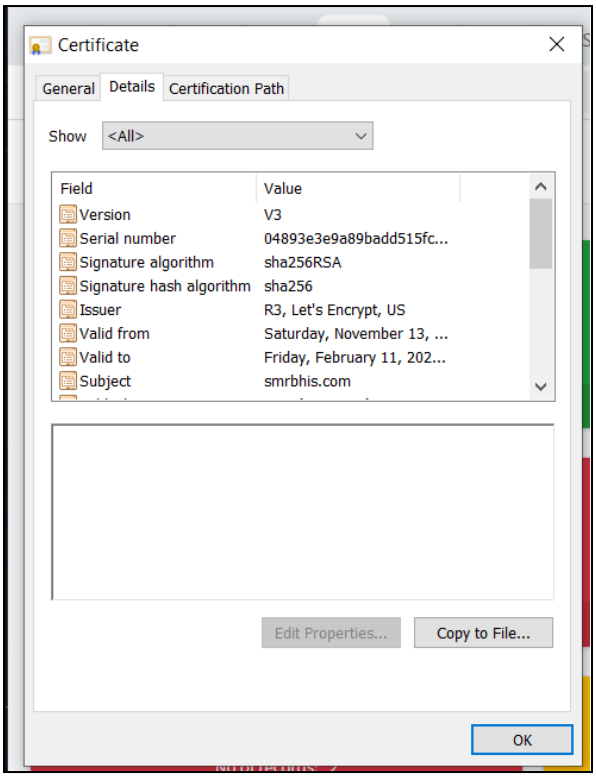


Figure 33: SMRBHIS Security Report Information

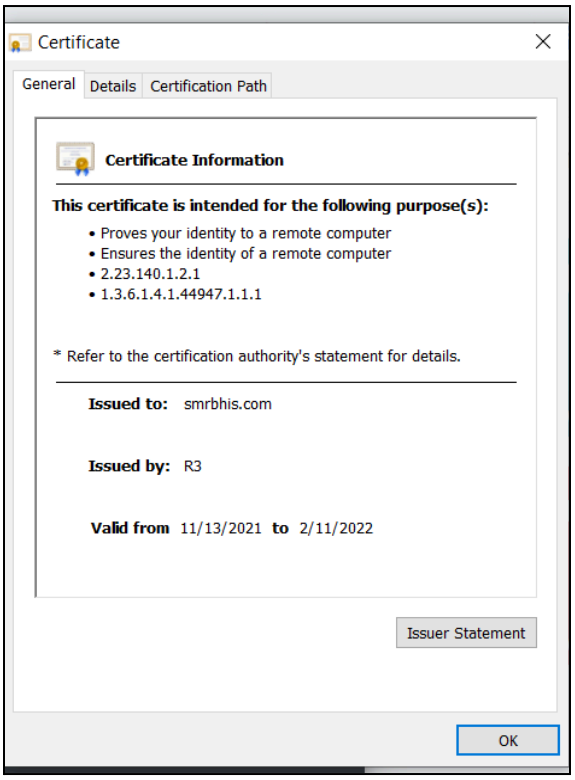


Figure 34: SMRBHIS Certificate

Figure 33 and Figure 34 provides the Certificate Information of the proposed system. It displayed SHA256 which is a more complex hashing algorithm (Ivanti, 2021). In evidence authentication, it is said that SHA256 is much more secure than the other common hashing algorithms considering that it is under the industry standard that is trusted by leading public-sectors.

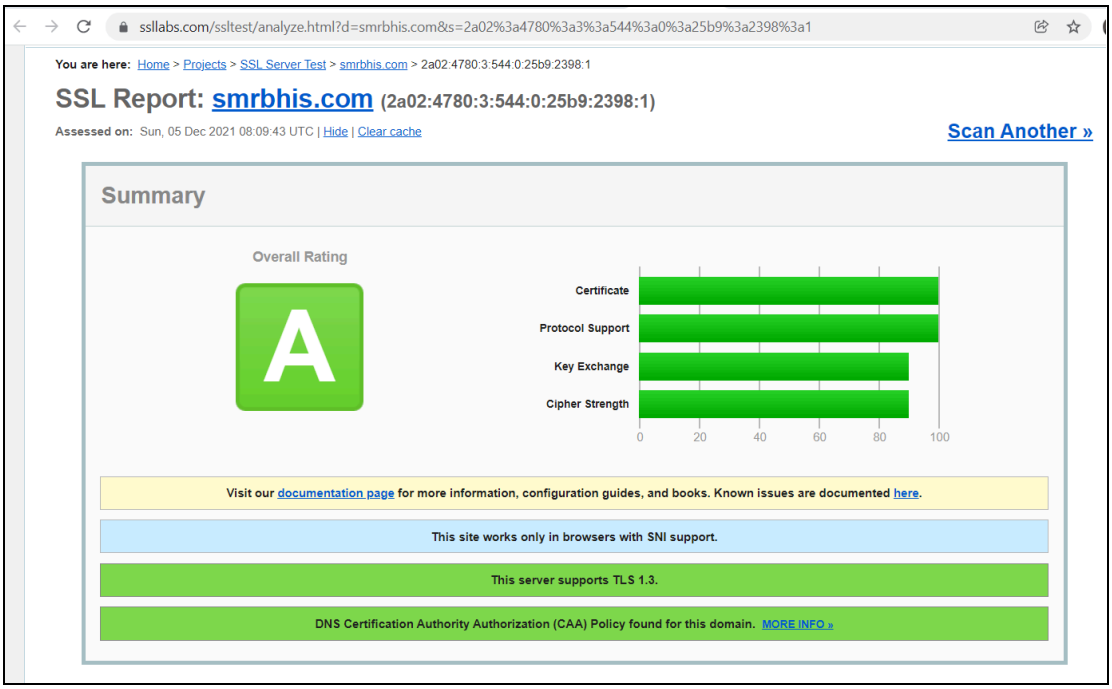


Figure 35: SMRBHIS SSL Report

As seen in the reports run through SSL Labs, the overall rating of SMRBHIS' ssl report is A. This also includes a report of the server support which is TLS 1.3 which is an upgraded version of SSL. Given this data, it is evident that the proposed system was able to coincide with the Health Privacy Code for Encryption and Decryption. See Appendix C for a more detailed report.

- Agreement

  Parallel with the bring your own device ideals, the barangay health officials also constructed a new policy through their internal documents. This will give health care providers the guidelines on the things needed to consider in bringing their own device at work.

- Training

  Uses of owned devices are also required to undergo annual training for the system
  updates and security.

CHAPTER 4

ANALYSIS AND INTERPRETATION OF DATA

This chapter discusses and interprets the responses of stakeholders to the series of testing and module evaluation for the proposed Adaptive Barangay Health Services Information System considering the success indicators: *portability, reliability, efficiency, usability, testability, understandability, modifiability and security* using the likert scale where 5 = Excellent, 4=Very Good,
3 =Good, 2 = Fair, and 1= Poor.

Barangay Guitnang Bayan 1 : Barangay Health Workers

Below are the tabulated responses of Barangay Health Workers from Barangay Guitnang Bayan 1, San Mateo, Rizal.

Table 8: Barangay Guitnang Bayan 1's response on Reliability

Reliability: Extent to which software performs according to requirements.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Save client information/profiles.	8	2				10
2. Record and update client vitals and diagnosis.	6	4				10
3. Help monitor patient history.	4	6				10
4. Lessen time spent in record finding and record keeping.	8	2				10
5. Allows ease of entry and exit of program, backup, change inputs and give commands.	7	3				10
6. Function as expected in multiple user levels.	5	5				10

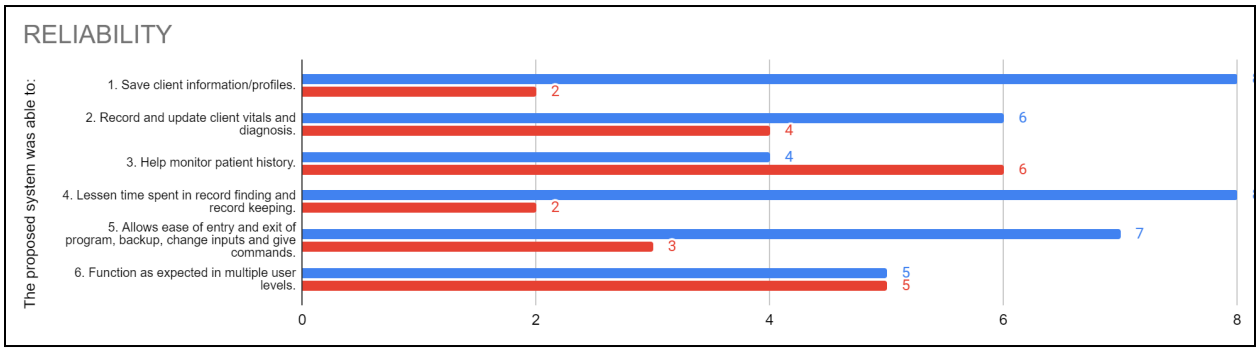


Chart 1: Barangay Guitnang Bayan 1's response on Reliability

Barangay Guitnang Bayan 1 rated reliability in the aspect of (1) saving of client information with majority on rating scale of 5, (2) record and updates vitals and diagnoses with majority on rating 5, (3) monitors patient history with majority of rating on 5, (4) lessens time in finding and record keeping with a rating of 5 from 8 participants, (5) ease of entry and program work around where majority of the participants chose a rating of 5 while on (6) functions for multi level users, there is a mutual set of observation for excellent and very good. For this table, the weighted mean is 4.63.

Table 9: Barangay Guitnang Bayan 1's response on Usability

Usability (Human Engineering): Extent of effort required to learn, operate and understand functions of the software.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Process requests as per transaction.	2	5	3			10
2. Generate reports per user needs.	10					10
3. Provides information useful in instruction.	6	2	2			10
4. Works efficiently with less/no assistance from the IT personnel.	1	6	3			10
5. Takes not much of a time for features familiarization.	5	5				10
6. Incorporate features that are essential in creating reports to be submitted to the Municipal Health Office.	8	2				10

7. Follow and improve the physical forms used on the current system.		6	4			10
8. Match the data required for consultations and patient endorsement to the Super Health Center.	2	6	2			10
9. Provide options for change in transaction such as child vaccination, maternity consultation etc.	10					10
10. Provide easy navigation to different pages on the online system based on the defined level of access.	10					10

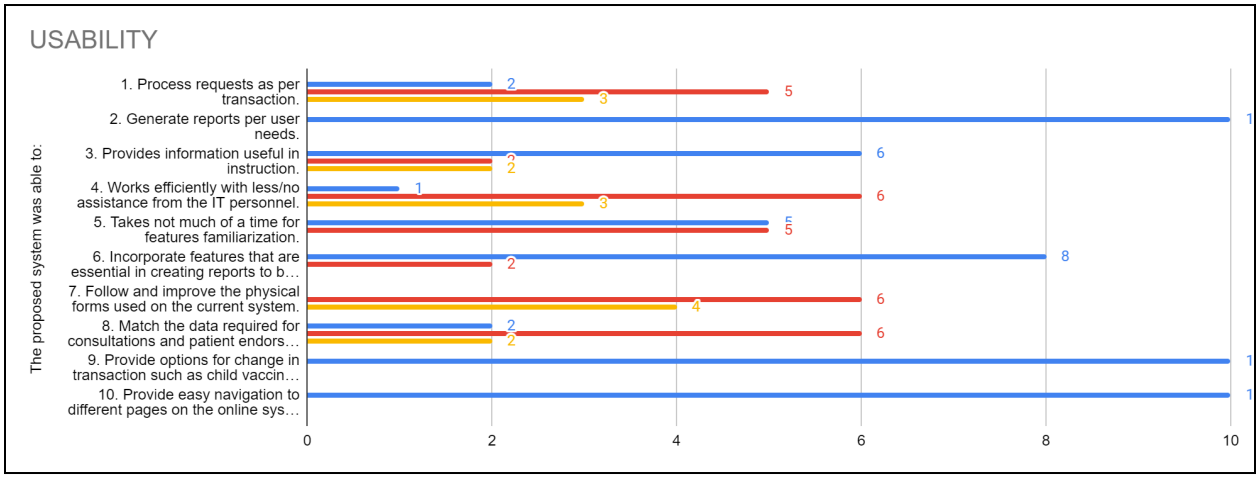


Chart 2: Barangay Guitnang Bayan 1's response on Usability

Barangay Guitnang Bayan 1's response on Usability were divided into ten (10) consideration where: (1) the process requests as per transaction's highest rating falls under 4 or *Very Good* with 5 participants; (2) Generate reports per user needs had a complete response of ten participants under the *Excellent* rating; (3) Provides information useful in instruction has majority of its responses on *Excellent* with 6 participants; (4) Works efficiently with less/no assistance from the IT personnel has six participants on the *Very Good* rating; (5) Takes not much of a time for features familiarization has an equal value of five for both *Excellent* and *Very Good*; (6) Incorporate features that are essential in creating reports to be submitted to the Municipal Health Office had eight participants on *Excellent;* (7) Follow and improve the physical forms used on the current system with six participants identifying it as *Very Good*; (8) Match the data required for consultations and patient endorsement to the Super Health Center with six participants on *Very Good* rating; (9) provide option for change in transaction

such as child vaccination, maternity consultation, etc. with all respondents under *Excellent;* (10) Provide easy navigation to different pages on the online system based on the defined level of access with ten respondents on *Excellent.* For this table, the weighted mean is 4.4.

Table 8: Barangay Guitnang Bayan 1's response on Usability

Understandability: Effort required for a user to recognize logical concepts and its applicability.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Evidently present functions separate for users (patient/barangay health workers).	10					10
2. Posts updates understandable both for patients and barangay health workers.	2	6	2			10
3. Procedures were relatively intuitive and understandable considering the features of the current system.	5	5				10
4. Training for system familiarity is appropriate and efficient enough.	5	5				10
5. The medium of instructions for steps in completing a task within the system is simple and accurate.	3	5	2			10

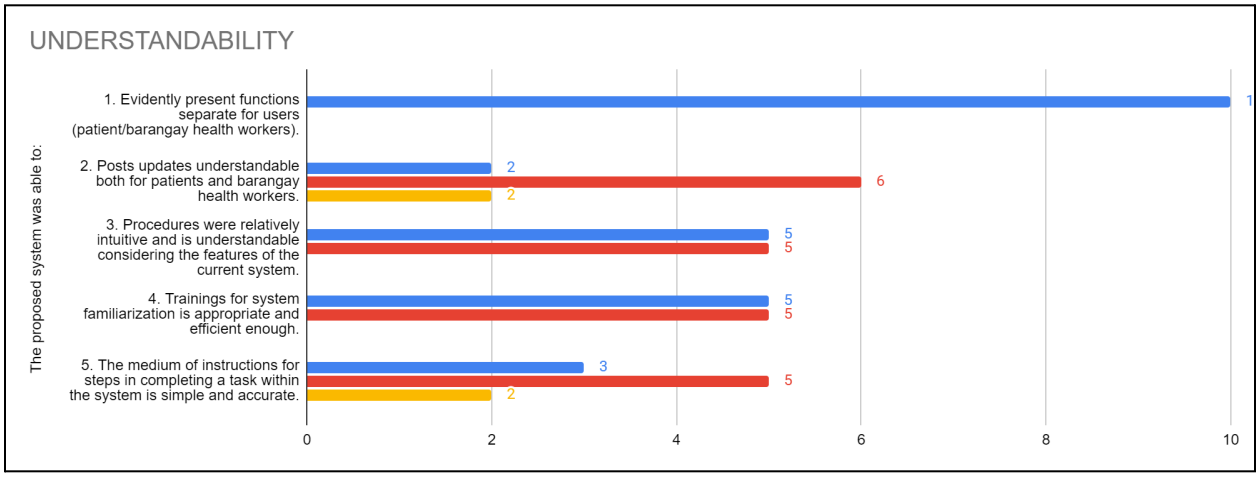


Chart 3: Barangay Guitnang Bayan 1's response on Understandability

Based on the response of Barangay Guitnang Bayan 1's Health workers, the effort required for a user to recognize logical concepts and the proposed system's applicability were: (1) Evidently presente functions separate for users (barangay health workers) had all responses under *Excellent*; (2) Posts update understandable both for patients and barangay health workers had majority of the participants on *Very Good* with six respondents; (3) procedures were relatively intuitive and understandable considering the features of the current system had an equal response for *Very Good* and *Excellent* with 5 respondents each; (4) Training for system familiarity is appropriate and efficient enough has the same result for *Very Good* and *Excellent* with a value of five; (6) the medium of instructions for steps in completing a task within the system is simple and accurate with a response of five under *Very Good*. For this table, the weighted mean is 4.18.

Table 10: Barangay Guitnang Bayan 1's response on Security

Security: Encompasses a holistic restriction based on access in information from authentication to modification both internally and externally.	5	4	3	2	1	
The proposed system/proponent was able to:	CRITERIA					TOTAL
1. Allow users to access the system based on the registered account and password.	10					10
2. Restrict access based on the list of barangays provided.	4	5	1			10
3. Limit the functionalities available per user.	5	5				10
4. Provide the notification based on the Data Sharing Agreement (DSA) for patients and Municipal Hall.	5	5				10
5. Follow the standard requirement for Online Implementation security based on the Barangay Health Services administrative personnel.	5	5				10

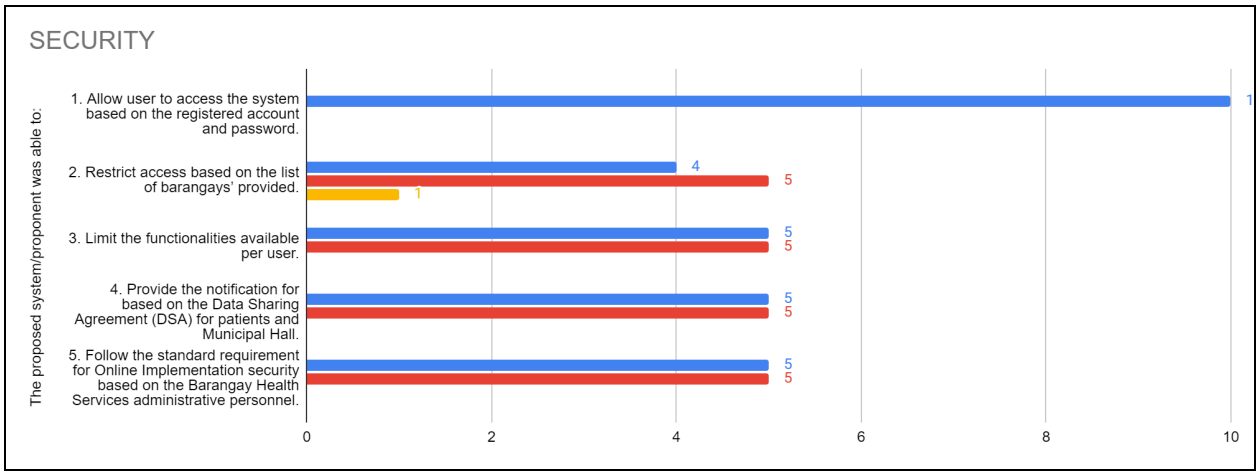


Chart 4: Barangay Guitnang Bayan 1's response on Security

For Security, the health workers of Barangay Guitnang Bayan 1 had the following responses per category. (1) Allow users to access the system based on the registered account and password with ten responses under *Excellent;* (2) Restrict access based on the list of barangays provided had five responses under *Very Good;* (3) Limit the functionalities available per user had an equal response both for *Very Good* and *Excellent* responses; (4) Provide the notification based on the Data Sharing Agreement for patient and Municipal hall has a response of five both for *Excellent* and *Very Good;* (5) Follow the standard requirements for Online Implementation security based on the Barangay Health Services administrative personnel both have the response of five for *Very Good* and *Excellent.* For this table, the weighted mean is 4.5.

Table 11: Barangay Guitnang Bayan 1's response on Support

Support: Covers support for users of the proposed system in different levels in formats such as written materials, training and cost.	5	4	3	2	1	
The proposed system/proponent was able to:	CRITERIA					TOTAL
1. Provide manuals and explain the content and functionalities of the system.	8	2				10
2. Facilitate an on-site training for administrative successors of the system.	5	5				10

3. Facilitate reassignment of user level based on the ability of manpower in the barangay health center.	4	6				10

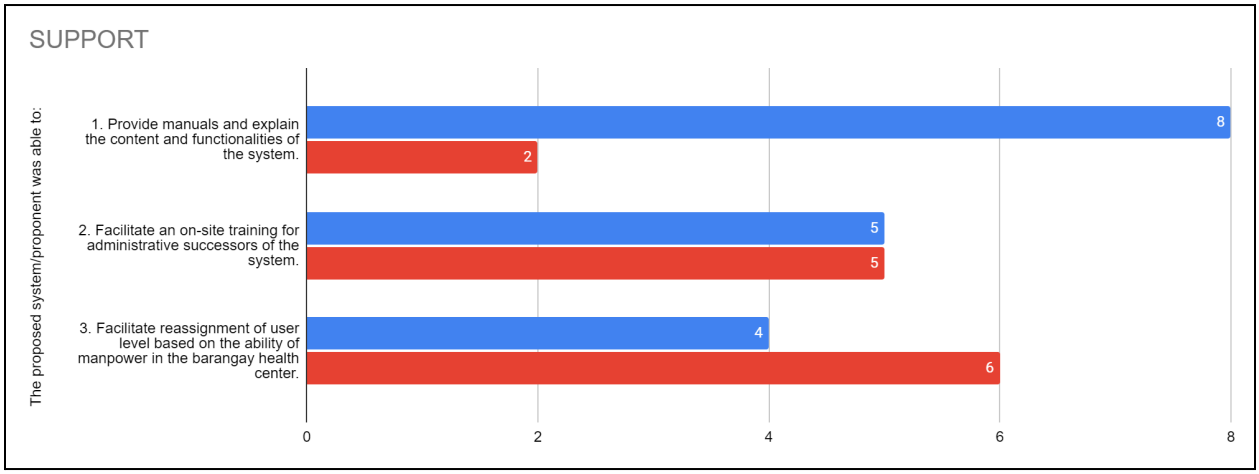


Chart 5: Barangay Guitnang Bayan 1's response on Support

Support for the Barangay Guitnang Bayan 1 were divided into different categories as well with the following results: (1) Provide manuals and explain the content and functionalities of the system with eight respondents on *Excellent;* (2) Facilitate an on-site training for administrative successors of the system with both five (5) on *Excellent* and *Very Good;* (3) Facilitate reassignment of user level based on the ability of manpower in the barangay health center had six respondents on a *Very Good* rating. For this table, the weighted mean is 4.57.

Table 12: Barangay Guitnang Bayan 1's Mean Average

Barangay Guitnang Bayan 1		
CRITERIA	MEAN	INTERPRETATION
Reliability	4.63	Very Adaptable
Usability	4.4	Very Adaptable
Understandability	4.18	Very Adaptable
Security	4.5	Very Adaptable
Support	4.57	Very Adaptable

The mean responses of Barangay Guitnang Bayan 1's barangay health services staff were interpreted as Very Adaptable based on the categories provided.

Barangay Sta. Ana : Barangay Health Workers

Table 13: Barangay Sta. Ana's response on Support

Reliability: Extent to which software performs according to requirements.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Save client information/profiles.	5	5				10
2. Record and update client vitals and diagnosis.	4	6				10
3. Help monitor patient history.	4	6				10
4. Lessen time spent in record finding and record keeping.	5	5				10
5. Allows ease of entry and exit of program, backup, change inputs and give commands.	3	7				10
6. Function as expected in multiple user levels.	4	4	2			10

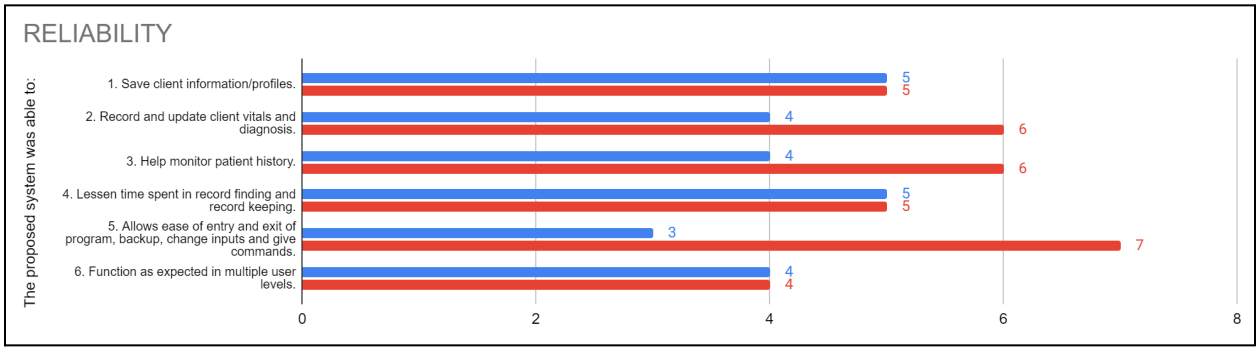


Chart 6: Barangay Sta. Ana's response on Reliability

Barangay Sta. Ana's response on Reliability has the result of (1) Save client information/profiles with equal rating for *Excellent* and *Very Good*, (2) Record and update

client vitals and diagnosis, (3) help monitor patient history with six participants for *Very Good* (4) Lessen time spent in record finding and record keeping, with equal ratings of five on *Excellent* and *Very Good,* (5) Allows ease of entry and exit of program, backup, change inputs and give commands with a rating of seven for *Very Good,* the majority of participants, (6) Function as expected in multiple user levels, with an equal value of four both for *Excellent* and *Very Good.*

Table 14: Barangay Sta. Ana's response on Usability

Usability (Human Engineering): Extent of effort required to learn, operate and understand functions of the software.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Process requests as per transaction.	3	5	2			10
2. Generate reports per user needs.	10					10
3. Provides information useful in instruction.	4	4	2			10
4. Works efficiently with less/no assistance from the IT personnel.	3	3	4			10
5. Takes not much of a time for features familiarization.	4	4	2			10
6. Incorporate features that are essential in creating reports to be submitted to the Municipal Health Office.	5	5				10
7. Follow and improve the physical forms used on the current system.	5	4	1			10
8. Match the data required for consultations and patient endorsement to the Super Health Center.	5	3	2			10
9. Provide options for change in transaction such as child vaccination, maternity consultation etc.	10					10
10. Provide easy navigation to different pages on the online system based on the defined level of access.		5	5			10

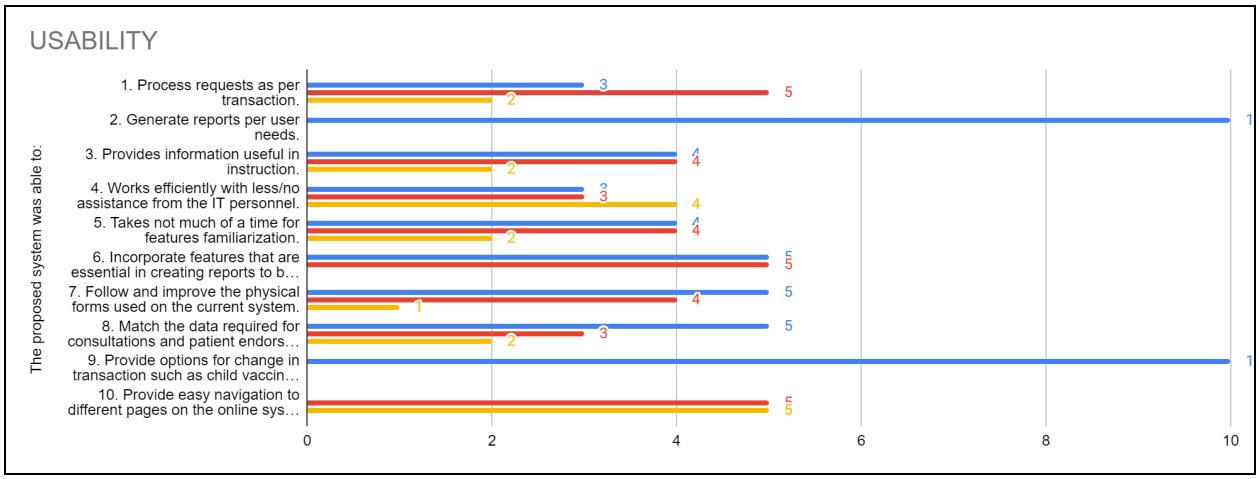


Chart 7: Barangay Sta. Ana's response on Usability

Barangay Sta. Ana's response to Usability had the result of (1) Process requests as per transaction, with a result of five for *Very Good,* (2) Generate reports per user needs, with ten participants for *Excellent,* (3) Provides information useful in instructions with an equal rating of four for both *Excellent* and *Very Good,* (4) Works efficiently with less/no assistance from the IT personnel (5) Takes not much of a time for features familiarization, with a response of four as the highest rating under *Good* , (6) Incorporate features that are essential in creating reports to be submitted to the Municipal Health Office with a result of five both for *Excellent* and *Very Good.* (7) Follow and improve the physical forms used on the current system, where most responses landed on *Excellent* with five participants, (8) Match the data required for consultations and patient endorsement to the Super Health Center, where five participants chose *Excellent* as a rating, (9) Provide options for change in transaction such as child vaccination, maternity consultation, etc. with a result of ten under *Excellent,* (10) Provide easy navigation to different pages on the online system based on the defined level access, participants chose *Very Good* and *Good* with both 5 participants each.

Table 15: Barangay Sta. Ana's response on Understandability

Understandability: Effort required for a user to recognize logical concepts and its applicability.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Evidently present functions separate for users (patient/barangay health workers).	10					10
2. Posts updates understandable both for patients and barangay health workers.	3	5	2			10
3. Procedures were relatively intuitive and is understandable considering the features of the current system.	5	5				10
4. Trainings for system familiarization is appropriate and efficient enough.	3	7				10
5. The medium of instructions for steps in completing a task within the system is simple and accurate.	8	2				10

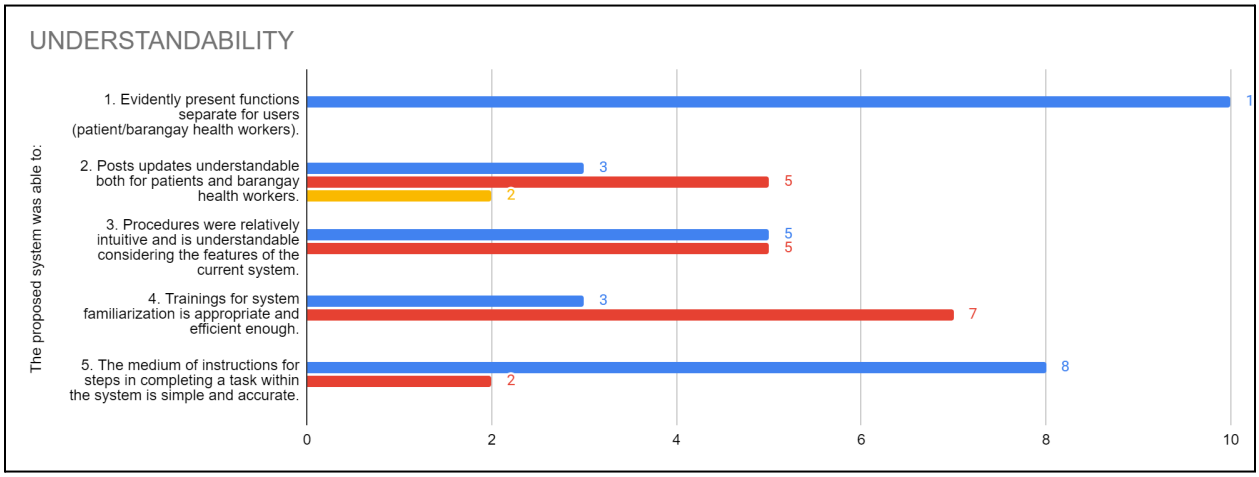


Chart 8: Barangay Sta. Ana's response on Understandability

Barangay Sta. Ana's response on understandability are as follows: (1) Evidently present functions separate for users (patient/barangay health workers) has the rating of *Excellent* with ten participants, (2) Post updates understandable both for patients and barangay health workers with a rating of five under *Very Good*, (3) Procedures were relatively intuitive and is understandable considering the features of the current system where participants have equal

response for *Excellent* and *Very Good*. (4) Training for system familiarization is appropriate and efficient enough with a rating of seven for *Very Good*, (5) the medium of instructions for steps in completing a task within the system is simple and accurate with a rating of *Excellent* based on eight participants.

Table 16: Barangay Sta. Ana's response on Security

Security: Encompasses a holistic restriction based on access in information from authentication to modification both internally and externally.	5	4	3	2	1	
The proposed system/proponent was able to:	CRITERIA					TOTAL
1. Allow user to access the system based on the registered account and password.	10					10
2. Restrict access based on the list of barangays' provided.	6	3	1			10
3. Limit the functionalities available per user.	5	5				10
4. Provide the notification based on the Data Sharing Agreement (DSA) for patients and Municipal Hall.	7	3				10
5. Follow the standard requirement for Online Implementation security based on the Barangay Health Services administrative personnel.	5	5				10

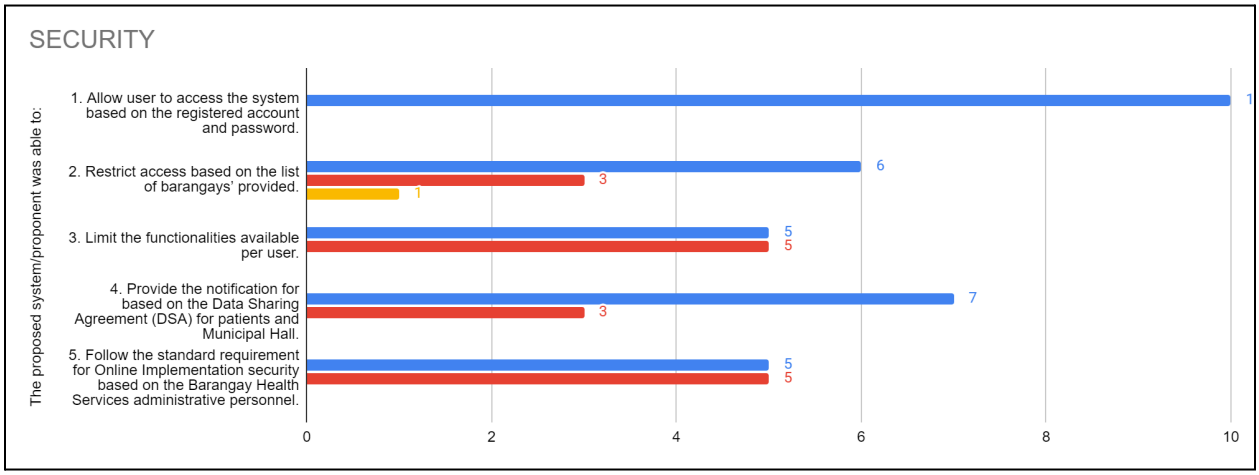


Chart 9: Barangay Sta. Ana's response on Security

Barangay Sta. Ana responded on the Security category of this system based on (1) Allow users to access the system based on the registered account and password, with a response of ten under *Excellent* (2) Restrict access based on the list of barangays provided had a result of six participants under *Excellent,* (3) Limit the functionalities available per user with a response of five both for *Excellent* and *Very Good,* (4) Provide the notification based on the Data Sharing Agreement (DSA) for patients and Municipal Hall with a response of seven under *Excellent,* (5) Follow the standard requirement for Online Implementation security based on the Barangay Health Services administrative personnel with a response of both 5 participants for *Excellent* and *Very Good.*

Table 17: Barangay Sta. Ana's response on Support

Support: Covers support for users of the proposed system in different levels in formats such as written materials, training and cost.	5	4	3	2	1	
The proposed system/proponent was able to:	CRITERIA					TOTAL
1. Provide manuals and explain the content and functionalities of the system.	6	4				10
2. Facilitate an on-site training for administrative successors of the system.	3	7				10
3. Facilitate reassignment of user level based on the ability of manpower in the barangay health center.	6	3	1			10

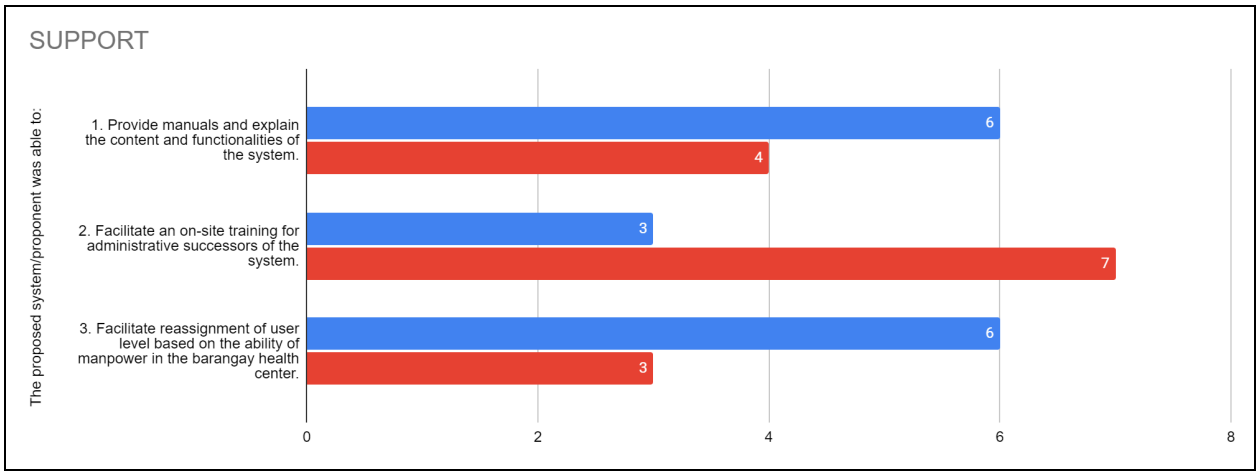


Chart 10: Barangay Sta. Ana's response on Support

Barangay Sta. Ana's response on Support are as follows: (1) Provide manuals and explain the content and functionalities of the system has a response of six for *Excellent* (2) Facilitates an on-site training for administrative successors of the system, where seven participants chose *Very Good,* (3) Facilitate reassignment of user level based on the ability of manpower in the Barangay Health Center, with a response of six participants for *Excellent.*

Table 18: Barangay Sta. Ana's Mean Average

Barangay Sta. Ana		
CRITERIA	MEAN	INTERPRETATION
Reliability	4.38	Very Adaptable
Usability	4.31	Very Adaptable
Understandability	4.42	Very Adaptable
Security	4.58	Very Adaptable
Support	4.37	Very Adaptable

The mean responses of Barangay Sta. Ana's barangay health services staff were interpreted as Very Adaptable based on the categories provided.

Table 19: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Reliability

Reliability: Extent to which software performs according to requirements.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Save client information/profiles.	13	7				20
2. Record and update client vitals and diagnosis.	10	10				20
3. Help monitor patient history.	10	10				20
4. Lessen time spent in record finding and record keeping.	13	7				20
5. Allows ease of entry and exit of program, backup, change inputs and give commands.	10	10				20
6. Function as expected in multiple user levels.	9	9	2			20

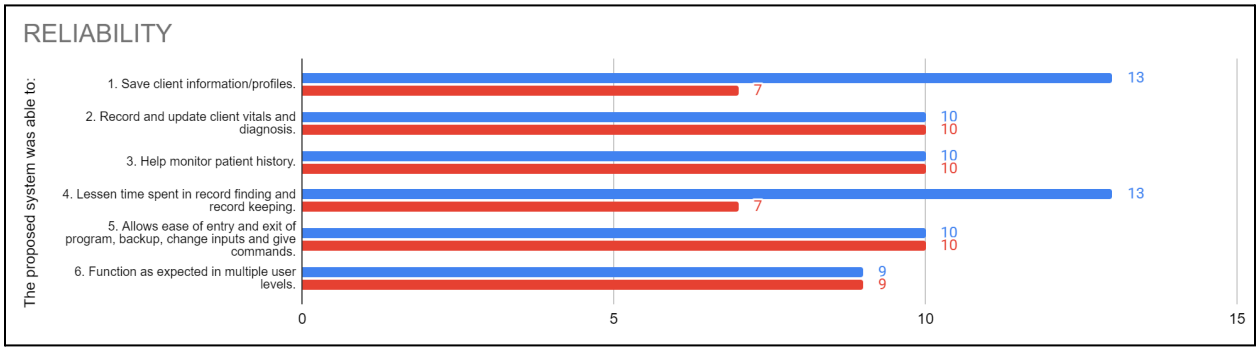


Chart 11: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Reliability

The combined response of Barangay Guitnang Bayan 1 and Barangay Sta. Ana on Reliability to express the extent of software performance according to requirements expressed that both barangay agreed as manifested on the number of participants leaning under the rating of *Excellent* and *Very Good*.

Table 20: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Usability

Usability (Human Engineering): Extent of effort required to learn, operate and understand functions of the software.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Process requests as per transaction.	5	10	5			20
2. Generate reports per user needs.	20					20
3. Provides information useful in instruction.	10	6	4			20
4. Works efficiently with less/no assistance from the IT personnel.	4	9	7			20
5. Takes not much of a time for features familiarization.	9	9	2			20
6. Incorporate features that are essential in creating reports to be submitted to the Municipal Health Office.	13	7				20
7. Follow and improve the physical forms used on the current system.	5	10	5			20
8. Match the data required for consultations and patient endorsement to the Super Health Center.	7	9	4			20
9. Provide options for change in transaction such as child vaccination, maternity consultation etc.	20					20
10. Provide easy navigation to different pages on the online system based on the defined level of access.	10	5	5			20

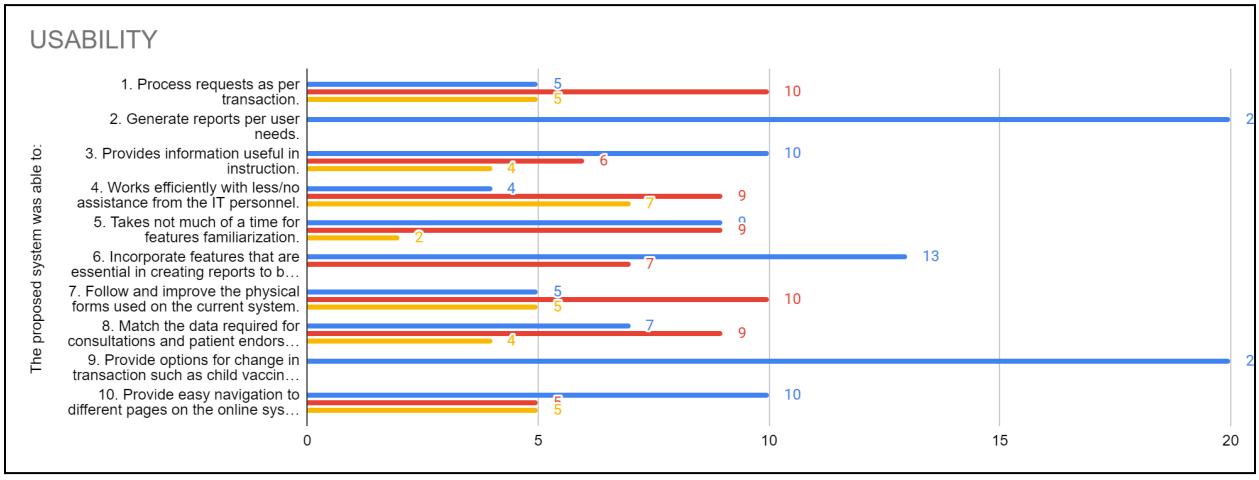


Chart 12: Barangay Guitnang Bayan 1 and Barangay Sta. Ana's Response to the proposed system's Usability

The joint responses of Barangay Guitnang Bayan 1 and Barangay Sta. Ana under the Usability category manifested different views. Unlike reliability, usability has provided a perspective where participants chose *Good* at some aspects, formulating varied points of view.

Table 21: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Understandability

Understandability: Effort required for a user to recognize logical concepts and its applicability.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Evidently present functions separate for users (patient/barangay health workers).	20					20
2. Posts updates understandable both for patients and barangay health workers.	5	11	4			20
3. Procedures were relatively intuitive and is understandable considering the features of the current system.	10	10				20
4. Trainings for system familiarization is appropriate and efficient enough.	8	12				20
5. The medium of instructions for steps in completing a task within the system is simple and accurate.	11	7	2			20

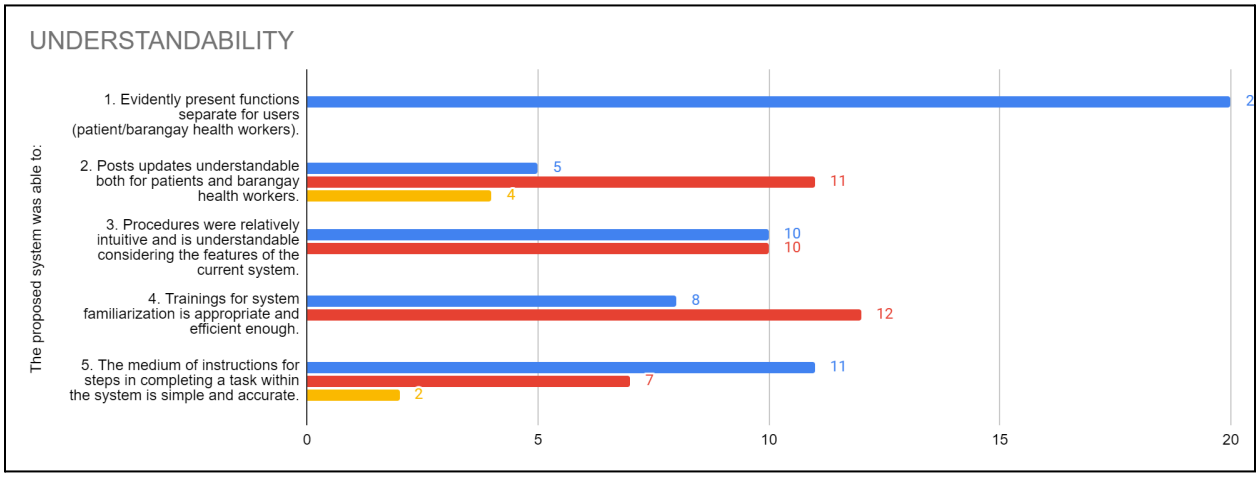


Chart 13: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Understandability

Based on the responses for both Barangay Guitnang Bayan 1 and Barangay Sta. Ana, both were able to provide a wide range of perspectives on how they understand the system. Both had the majority of responses on *Excellent* and *Very Good* while there were six responses that landed on *Good*.

Table 22: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Security

Security: Encompasses a holistic restriction based on access in information from authentication to modification both internally and externally.	5	4	3	2	1	
The proposed system/proponent was able to:	CRITERIA					TOTAL
1. Allow user to access the system based on the registered account and password.	20					20
2. Restrict access based on the list of barangays' provided.	9	8	2			19
3. Limit the functionalities available per user.	10	10				20
4. Provide the notification for based on the Data Sharing Agreement (DSA) for patients and Municipal Hall.	12	8				20
5. Follow the standard requirement for Online Implementation security based on the Barangay Health Services administrative personnel.	10	10				20

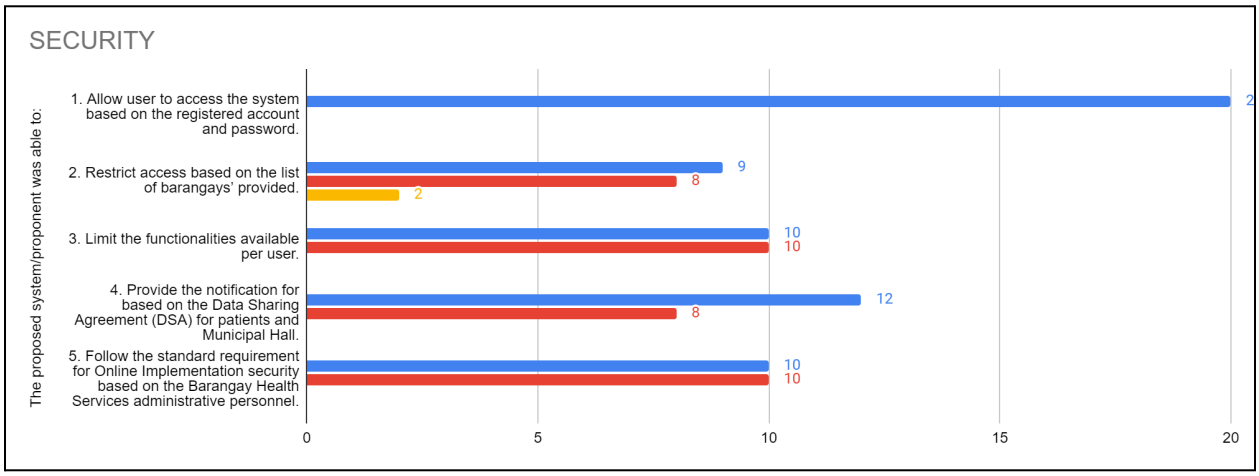


Chart 14: Barangay Guitnang Bayan 1 and Barangay Sta. Ana's Response to the proposed system's Security

The responses for both Barangay Guitnang Bayan 1 and Barangay Sta. Ana manifested that participants agreed on the appropriateness of Security rendered by the proposed system in the context of health services where the majority chose *Excellent* and *Very Good.*

Table 23: Barangay Guitnang Bayan 1 and Barangay Sta. Ana's Response to the proposed system's Support

Support: Covers support for users of the proposed system in different levels in formats such as written materials, training and cost.	5	4	3	2	1	
The proposed system/proponent was able to:	CRITERIA					TOTAL
1. Provide manuals and explain the content and functionalities of the system.	14	6				20
2. Facilitate an on-site training for administrative successors of the system.	8	12				20
3. Facilitate reassignment of user level based on the ability of manpower in the barangay health center.	10	9	1			20

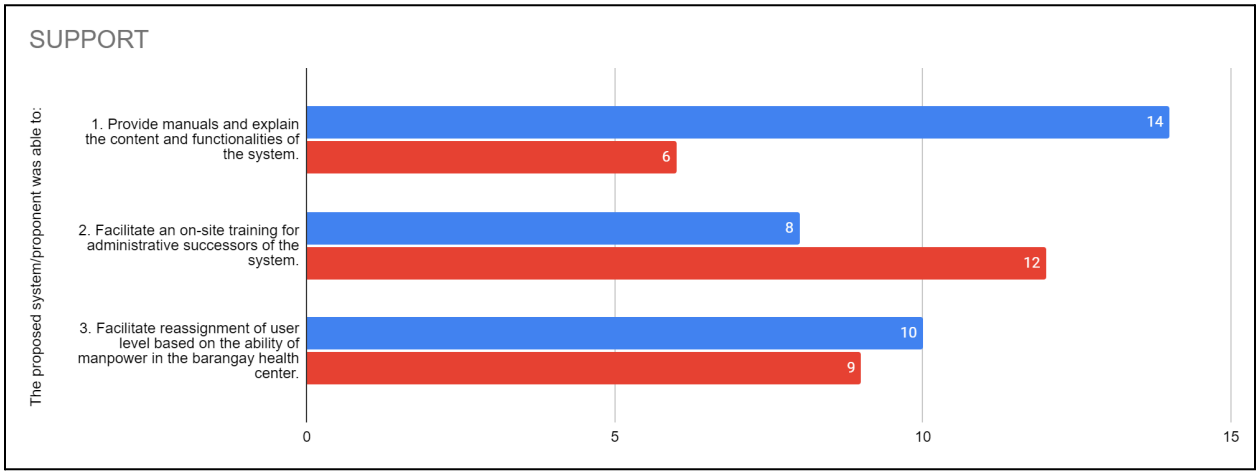


Chart 15: Barangay Guitnang Bayan and Barangay Sta. Ana's Response to the proposed system's Security

Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Support which covers assistance to help participants and end-users in using the system manifested substantially. The results were presented on *Excellent* and *Very Good.*

Table 24: Barangay Guitnang Bayan 1 and Barangay Sta. Ana's Combined Mean Average

Guitnang Bayan 1 and Sta. Ana		
CRITERIA	MEAN	INTERPRETATION
Reliability	4.53	Very Adaptable
Usability	4.36	Very Adaptable
Understandability	4.48	Very Adaptable
Security	4.55	Very Adaptable
Support	4.52	Very Adaptable

The combined mean evaluation of both Barangay Guitnang Bayan 1 and Barangay Sta. Ana presented that the system is adaptable based on their assessment.

Citizens of Barangay Guitnang Bayan 1

Table 25: Citizens of Barangay Guitnang Bayan 1's response on Reliability

Reliability: Extent to which software performs according to requirements.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Save client information/profiles.	10	12	28			50
2. Lessen time spent in record finding and record keeping.	23	5	22			50
3. Allows ease of entry and exit of program, backup, change inputs and give commands.	26	14	10			50

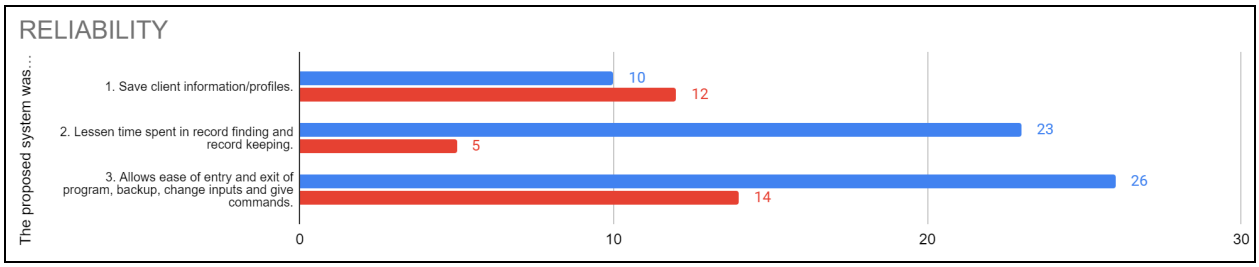


Chart 16: Citizens of Barangay Guitnang Bayan 1's response on Reliability

The citizens of Barangay Guitnang Bayan 1 has responded on reliability where (1) Save client information/profiles function of the citizen were rated with *Good* based on twenty eight participants (2) twenty-three responded *Excellent* for lesser time spent in record finding and record keeping, while (3) twenty-six responded on *Excellent* for ease of entry and exit of program, backup, change of inputs and giving of commands.

Table 26: Citizens of Barangay Guitnang Bayan 1's response on Usability

Usability (Human Engineering): Extent of effort required to learn, operate and understand functions of the software.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Process requests as per transaction.	14	33	3			50
2. Provides information useful in instruction.	24	16	10			50
3. Works efficiently with less/no assistance from the IT personnel.	15	12	20			47
4. Takes not much of a time for features familiarization.	25	10	15			50
5. Match the data required for consultations and patient endorsement to the Super Health Center.	16	20	14			50
6. Provide options for change in transaction such as child vaccination, maternity consultation etc.	20	10	20			50

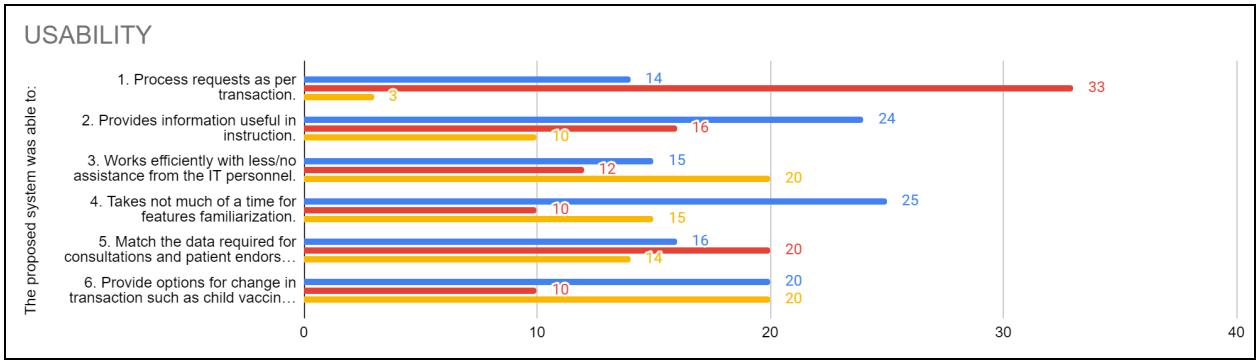


Chart 17: Citizens of Barangay Guitnang Bayan 1's response on Usability

For usability, the citizens of Guitnang Bayan 1 responded where (1) thirty-three participants chose *Very Good* for process of requests as per transaction, (2) twenty-four respondents chose *Excellent* for the provision of information useful in instruction, for (3) works efficiently with less/no assistance from the IT personnel, twenty participants chose *Good,* (4) takes not much of a time for features familiarization, twenty-five participants answered *Excellent,* (5) match the data required for consultation and patient endorsement to the Super Health Center got twenty residents on *Very Good,* (6) provision of options for change in transaction such as child vaccination, maternity consultatio, etc. got an equal response of twenty both for *Excellent* and *Very Good.*

Table 27: Citizens of Barangay Guitnang Bayan 1's response on Understandability

Understandability: Effort required for a user to recognize logical concepts and its applicability.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Posts updates understandable both for patients and barangay health workers.	33	10	6	1		50
2. Procedures were relatively intuitive and is understandable considering the features of the current system.	10	25	15			50
3. Trainings for system familiarization is appropriate and efficient enough.	28	12	10			50
4. The medium of instructions for steps in completing a task within the system is simple and accurate.	18	22	8			48

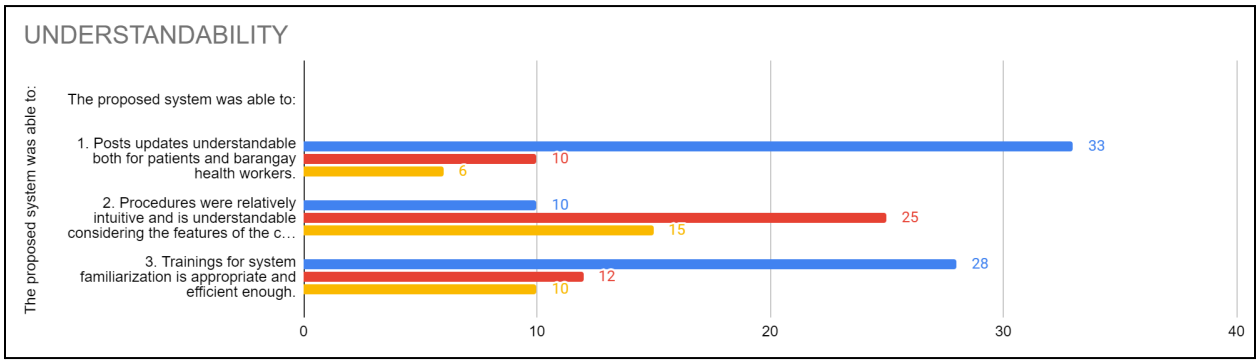


Chart 18: Citizens of Barangay Guitnang Bayan 1's response on Understandability

For understandability, citizens of Barangay Guitnang Bayan 1's responses were (1) posts updates understandable for patients and barangay health workers, got thirty three participants choosing *Excellent,* (2) procedures were relatively intuitive and is understandable considering the features of the current system got twenty eight responses for *Excellent,* (3) the medium of instructions for step in completing a task within the system is simple and accurate got twenty two for *Very Good.*

Table 28: Citizens of Barangay Guitnang Bayan 1's response on Security

Security: Encompasses a holistic restriction based on access in information from authentication to modification both internally and externally.	5	4	3	2	1	
The proposed system/proponent was able to:	CRITERIA					TOTAL
1. Allow users to access the system based on the registered account and password.	50	0	0	0		50
2. Restrict access based on the list of barangays' provided.	30	20	0			50
3. Provide the notification based on the Data Sharing Agreement (DSA) for patients and Municipal Hall.	16	34	0			50

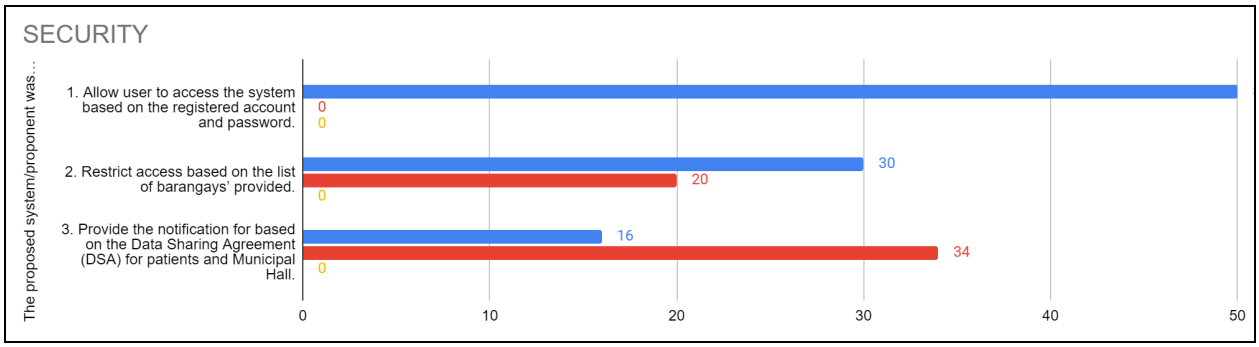


Chart 19: Citizens of Barangay Guitnang Bayan 1's response on Security

The citizens of Barangay Guitnang Bayan 1 responded to the security feature questions of the system where (1) access of users to the system was based on the account and password got fifty participants on *Excellent,* (2) restricted access based on the list of barangays provided got thirty participants on *Excellent,* (3) provision of notification based on the Data Sharing Agreement (DSA) for patients and Municipal Hall.

Table 29: Citizens of Barangay Guitnang Bayan 1's Average Mean

Citizens of Barangay Guitnang Bayan 1		
CRITERIA	MEAN	INTERPRETATION
Reliability	3.99	Very Adaptable
Usability	4.07	Very Adaptable
Understandability	4.20	Very Adaptable
Security	4.64	Very Adaptable

The average means for each category as per the response of Barangay Guitnang Bayan 1 citizens were identified as Very Adaptable based on the user module.

Table 30: Citizens of Barangay Sta. Ana's response on Reliability

Reliability: Extent to which software performs according to requirements.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Save client information/profiles.	32	14	3	1		50
2. Lessen time spent in record finding and record keeping.	18	20	12			50
3. Allows ease of entry and exit of program, backup, change inputs and give commands.	27	12	6	5		50

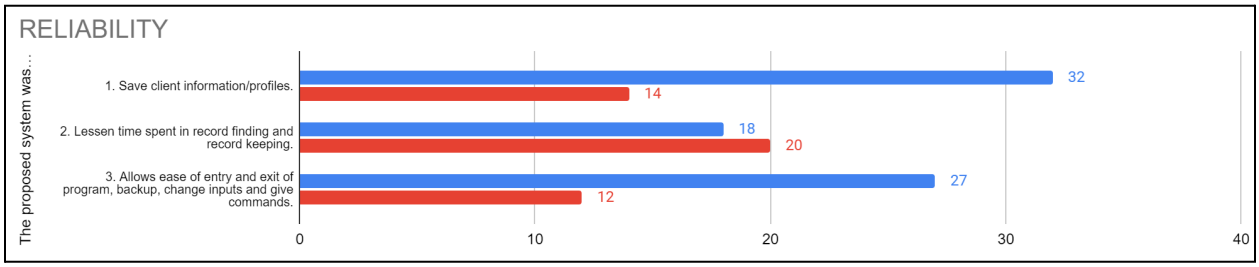


Chart 20: Citizens of Barangay Barangay Sta. Ana's response on Reliability

The citizens of barangay Sta. Ana responded on system reliability where (1) saving of client information / profiles got thirty two participants under *Excellent,* (2) lessen time spent in record finding and record keeping got twenty participants on *Very Good* and eighteen on *Excellent,* (3) ease of entry and exit of program, backup, change inputs and give commands got twenty seven participants lean on *Excellent.*

Table 31: Citizens of Barangay Barangay Sta. Ana's response on Usability

Usability (Human Engineering): Extent of effort required to learn, operate and understand functions of the software.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Process requests as per transaction.	12	25	11	2		50

2. Provides information useful in instruction.	16	20	13	1		50
3. Works efficiently with less/no assistance from the IT personnel.	31	16	3			50
4. Takes not much of a time for features familiarization.	15	20	10	5		50
5. Match the data required for consultations and patient endorsement to the Super Health Center.	22	16	10	2		50
6. Provide options for change in transaction such as child vaccination, maternity consultation etc.	20	25	5			50

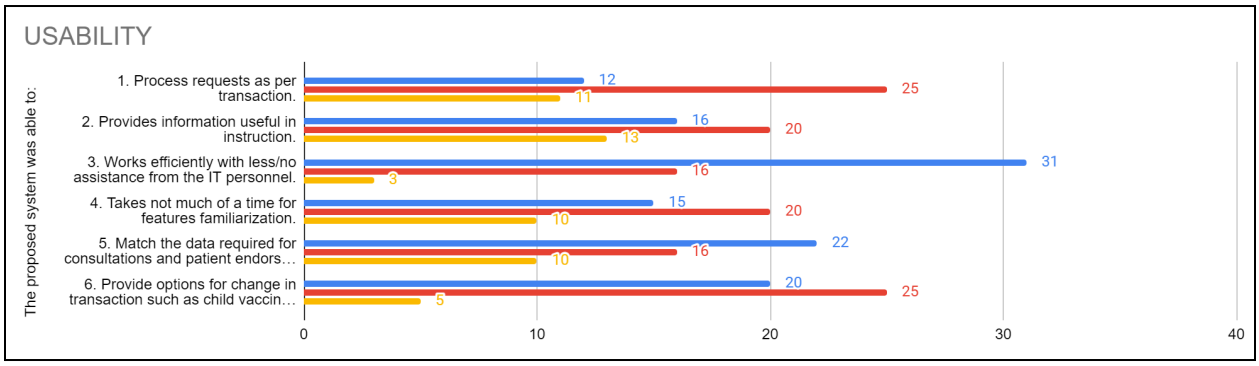


Chart 21: Citizens of Barangay Barangay Sta. Ana's response on Usability

For the citizens of Barangay Sta. Ana, the usability of the proposed system based (1) process requests as per transaction, where twelve participants chose *Excellent,* twenty five participants chose *Very Good,* (2) provision of information useful for transaction where sixteen participants stated that it is *Excellent,* twenty chose *Very Good* and thirteen on *Good,* (3) works efficiently with less/no assistance from the IT personnel got thirty one responses for *Excellent* and sixteen on *Very Good,* (4) takes not much of a time for familiarization to different system features got fifteen responses for *Excellent* and twenty on *Very Good,* (5) match the data required for consultation and patient endorsement to the Super Health Center got a response of twenty two for *Excellent,* (6) provision of options for change in transaction such as a child vaccination, maternity consultation etc. got twenty responses on *Excellent* and twenty five for *Very Good.*

Table 32: Citizens of Barangay Barangay Sta. Ana's response on Understandability

Understandability: Effort required for a user to recognize logical concepts and its applicability.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Posts updates understandable both for patients and barangay health workers.	32	6	12			50
2. Procedures were relatively intuitive and understandable considering the features of the current system.	15	25	10			50
3. Training for system familiarization is appropriate and efficient enough.	16	9	25			50
4. The medium of instructions for steps in completing a task within the system is simple and accurate.	23	15	12			50

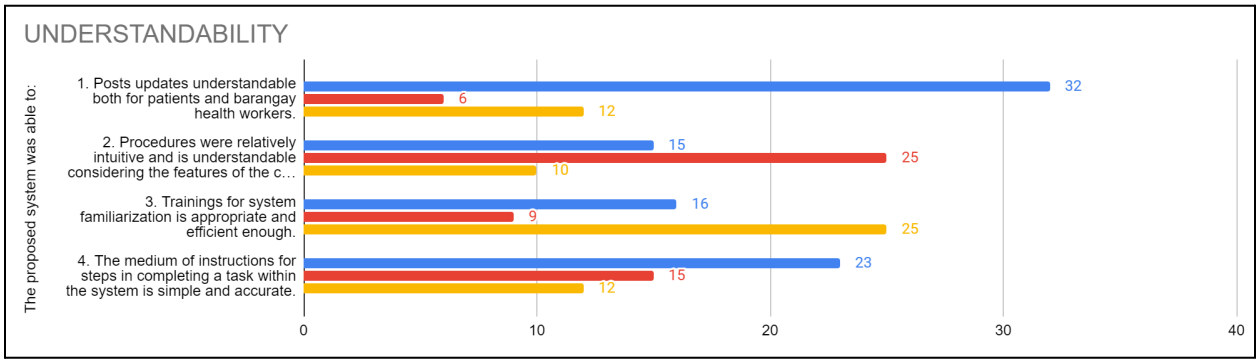


Chart 22: Citizens of Barangay Barangay Sta. Ana's response on Understandability

For Understandability, the citizens of Barangay Sta. Ana manifested (1) posts updates understandable both for patients and barangay health workers with thirty two participants on *Excellent,* six on *Very Good* and twelve on *Good.* (2) for procedures were relatively intuitive and understandable considering the features of the current system, fifteen chose *Excellent,* twenty five chose *Very Good* and ten chose *Good* (3) training for system familiarization is appropriate and efficient enough with sixteen participants on *Excellent,* fifteen on *Very Good* and twelve on *Good.*

Table 33: Citizens of Barangay Barangay Sta. Ana's response on Security

Security: Encompasses a holistic restriction based on access in information from authentication to modification both internally and externally.	5	4	3	2	1	
The proposed system/proponent was able to:	CRITERIA					TOTAL
1. Allow users to access the system based on the registered account and password.	45	5				50
2. Restrict access based on the list of barangays' provided.	24	26				50
3. Provide the notification based on the Data Sharing Agreement (DSA) for patients and Municipal Hall.	24	12	14			50

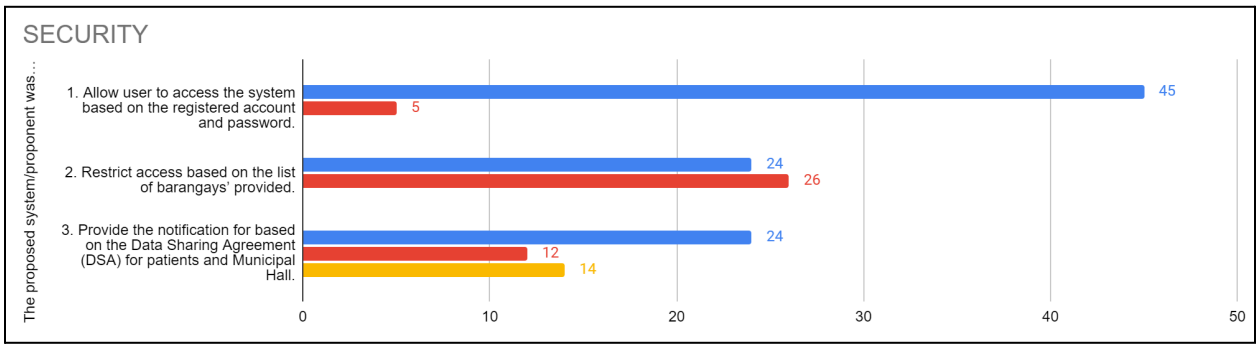


Chart 23: Citizens of Barangay Sta. Ana's response on Security

For Security, the citizens of Barangay Sta. Ana had their responses as, (1) Allow users to access the system based on the registered account and password has forty five participants on *Excellent* and five on *Good,* (2) for restrict access based on the list of barangays provided, twenty four participants chose *Excellent,* and twenty six participants on *Very good,* (3) provides the notifications based on the DSA for patient and municipal Hall, twenty four participants chose *Excellent,* twelve on *Very Good* and fourteen on *Good.*

Table 34: Citizens of Barangay Barangay Sta. Ana's Average Mean

Citizens of Brgy. Sta. Ana		
CRITERIA	MEAN	INTERPRETATION
Reliability	4.29	Very Adaptable
Usability	4.15	Very Adaptable
Understandability	4.14	Very Adaptable
Security	4.53	Very Adaptable

Based on the average mean, participants or citizens of Barangay Sta. Ana has rated the proposed system as very adaptable considering Reliability, Usability, Understandability and Security.

Citizens of Barangay Guitnang Bayan 1 and Barangay Sta. Ana

Table 35: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Reliability

Reliability: Extent to which software performs according to requirements.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Save client information/profiles.	42	26	31	1		100
2. Lessen time spent in record finding and record keeping.	41	25	34			100
3. Allows ease of entry and exit of program, backup, change inputs and give commands.	53	26	16	5		100

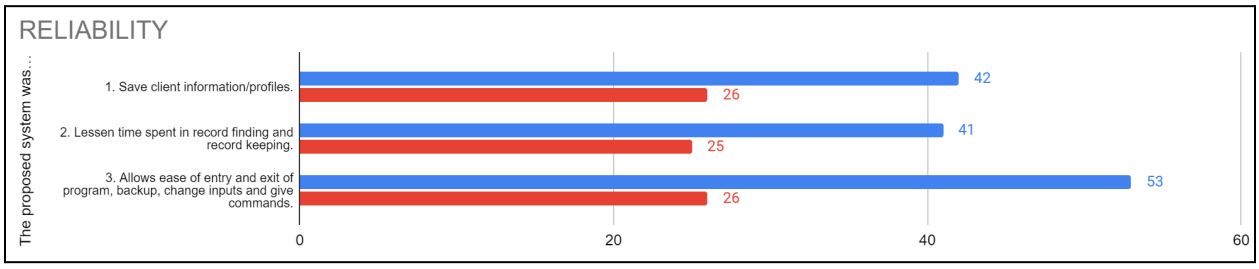


Chart 24: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Reliability

For reliability based on the citizens of Barangays Guitnang Bayan 1 and Sta. Ana, it is manifested that the majority referred to the saving of client information and profiles as Excellent with 42 responses, while there's 26 for *Very Good,* 31 for *Good* and one for *Fair*. On the second item, which talks about how to lessen time spent in record finding and record keeping, the majority responded on *Excellent* while the rest falls under *Very good* and *Good*. For the last item in this category, the majority of the participants chose *Excellent* for ease of entry and exit of program, backup, change inputs and command inputs.

Table 36: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Usability

Usability (Human Engineering): Extent of effort required to learn, operate and understand functions of the software.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Process requests as per transaction.	26	58	14	2		100
2. Provides information useful in instruction.	40	36	23	1		100
3. Works efficiently with less/no assistance from the IT personnel.	46	30	23			99
4. Takes not much of a time for features familiarization.	40	30	25	5		100
5. Match the data required for consultations and patient endorsement to the Super Health Center.	38	36	24	2		100
6. Provide options for change in transaction such as child vaccination, maternity consultation etc.	40	35	25			100

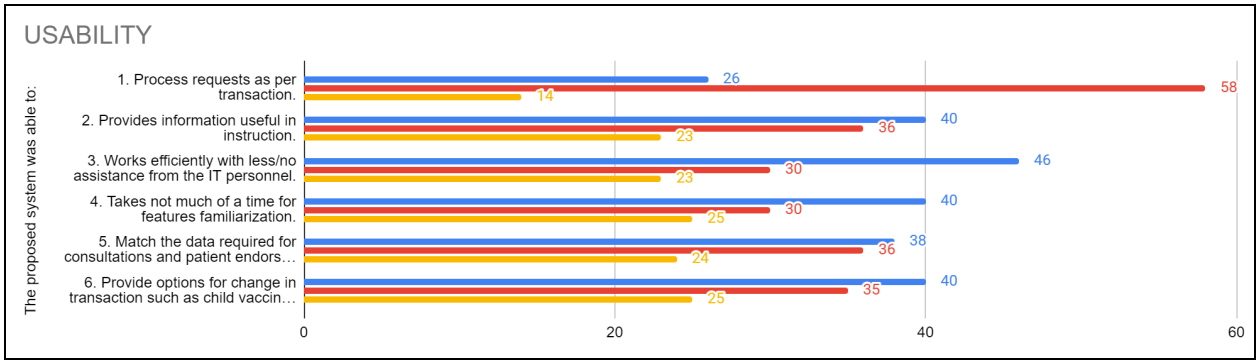


Chart 25: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Usability

For usability as referred by the citizens of Guitnang Bayan 1 and Sta. Ana, majority of the population responded for *Excellent* except for process requests as per transaction which has fifty-eight responses.

Table 37: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Understandability

Understandability: Effort required for a user to recognize logical concepts and its applicability.	5	4	3	2	1	
The proposed system was able to:	CRITERIA					TOTAL
1. Procedures were relatively intuitive and understandable considering the features of the current system.	25	50	25			100
2. Training for system familiarization is appropriate and efficient enough.	44	21	35			100
3. The medium of instructions for steps in completing a task within the system is simple and accurate.	41	37	20			98

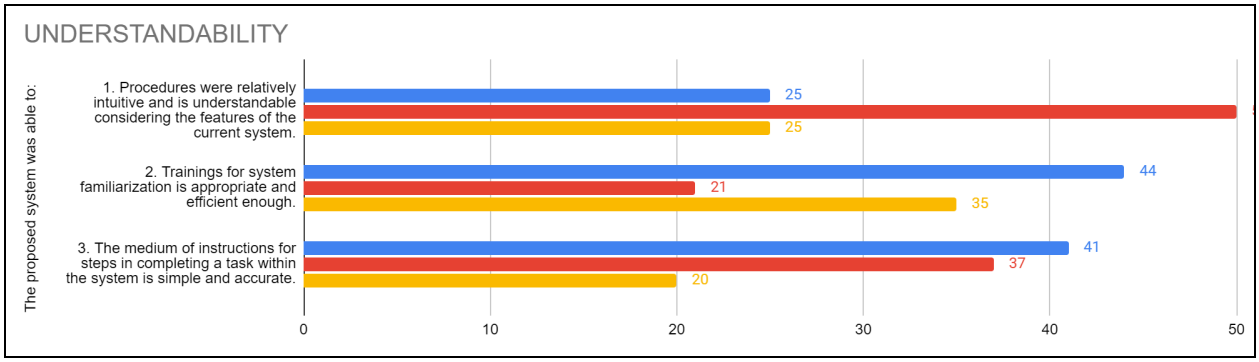


Chart 26: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Understandability

For understandability on both barangays,more than 40 respondents chose *Excellent* for training for system familiarization and medium of instruction. While on the first item, which are procedures that were relatively intuitive and understandable considering the features of the current system, most participants selected *Very Good*.

Table 38: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Security

Security: Encompasses a holistic restriction based on access in information from authentication to modification both internally and externally.	5	4	3	2	1	
The proposed system/proponent was able to:	CRITERIA					TOTAL
1. Allow user to access the system based on the registered account and password.	95	5				100
2. Restrict access based on the list of barangays' provided.	54	46				100
3. Provide the notification for based on the Data Sharing Agreement (DSA) for patients and Municipal Hall.	40	46	14			100

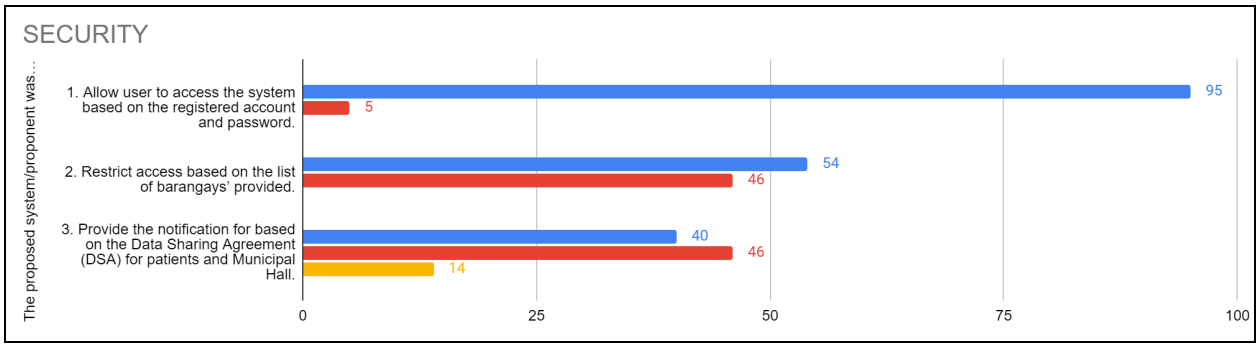


Chart 27: Citizens of Barangay Barangay Guitnang Bayan 1 and Barangay Sta. Ana's response on Security

According to the citizens of both barangays, the security feature that allowed them to access the system based on the registered account and password is *Excellent* with a response value of 95. The restricted access has 54 responses on *Excellent,* and the notification based on Data Sharing Agreement for patient and municipal hall got the highest points on *Very Good.*

User-System Adaptability

The proponent referred to Zeng, et.al (2016) in evaluating the user-system adaptability for the proposed system's functionalities.

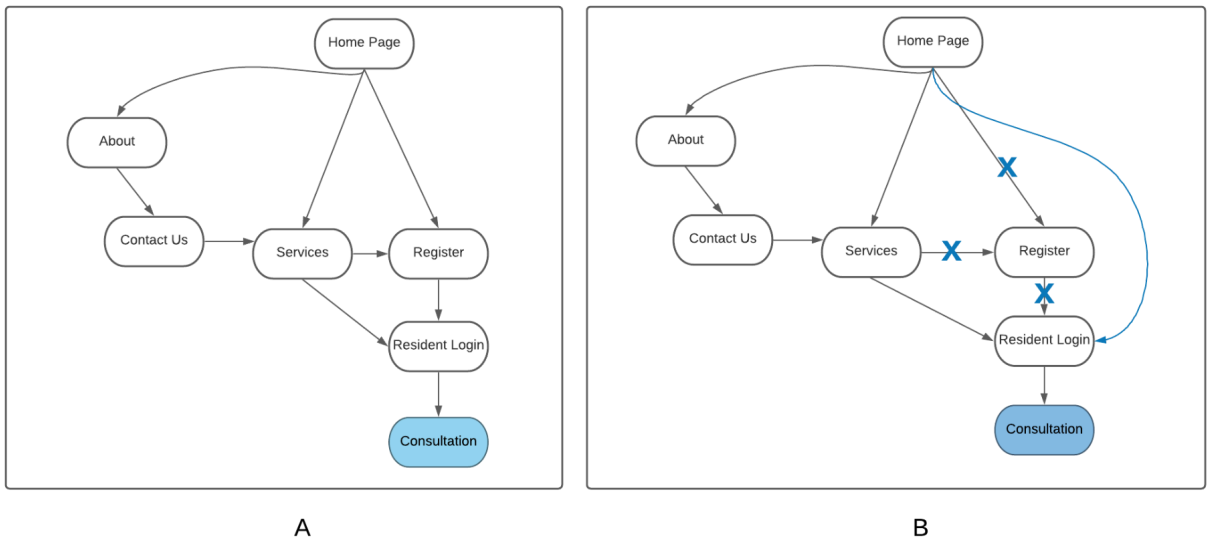


Figure 36: Use of Access Efficiency Model to evaluate the U-S Adaptability in Adaptation Process for user login and consultation for patients

To start, the proponent indicated in figure __ the adaptive procedures held for user login and consultation. At the first phase of user access, there are two (2) inlinks that should be crossed before reaching the desired page as indicated in box A. Then, with constant consultation and observation, the proponent adjusted the system to another clickstream for users to adapt firmly in the process that is cognitively fitted with the system process.

The users were able to navigate through the login process of the system more efficiently and with ease, as observed, since it doesn't ask them to cross the registration path once again even if they already have their login credentials. As presented in box B of figure 36, the registration and service inlinks can be bypassed to reach the consultation page.

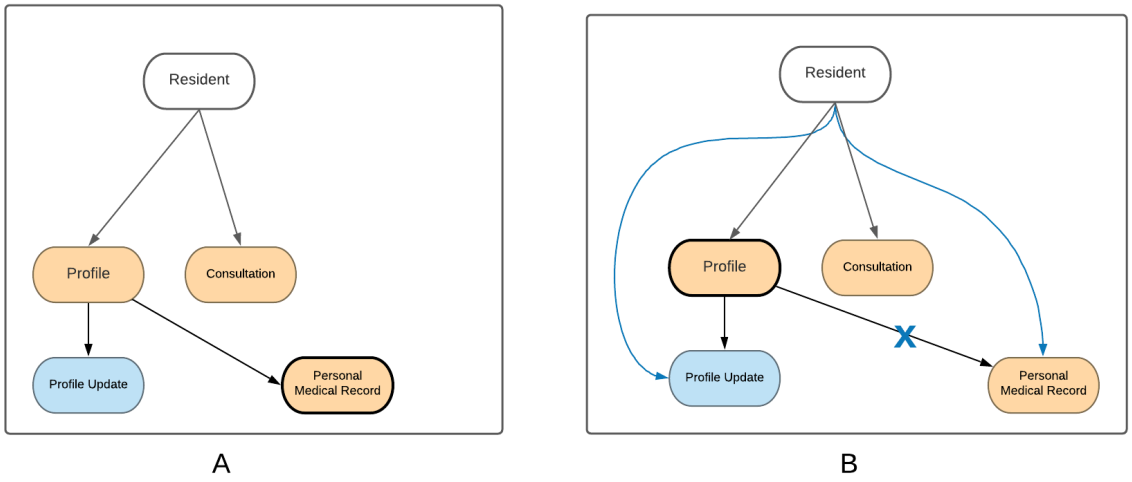


Figure 37: Use of Access Efficiency Model to evaluate the U-S Adaptability in Adaptation Process for access to medical records and profile update for patients

Box A of figure 37 displayed the initial procedure that users had to undergo before accessing the personal medical record and updating their profile. It was observed that most patients are challenged in viewing their medical record right away without passing through the profile path first. This is also manifested in the profile update path.

To address the concern and properly set up the structure of the proposed system in the most organized way possible, the proponent shortened the inlinks path and provided the link options in the page easily accessible for patients as system users - presented in Box B of figure 37.

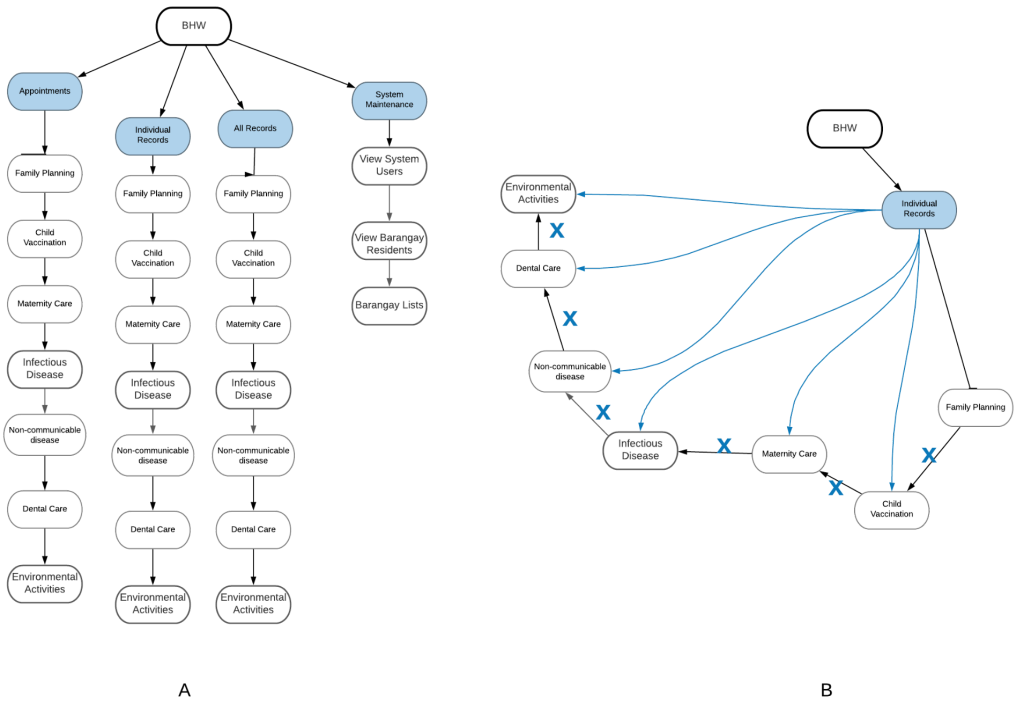


Figure 38: Use of Access Efficiency Model to evaluate the U-S Adaptability in Adaptation Process for access to individual records for Barangay Health Workers

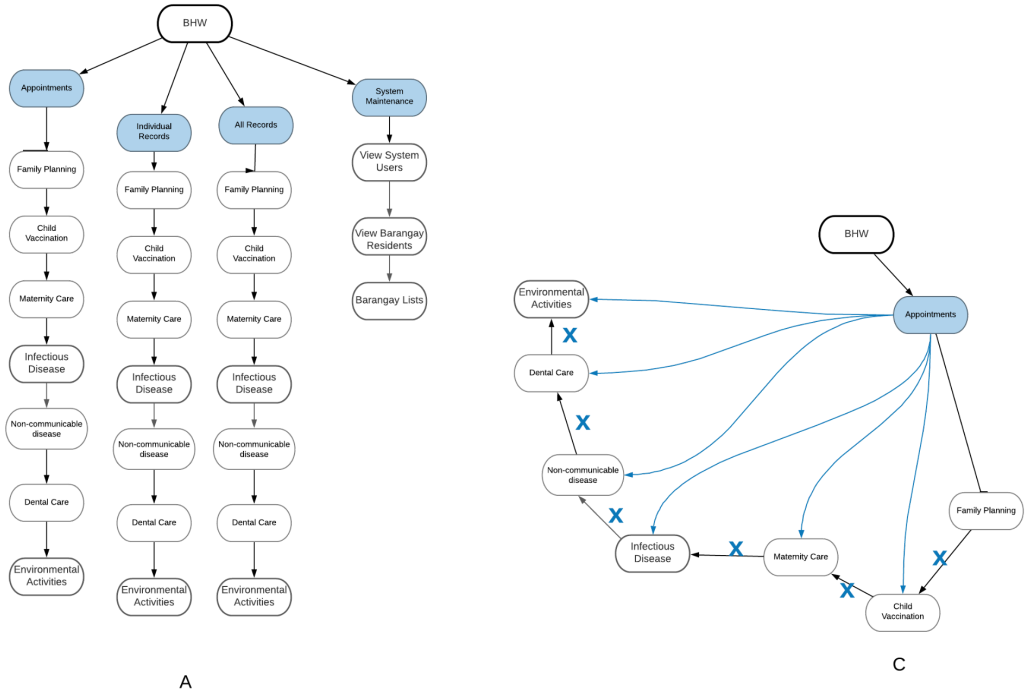


Figure 39: Use of Access Efficiency Model to evaluate the U-S Adaptability in Adaptation Process for access to appointment records for Barangay Health Workers

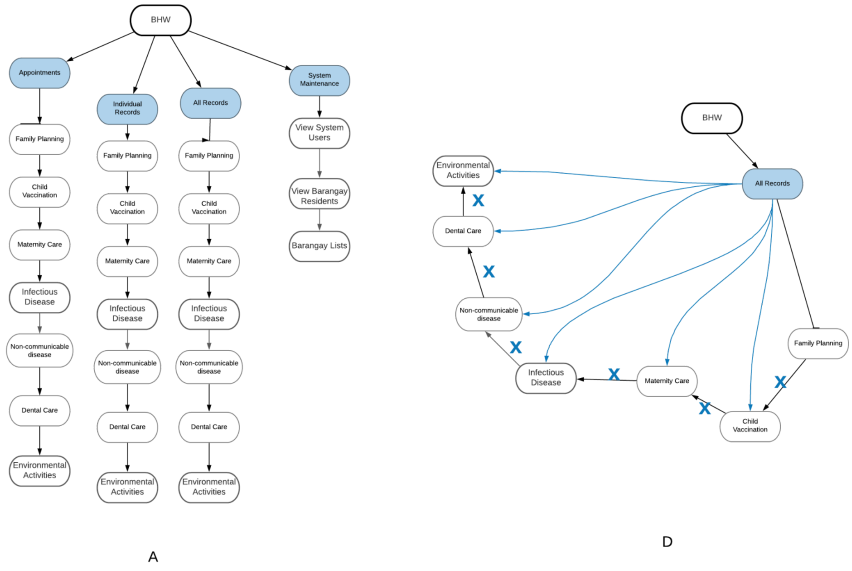


Figure 40: Use of Access Efficiency Model to evaluate the U-S Adaptability in Adaptation Process for access to all records for Barangay Health Workers

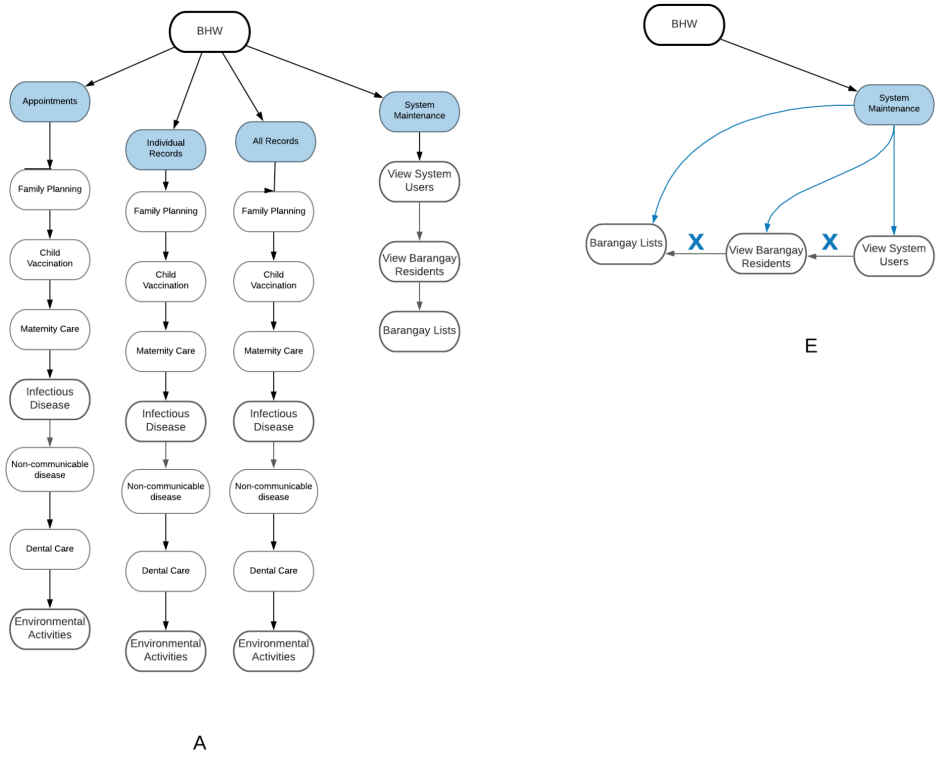


Figure 41: Use of Access Efficiency Model to evaluate the U-S Adaptability in Adaptation Process for access to all records for Barangay Health Workers

Figures 39, 40 and 41 display the transition of user access from the original layout to its different forms to help barangay health workers adapt easily to the access path that will match their workflow.

In the existing system, the patients visit their barangay health centers with different concerns. Some patients would come for child vaccination and would not want maternity consultations and the like. Therefore, it is very important that the system will be able to display the transactions on the SMRBHIS dashboard to shorten the search time and provide health workers with ease of access.

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

This chapter discusses the finalization of findings and provision of recommendations.

Conclusion

The opportunities identified prior to the development and implementation of an online health services system for barangays as an adaptive health information system proves to be achievable based on the responses of the clients/patients. Both barangays in San Mateo Rizal: Guitnang Bayan 1 and Sta. Ana were able to test and attest the functionalities of the proposed system. The respondents were able to manipulate the system with less supervision and provide reports based on the Field Health Services Information System (FHSIS) as per the policy and guidelines of the department of health.

The automated computation of transactions per category per day were able to provide reports per month. Correct and accurate numbers appeared as observed based on the number of patient consultations that increases reliability.

Nurses, midwives and the barangay health workers, were able to use the system based on the credentials (username and password) they were registered into and maximize the level of usability as per security permission.

Given the bulk of workload, the proposed system will be able to provide ease of use and lessen rework for the services providers of the barangay health centers considering its functions.

Security was highlighted in the development of the system by initially providing a Data Sharing Agreement for the barangay health workers and the municipal health office and patients. This gives both parties reliable documentation that there is a clear understanding regarding the transition of data from the patient to the barangay health service providers and the municipal health office. Also, each user was given strict credentials whose validity primarily extends up to the enlistment of their barangay. The super admin also has the privilege of limiting access within the users, managing/removing suspicious accounts and degree of qualifications in the disposal of inactive health records.

In this context, the success indicators, whose purpose is to address the different applications for adaptability were achieved.

As per the user-system adaptability study of Zeng and Yen which mainly for the navigational features and usability of the proposed system both agreed to be very evident in the context of the user/respondents. Then, the electronic health record adaptability and interoperability adhering to knowledge management were also established throughout the development and implementation of the system. This is through constant consultation and feedback - agile method, knowledge management – by educating the users through training, and creation of learning modules, and addressing multiple *How's* in the system functionalities.

These aspects were encapsulated through the quality model by B.W. Boehm. Having satisfactorily reached a very acceptable mean for each item on the success indicator and the ability of users coming from two (2) barangays to use the proposed Online Health Services System including patients has proven the user-system adaptability.

Recommendations

The proposed system for the barangays of San Mateo, Rizal was developed and implemented and was accepted by the barangays' administration and citizens. In this context, limitations were identified and may provide room for improvement and recommendations as the following:

- *For the Barangay Health Workers.* The system may further be effective if barangay health workers will also have further rigid training in technology for them to also extend help to walk in patients.

- *A Data Center.* During the public presentation of the barangay health information system, it is recommended that it be connected to a data center to further provide security and expanded services to the members of the barangay.

- *Signature Hash Algorithm.* According to pyCryptodome.io (PyCryptodome, 2016) SHA 384 which is a part of the SHA2 family is 50% faster than SHA256 on a 64-bit machine and produces a 384 bit digest of a message. In this context, future developers may improve the system and may improve its signature hash algorithm.

- *Improved involvement of the Super Health Center.* The barangays of San Mateo, Rizal provide reports to the main health center which is called the Super Health Center. This office provides extensive medical care for the citizens in its area and has more medical practitioners. It is recommended that this facility be introduced to the system to fast track the implementation of a region-wide barangay health information system.

- *Logistics.* The expanded implementation of the proposed system may be better if logistics support is provided. Like, computer units for kiosks to be installed outside

barangay health centers to help more citizens register and avail the benefits of their online health system, internet subsidies and the like.

- *Barangay Administrators.* The barangay administrators such as *Barangay Health Secretariat* and *Barangay Captains* of Barangays Sta. Ana and Guitnang Bayan 1 may also look into influencing other barangays in San Mateo Rizal to use the proposed system to help benefit from its features and functionalities.
- *Implementation to Other Barangays.* Given that two amongst the fifteen barangays in San Mateo Rizal accepted the developed and implemented Barangay Health Information System, the remaining barangays can also implement the same system given the endorsement from the Municipality of San Mateo, Rizal. The municipality leaders may refer to the table below:

Table 39: Implementation Plan of the Proposed System

Activities	Timeline	In-charge
1. Endorsement from the Municipality of San Mateo Rizal / Mayor's Office	2 months	Proponent and Mayor's office
2. Meetings with the Barangay Captains	1 month	Proponent and Barangay Captains
3. Establishment of a Data Center	2 years	Proponent and In-charge of IT in San Mateo Municipal hall
4. Endorsement to Barangay Health Workers	6 months	Proponent and Barangay Captains
5. Trainings on use of the System	2 months	Proponent and Barangay Health workers
6. Education on Data Privacy	2 months	Data Privacy officer of the Municipality of San Mateo Rizal
7. Advertisement of the System to the citizens of each barangays.	6 months	Barangay Health Officers
8. Registration of citizens.	to be determined	Citizens/Patients to be assisted by Barangay Health Workers
9. Use of services.	to be determined	Citizens of each Barangays.

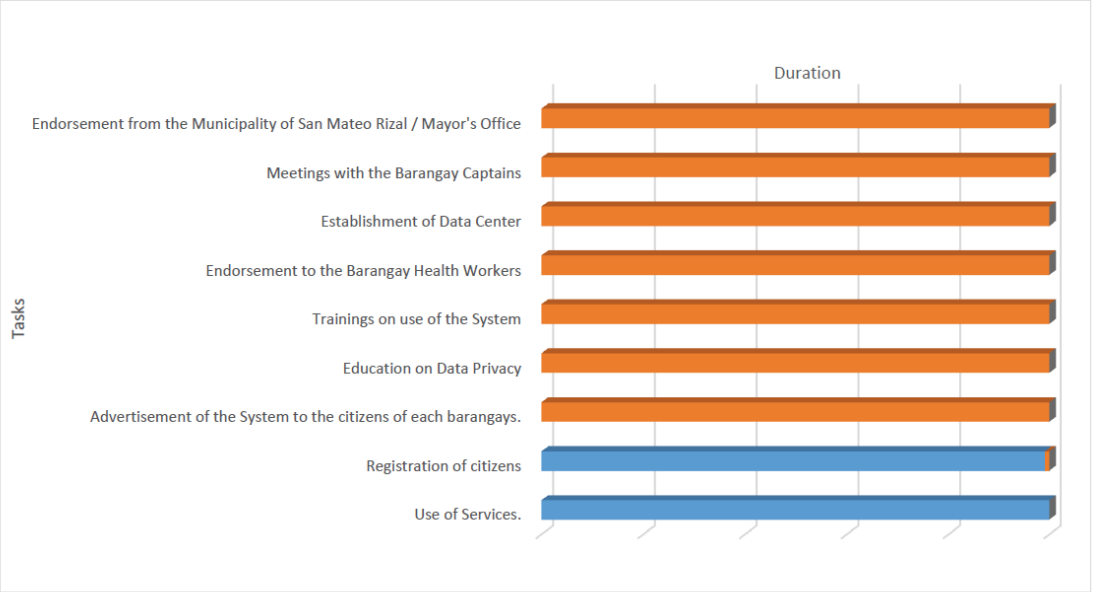


Figure 42: Gantt Chart for the Implementation of the Proposed System

- *Data Privacy Implementation.* The data privacy officer stationed in the municipality health center's office of the Municipality of San Mateo may help in the implementation and guidance of both administrators and citizens involved in the process needed for Barangay Health Center Information System. Refer to the Designation of Data Protection Officers pursuant to Republic Act no. 10173, titled, Data Privacy Act of 2012 Memorandum Circular No. 2018-36 *(See Appendix B: Department of the Interior and Local Government, 2018, 1-3)*

- *To other Researchers.* The proponent was able to conceptualize the proposed system in line with experiences and hope to have a better health services in rural areas such as San Mateo Rizal which is located in the outskirts of Marikina City and Quezon City. Being able to develop this capstone project is not easy and took a lot of coordination with the barangays health centers and barangay health workers. If there will be someone who will have the interest of creating the same health information system for barangays to further improve the government services, make sure to find some time to conceptualize and consider all aspects involving hardware, software and most importantly, peopleware. For more information on the procedure, challenges and different aspects for the development and implementation of this system refer to *Appendix I* for contact details.

References:

Republic Act No. 7160 | GOVPH. (2021). Retrieved 2 December 2021, https://www.officialgazette.gov.ph/1991/10/10/republic-act-no-7160/

Philippines Field Health Services Information System (FHSIS) | GHDx. (2021). December 2021, from http://ghdx.healthdata.org/series/philippines-field-health-services-information-system-fhsis

Philippine National Formulary Manual for Primary Healthcare. (2021). [Ebook] (8th ed.). April 2021, from https://caro.doh.gov.ph/wp-content/uploads/2018/04/PNF-Manual-for-Primary-Healthcare.pdf.

Implementation of the National eHealth Electronic Medical Record System Validation for National Health Data Reporting Requirements http://ehealth.doh.gov.ph/images/eHealthPDF/AOEMRSValidationMaster05122016.pdf

Kashefi, A., Abbott, P., & Ayoung, A. (2015). User IT Adaptation Behaviors: What have we learned and why does it matter? [Ebook] (p. 6). https://core.ac.uk/download/pdf/301367873.pdf.

Lippeveld, T., Sauerborn, R., & Bodart, C. (2000). Design and Implementation of Health Information System. https://www.researchgate.net/publication/242520887_Design_and_Implementation_of_Health_Information_Systems.

Estinar, A., Grefiel, L., Lu, L., Libre, L. And Tangkeko, M., 2018. Pampanga's Barangay Health Information System (Pbhis): A Decision Support & Health Information System For Rural Health Unit 1. In: Dlsu Research Congress. Manila: Dlsu Research Congress, Pp.1-6.

Kendall, K. and Kendall, J., 2010. Systems Analysis and Design. 8th ed. Jurong, Singapore: Pearson, p.183.

Bergmann, E. and Hamilton, A., 2013. Agile-Waterfall Hybrid.

Lucidchart.com. 2021. Agile-Waterfall Hybrid: Is It Right for Your Team? | Lucidchart Blog. [online] Available at: <https://www.lucidchart.com/blog/is-agile-waterfall-hybrid-right-for-your-team>

Five methods used for quantitative data collection | QuestionPro. (2021) https://www.questionpro.com/blog/quantitative-data-collection-methods/

Imus, J., Magleo, E., Soriano, M. and Olalia, R., 2018. Barangay Management Information System for Cities and Municipalities in the Philippines. International Journal of Computer Applications, 180(19), p.34.

Non-Probability Sampling - AAPOR. Aapor.org. (2021). https://www.aapor.org/Education-Resources/Reports/Non-Probability-Sampling.aspx.

Probability/Non-Probability Sampling: Baker, R., & Brick, J. (2013). Non-Probability Sampling - AAPOR. Aapor.org. https://www.aapor.org/Education-Resources/Reports/Non-Probability-Sampling.aspx.

DESSEL, G. (2013). How to determine population and survey sample size?. CheckMarket. https://www.checkmarket.com/blog/how-to-estimate-your-population-and-survey-sample-size/.

Mcleod, S. (2021). Likert Scale Definition, Examples and Analysis | Simply Psychology. Simplypsychology.org. https://www.simplypsychology.org/likert-scale.html.

Lederer, A. L., Maupin, D. J., Sena, M. P., & Zhuang, Y. (2000). The technology acceptance model and the World Wide Web. Decision support systems, 29(3), 269-282.

Technology acceptance model - Wikipedia. En.wikipedia.org. (2021). https://en.wikipedia.org/wiki/Technology_acceptance_model.

Kashefi, A., Abbott, P., & Ayoung, A. (2015). User IT Adaptation Behaviors: What have we learned and why does it matter? [Ebook] (p. 6). https://core.ac.uk/download/pdf/301367873.pdf.

Koenig, M. (2018). What is KM? Knowledge Management Explained. KMWorld. https://www.kmworld.com/About/What_is_Knowledge_Management.

Razzaque, A., & Jalal-Karim, A. (2010). Conceptual Healthcare Knowledge Management Model For Adaptability And Interoperability Of Ehr [Ebook]. http://Emcis.Eu/Emcis_archive/Emcis/Emcis2010/Proceedings/Accepted%20refereed%20papers/C68.Pdf.

Boehm's Software Quality Model - GeeksforGeeks. GeeksforGeeks. (2021). https://www.geeksforgeeks.org/boehms-software-quality-model/.

(2021). [Ebook]. http://ehealth.doh.gov.ph/images/HealthPrivacyCode.pdf.

What Is Risk Management in Healthcare?. Catalyst.nejm.org. (2021). https://catalyst.nejm.org/doi/full/10.1056/CAT.18.0197.

PHP: Hypertext Preprocessor. Php.net. (2021). https://www.php.net/.

Health Privacy Code (2021). [Ebook]. http://ehealth.doh.gov.ph/images/HealthPrivacyCode.pdf.

Rosencrance, L. (2021). *What is Role-Based Access Control (RBAC)? Definition from SearchSecurity*. SearchSecurity. Retrieved 5 December 2021, from

https://www.techtarget.com/searchsecurity/definition/role-based-access-control-RBAC.

What is SSL, TLS and HTTPS?. What is SSL, TLS and HTTPS?. (2021). https://www.websecurity.digicert.com/security-topics/what-is-ssl-tls-https.

Signature Hashing. Help.ivanti.com. (2021). https://help.ivanti.com/ap/help/en_US/am/2020/Content/Application_Manager/Signature_Hashing.htm#About_Signature_Hashing.

Callaghan, P. (2021). *Why You Should Use SHA-256 in Evidence Authentication*. Blog.pagefreezer.com. https://blog.pagefreezer.com/sha-256-benefits-evidence-authentication.

Villafuerte, Jr., H. (2016). *House Bill 1339* [Ebook]. https://www.congress.gov.ph/legisdocs/basic_17/HB01339.pdf.

Cognitive fit theory - IS Theory. (2021). https://is.theorizeit.org/wiki/Cognitive_fit_theory

Zeng, B., & Yen, B. (2016). Evaluating User-System Adaptability in the Use of Web-based Information Systems through the Model of Access Efficiency. *Pacific Asia Journal Of The Association For Information Systems*, *8*, 1-16.

SHA-384 — PyCryptodome 3.12.0 documentation. (2021). https://pycryptodome.readthedocs.io/en/latest/src/hash/sha384.html

Sligo, J., Gauld, R., Roberts, V., & Villa, L. (2016). A literature review for large-scale health information system project planning, implementation and evaluation. *International Journal Of Medical Informatics*, *97*. https://www.sciencedirect.com/science/article/abs/pii/S1386505616302003?via%3Dihub

Imus, J. K. P. (2018, February). Barangay Management Information System (BMIS) for Cities and Municipalities in the Philippines. *International Journal of Computer Applications*, *180*(19), 1-4. chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.ijcaonline.org/archives/volume180/number19/imus-2018-ijca-916441.pdf

Department of the Interior and Local Government. (2018). *Designation of Data Protection Officers pursuant to Republic Act no. 10173, titled, Data Privacy Act of 2012*. chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.dilg.gov.ph/PDF_File/issuances/memo_circulars/dilg-memocircular-2018319_982ed4d4b0.pdf

Department of Health. (2019). *Annual Report (2019): Field Service Health Information System*. 2019 Annual Report: FSHIS.

chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://doh.gov.ph/sites/default/files/publications/FHSIS%202019.pdf

User, S. (2021). DILG Calabarzon. http://calabarzon.dilg.gov.ph/

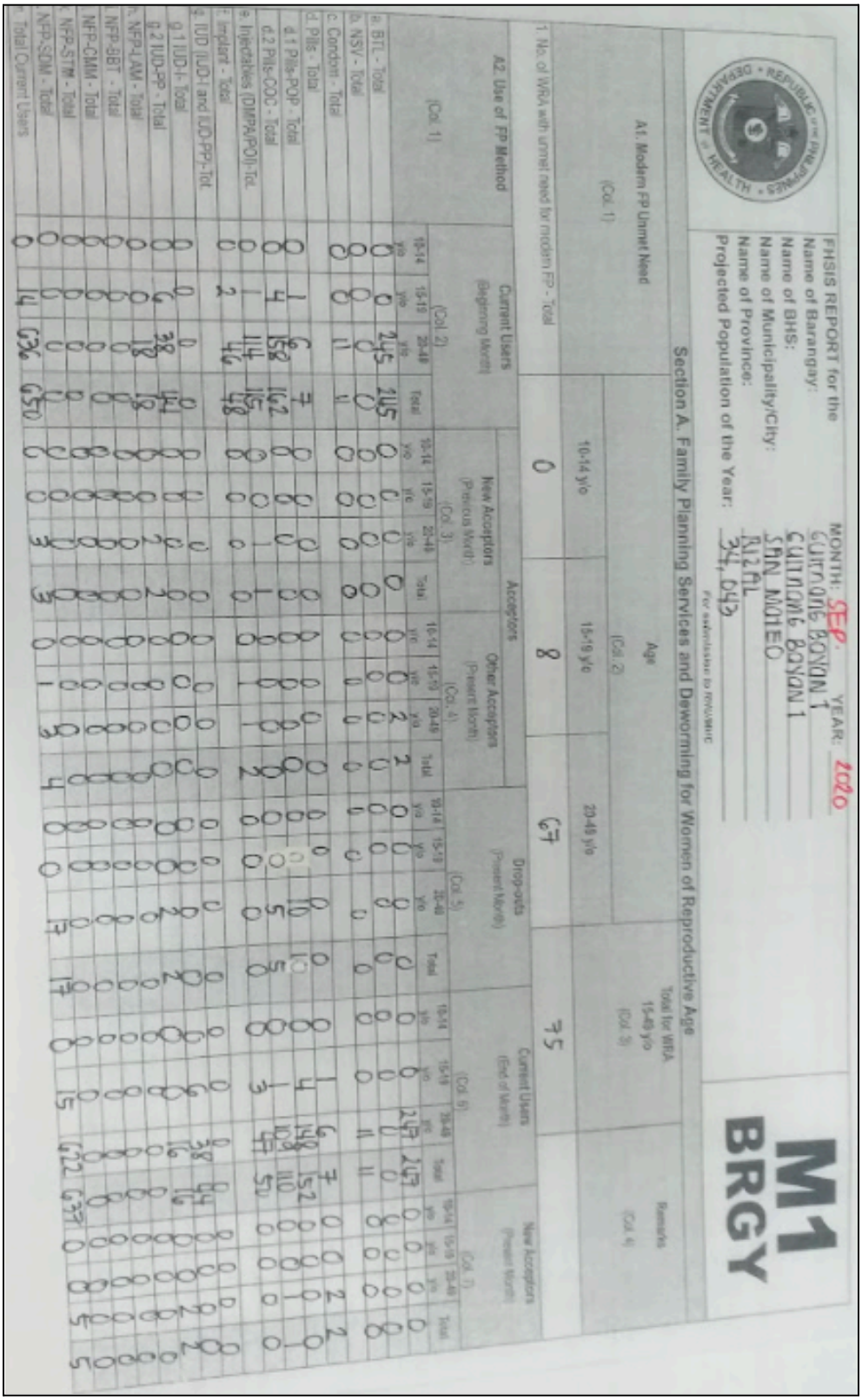
San Mateo, Rizal - Wikipedia. En.wikipedia.org. https://en.wikipedia.org/wiki/San_Mateo,_Rizal#Barangays.

Philippine Health System at a Glance. (2021). https://doh.gov.ph/sites/default/files/basic-page/chapter-one.pdf

Appendices

Appendix A

Field Service Health Information System
Entity Relationship Diagrams
Gantt Charts



FHSIS REPORT for the
Name of Barangay: CUMORO BOYON 1
Name of BHS: CUMORO BOYON 1
Name of Municipality/City: SAN MATEO
Name of Province: RIZAL
Projected Population of the Year: 34,043
MONTH: SEP. YEAR: 2020

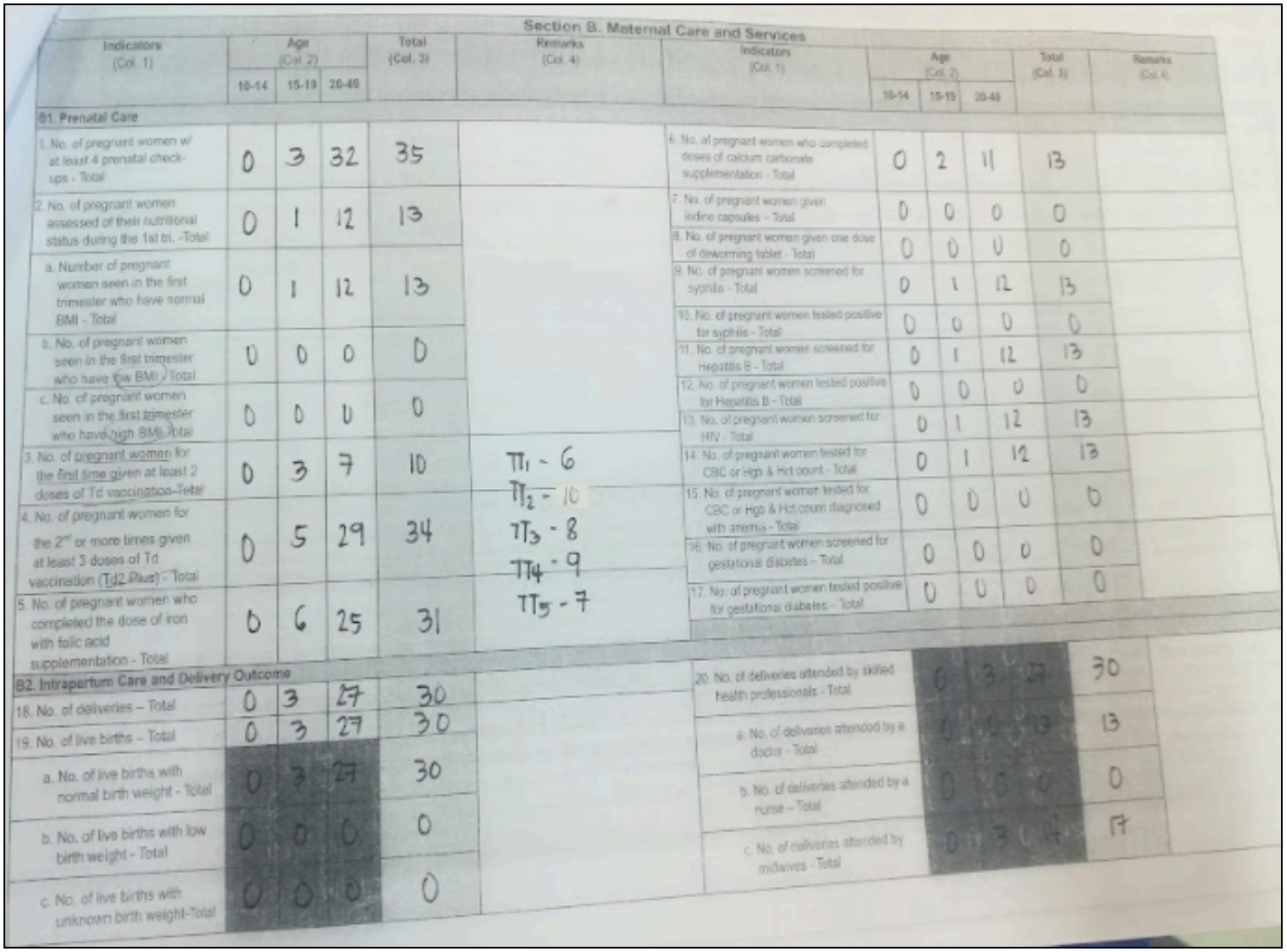
M1
BRGY

Section A. Family Planning Services and Deworming for Women of Reproductive Age

A1. Modern FP Unmet Need	Age			Total for WRA 15-49 yo (Col. 3)	Remarks (Col. 4)
(Col. 1)	10-14 yo (Col. 2)	15-19 yo (Col. 2)	20-49 yo (Col. 2)		
1. No. of WRA with unmet need for modern FP - Total	0	8	67	75	

A2. Use of FP Method	Current Users (Beginning Month)				New Acceptors (Previous Month)				Other Acceptors (Present Month)				Drop-outs (Present Month)				Current Users (End of Month)				New Acceptors (Present Month)			
(Col. 1)	(Col. 2)				(Col. 3)				(Col. 4)				(Col. 5)				(Col. 6)				(Col. 7)			
	10-14 yo	15-19 yo	20-49 yo	Total	10-14 yo	15-19 yo	20-49 yo	Total	10-14 yo	15-19 yo	20-49 yo	Total	10-14 yo	15-19 yo	20-49 yo	Total	10-14 yo	15-19 yo	20-49 yo	Total	10-14 yo	15-19 yo	20-49 yo	Total
a. BTL - Total	0	6	245	245	0	0	0	0	0	2	2	2	0	0	0	0	0	8	247	247	0	0	0	0
b. NSV - Total	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
c. Condom - Total	8	8	11	11	0	0	0	0	0	0	0	0	0	0	0	0	0	8	11	11	0	0	0	0
d. Pills - Total	0	6	152	152	0	0	0	0	0	0	0	0	0	10	10	10	0	4	148	152	0	0	0	0
d.1 Pills-POP - Total	0	4	152	152	0	0	0	0	0	0	0	0	0	5	5	5	0	3	109	110	0	0	0	0
d.2 Pills-COC - Total	0	1	114	115	0	0	0	0	0	1	1	1	0	0	0	0	0	1	45	50	0	0	0	0
e. Injectables (DMPA/POI)-Tot.	0	2	46	48	0	0	0	0	0	0	2	2	0	0	0	0	0	0	0	0	0	0	0	0
f. Implant - Total	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
g. IUD (IUD-I and IUD-PP)-Tot	0	6	38	44	0	0	0	0	0	0	0	0	0	0	0	0	0	6	38	44	0	0	0	0
g.1 IUD-I - Total	0	6	38	44	0	0	0	0	0	0	0	0	0	0	0	0	0	6	38	44	0	0	0	0
g.2 IUD-PP - Total	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
h. NFP-LAM - Total	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
i. NFP-BBT - Total	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
j. NFP-CMM - Total	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
k. NFP-STM - Total	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
l. NFP-SDM - Total	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
m. Total Current Users	0	14	636	650	0	0	0	0	0	3	3	3	0	0	0	0	0	17	622	637	0	0	0	0

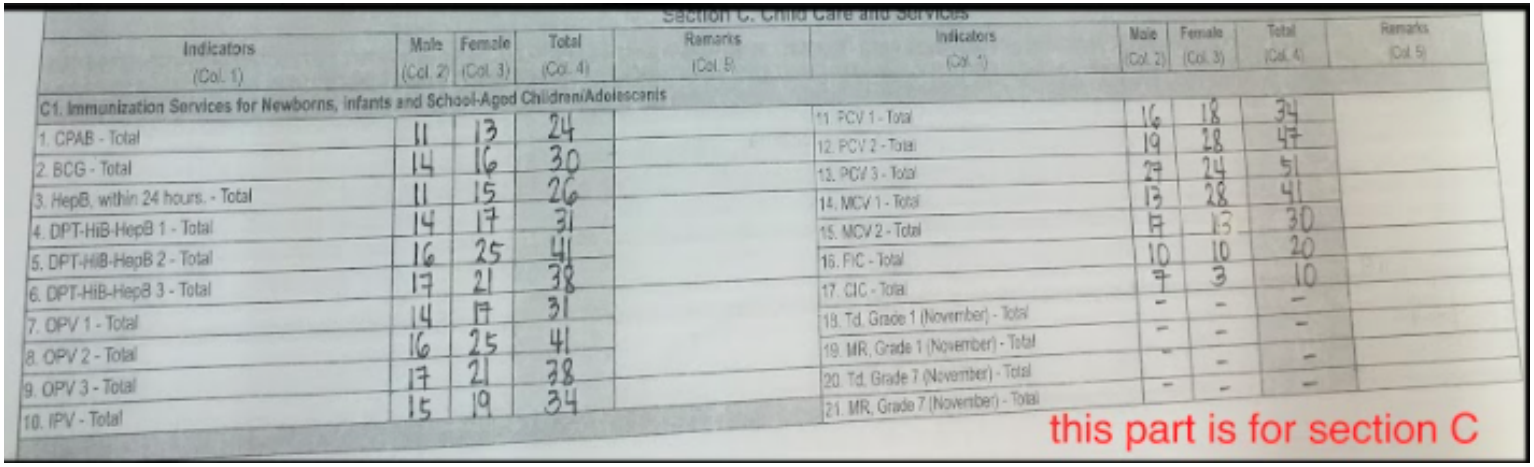
FHSIS: Family Planning



Section B. Maternal Care and Services

Indicators (Col. 1)	Age (Col. 2) 10-14	15-19	20-49	Total (Col. 3)	Remarks (Col. 4)
B1. Prenatal Care					
1. No. of pregnant women w/ at least 4 prenatal check-ups - Total	0	3	32	35	
2. No. of pregnant women assessed of their nutritional status during the 1st tri. -Total	0	1	12	13	
a. Number of pregnant women seen in the first trimester who have normal BMI - Total	0	1	12	13	
b. No. of pregnant women seen in the first trimester who have low BMI - Total	0	0	0	0	
c. No. of pregnant women seen in the first trimester who have high BMI - Total	0	0	0	0	
3. No. of pregnant women for the first time given at least 2 doses of Td vaccination-Total	0	3	7	10	TT1 - 6 TT2 - 10 TT3 - 8 TT4 - 9 TT5 - 7
4. No. of pregnant women for the 2nd or more times given at least 3 doses of Td vaccination (Td2 Plus) - Total	0	5	29	34	
5. No. of pregnant women who completed the dose of iron with folic acid supplementation - Total	0	6	25	31	
6. No. of pregnant women who completed doses of calcium carbonate supplementation - Total	0	2	11	13	
7. No. of pregnant women given iodine capsules – Total	0	0	0	0	
8. No. of pregnant women given one dose of deworming tablet - Total	0	0	0	0	
9. No. of pregnant women screened for syphilis - Total	0	1	12	13	
10. No. of pregnant women tested positive for syphilis - Total	0	0	0	0	
11. No. of pregnant women screened for Hepatitis B - Total	0	1	12	13	
12. No. of pregnant women tested positive for Hepatitis B - Total	0	0	0	0	
13. No. of pregnant women screened for HIV - Total	0	1	12	13	
14. No. of pregnant women tested for CBC or Hgb & Hct count - Total	0	1	12	13	
15. No. of pregnant women tested for CBC or Hgb & Hct count diagnosed with anemia - Total	0	0	0	0	
16. No. of pregnant women screened for gestational diabetes – Total	0	0	0	0	
17. No. of pregnant women tested positive for gestational diabetes – Total	0	0	0	0	
B2. Intrapartum Care and Delivery Outcome					
18. No. of deliveries – Total	0	3	27	30	
19. No. of live births – Total	0	3	27	30	
a. No. of live births with normal birth weight - Total	0	3	27	30	
b. No. of live births with low birth weight - Total	0	0	0	0	
c. No. of live births with unknown birth weight-Total	0	0	0	0	
20. No. of deliveries attended by skilled health professionals - Total	0	3	27	30	
a. No. of deliveries attended by a doctor - Total	0	0	13	13	
b. No. of deliveries attended by a nurse – Total	0	0	0	0	
c. No. of deliveries attended by midwives - Total	0	3	14	17	

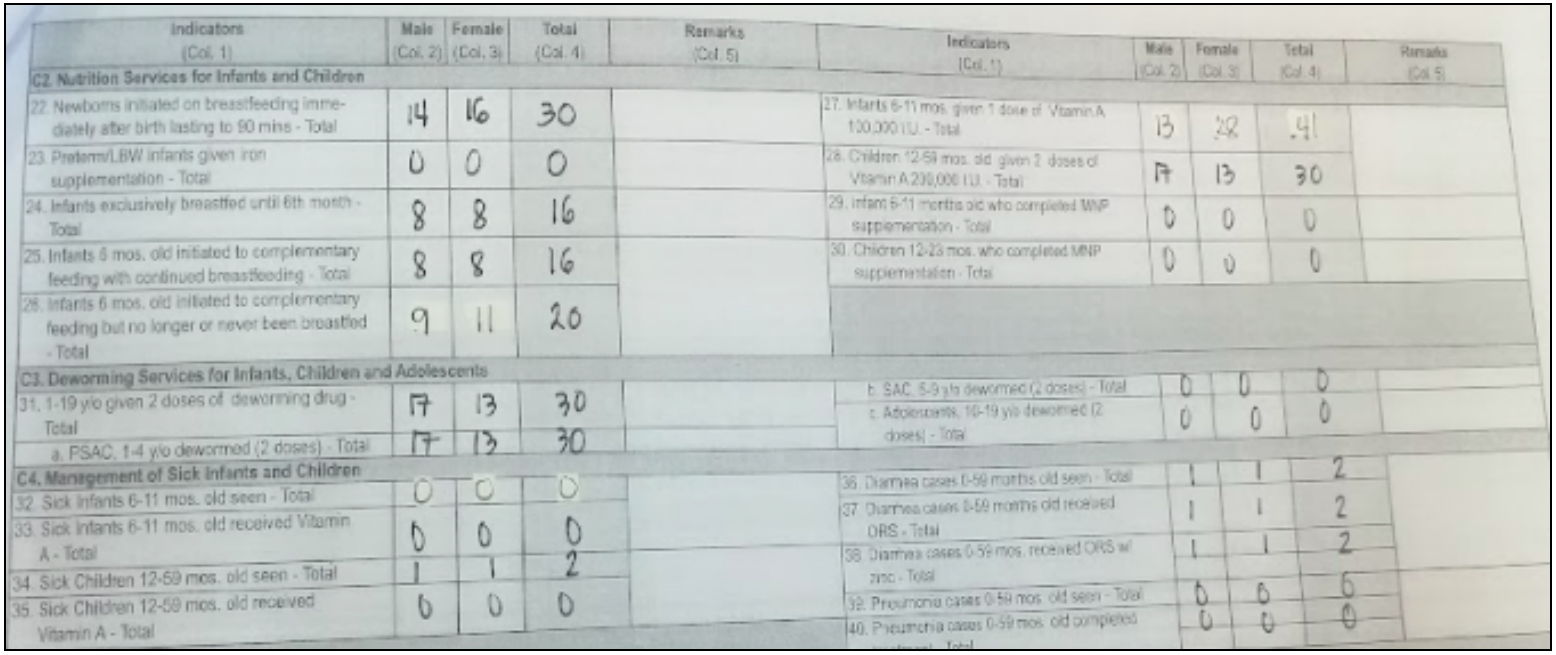
FHSIS: Maternity Services



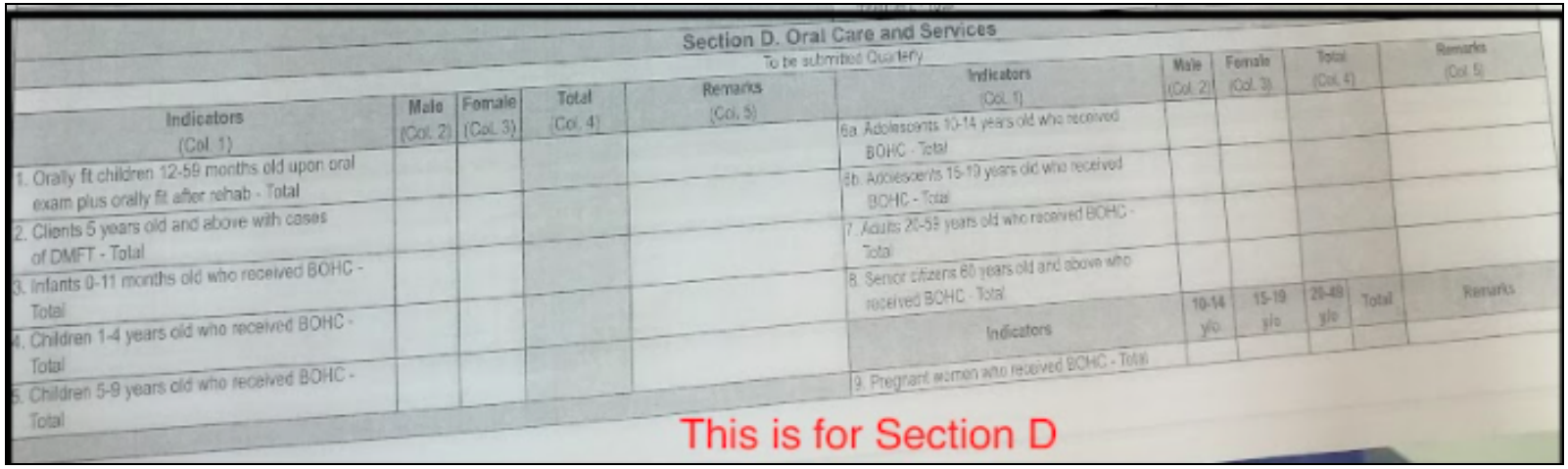
Section C. Child Care and Services

Indicators (Col. 1)	Male (Col. 2)	Female (Col. 3)	Total (Col. 4)	Remarks (Col. 5)	Indicators (Col. 1)	Male (Col. 2)	Female (Col. 3)	Total (Col. 4)	Remarks (Col. 5)
C1. Immunization Services for Newborns, Infants and School-Aged Children/Adolescents									
1. CPAB - Total	11	13	24		11. PCV 1 - Total	16	18	34	
2. BCG - Total	14	16	30		12. PCV 2 - Total	19	28	47	
3. HepB, within 24 hours. - Total	11	15	26		13. PCV 3 - Total	27	24	51	
4. DPT-HiB-HepB 1 - Total	14	17	31		14. MCV 1 - Total	13	28	41	
5. DPT-HiB-HepB 2 - Total	16	25	41		15. MCV 2 - Total	17	13	30	
6. DPT-HiB-HepB 3 - Total	17	21	38		16. FIC - Total	10	10	20	
7. OPV 1 - Total	14	17	31		17. CIC - Total	7	3	10	
8. OPV 2 - Total	16	25	41		18. Td, Grade 1 (November) - Total	-	-	-	
9. OPV 3 - Total	17	21	38		19. MR, Grade 1 (November) - Total	-	-	-	
10. IPV - Total	15	19	34		20. Td, Grade 7 (November) - Total	-	-	-	
					21. MR, Grade 7 (November) - Total	-	-	-	

this part is for section C



Indicators (Col. 1)	Male (Col. 2)	Female (Col. 3)	Total (Col. 4)	Remarks (Col. 5)	Indicators (Col. 1)	Male (Col. 2)	Female (Col. 3)	Total (Col. 4)	Remarks (Col. 5)
C2. Nutrition Services for Infants and Children									
22. Newborns initiated on breastfeeding immediately after birth lasting to 90 mins - Total	14	16	30		27. Infants 6-11 mos. given 1 dose of Vitamin A 100,000 I.U. - Total	13	28	41	
23. Preterm/LBW infants given iron supplementation - Total	0	0	0		28. Children 12-59 mos. old given 2 doses of Vitamin A 200,000 I.U. - Total	17	13	30	
24. Infants exclusively breastfed until 6th month - Total	8	8	16		29. Infant 6-11 months old who completed MNP supplementation - Total	0	0	0	
25. Infants 6 mos. old initiated to complementary feeding with continued breastfeeding - Total	8	8	16		30. Children 12-23 mos. who completed MNP supplementation - Total	0	0	0	
26. Infants 6 mos. old initiated to complementary feeding but no longer or never been breastfed - Total	9	11	20						
C3. Deworming Services for Infants, Children and Adolescents									
31. 1-19 y/o given 2 doses of deworming drug - Total	17	13	30		b. SAC, 5-9 y/o dewormed (2 doses) - Total	0	0	0	
a. PSAC, 1-4 y/o dewormed (2 doses) - Total	17	13	30		c. Adolescents, 10-19 y/o dewormed (2 doses) - Total	0	0	0	
C4. Management of Sick Infants and Children									
32. Sick Infants 6-11 mos. old seen - Total	0	0	0		36. Diarrhea cases 0-59 months old seen - Total	1	1	2	
33. Sick Infants 6-11 mos. old received Vitamin A - Total	0	0	0		37. Diarrhea cases 0-59 months old received ORS - Total	1	1	2	
34. Sick Children 12-59 mos. old seen - Total	1	1	2		38. Diarrhea cases 0-59 mos. received ORS w/ zinc - Total	1	1	2	
35. Sick Children 12-59 mos. old received Vitamin A - Total	0	0	0		39. Pneumonia cases 0-59 mos. old seen - Total	0	0	0	
					40. Pneumonia cases 0-59 mos. old completed [illegible] - Total	0	0	0	



Section D. Oral Care and Services

To be submitted Quarterly

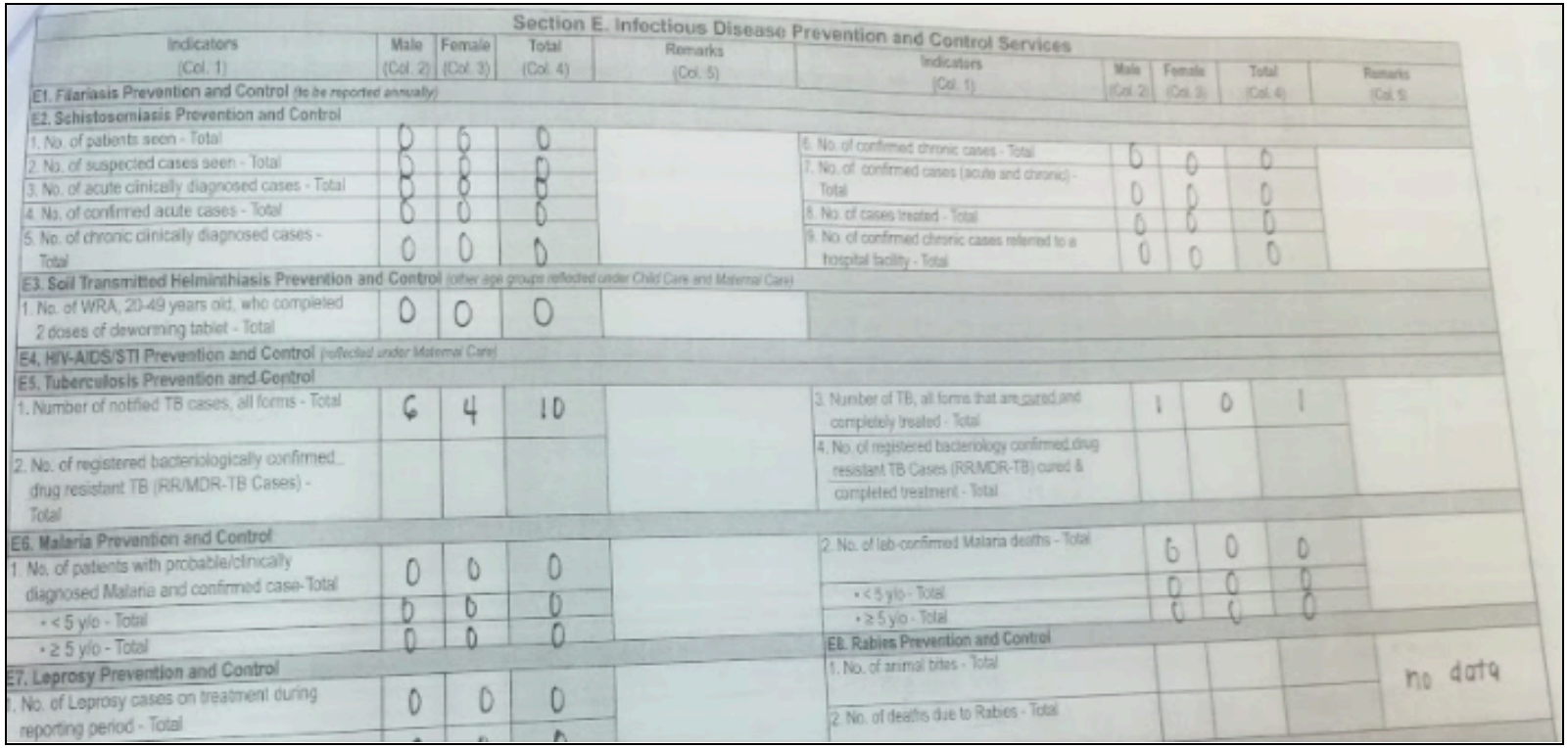
Indicators (Col. 1)	Male (Col. 2)	Female (Col. 3)	Total (Col. 4)	Remarks (Col. 5)
1. Orally fit children 12-59 months old upon oral exam plus orally fit after rehab - Total				
2. Clients 5 years old and above with cases of DMFT - Total				
3. Infants 0-11 months old who received BOHC - Total				
4. Children 1-4 years old who received BOHC - Total				
5. Children 5-9 years old who received BOHC - Total				

Indicators (Col. 1)	Male (Col. 2)	Female (Col. 3)	Total (Col. 4)	Remarks (Col. 5)
6a. Adolescents 10-14 years old who received BOHC - Total				
6b. Adolescents 15-19 years old who received BOHC - Total				
7. Adults 20-59 years old who received BOHC - Total				
8. Senior citizens 60 years old and above who received BOHC - Total				

Indicators	10-14 y/o	15-19 y/o	20-49 y/o	Total	Remarks
9. Pregnant women who received BOHC - Total					

This is for Section D

FHSIS: Oral Care Services

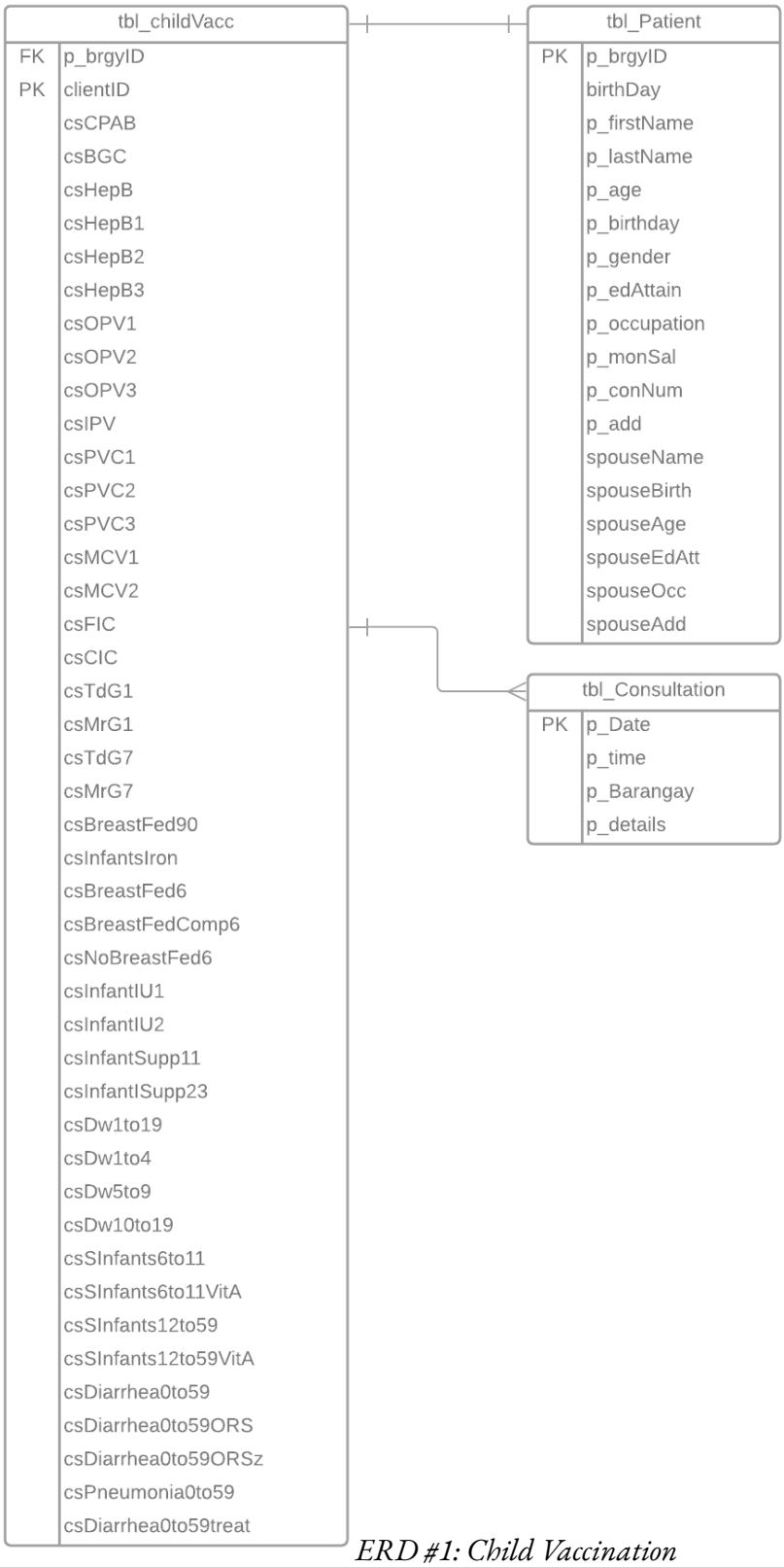


Section E. Infectious Disease Prevention and Control Services

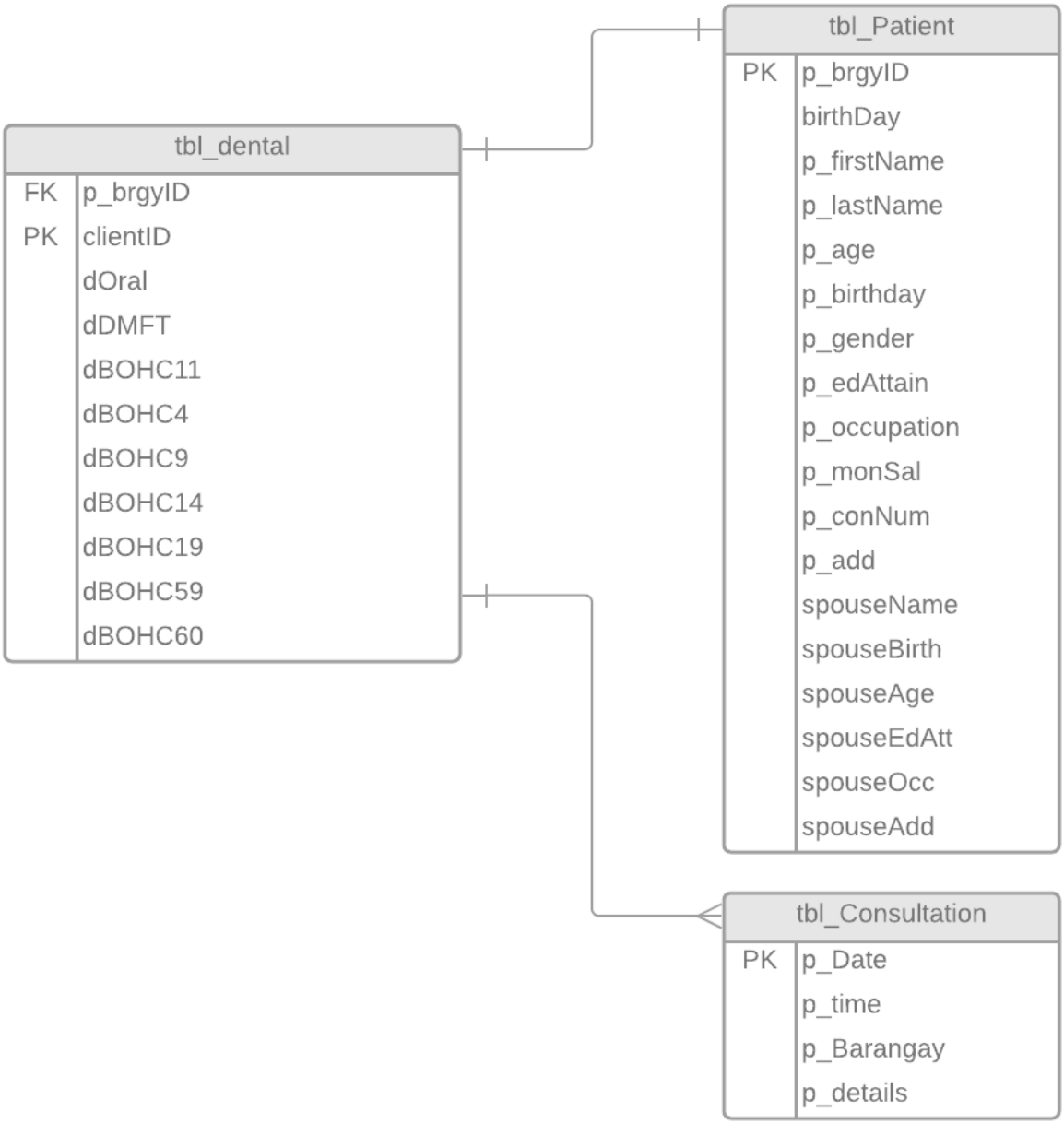
Indicators (Col. 1)	Male (Col. 2)	Female (Col. 3)	Total (Col. 4)	Remarks (Col. 5)	Indicators (Col. 1)	Male (Col. 2)	Female (Col. 3)	Total (Col. 4)	Remarks (Col. 5)
E1. Filariasis Prevention and Control *(to be reported annually)*									
E2. Schistosomiasis Prevention and Control									
1. No. of patients seen - Total	0	0	0		6. No. of confirmed chronic cases - Total	0	0	0	
2. No. of suspected cases seen - Total	0	0	0		7. No. of confirmed cases (acute and chronic) - Total	0	0	0	
3. No. of acute clinically diagnosed cases - Total	0	0	0		8. No. of cases treated - Total	0	0	0	
4. No. of confirmed acute cases - Total	0	0	0		9. No. of confirmed chronic cases referred to a hospital facility - Total	0	0	0	
5. No. of chronic clinically diagnosed cases - Total	0	0	0						
E3. Soil Transmitted Helminthiasis Prevention and Control *(other age groups reflected under Child Care and Maternal Care)*									
1. No. of WRA, 20-49 years old, who completed 2 doses of deworming tablet - Total	0	0	0						
E4. HIV-AIDS/STI Prevention and Control *(reflected under Maternal Care)*									
E5. Tuberculosis Prevention and Control									
1. Number of notified TB cases, all forms - Total	6	4	10		3. Number of TB, all forms that are cured and completely treated - Total	1	0	1	
2. No. of registered bacteriologically confirmed drug resistant TB (RR/MDR-TB Cases) - Total					4. No. of registered bacteriology confirmed drug resistant TB Cases (RR/MDR-TB) cured & completed treatment - Total				
E6. Malaria Prevention and Control									
1. No. of patients with probable/clinically diagnosed Malaria and confirmed case-Total	0	0	0		2. No. of lab-confirmed Malaria deaths - Total	0	0	0	
• < 5 y/o - Total	0	0	0		• < 5 y/o - Total	0	0	0	
• ≥ 5 y/o - Total	0	0	0		• ≥ 5 y/o - Total	0	0	0	
E7. Leprosy Prevention and Control					**E8. Rabies Prevention and Control**				
1. No. of Leprosy cases on treatment during reporting period - Total	0	0	0		1. No. of animal bites - Total				no data
					2. No. of deaths due to Rabies - Total				

FHSIS: Infectious Disease

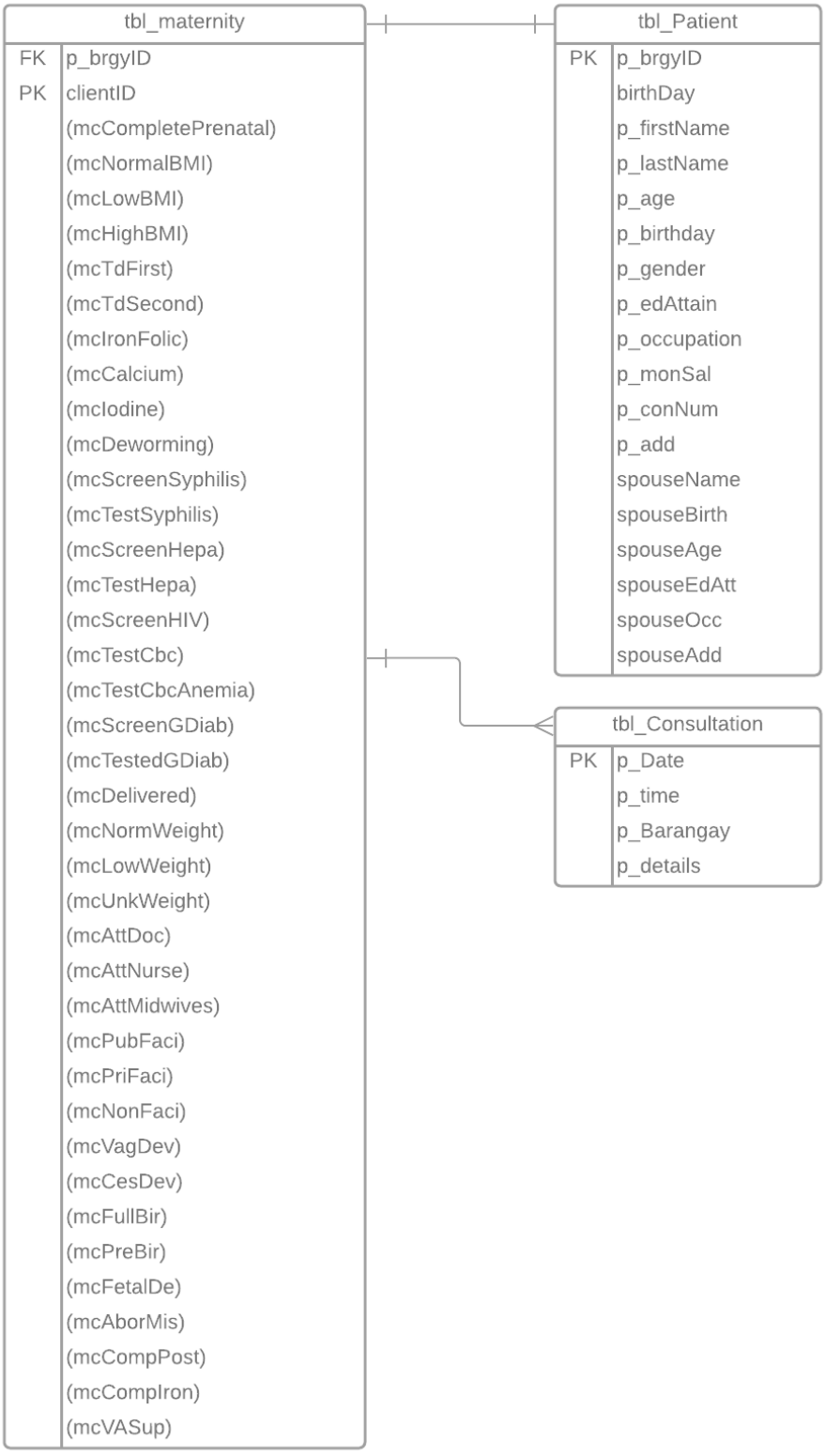
Entity Relationship Diagrams



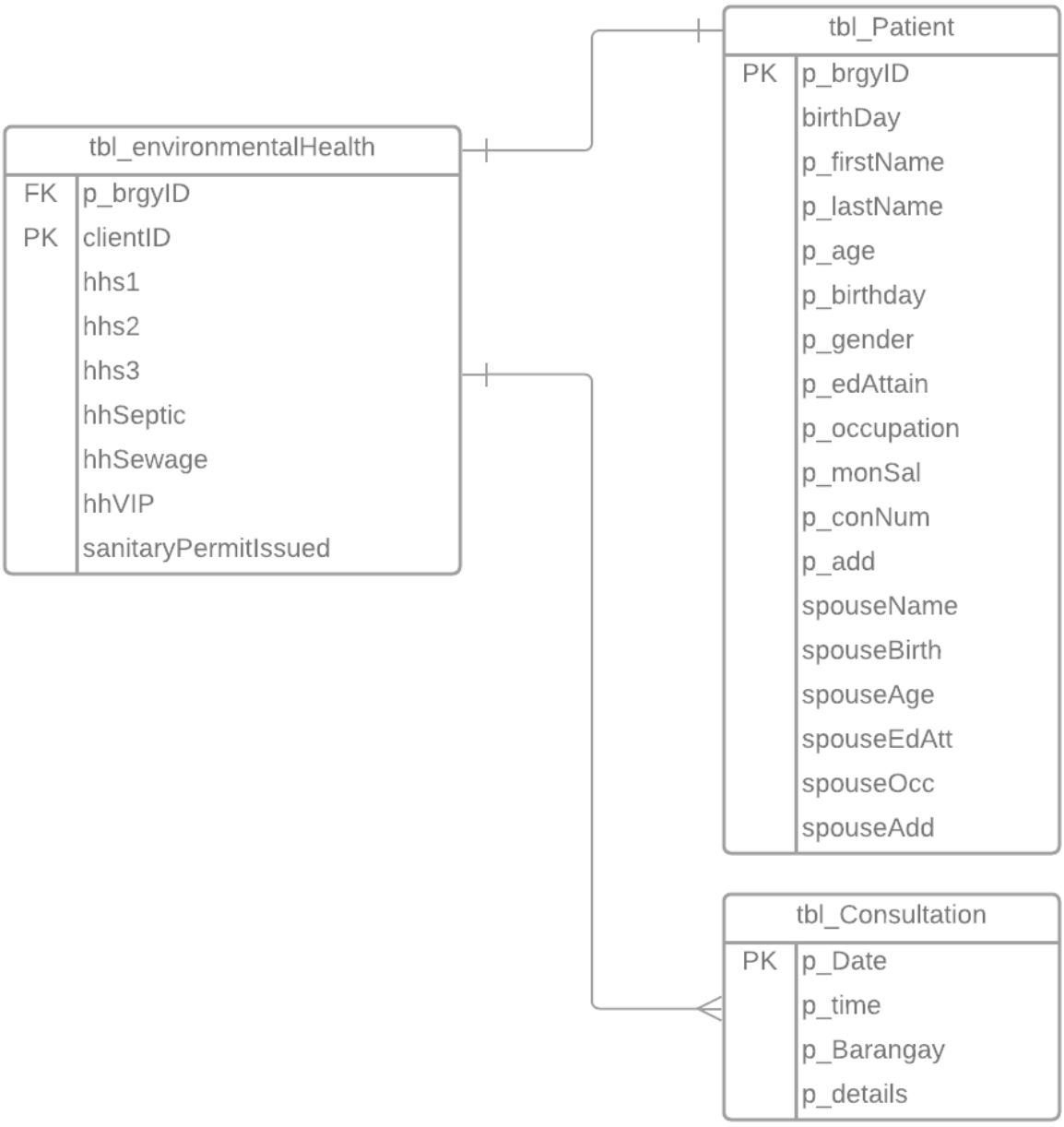
ERD #1: Child Vaccination



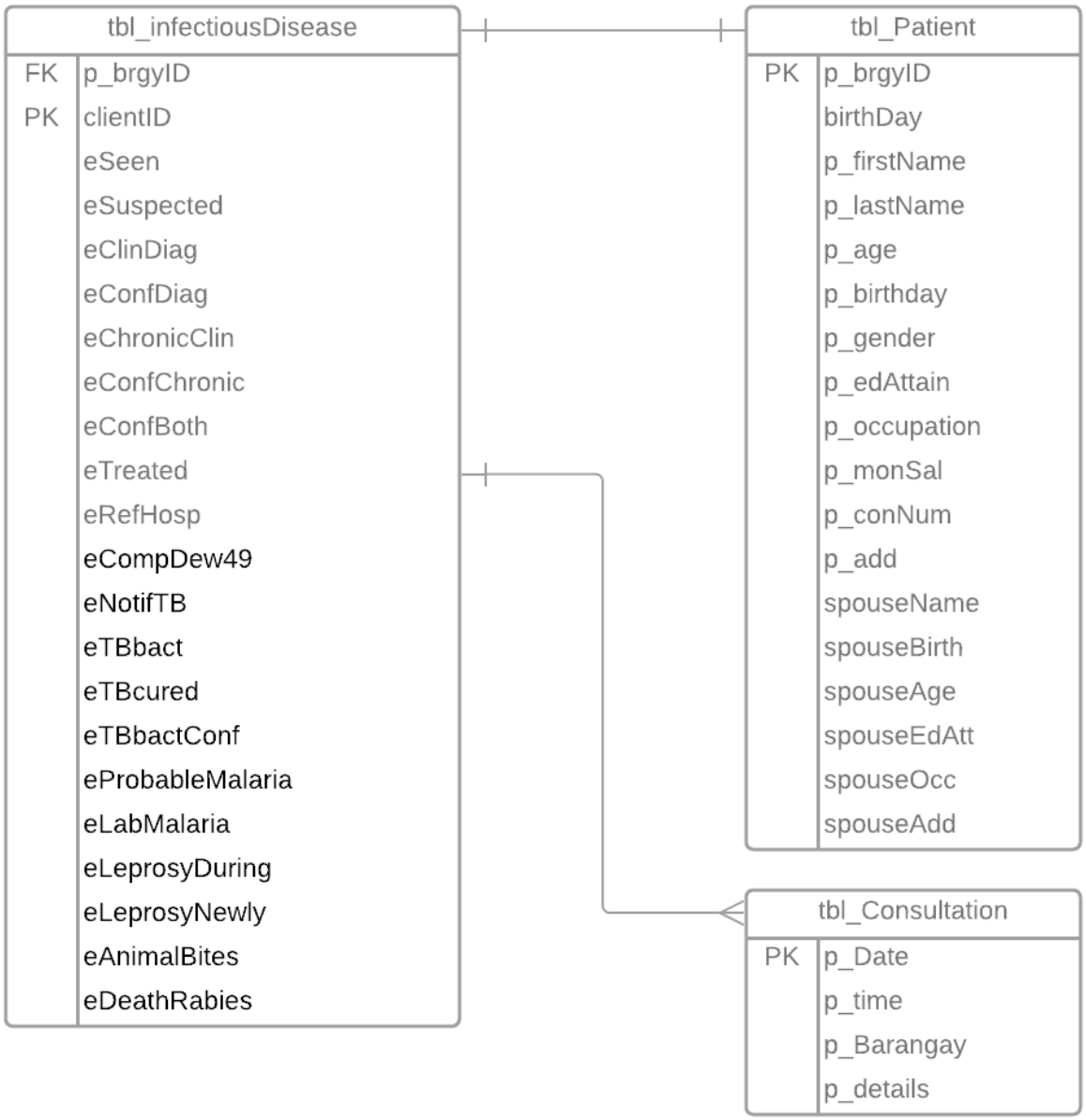
ERD #2: Dental Checkup



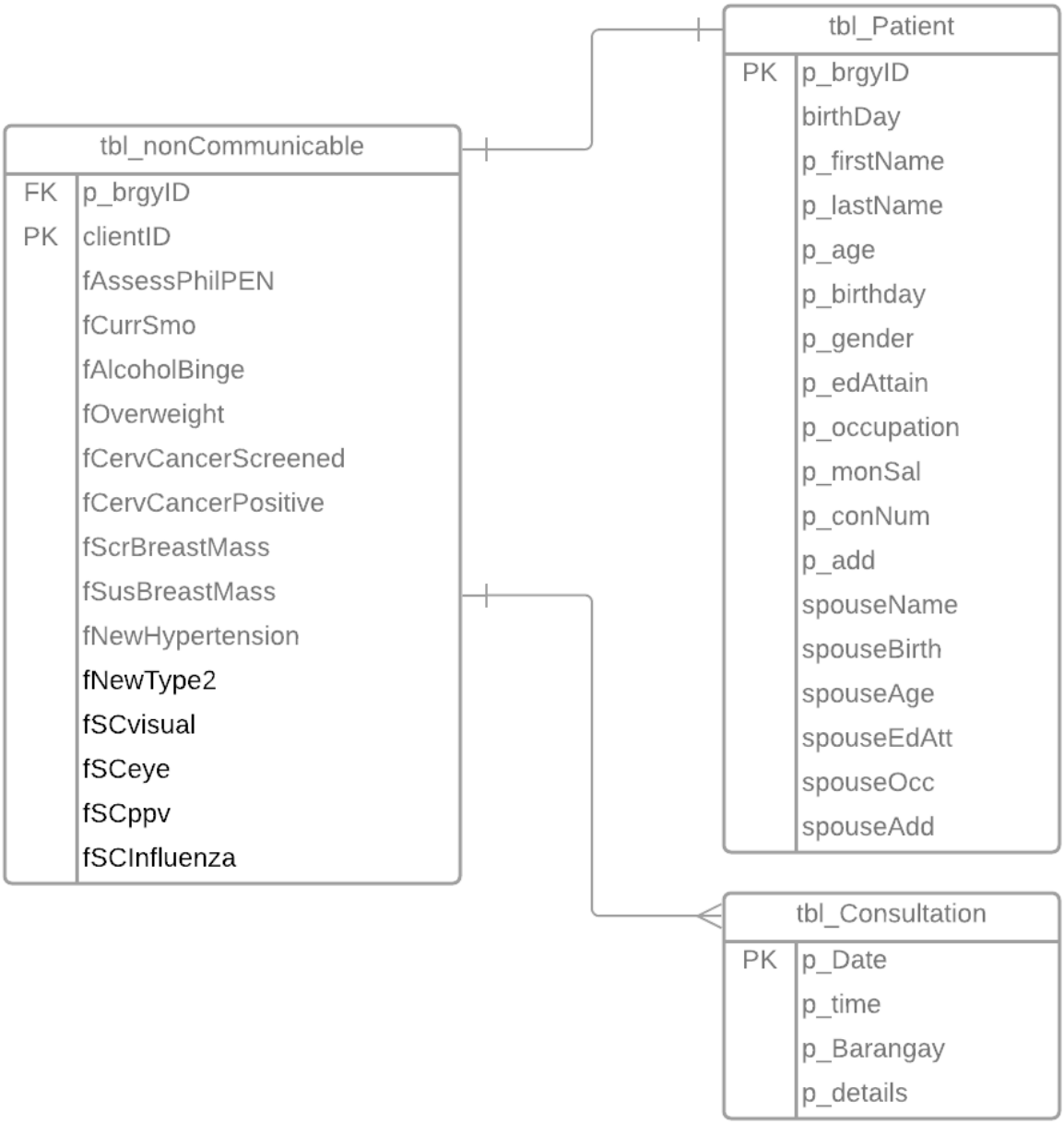
ERD #3: Child Vaccination



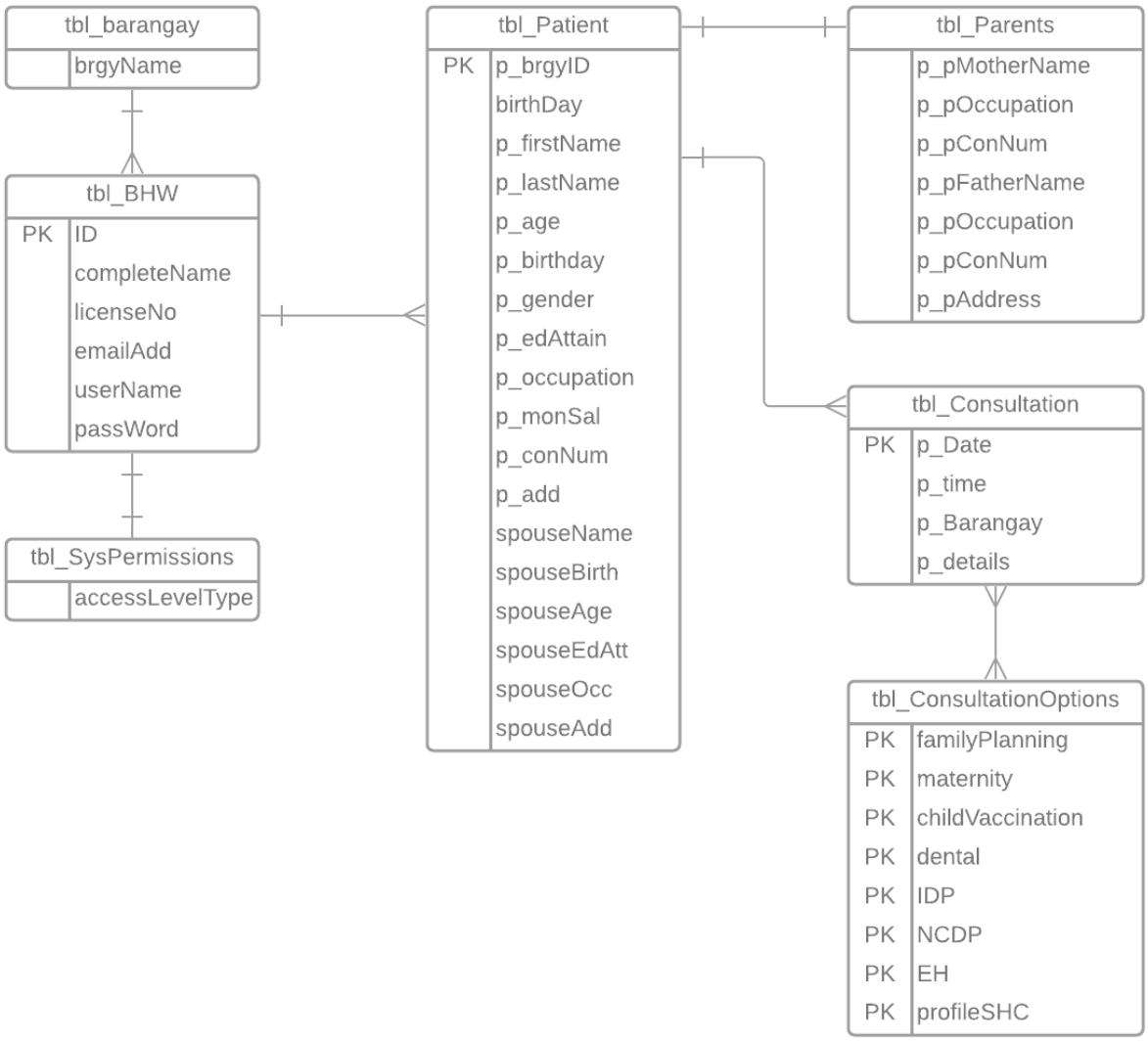
ERD #4: Child Vaccination



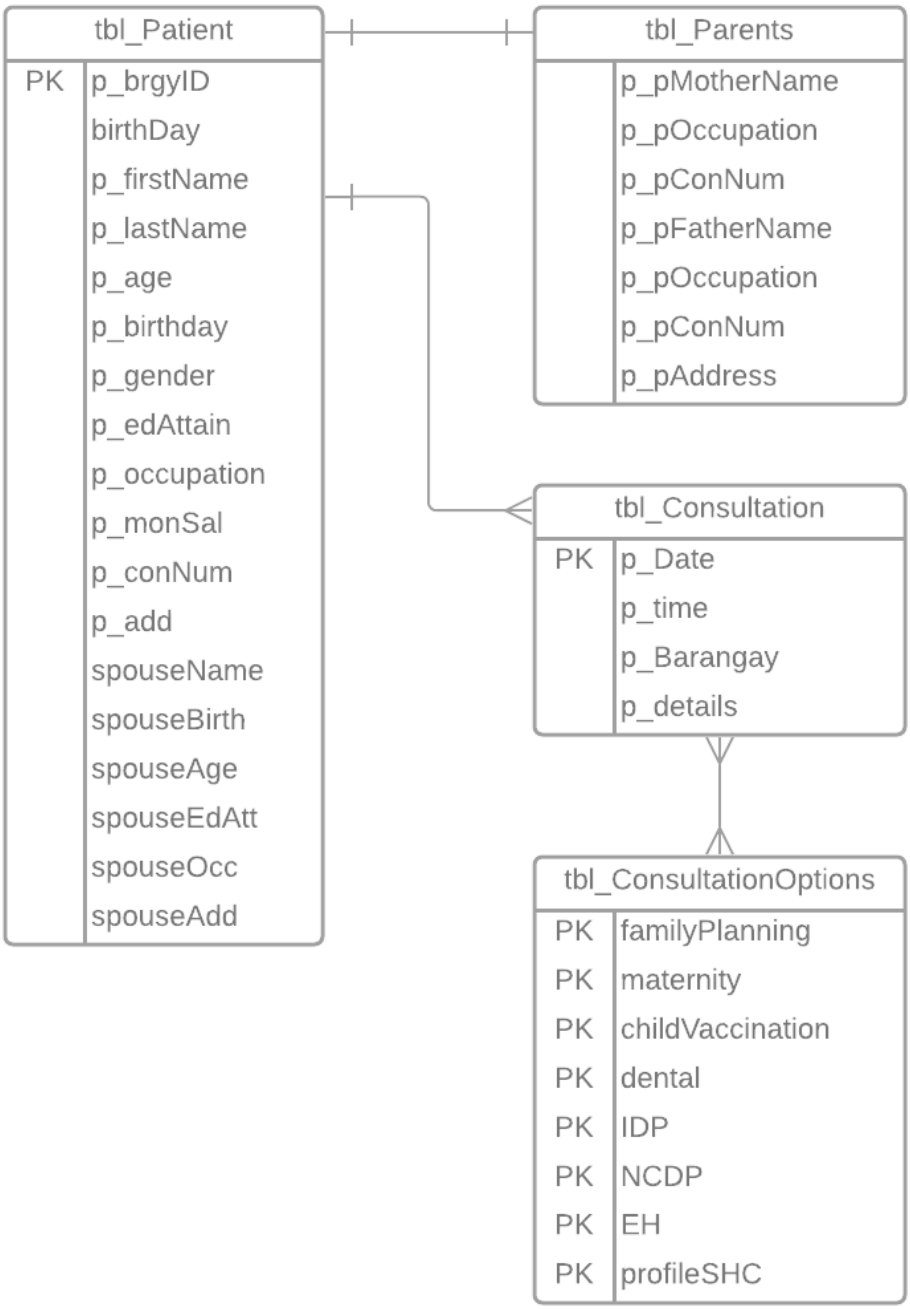
ERD #5: Child Vaccination



ERD #6: Non-communicable Disease

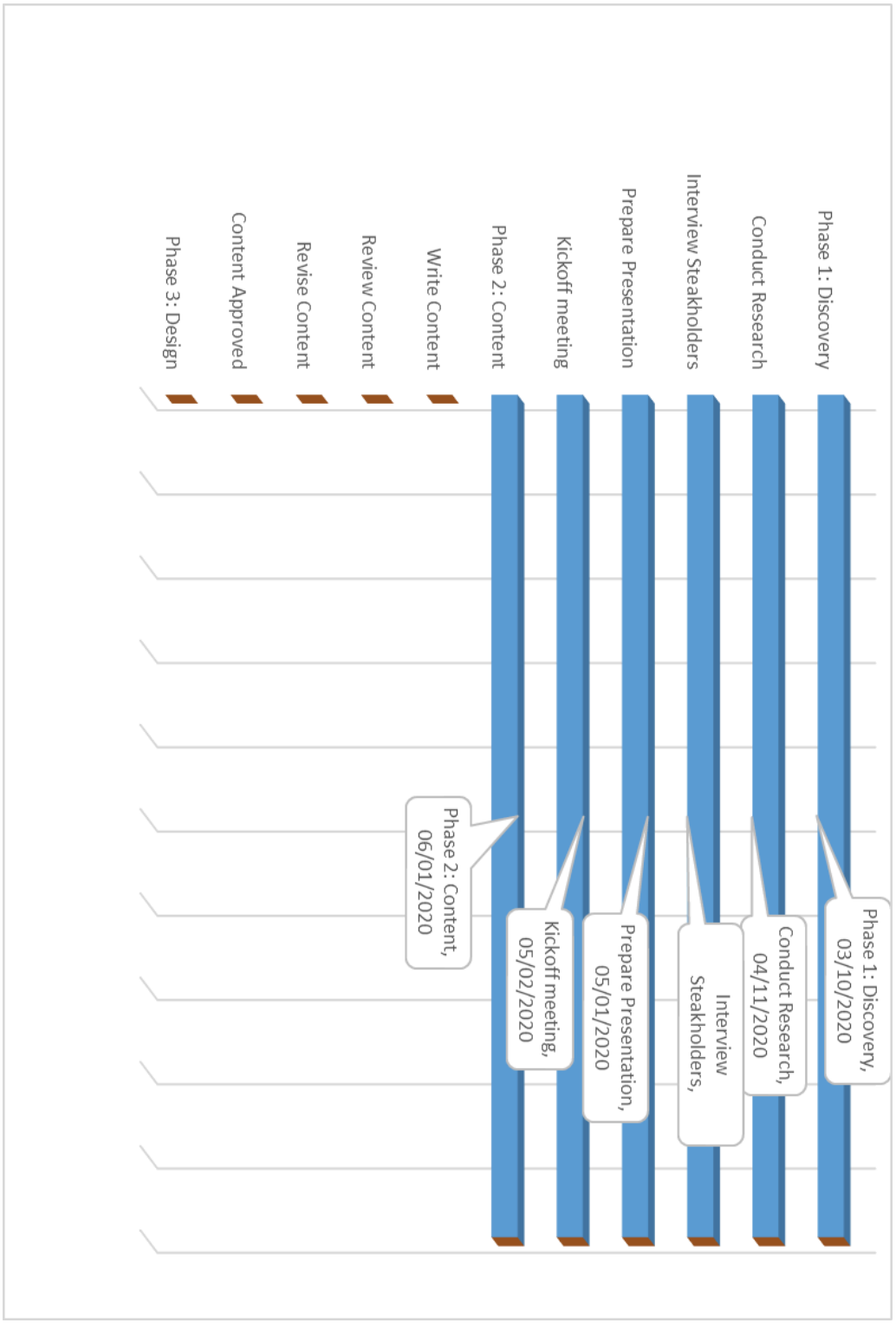


ERD #7: Patient & Barangay Health Worker

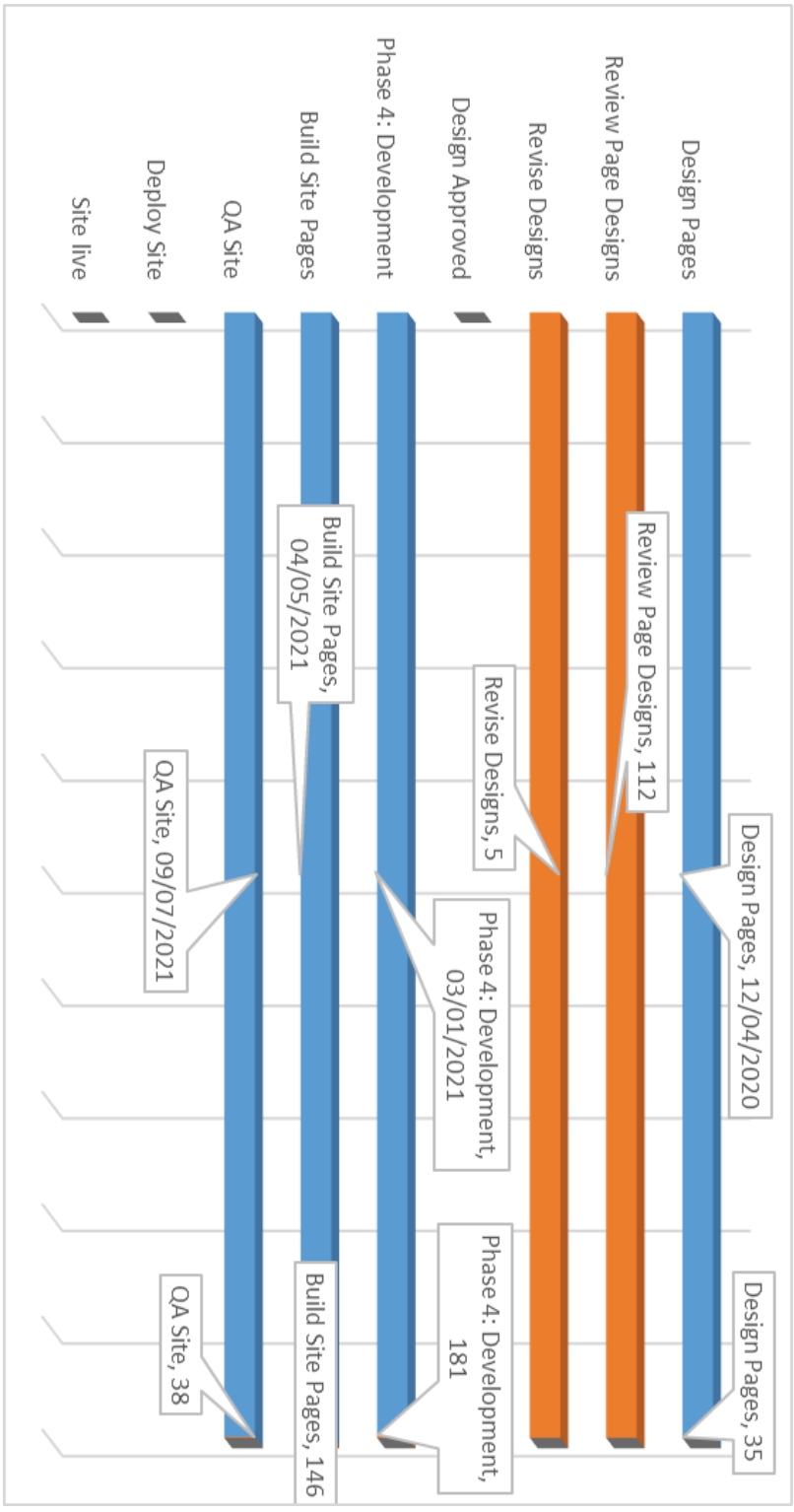


ERD #8: Patient

ERD #9: Family Planning



Ganttt Chart 1: Completed Phases 1 and 2



Ganttt Chart 2: Completed Phases 3 and 4

Appendix B

Supporting Documentation for Related Literature

I. Privacy Guidelines for Implementation of Philippine Health Information Exchange

According to the Health Privacy Code Implementing the joint administrative order no. 2016-0002[27] "Privacy Guidelines for the Implementation of the Philippine Health Information Exchange", health information privacy rules has:

Rule #1: Collecting and Processing of Health Information

- Collection and Processing of Health Information
    - Where the consent shall conform to the requirements or characteristics of a valid informed consent which consist of a sound mind, at least 18 years old, and not under the influence of drugs or liquor;
    - Where amount and accuracy of information has relevant factual data about a procedure and/or treatments, its benefits, risks, and possible complications or outcomes;
    - Where patient understanding covers education, language or dialect;
    - Where voluntariness makes an autonomous decision without force or intimidation, and understands that he/she can withdraw consent anytime without consequences.
- Point of Collection of Information
- Processing of Information
- Patient Identifier
- Point of de-identification
- Highly Communicable Disease and Special Conditions
- Authorized personnel to amend data if required.
- Reportorial Requirements
- Information to be Shared

- Filing and Storage

Rule #2: Access of Health Information

- Access of Healthcare Providers
- Accessible Information for Health Care Providers
- Approval of Access
- Access of Patient or Client (minors, incapacitated)
- Access of Third Party
- Third Party Use and Disclosure

Rule #3: Use and Disclosure of Health Information

- Use and Disclosure (Deceased Individuals, Medico-legal cases, legal authorities and/or Government Agencies)
- Privilege Communication
- Training Hospitals and Academic or Clinical Requirements Purposes

Rule #4: Organizational Security Measures

- Policies and Procedures
- Contract with Third Party
- Authorization and Document Retention
- The Information Technology Personnel
- The Medical Records Officer

Rule #5: Physical Security

- Inventory of Information Technology Physical Devices
- Access to Physical Infrastructure (Server Access, Computer Access, Computer Loss)
- Bringing of devices outside the health facility
- Bring-your-own-device (BYOD) (Agreement, Training, Configuration, Device Requirements)
- Business continuity and Disaster recovery (Physical Backup, Business Continuity)

Rule #6: Technical Safeguards

- Access Controls (Information Access Management, User Identification, Emergency Access Procedure, Automatic Logoff, Encryption and Decryption, Multi-factor Authentication)
- Audit Controls (Recording of Information, Audit Data Life Span, Access to Audit Data)
- Integrity Controls (Mechanism to authenticate electronic health information, digital signatures, sum verification, data storage encryption, proper handling of mechanical components, offline modes and caching, interface integration of information systems)
- Transmission Security
- Identity Authentication
- Storage Security

Rule #7: Cloud Services

- Cloud Services
- Contract between health care provider and cloud service provider
- Composite Services

Rule #8: Use of Social Media

- Definition of Social Media
- Unauthorized posting of personal data
- Administrative Responsibilities
- Responsible Social Media Use of Health Care Professionals
- Health Education and Promotion
- Professional Guidelines for Social media Use for Persons Involved in the PHIE

Rule #9: Human Resources

- On-boarding of employees of the health care facilities
- Employment Period (management responsibilities, awareness and training, disciplinary process)

- Termination or Off-boarding of Employees (Return of Assets, Access Rights)

Rule #10: Health Research

- Research Subject
- Research Protocol
- Research Projects
- Research Data

Rule #11: Patient Registries

- Registry Design
- Informed Consent
- Registry Data
- Data Collection
- Registry Data for Research
- Registries for Vulnerable Population
- Linking of Registries

Rule #12: Publication and Public Communication

- Publication of Privacy Policy
- Site Privacy Policy

Rule #13: Health Privacy Board

- Rationale
- Composition
- Competencies and Qualifications
- General Roles and Functions
- Board Member for Training and Capacity Building
- Board Member for Privacy Compliance and Planning

Rule #14: The Privacy Team of a Health Facility

- Rationale
- Appointment

- Qualifications
- Roles and Functions
- Staff

Rule #15: Compliance, Incident Reporting, Response

- Compliance
- Incident Reporting and Response
- Notification in the Case of Breach (Forms of Notification, Content of Notification, Delay of Notification

Rule #16: Procedures in the Investigation of Complaints Filed before the Health Privacy Board

- General Principles
- Procedure and Complaint and Investigation
- Resolution

Rule #17: Penalty Clause

- Penalties

Designation of Data Protection Officers Pursuant to Republic Act No. 10173, Titled, Data Privacy Act of 2012

Republic of the Philippines
DEPARTMENT OF THE INTERIOR AND LOCAL GOVERNMENT
DILG-NAPOLCOM Center, EDSA corner Quezon Avenue, West Triangle, Quezon City
http://www.dilg.gov.ph

DESIGNATION OF DATA PROTECTION OFFICERS PURSUANT TO REPUBLIC ACT NO. 10173, TITLED, DATA PRIVACY ACT OF 2012

Memorandum Circular No. 2018-36
March 19, 2018

1.0 Background

1.1. The National Privacy Commission (NPC) is a government agency mandated to administer and implement the Data Privacy Act of 2012 (R.A. No. 10173), to monitor and ensure compliance of entities that process personal data with the standards set for data protection in government and the private sector.

1.2. The NPC has previously issued NPC Advisory NO. 2017-01 dated March 14, 2017 relative on the designation of Data Protection Officers (DPO).

2.0 Purpose

2.1 The purpose of this Circular is to ensure compliance with the provisions of the Data Privacy Act of 2012 including its Implementing Rules and Regulations (IRR) and Section 4 of NPC Circular 2016-01, which provides that a government agency engaged in the processing of personal data shall, through its head of agency, designate a DPO.

3.0 Legal Compliance

3.1. Pursuant to Section 7 of Republic Act No.10173, also known as the Data Privacy Act of 2012 (DPA), the National Privacy Commission (NPC) is charged with the administration and implementation of the provision of the law, which includes ensuring compliance with the provisions of the DPA and with international standards for data protection, and carrying out efforts to formulate and implement plans and policies that strengthen the protection of personal information in the country, in coordination with other government agencies and the private sector.

3.2. Under Section 26(a) of the IRR of the same law, any natural or juridical person or other body involved in the processing of personal data shall designate an individual or individuals who shall function as data protection officer (DPO), compliance officer, or shall otherwise be accountable for ensuring compliance with applicable laws and regulations for the protection of data privacy and security.

4.0 Scope/Coverage

All Provincial Governors, City and Municipal Mayors, Punong Barangays, DILG Regional Directors, ARMM Regional Governor, and all others concerned.

5.0 Policy Content and Guidelines

5.1. Local Chief Executives are hereby reminded to ensure compliance to the provisions of laws and issuances relative to privacy and data protection:

5.1.1. Each LGU shall designate a DPO. A component city, municipality, or barangay is allowed to designate a Compliance Officer for Privacy (COP), which shall be under the supervision of the DPO of the province of such component city, municipality or barangay.

5.1.2. Local Chief Executives are advised to download a copy of NPC Advisory 17-01 through their website, privacy.gov.ph, for guidance on the designation of a DPO as well as the qualifications necessary to be an effective DPO/COP for the LGU.

5.2. All DILG Regional Directors and the ARMM Regional Governor are hereby directed to:

5.2.1. Download more information about the DPA and NPC as well as the form for the Data Protection Officer through their website, www. privacy.gov.ph; and

5.2.2. Cause the immediate and widest dissemination of this Memorandum Circular to all local government units within their respective regional jurisdictions.

6.0 Sanctions

6.1. Non-compliance to this Memorandum Circular may be a ground for the erring official/s of the local government unit to disciplinary actions:

6.1.1. By the President, in case of elective official of a province, a highly urbanized or an independent component city;

6.1.2. By the governor, if the respondent is an elective official of a component city or municipality; or

6.1.3. By the mayor, if the respondent is an elective official of the barangay.

7.0 References

7.1. Republic Act No.10173, also known as the Data Privacy Act of 2012 (DPA)
7.2. IRR of RA No. 10173
7.3. NPC Advisory No. 2017-01 dated March 14, 2017

8.0 Repealing Clause

All DILG Memorandum Circulars inconsistent herewith in part or in full, are hereby modified, revoked, or repealed accordingly.

9.0 Effectivity

This Memorandum Circular shall take effect immediately.

10.0 Approving Authority

USEC. EDUARDO M. AÑO
Officer-in-Charge

11.0 Feedback

For related queries, kindly contact Atty. Vida Zora Bocar of the National Privacy Commission at Tel Nos. (02) 517-7810 or (02) 920-0101 loc. 7041 or at email address at vida.bocar@privacy.gov.ph.

Appendix C

Consent Form

SSL Labs SSL Report

Privacy Impact Assessment

Consent Form

SAN MATEO RIZAL BARANGAY HEALTH INFORMATION SYSTEM

PATIENT CONSENT FORM

Patient Name :
Location of Patient :
Date of Birth :
Barangay :
Barangay Health Worker-in-Charge :

Patient ID #:
Date of Consent Discussed:

FORM: *For patients seeking out-patient services*

I am asking for medical care and treatment at the Barangay Health Center of _____ and agree to accept services which may diagnose a medical condition, procedures to treat my condition and routine dental and medical care, including vaccination. I understand that these services will be provided to me by physicians, dentists, nurse practitioners, midwives, physician assistants and barangay health workers, some of whom may be in training. I have not been given any guarantees as to the results of the services I will receive.

I understand that my agreement to accept these services will remain in effect unless I say that I no longer want these services or until my treatment is completed.

I understand that my agreement to accept these services is called a General Consent and that it includes any routine procedure(s) or treatment(s) physical examination, administration of medication and other non-invasive procedures.

_____ _____
Signature of Health Care Agent/Legal Guardian — Date & Time
(Place a copy of the authorizing document in the medical record)

_____ _____
Signature and Relation of Surrogate — Date & Time
(Place a copy of the authorizing document in the medical record)

WITNESS:

I,_____ am a staff member who is not the patient's physician or authorized health care provider and I have witnessed the patient or other appropriate person voluntarily sign this form.

_____ _____
Signature of Health Care Agent/Legal Guardian — Date & Time

INTERPRETER/TRANSLATOR: (To be signed by the interpreter/translator if the patient required such assistance)

To the best of my knowledge the patient understood what was interpreted/translated and voluntarily signed this form.

_____ _____
Signature of Interpreter/Translator — Date & Time

Privacy Impact Assessment

I. Project Description

Barangay Health Services whether in an Urban or Rural area serves as the first correspondence on health concerns in a community. It cater to all walks of life providing free medical services on generic patient concerns such as child vaccination, maternal care, non-communicable diseases, etc.

The San Mateo Barangay Health Information System is implemented to provide automated record keeping on patients, online consultation scheduling with role-based access covering health services of the barangay. It is also adaptive to the current barangay health information system procedures following the Field Health Service Information System records from the Department of Health.

II. PIA Scope

The San Mateo Barangay Health Information System shall cover patient and health care provider's personal and health records residing at the identified barangays in San Mateo, Rizal – Philippines.

III. ***Personal Data Flow***

ASPECT	DETAILS
Collect	***Internal Use or Proprietary Information*** (Barangay Health Center Policies) ***Personal Information*** Patient's name ***Personal Confidential Information*** Patient's name, Contact Number, Barangay, Educational Attainment, Occupation, Monthly Salary, Contact number, Address, parent details, spouse details. ***Derivative Data*** (Barangay Health Center's Operating Procedure) ***Sensitive Personal Information (Enrolment)*** (Race, ethnic origin, medical, psychological beliefs, civil records, legal liabilities, court orders) ***Patients: Processing of Data*** Upon confirmation of registration into the Barangay Health Information System, patients, and their legal guardians (for minor) should undertake an agreement to share patient data between themselves. Patient records may be released to legal guardians when children are still minors without the need for such an agreement.
Store	Patient's Personal Information, Medical Record based on the chosen transaction. Patients Communications to the patients will be managed by the barangay health workers.

	The assigned barangay health workers may refer to the stored communication should there be an instance of validation. For the registration module, physical records must be kept in secure confines that restrict access only to the aforementioned authorized personnel. Medical passwords should be password protected with access restricted to the physicians and other authorized personnel.
Use	Patient's Personal Information, Medical Record based on the chosen transaction. ***Barangay Health Workers*** Only duly authorized personnel shall be entrusted with receiving and processing of student data and communication. Use of student data for patient access is justified when its access or use is compatible with the purpose and proportionate to the seriousness of the matter under investigation for grievance and other derivative records. ***Patients*** Patients
Retain	Patient's Personal Information, Medical Records
Transfer	Patient's Personal Information and Medical Record
Dispose	None

IV. ***Privacy Impact Analysis***

Transparency	Yes	No	Not Applicable / Remarks
Are data subjects aware of the nature, purpose, and extent of the processing of his or her personal data?	✓		
Are data subjects aware of the risks and safeguards involved in the processing of his or her personal data?	✓		
Are data subjects aware of his or her rights as a data subject and how these can be exercised? Below are the rights of the data subjects: a. Right to be informed b. Right to object c. Right to access d. Right to correct e. Right for erasure or blocking f. Right to file a complaint g. Right to damages h. Right to data portability	✓		
Is there a document available for public review that sets out the policies for the management of personal data?		✓	
Are there steps in place to allow an individual to know what personal data it holds about them and its purpose of collection, usage and disclosure?	✓		

Are the data subjects aware of the identity of the personal information controller or the organization/entity processing their personal data?	✓		
Are the data subjects provided information about how to contact the organization's Data Protection Officer (DPO)?	✓		

Legitimate Purpose	Yes	No	Not Applicable / Remarks
Is the processing of personal data compatible with a declared and specified purpose cwhich are not contrary to law, morals, or public policy?	✓		
Is the processing of personal data authorized by a specific law or regulation, or by the individual through express consent?	✓		

Proportionality	Yes	No	Not Applicable / Remarks
Is the processing of personal data adequate, relevant, suitable, necessary and not excessive in relation to a declared and specified purpose?	✓		
Is the processing of personal data necessary to fulfill the purpose of the processing and no other means are available?	✓		

Data Security	Yes	No	Not Applicable / Remarks
Please identify all steps taken to ensure that all data that is collected, used or disclosed will be accurate, complete and up to date:			
a. The APIs have pre-defined parameters that must be properly filled to pass the built-in validations.	✓		
b. The system is regularly tested for accuracy	✓		
c. Periodic reviews of the information	✓		
d. A disposal schedule in place that deletes information that is over the retention period	✓		
e. Staff are trained in the use of the tools and receive periodic updates f. Reviews of audit trails are undertaken regularly.	✓		
g. Independent oversight (i.e. Internal Audit)	✓		
h. Incidents are reviewed for lessons learnt and systems/processes updated appropriately.	✓		

Data Security	Yes	No	Not Applicable / Remarks
Do you have appropriate and reasonable organizational, physical and technical security measures in place?			

Organizational measures - refer to the system's environment, particularly to the individuals carrying them out. Implementing the organizational data protection policies aim to maintain the availability, integrity, and confidentiality of personal data against any accidental or unlawful processing (i.e. access control policy, employee training, surveillance, etc.,)	✓		
Physical measures – refers to policies and procedures shall be implemented to monitor and limit access to and activities in the room, workstation or facility, including guidelines that specify the proper use of and access to electronic media (i.e. locks, backup protection, workstation protection, etc.,)	✓		
Technical measures - involves the technological aspect of security in protecting personal information (i.e. encryption, data center policies, data transfer policies, etc.,)	✓		

Organizational Security	Yes	No	Not Applicable / Remarks
Have you appointed a data protection officer or compliance officer?	✓		
Are there any data protection and security measure policies in place?	✓		
Do you have an inventory of processing systems? Will you include this project/system?	✓		
Are the users/staffs that will process personal data through this project/system under strict confidentiality if the personal data are not intended for public disclosure?	✓		
If the processing is delegated to a Personal Information Processor, have you reviewed the contract with the personal information processor?	✓		

Physical Security	Yes	No	Not Applicable / Remarks
Are there policies and procedures to monitor and limit the access to this project/system?	✓		
Are the duties, responsibilities and schedule of the individuals that will handle the personal data processing clearly defined?	✓		
Do you have an inventory of processing systems? Will you include this project/system? (may inventory of IT assets: IT asset inventory)		✓	

Technical Security	Yes	No	Not Applicable / Remarks
Is there a security policy with respect to the processing of personal data?	✓		

Do you have policies and procedures to restore the availability and access to personal data when an incident happens?	✓		
Do/Will you regularly test, assess and evaluate the effectiveness of the security measures of this project/ system?	✓		
Are the personal data processed by this project/system encrypted while in transit or at rest?	✓		
The program has taken reasonable steps to protect the personal data it holds from misuse and loss and from unauthorized access, modification or disclosure?	✓		
If yes, which of the following has the program undertaken to protect personal data across the information lifecycle: a. Identifying and understanding information types b. Assessing and determining the value of the information c. Identifying the security risks to the information d. Applying security measures to protect the information e. Managing the information risks.	 ✓ ✓ ✓ ✓ ✓		
Disposal	**Yes**	**No**	**Not Applicable / Remarks**
The program will take reasonable steps to destroy or de-identify personal data if it is no longer needed for any purpose. If YES, please list the steps.	✓		
Cross-border Data Flows (if Applicable)	**Yes**	**No**	**Not Applicable / Remarks**
The program will transfer personal data to an organization or person outside of the Philippines If YES, please describe			**NOT APPLICABLE**
Personal data will only be transferred to someone outside of the Philippines if any of the following apply: a. The individual consents to the transfer b. The organization reasonably believes that the recipient is subject to laws or a contract enforcing information handling principles substantially similar to the DPA of 2012 c. The transfer is necessary for the performance of a contract between the individual and the organization d. The transfer is necessary as part of a contract in the interest of the individual between the organization and a third party e. The transfer is for the benefit of the individual			 **NOT APPLICABLE** **NOT APPLICABLE** **NOT APPLICABLE** **NOT APPLICABLE**

The organization has taken reasonable steps so that the information transferred will be stored, used, disclosed and otherwise processed consistently with the DPA of 2012 If YES, please describe			**NOT APPLICABLE**

Risk Estimation

Risk Estimation is computed based on the probability and level of impact of Personal Information being inappropriately accessed, used or disclosed depends on the exposure of PI data across various sub processes, systems and manual interventions. The PI impact level indicates the potential harm that could result to the subject individuals and/or the organization of PI were inappropriately accessed, used or disclosed.

Impact – severity of the injuries / harm that might arise if the event does occur.		
IMPACT RATING		
RATING	**DESCRIPTION**	**EXAMPLE**
1 Negligible	Either not be affected or may encounter a few inconveniences	Re-entering data, irritations, aggravations
2 Limited	Significant inconveniences	Additional costs, inaccessibility, minor physical ailments, stress
3 Significant	Significant inconveniences with serious difficulties	Misuse of funds, unemployment, subpoena, damage to property
4 Maximum	Significant inconveniences which may not overcome	Unserviceable debt, long term psychological or physical ailments, deaths

Probability – chance or probability of something happening.		
IMPACT RATING		
RATING	**DESCRIPTION**	**EXAMPLE**
1 Unlikely	Not Expected	Data room protected by badge reader and lock
2 Possible	Casual Occurrence	Data room protected by badge or lock
3 Likely	Frequent Occurrence	Has a security lobby
4 Very Likely	Very Likely	No security measure

PRIVACY RISK MAPPING AND RATING

1 Negligible	IMPACT
2 Limited	
3 Significant	
4 Maximum	

MAXIMUM				
SIGNIFICANT				
LIMITED				
NEGLIGIBLE				
	NEGLIGIBLE	**POSSIBLE**	**LIKELY**	**ALMOST CERTAIN**

Formula: Risk Rating = Impact x Probability

Rating	Types
1	Negligible
2 to 4	Low Risk
6 to 9	Medium Risk
10 to 16	High Risk

The threat/s vulnerabilities were considered for the Enrolment and Text Blast feature of the system.

Ref #	Threats / Vulnerabilities	Impact				Probability				Risk Rating	
1	**Customer Impact:** Complete failure of service across multiple lines of business > or = 5 minutes.	1	2	3	4	1	2	3	4	4	LOW RISK
2	**Staff Impact:** Actual or high risk of death or injury	1	2	3	4	1	2	3	4	8	MEDIUM RISK
3	**Brand Impact:** Company name becomes a byword for corporate misconduct or misadventure	1	2	3	4	1	2	3	4	4	LOW RISK
4	**Media Impact:** International, long-term media coverage	1	2	3	4	1	2	3	4	8	MEDIUM RISK

Appendix D

Survey Forms

Software Evaluation Criteria for Patients/Citizens of the Barangay
Development and Implementation of Online Health Services System

Privacy Statement:

To help further evaluate the ease of use and efficiency of the proposed system for the development and implementation of Online Health Services System, we would like to request your honest response in identifying the criteria for each category.

The information collected will only be used for this capstone project's purpose and all inputs will be treated with utmost confidentiality. The proponent and the institution affiliated respects your trust and protect your privacy, and therefore will never sell or share the raw data to any third parties.

TRANSACTION: **DATE:** _____

Please check the box.

	Scheduled Checkup		IDPCS
	Vaccination		NCDPC
	Family Planning		Environmental Health and Sanitation
	Maternity Checkup		Others: _____

Direction: Select from the given criteria below.

5 – Outstanding *4 – Good* *3 – Satisfactory* *2 – Poor* *1 – Unsatisfactory*

Reliability: Extent to which software performs according to requirements.

The proposed system was able to:	CRITERIA				
1. Save client information/profiles.	5	4	3	2	1
2. Lessen time spent in lining up for checkup/appointment.	5	4	3	2	1
3. Allows ease of entry and exit of program, backup, change inputs and give commands.	5	4	3	2	1
4. Allow off-site communication with the BHW.	5	4	3	2	1

Usability (Human Engineering): Extent of effort required to learn, operate and understand functions of the software.

The proposed system was able to:	CRITERIA				
1. Process requests as per transaction.	5	4	3	2	1
2. Provides information useful in instruction.	5	4	3	2	1
3. Works efficiently with less/no assistance from the BHW.	5	4	3	2	1
4. Takes not much of a time for features familiarization.	5	4	3	2	1
5. Match the data required for consultations and patient endorsement to the Super Health Center.	5	4	3	2	1

6. Provide options for change in transaction such as child vaccination, maternity consultation etc.	5	4	3	2	1

Understandability: Effort required for a user to recognize logical concepts and its applicability.

The proposed system was able to:	CRITERIA				
1. Posts updates understandable for patients.	5	4	3	2	1
2. Procedures were relatively intuitive and understandable considering the features of the current system.	5	4	3	2	1
3. Training for system familiarization is appropriate and efficient enough. (Video tutorials)	5	4	3	2	1
4. The medium of instructions for steps in completing a task within the system is simple and accurate.	5	4	3	2	1

Security: Encompasses a holistic restriction based on access in information from authentication to modification both internally and externally.

The proposed system/proponent was able to:	CRITERIA				
1. Allow users to access the system based on the registered account and password.	5	4	3	2	1
2. Restrict access based on the list of barangays' provided.	5	4	3	2	1
3. Provide the notification based on the Data Sharing Agreement (DSA) for patients.	5	4	3	2	1

Software Evaluation Criteria for Barangay Health Workers
Development and Implementation of Online Health Services System

POSITION: _____ DATE: _____

Privacy Statement:

To help further evaluate the ease of use and efficiency of the proposed system for the development and implementation of Online Health Services System, we would like to request your honest response in identifying the criteria for each category.

The information collected will only be used for this capstone project's purpose and all inputs will be treated with utmost confidentiality. The proponent and the institution affiliated respects your trust and protect your privacy, and therefore will never sell or share the raw data to any third parties.

Direction: Select from the given criteria below. Encircle your answer.

5 – Outstanding *4 – Good* *3 – Satisfactory* *2 – Poor* *1 – Unsatisfactory*

Reliability: Extent to which software performs according to requirements.

The proposed system was able to:	CRITERIA				
1. Save client information/profiles.	5	4	3	2	1
2. Record and update client vitals and diagnosis.	5	4	3	2	1
3. Help monitor patient history.	5	4	3	2	1
4. Lessen time spent in record finding and record keeping.	5	4	3	2	1
5. Allows ease of entry and exit of program, backup, change inputs and give commands.	5	4	3	2	1
6. Function as expected in multiple user levels.	5	4	3	2	1

Usability (Human Engineering): Extent of effort required to learn, operate and understand functions of the software.

The proposed system was able to:	CRITERIA				
1. Process requests as per transaction.	5	4	3	2	1
2. Generate reports per user needs.	5	4	3	2	1
3. Provides information useful in instruction.	5	4	3	2	1
4. Works efficiently with less/no assistance from the IT personnel.	5	4	3	2	1
5. Takes not much of a time for features familiarization.	5	4	3	2	1
6. Incorporate features that are essential in creating reports to be submitted to the Municipal Health Office.	5	4	3	2	1
7. Follow and improve the physical forms used on the current system.	5	4	3	2	1
8. Match the data required for consultations and patient endorsement to the Super Health Center.	5	4	3	2	1

9. Provide options for change in transaction such as child vaccination, maternity consultation etc.	5	4	3	2	1
10. Provide easy navigation to different pages on the online system based on the defined level of access.	5	4	3	2	1

Understandability: Effort required for a user to recognize logical concepts and its applicability.

The proposed system was able to:	CRITERIA				
1. Evidently present functions separate for users (patient/barangay health workers).	5	4	3	2	1
2. Posts updates understandable both for patients and barangay health workers.	5	4	3	2	1
3. Procedures were relatively intuitive and understandable considering the features of the current system.	5	4	3	2	1
4. Training for system familiarization is appropriate and efficient enough.	5	4	3	2	1
5. The medium of instructions for steps in completing a task within the system is simple and accurate.	5	4	3	2	1

Security: Encompasses a holistic restriction based on access in information from authentication to modification both internally and externally.

The proposed system/proponent was able to:	CRITERIA				
1. Allow users to access the system based on the registered account and password.	5	4	3	2	1
2. Restrict access based on the list of barangays' provided.	5	4	3	2	1
3. Limit the functionalities available per user.	5	4	3	2	1
4. Provide the notification based on the Data Sharing Agreement (DSA) for patients and Municipal Hall.	5	4	3	2	1
5. Follow the standard requirement for Online Implementation security based on the Barangay Health Services administrative personnel.	5	4	3	2	1

Support: Covers support for users of the proposed system in different levels in formats such as written materials, training and cost.

The proposed system/proponent was able to:	CRITERIA				
1. Provide manuals and explain the content and functionalities of the system.	5	4	3	2	1
2. Facilitate an on-site training for administrative successors of the system.	5	4	3	2	1
3. Facilitate reassignment of user level based on the ability of manpower in the barangay health center.	5	4	3	2	1

Appendix E

Product Information
System How To's

Product Information

Product Name	:	Barangay Health Information System for San Mateo Rizal
Product Costings	:	Estimate: Php 132, 820.00
Product Specifications following	:	The system runs in a cloud based environment with the specifications:

>Disk Space 30 GB
>RAM 68MB
>CPU Cores (1)
>Inodes 200000
>Bandwidth 100GB
>Website: http://smrbhis.com

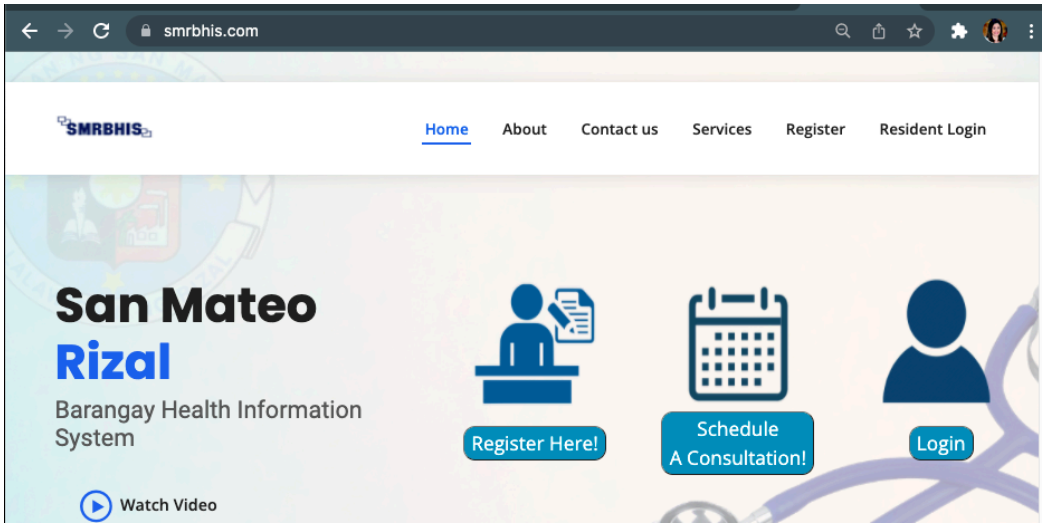
Product Functions:

The system provides the following features/functions:

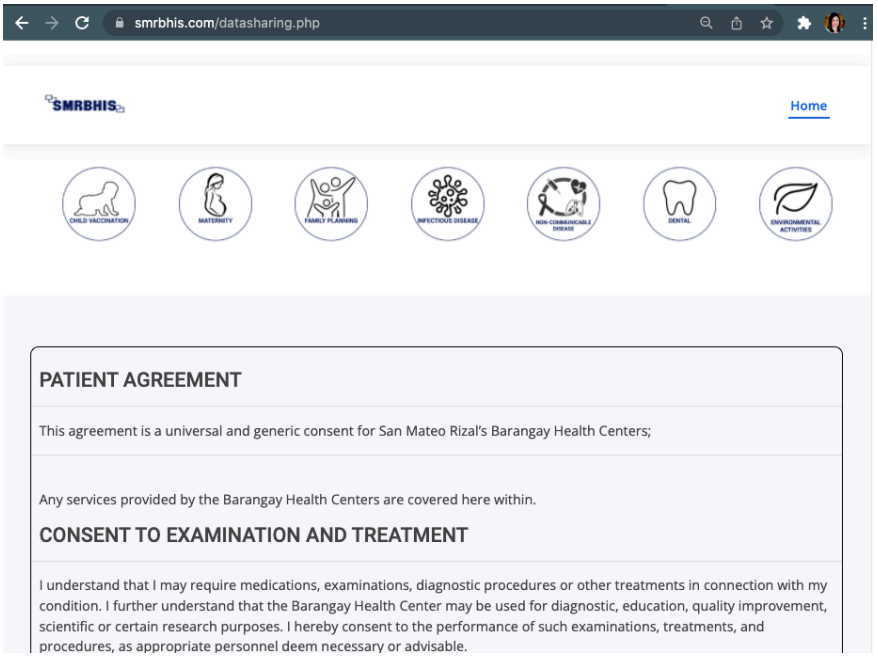
A. Patient Profiling
B. Consultation Appointment
C. Medical Health Records Keeping
D. Role Access based login
E. Patient Record updates
F. Content Management
G. Patient online registration, record access and medical records
H. Barangay Health Center access

Register as Patient

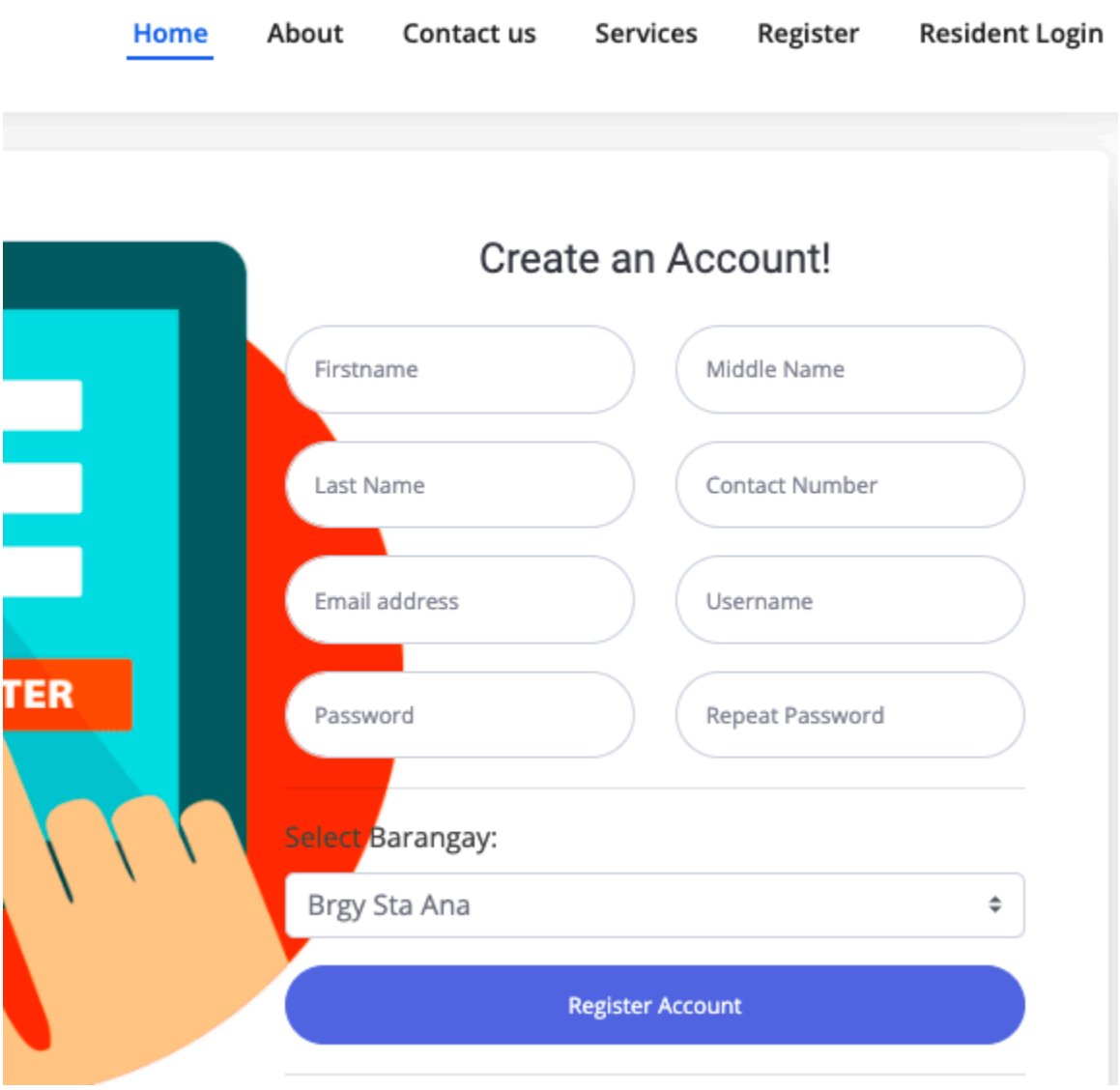
1. Launch your browser.
2. Visit www.smrbhis.com
3. Click the REGISTER button.



4. Read the Patient Agreement page.
5. Click Agree.

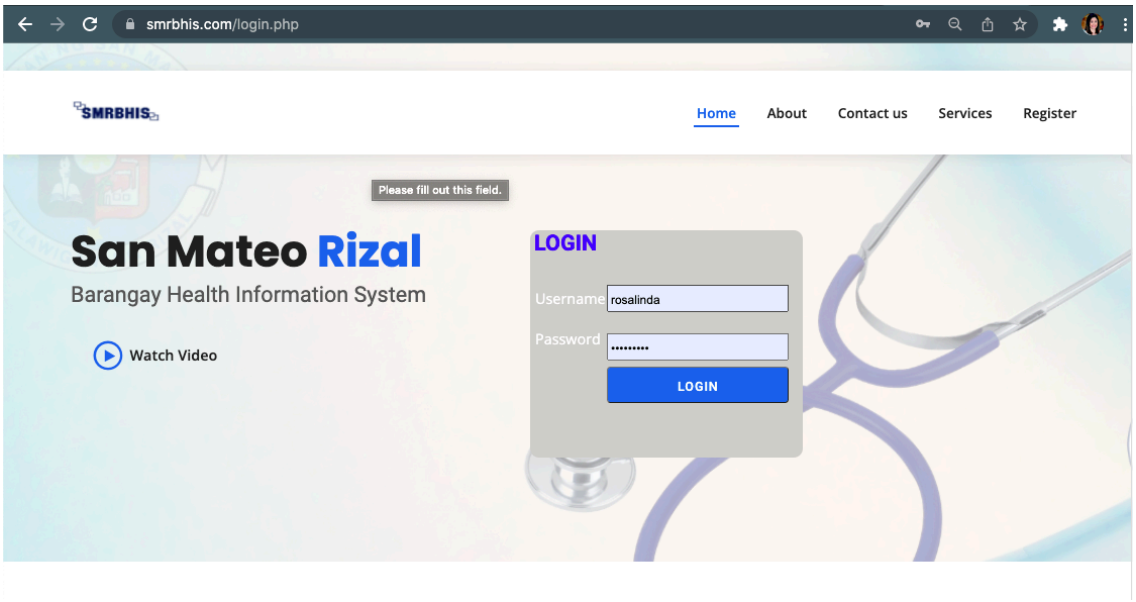


6. Register with the details required.
7. Enter a safe and secure password.
8. Choose your barangay.
9. Click Register Account.



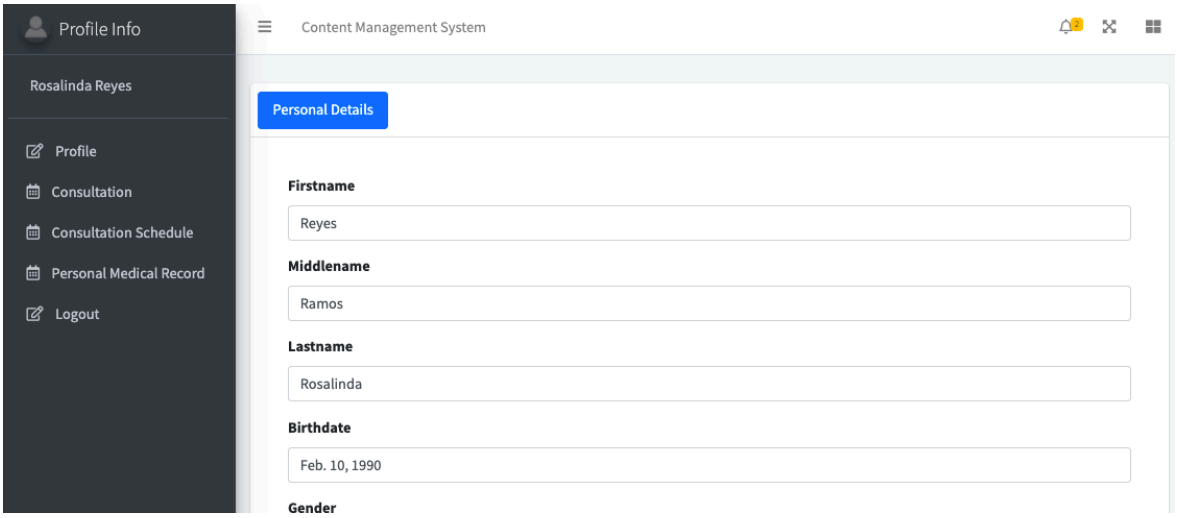
Log-in as Patient

1. On the smrbhis.com website, click LOGIN.
2. Input your username and password.
3. Click Login.

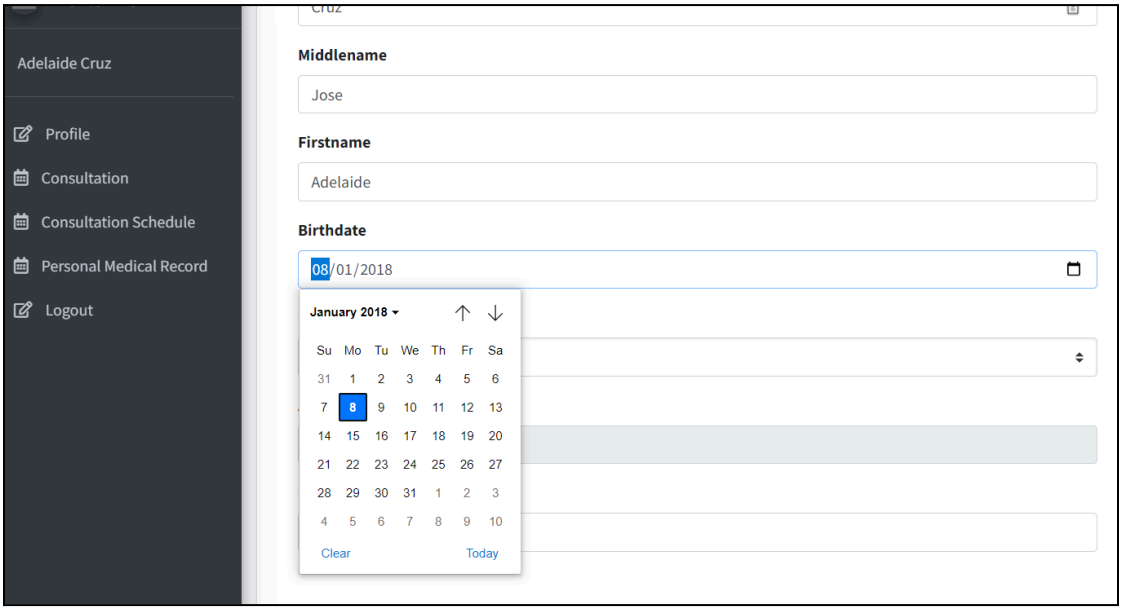


Patient Profile Update

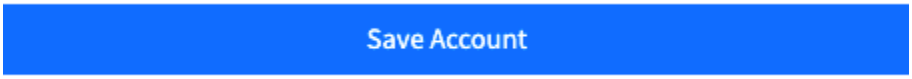
1. Click the Update Profile link on the upper right corner of your screen.
2. Input the details that you need to update.



4. For birthdates, the calendar option will appear. It will automatically compute for your age once the changes are saved.

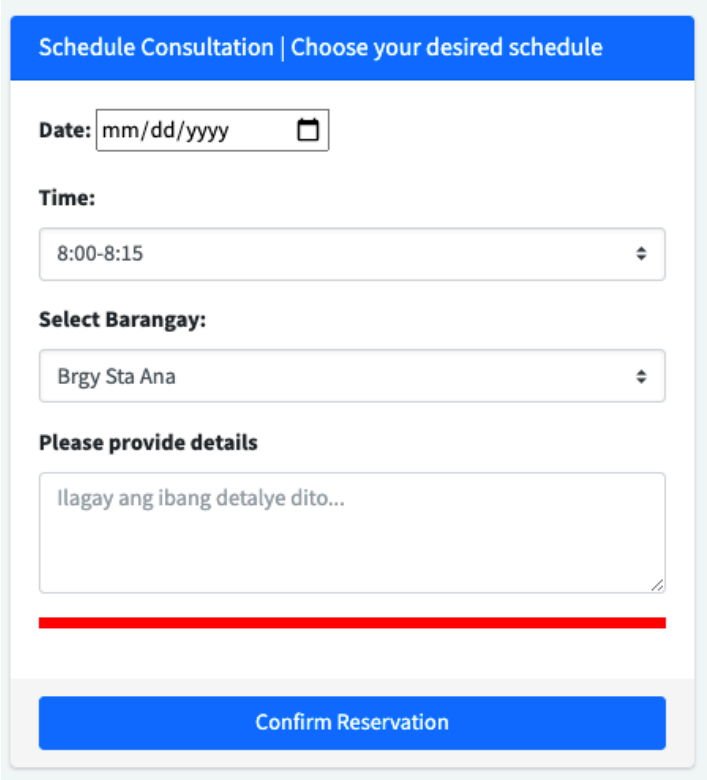


5. Click Save Account.



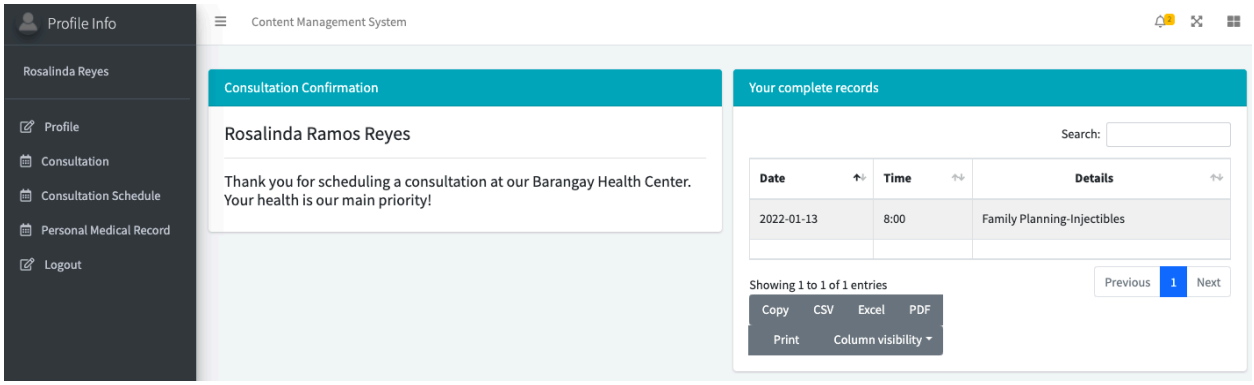
Consultation Schedule

1. To set a consultation appointment, click the Consultation tab on the left side of your screen.
2. On the right side, for schedule consultation, choose the date that you prefer.
3. Choose the time that you prefer.
4. Indicate any further details about your consultation.
5. Click the Confirm Reservation button.



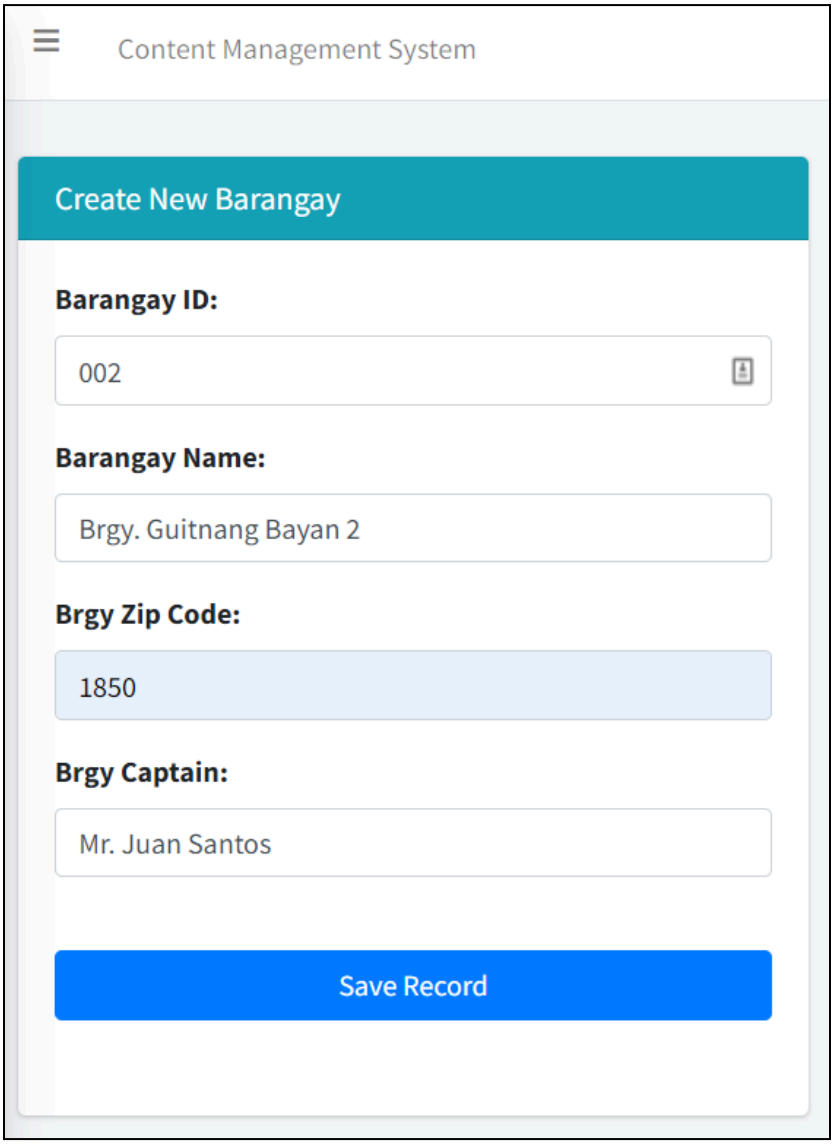
Confirm your consultation schedule.

1. To confirm your consultation schedule, click the Consultation Schedule link.



Login as Administrator | Add New Barangay

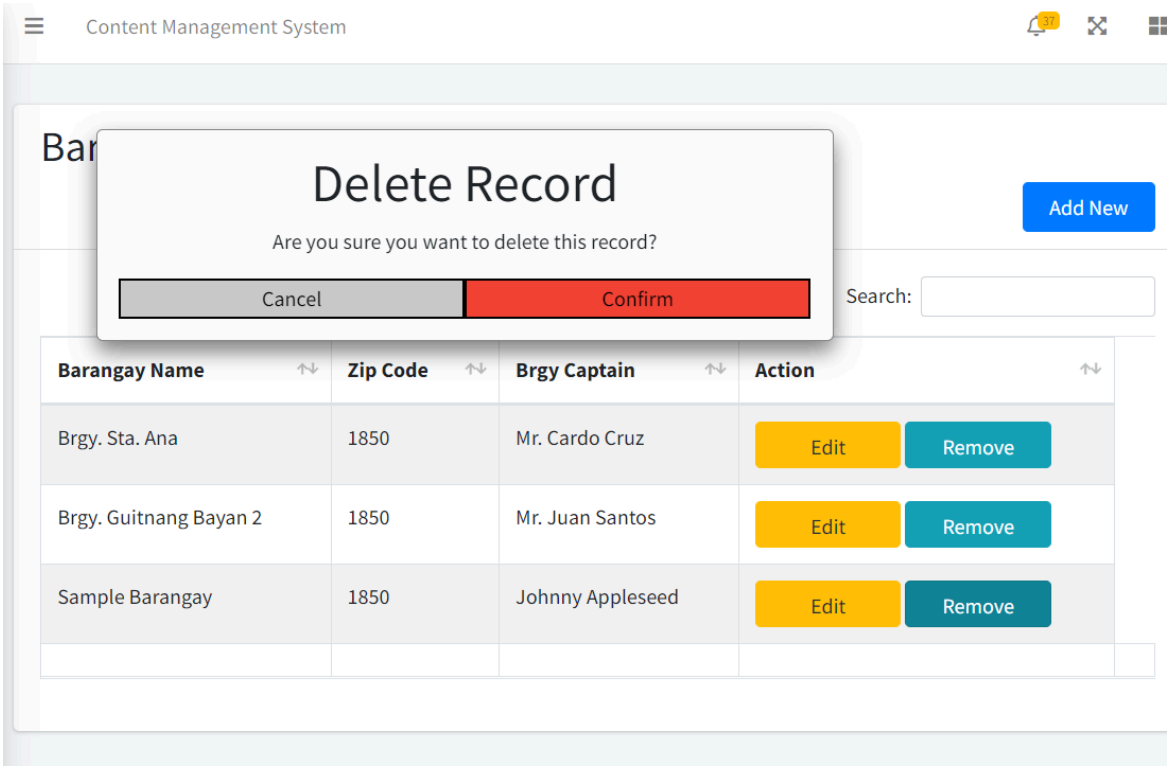
1. To add a new barangay on the system, click System Maintenance and ADD NEW.



2. Click Save Record.
3. Notice that a new Barangay is added.

Login as Administrator | Remove Barangay from the List

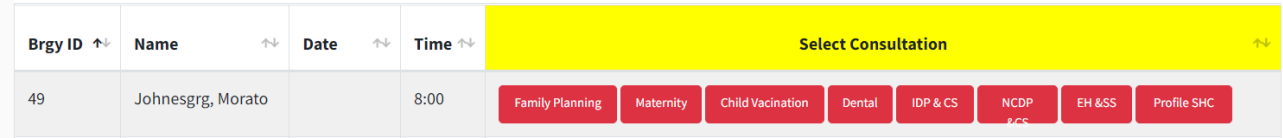
1. Click Remove
2. Confirm or cancel command.



3. Wait for the barangay to be deleted from the list.

Administrator | Confirm Appointments

1. Go to Appointments tab.
2. Click consultation.
3. Search for the patient's record.
4. Choose transaction / record to access.
5. Proceed to consultation.



Appendix F

Certifications from Clients

Presentation Certificates

Certificate of System Acceptance from Barangay Sta. Ana

CERTIFICATION

This is to certify that the proposed capstone project presented by the proponent Sarah Andrea B. Viana entitled **"Development and Implementation of On-Line Health Services System: An Adaptive Barangay Health Information System"**, has been carefully tested by the concerned barangay health services providers. The result of the testing phase showed that the system mentioned is working efficiently and is approved by this office for implementation.

This certification is being issued to Ms. Viana in partial fulfillment of the requirements for the subject Capstone 2, leading to the degree Master in Information Technology (MIT) in the Department of Information Sciences and Technology, University of Asia and the Pacific.

Issued this 7th of July 2021 at Barangay Sta. Ana, San Mateo Rizal.

Lorenzo D. Buenviaje

RURAL HEALTH MIDWIFE
STA. ANA HEALTH CENTER
SAN MATEO, RIZAL

Certificate of System Acceptance from Guitnang Bayan 1

CERTIFICATION

This is to certify that the proposed capstone project presented by the proponent Sarah Andrea B. Viana entitled **"Development and Implementation of On-Line Health Services System: An Adaptive Barangay Health Information System"**, has been carefully tested by the concerned barangay health services providers. The result of the testing phase showed that the system mentioned is working efficiently and is approved by this office for implementation.

This certification is being issued to Ms. Viana in partial fulfillment of the requirements for the subject Capstone 2, leading to the degree Master in Information Technology (MIT) in the Department of Information Sciences and Technology, University of Asia and the Pacific.

Issued this 7th of July 2021 at Barangay Guitnang Bayan 1, San Mateo Rizal.

MRS. NATIVIDAD R. CABAÑA
RURAL HEALTH MIDWIFE
GUITNANG BAYAN I
HEALTH CENTER
SAN MATEO RIZAL

UNIVERSITY OF ASIA AND THE PACIFIC
SCHOOL OF SCIENCES AND ENGINEERING
INFORMATION SCIENCE AND TECHNOLOGY DEPARTMENT

This

Certificate of Recognition

is presented to

Sarah Andrea B. Viana

University of Asia and the Pacific

as Author and Research Presenter of the paper entitled

Development and Implementation of Online Health Services Systems:
An Adaptive Barangay Information System

during the Seventh International Research Conference on Emerging Information Technology Trends in Asia and the Pacific 2021 with the theme

"Information Technology as a sustainable channel to business continuity, information security and green environment amidst the COVID-19 pandemic"

held on April 14, 2021 at the University of Asia and the Pacific, Pearl Drive, Ortigas Center, Pasig City

Florencio O. Gaa, PhD
Dean, School of Sciences and Engineering

Cyrus Paolo M. Buenafe
Department Chair, Information Science and Technology

Rommel Haber Dacoris
Conference Chair

Certificate of Presentation at the 7th International Research Conference (ITRC7)

Certificate of Presentation at the 1st Asian Training Summit and Research Congress on Information Technology (ATSRCIT 2021)

Republic of the Philippines
NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY
Cabanatuan City, Nueva Ecija, Philippines
ISO 9001:2015 CERTIFIED

COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
AND UNIVERSITY TRAINING DEPARTMENT & RESEARCH SERVICES DEPARTMENT

This

CERTIFICATE OF APPRECIATION

is hereby awarded to

SARAH ANDREA B. VIANA

for presenting her research paper entitled *"DEVELOPMENT AND IMPLEMENTATION OF ONLINE HEALTH SERVICES SYSTEM: AN ADAPTIVE BARANGAY INFORMATION SYSTEM"* during the

1st ASIAN TRAINING SUMMIT AND RESEARCH CONGRESS ON INFORMATION TECHNOLOGY (ATSRCIT 2021)

with the theme, **Leading the Future Through Data Analytics**

held on November 11, 2021.

Given this 11th day of November, in the year of our Lord, two thousand twenty one, at the Nueva Ecija University of Science and Technology, Cabanatuan City, Philippines.

ARNOLD DELA CRUZ, DIT
Dean, NEUST College of Information and Communications Technology

PROF. MARILENE C. HIPOLITO, RCh
Director, NEUST Research Department

JO NEIL T. PERIA, PhD, RMicro
Director, NEUST Training Department

RACHAEL R. MORALDE, PhD
Vice President for Research, Extension and Training

ENGR. FELICIANA P. JACOBA, EdD
University President

SCAN TO VERIFY
CICT2021-11-A0006

NEUST-TSD-F006
Rev. 00 (04.20.21)

ISO 9001:2015 Certified

"Transforming Communities through Science and Technology"

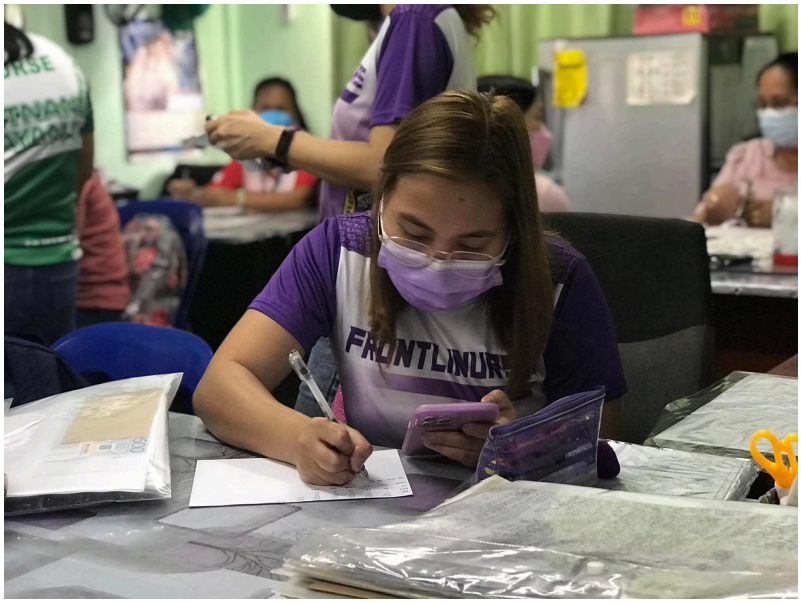
Contact No. (0927) 4778-392
Email: training@neust.edu.ph
www.neust.edu.ph

Appendix G

Photos

Child Vaccination Day at Guitnang Bayan 1

Midwife, Mrs. Natividad Cabana administering vitamin intake for children.



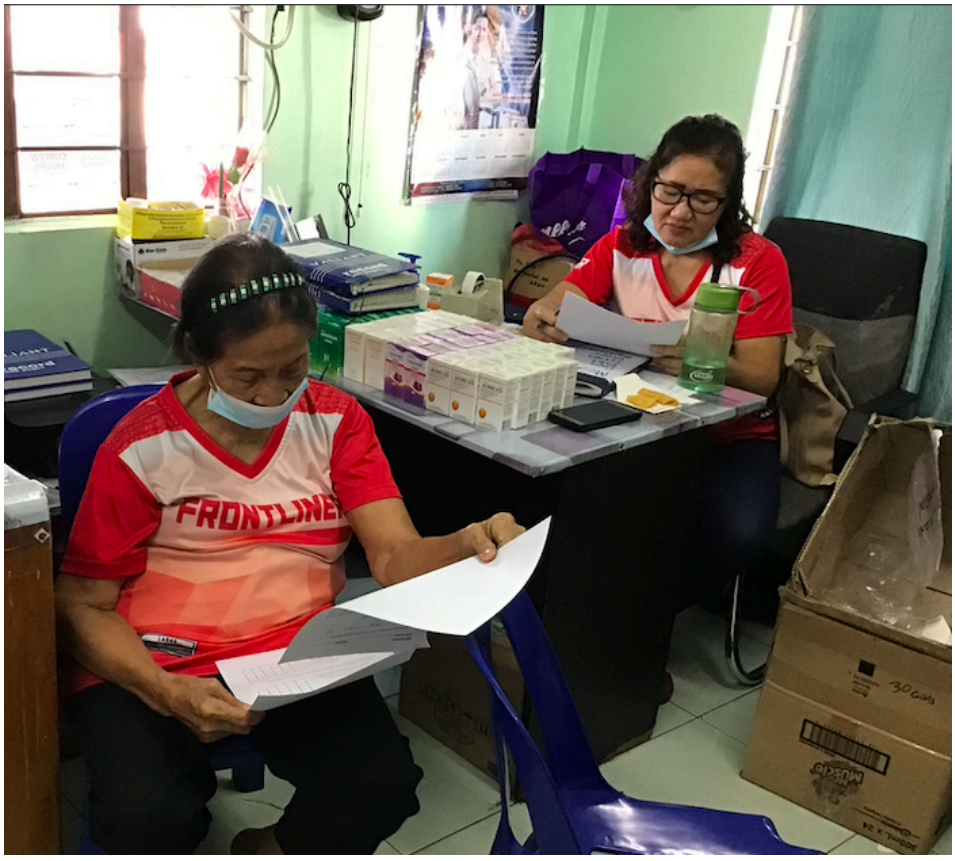
Nurse Marienell of Guitnang Bayan 1, updating the medical records.

User-System Evaluation - Brgy. Sta. Ana

Midwife, Mrs. Liwanag Buenviaje and her assistant barangay health worker

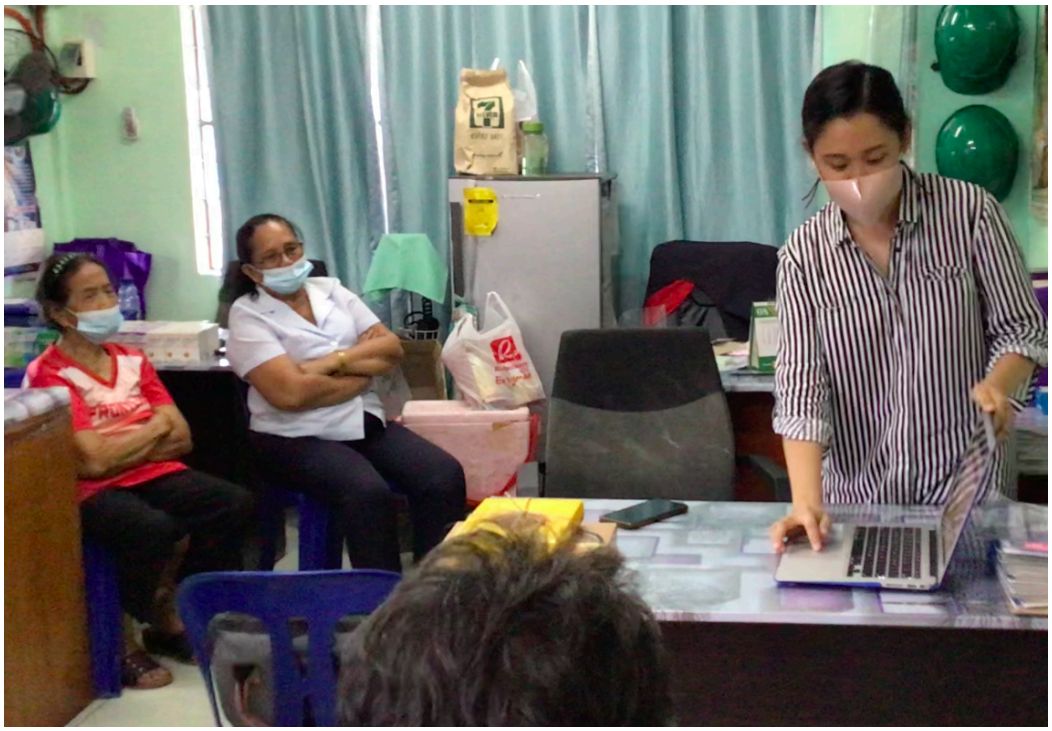
Barangay Health Workers of Sta. Ana

Midwife, Mrs. Cabana



Kagawad Sylvia Diesta and Barangay Health worker, Via Dioses

System Presentation at Guitnang Bayan 1



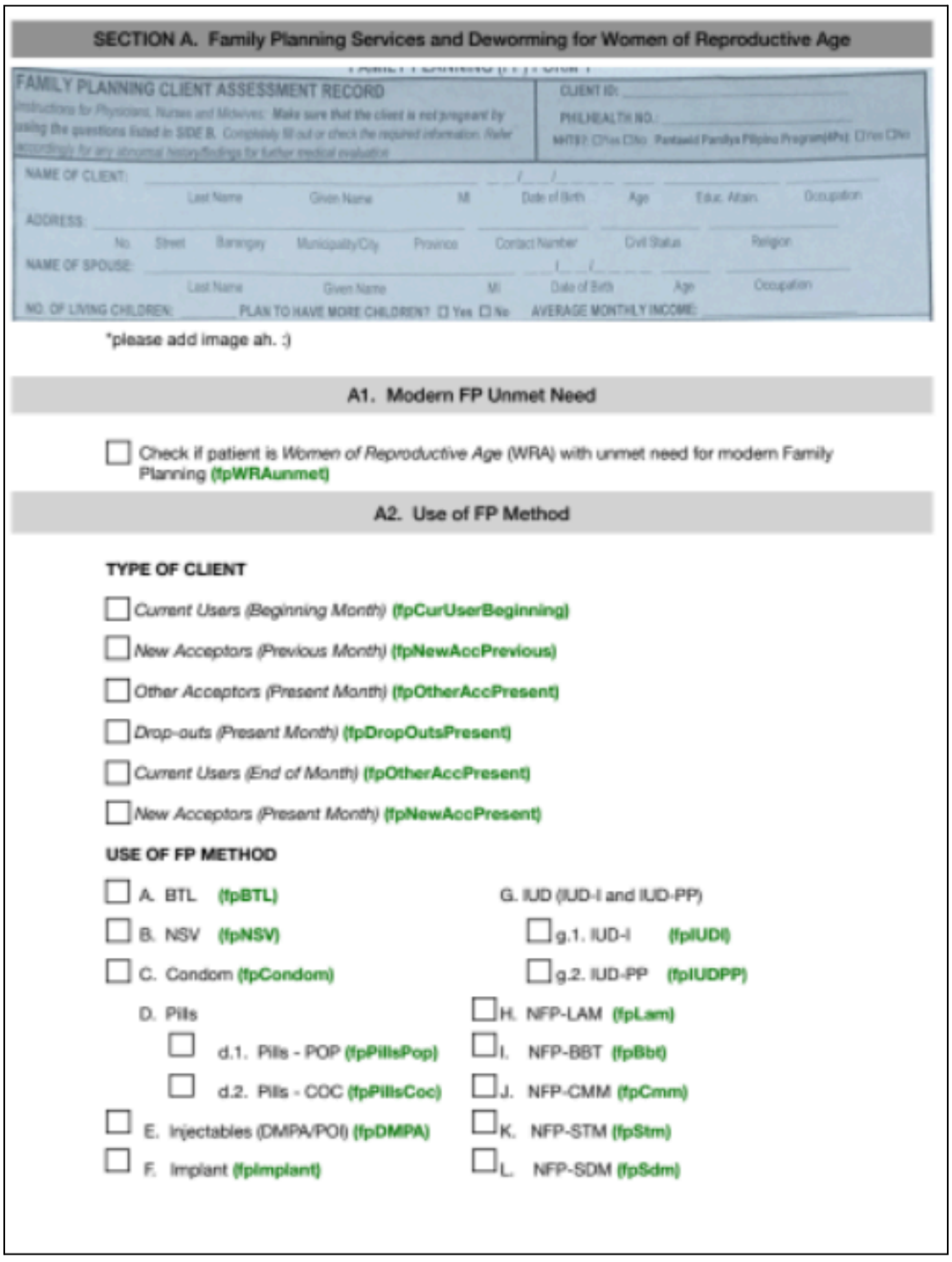
System Presentation at Sta. Ana

Appendix H

Documents Used for System Design

Initial System Page Designs

If the patient wants to do Family Planning, go to this form >> SECTION A



SECTION A. Family Planning Services and Deworming for Women of Reproductive Age

FAMILY PLANNING CLIENT ASSESSMENT RECORD

Instructions for Physicians, Nurses and Midwives: Make sure that the client is not pregnant by asking the questions listed in SIDE B. Completely fill out or check the required information. Refer accordingly for any abnormal history/findings for further medical evaluation

CLIENT ID:

PHILHEALTH NO.:

NHTS? ☐Yes ☐No Pantawid Pamilya Pilipino Program(4Ps)? ☐Yes ☐No

NAME OF CLIENT: ____ / ____ / ____ ____ ____ ____

Last Name | Given Name | MI | Date of Birth | Age | Educ. Attain. | Occupation

ADDRESS: ____ ____ ____ ____

No. | Street | Barangay | Municipality/City | Province | Contact Number | Civil Status | Religion

NAME OF SPOUSE: ____ ____ / ____ / ____ ____ ____

Last Name | Given Name | MI | Date of Birth | Age | Occupation

NO. OF LIVING CHILDREN: ____ PLAN TO HAVE MORE CHILDREN? ☐ Yes ☐ No AVERAGE MONTHLY INCOME: ____

*please add image ah. :)

A1. Modern FP Unmet Need

☐ Check if patient is *Women of Reproductive Age* (WRA) with unmet need for modern Family Planning **(fpWRAunmet)**

A2. Use of FP Method

TYPE OF CLIENT

☐ *Current Users (Beginning Month)* **(fpCurUserBeginning)**

☐ *New Acceptors (Previous Month)* **(fpNewAccPrevious)**

☐ *Other Acceptors (Present Month)* **(fpOtherAccPresent)**

☐ *Drop-outs (Present Month)* **(fpDropOutsPresent)**

☐ *Current Users (End of Month)* **(fpOtherAccPresent)**

☐ *New Acceptors (Present Month)* **(fpNewAccPresent)**

USE OF FP METHOD

☐ A. BTL **(fpBTL)**

☐ B. NSV **(fpNSV)**

☐ C. Condom **(fpCondom)**

D. Pills

☐ d.1. Pills - POP **(fpPillsPop)**

☐ d.2. Pills - COC **(fpPillsCoc)**

☐ E. Injectables (DMPA/POI) **(fpDMPA)**

☐ F. Implant **(fpImplant)**

G. IUD (IUD-I and IUD-PP)

☐ g.1. IUD-I **(fpIUDI)**

☐ g.2. IUD-PP **(fpIUDPP)**

☐ H. NFP-LAM **(fpLam)**

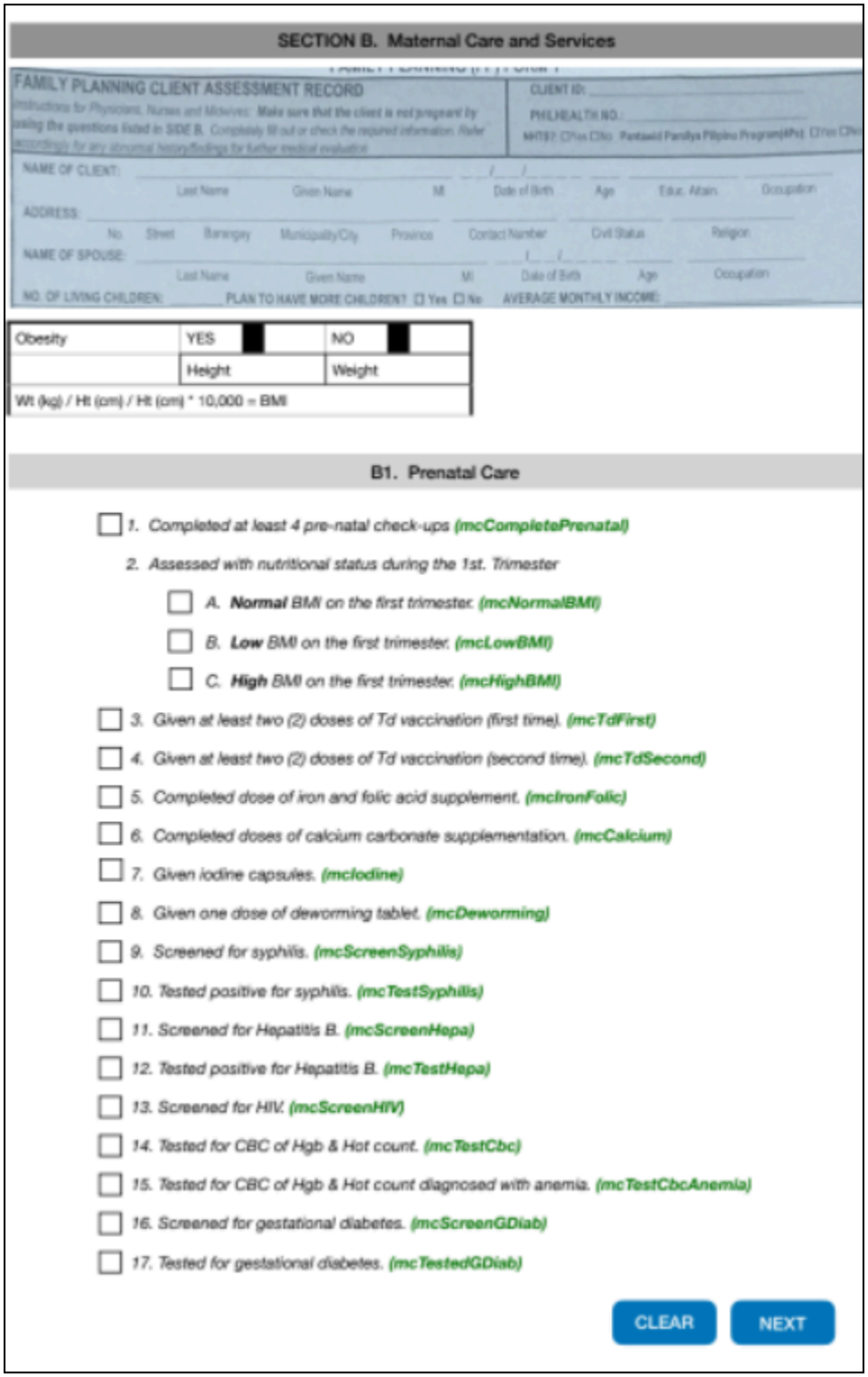
☐ I. NFP-BBT **(fpBbt)**

☐ J. NFP-CMM **(fpCmm)**

☐ K. NFP-STM **(fpStm)**

☐ L. NFP-SDM **(fpSdm)**

If the patient wants to do Maternity Checkup, go to this form >> SECTION B



SECTION B. Maternal Care and Services

FAMILY PLANNING CLIENT ASSESSMENT RECORD

Obesity	YES		NO
	Height		Weight
Wt (kg) / Ht (cm) / Ht (cm) * 10,000 = BMI			

B1. Prenatal Care

1. Completed at least 4 pre-natal check-ups (mcCompletePrenatal)
2. Assessed with nutritional status during the 1st. Trimester
   - A. **Normal** BMI on the first trimester. (mcNormalBMI)
   - B. **Low** BMI on the first trimester. (mcLowBMI)
   - C. **High** BMI on the first trimester. (mcHighBMI)
3. Given at least two (2) doses of Td vaccination (first time). (mcTdFirst)
4. Given at least two (2) doses of Td vaccination (second time). (mcTdSecond)
5. Completed dose of iron and folic acid supplement. (mcIronFolic)
6. Completed doses of calcium carbonate supplementation. (mcCalcium)
7. Given iodine capsules. (mcIodine)
8. Given one dose of deworming tablet. (mcDeworming)
9. Screened for syphilis. (mcScreenSyphilis)
10. Tested positive for syphilis. (mcTestSyphilis)
11. Screened for Hepatitis B. (mcScreenHepa)
12. Tested positive for Hepatitis B. (mcTestHepa)
13. Screened for HIV. (mcScreenHIV)
14. Tested for CBC of Hgb & Hct count. (mcTestCbc)
15. Tested for CBC of Hgb & Hct count diagnosed with anemia. (mcTestCbcAnemia)
16. Screened for gestational diabetes. (mcScreenGDiab)
17. Tested for gestational diabetes. (mcTestedGDiab)

CLEAR NEXT

B2. Intrapartum Care and Delivery Outcome

☐ *18. Delivered a childbirth.* ***(mcDelivered)***

19. Tested for gestational diabetes.

- ☐ *A. Normal Birth with normal birth weight.* ***(mcNormWeight)***
- ☐ *B. Live Birth with Low birth Weight.* ***(mcLowWeight)***
- ☐ *C. Live Birth with unknown birth weight.* ***(mcUnkWeight)***

20. Delivery attended by skilled health professional.

- ☐ *A. Attended by a doctor.* ***(mcAttDoc)***
- ☐ *B. Attended by a nurse.* ***(mcAttNurse)***
- ☐ *C. Attended by midwives.* ***(mcAttMidwives)***

21. Health facility-based deliveries.

- ☐ *A. Public Health Facility.* ***(mcPubFaci)***
- ☐ *B. Private Health Facility.* ***(mcPriFaci)***

☐ *22. Non-facility based deliveries.* ***(mcNonFaci)***

☐ *23a. Vaginal Delivery.* ***(mcVagDev)***

☐ *23b. Cesarian Section.* ***(mcCesDev)***

☐ *24a. Full-term births.* ***(mcFullBir)***

☐ *24b. Pre-term births.* ***(mcPreBir)***

☐ *24c. Fetal Death.* ***(mcFetalDe)***

☐ *24d. Abortion/miscarriage.* ***(mcAborMis)***

B3. Postpartum and Newborn Care

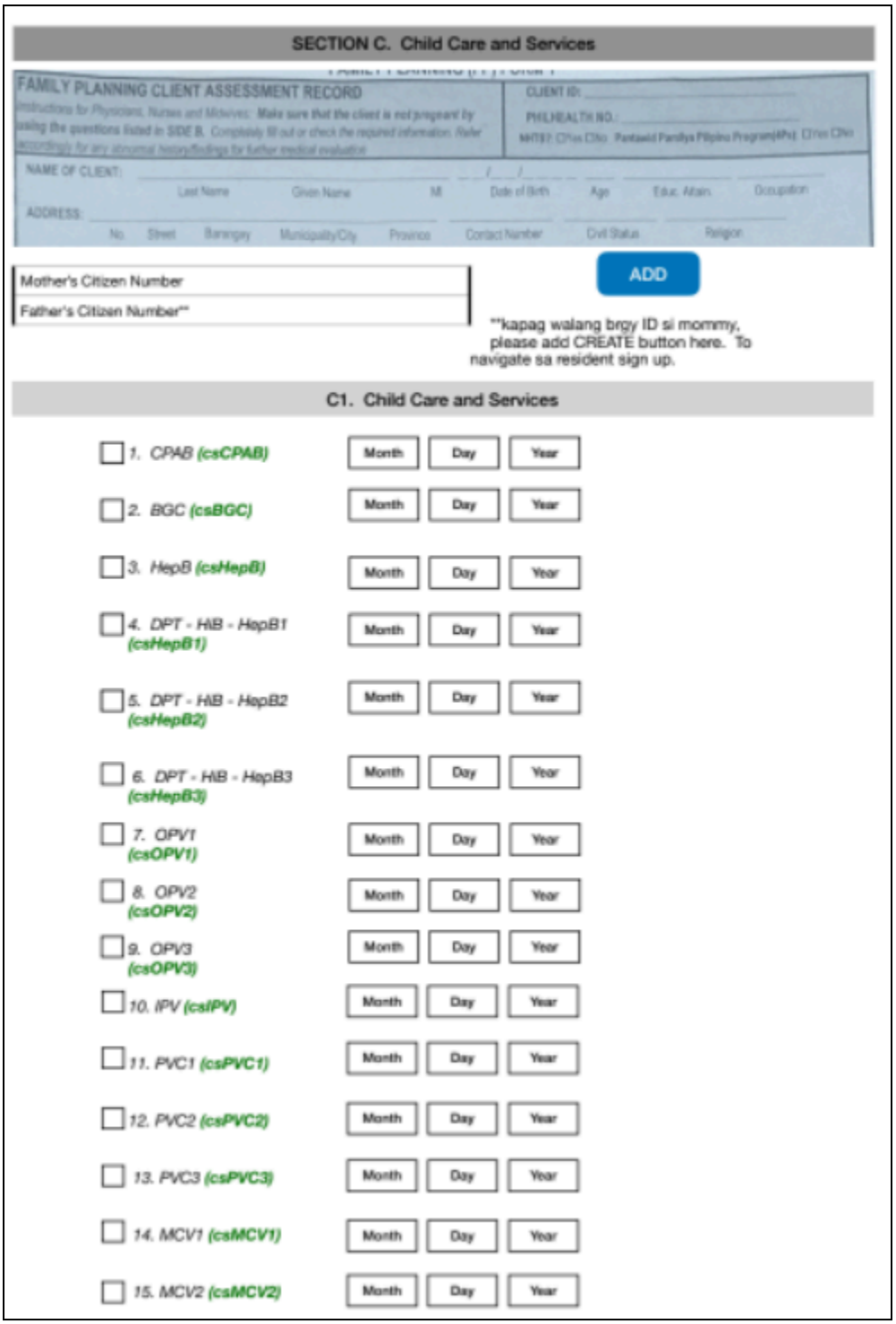
☐ *25. Women with their newborn completed at least 2 postpartum check-ups.* ***(mcCompPost)***

☐ *26. Women who completed iron with folic acid supplementation.* ***(mcCompIron)***

☐ *27. Postpartum women with Vitamin A supplementation.* ***(mcVASup)***

CLEAR NEXT

If the patient wants to do Child Vaccination, go to this form >> SECTION C



☐ 16. FIC (csFIC) Month Day Year

☐ 17. CIC (csCIC) Month Day Year

☐ 18. Td Grade 1 (csTdG1) Month Day Year

☐ 19. MR Grade 1 (csMrG1) Month Day Year

☐ 20. Td Grade 7 (csTdG7) Month Day Year

☐ 21. MR Grade 7 (csMrG7) Month Day Year

CLEAR NEXT

C2. Nutrition Services for Infants and Children

☐ 22. Newborns initiated on breastfeeding immediately after birth lasting to 90 mins. (csBreastFed90)

☐ 23. Preterm/LNW infants given iron supplementation. (csInfantsIron)

☐ 24. Infants exclusively breastfed until 6th month. (csBreastFed6)

☐ 25. Infants 6 mos. old initiated to complementary feeding with continued breastfeeding. (csBreastFedComp6)

☐ 26. Infants 6 months old initiated to complementary feeding but no long or never been breastfed. (csNoBreastFed6)

☐ 27. Infants 6-11 mos given 1 dose of Vitamin A 100,000 I.U. (csInfantIU1)

☐ 28. Children 12-59 mos. old given 2 doses of Vitamin A 200,000 IU. (csInfantIU2)

☐ 29. Infant 6-11 mos old who completed MNP supplementation. (csInfantSupp11)

☐ 30. Children 12-23 mos. who completed MNP supplementation. (csInfantISupp23)

C3. Deworming Services for Infants, Children and Adolescents

☐ 31. 1-19 y/o given 2 doses of deworming drug. (csDw1to19)

- ☐ A. PSAC, 1-4 y/o dewormed (2 doses). (csDw1to4)
- ☐ B. SAC, 5-9 y/o dewormed (2 doses). (csDw5to9)
- ☐ C. Adolescents, 10-19 y/o dewormed (2 doses). (csDw10to19)

C4. Management of Sick Infants and Children

☐ 32. Sick infants 6-11 mos old seen. (csSInfants6to11)

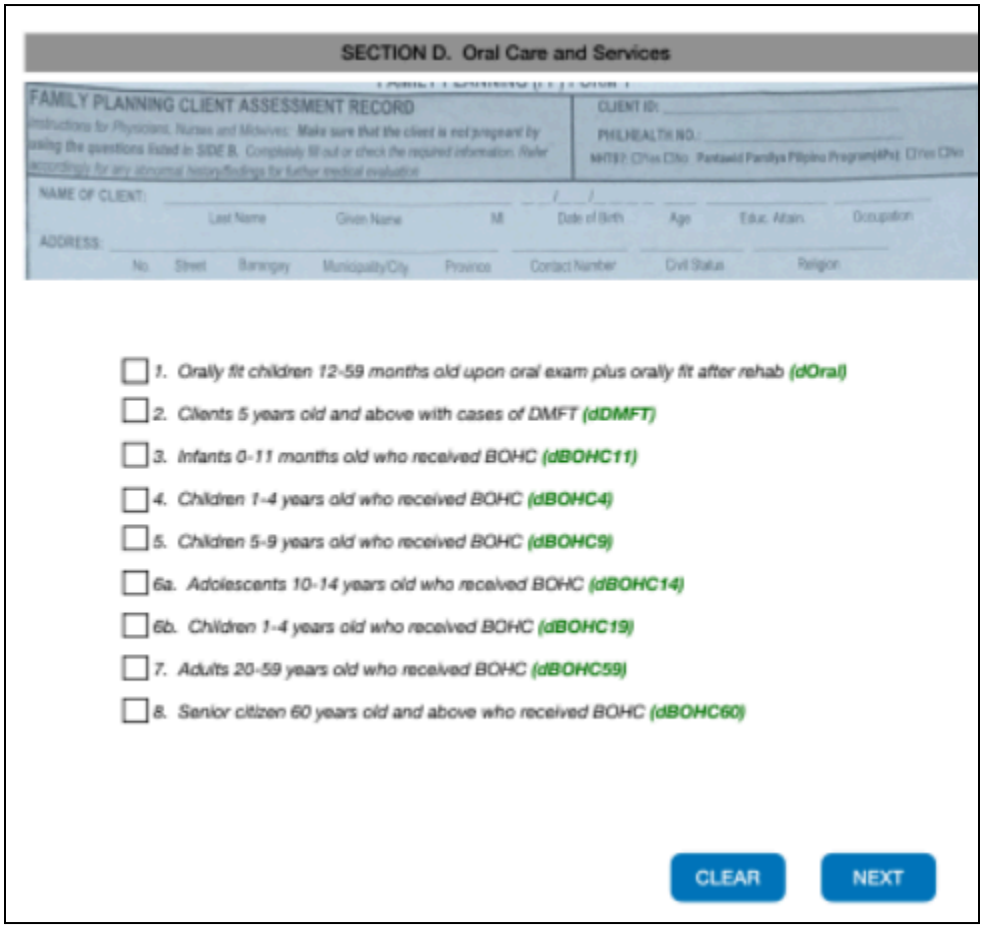
☐ 33. Sick infants 6-11 mos old received Vitamin A. (csSInfants6to11VitA)

☐ 34. Sick children 12-59 mos. old seen (csSInfants12to59)

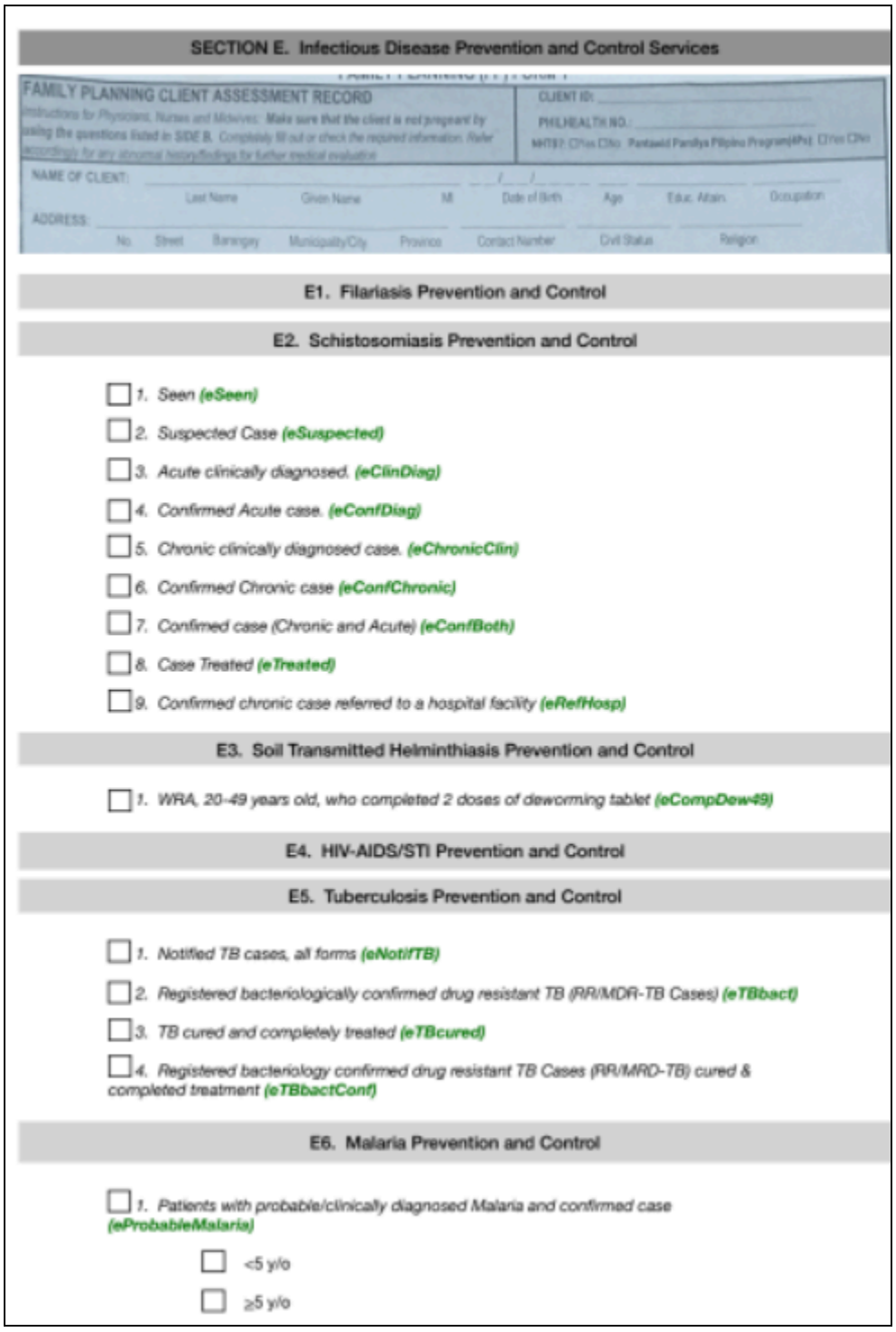
☐ 35. Sick children 12-59 mos. old received Vitamin A. (csSInfants12to59VitA)

36. Diarrhea cases 0-59 months old seen. (csDiarrhea0to59)
37. Diarrhea cases 0-59 months old received ORS. (csDiarrhea0to59ORS)
38. Diarrhea cases 0-59 months old received ORS with Zinc. (csDiarrhea0to59ORSz)
39. Pneumonia cases 0-59 months old seen. (csPneumonia0to59)
40. Diarrhea cases 0-59 months old completed treatment. (csDiarrhea0to59treat)
CLEAR
NEXT

If the patient wants to do Dental Checkup, go to this form >> SECTION D



If the patient's concern is about Infectious Disease Prevention and Control Services, go to this form >> SECTION E



SECTION E. Infectious Disease Prevention and Control Services

FAMILY PLANNING CLIENT ASSESSMENT RECORD

CLIENT ID:

PHILHEALTH NO.:

NAME OF CLIENT: / /

Last Name Given Name MI Date of Birth Age Educ. Attain. Occupation

ADDRESS:

No. Street Barangay Municipality/City Province Contact Number Civil Status Religion

E1. Filariasis Prevention and Control

E2. Schistosomiasis Prevention and Control

- ☐ 1. Seen **(eSeen)**
- ☐ 2. Suspected Case **(eSuspected)**
- ☐ 3. Acute clinically diagnosed. **(eClinDiag)**
- ☐ 4. Confirmed Acute case. **(eConfDiag)**
- ☐ 5. Chronic clinically diagnosed case. **(eChronicClin)**
- ☐ 6. Confirmed Chronic case **(eConfChronic)**
- ☐ 7. Confirmed case (Chronic and Acute) **(eConfBoth)**
- ☐ 8. Case Treated **(eTreated)**
- ☐ 9. Confirmed chronic case referred to a hospital facility **(eRefHosp)**

E3. Soil Transmitted Helminthiasis Prevention and Control

- ☐ 1. WRA, 20-49 years old, who completed 2 doses of deworming tablet **(eCompDew49)**

E4. HIV-AIDS/STI Prevention and Control

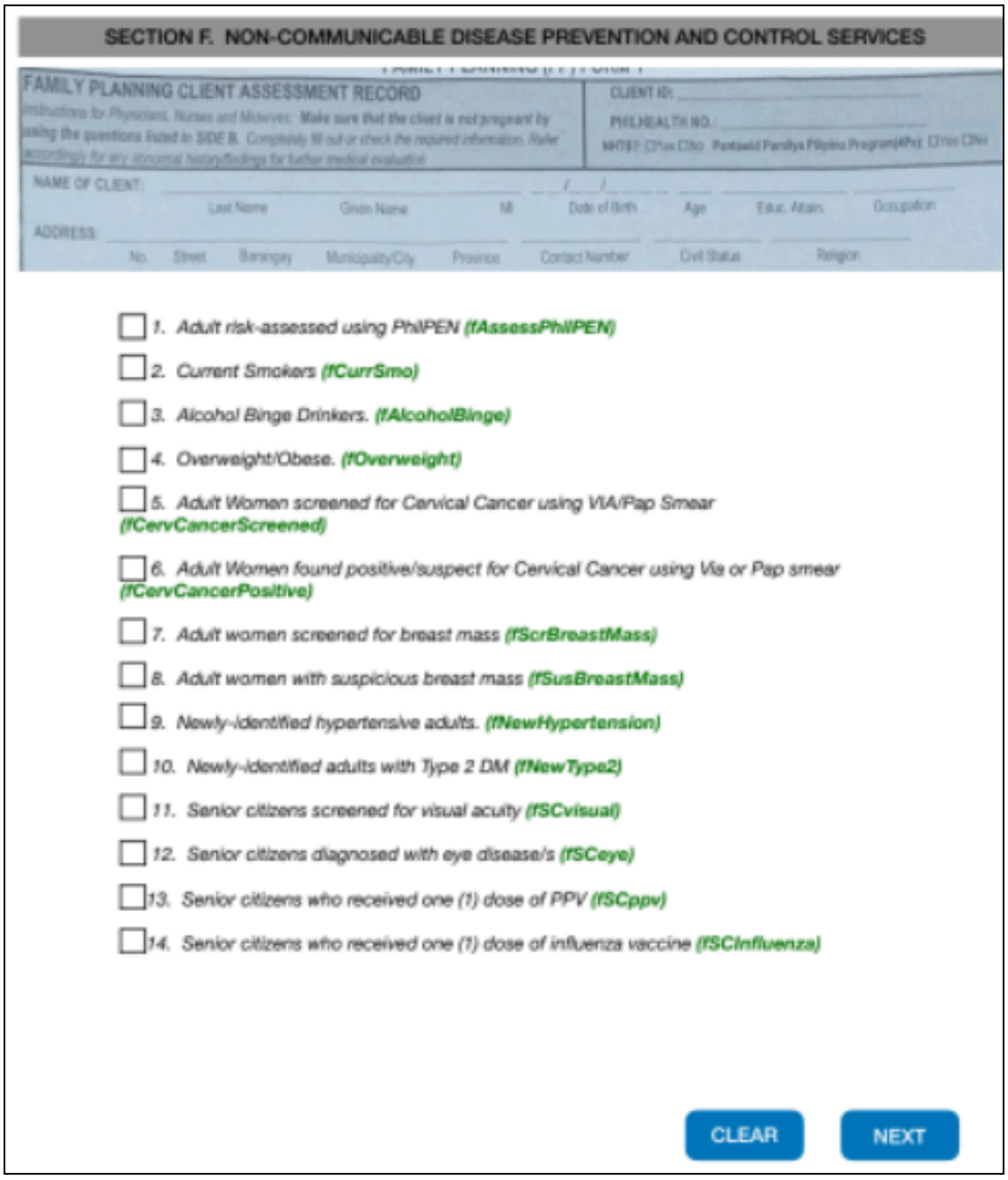
E5. Tuberculosis Prevention and Control

- ☐ 1. Notified TB cases, all forms **(eNotifTB)**
- ☐ 2. Registered bacteriologically confirmed drug resistant TB (RR/MDR-TB Cases) **(eTBbact)**
- ☐ 3. TB cured and completely treated **(eTBcured)**
- ☐ 4. Registered bacteriology confirmed drug resistant TB Cases (RR/MRD-TB) cured & completed treatment **(eTBbactConf)**

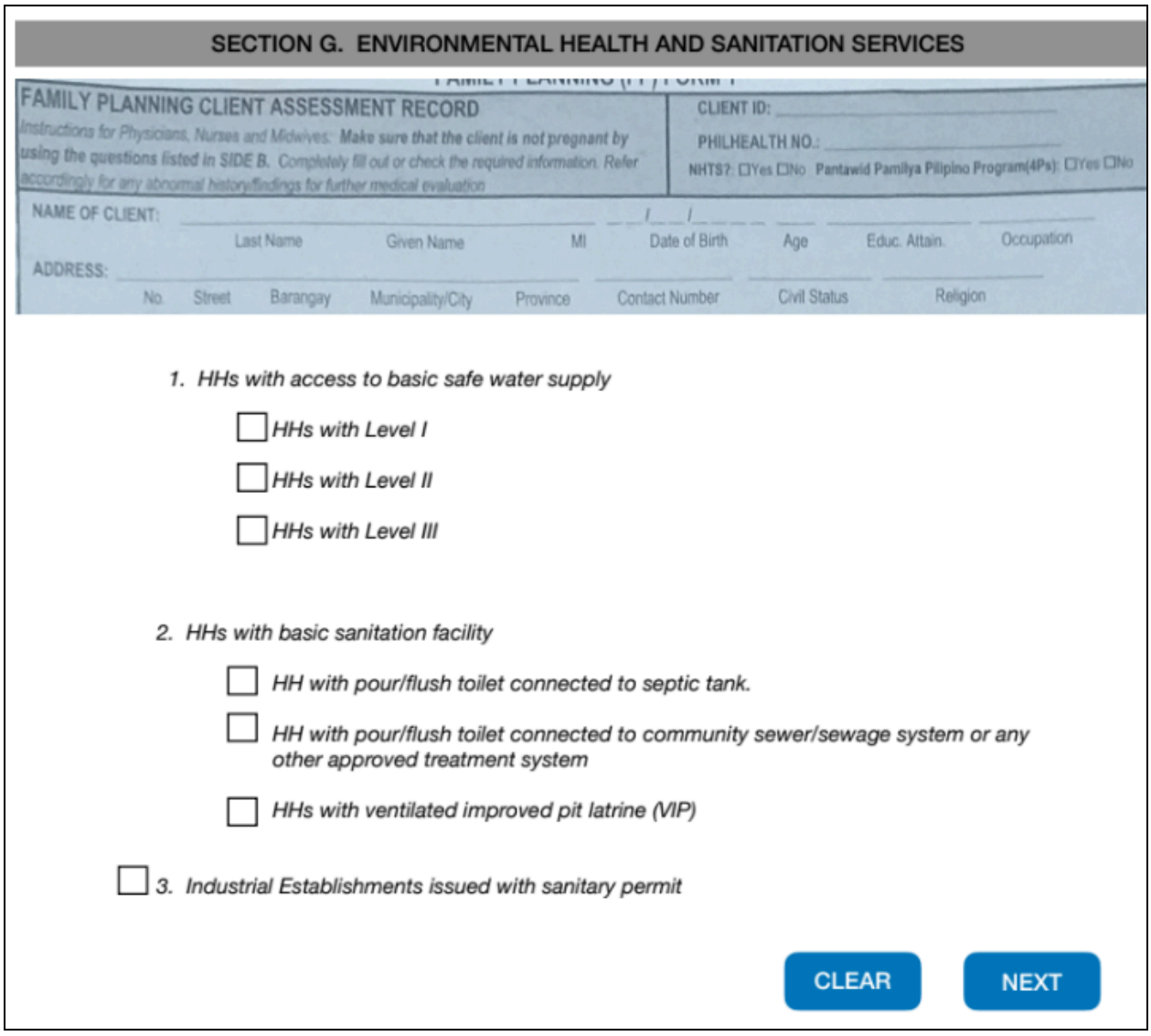
E6. Malaria Prevention and Control

- ☐ 1. Patients with probable/clinically diagnosed Malaria and confirmed case **(eProbableMalaria)**
  - ☐ <5 y/o
  - ☐ ≥5 y/o

For Non-Communicable Disease Prevention and Control Services >> SECTION F



For Environmental Health and Sanitation Services >> SECTION G



SECTION G. ENVIRONMENTAL HEALTH AND SANITATION SERVICES

FAMILY PLANNING CLIENT ASSESSMENT RECORD

Instructions for Physicians, Nurses and Midwives: **Make sure that the client is not pregnant by using the questions listed in SIDE B.** Completely fill out or check the required information. Refer accordingly for any abnormal history/findings for further medical evaluation

CLIENT ID: _____

PHILHEALTH NO.: _____

NHTS?: ☐Yes ☐No Pantawid Pamilya Pilipino Program(4Ps): ☐Yes ☐No

NAME OF CLIENT: _____ __/__/____ ____ _____ _____

Last Name Given Name MI Date of Birth Age Educ. Attain. Occupation

ADDRESS: _____ _____ _____ _____

No. Street Barangay Municipality/City Province Contact Number Civil Status Religion

1. *HHs with access to basic safe water supply*
   - ☐ *HHs with Level I*
   - ☐ *HHs with Level II*
   - ☐ *HHs with Level III*

2. *HHs with basic sanitation facility*
   - ☐ *HH with pour/flush toilet connected to septic tank.*
   - ☐ *HH with pour/flush toilet connected to community sewer/sewage system or any other approved treatment system*
   - ☐ *HHs with ventilated improved pit latrine (VIP)*

☐ 3. *Industrial Establishments issued with sanitary permit*

CLEAR NEXT

Home About Contact us Services Login

dental

DENTAL HEALTH SERVICES

DENTAL HEALTH PROGRAM

Oral disease continues to be a serious public health problem in the Philippines. The prevalence of dental caries on permanent teeth has generally remained above 90% throughout the years. About 92.4% of Filipinos have tooth decay (dental caries) and 78% have gum diseases (periodontal diseases) (DOH, NMEDS 1998). Although preventable, these diseases affect almost every Filipino at one point or another in his or her lifetime.

Table 1: Prevalence of the Two Most Common Oral Diseases by Year, Philippines

YEAR	Prevalence	
	Dental Caries	Periodontal Disease
1987	93.9%	65.5%
1992	96.3%	48.1%
1998	92.4%	78.3%

Home About Contact us Services Login

PROGRAMANG PANGKAPALIGIRAN

ENVIRONMENTAL HEALTH PROGRAMS

Vision

Environmental Health (EH) related diseases are prevented and no longer a public health problem in the Philippines *(based on on-going Strategic Plan 2019-2022)*

Mission

To guarantee sustainable Environmental Sanitation (ES) services in every community

Objectives

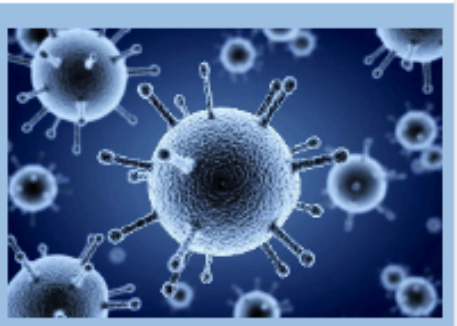
1. Expand and strengthen delivery of quality ES services
2. Institute supportive organizational, policy and management systems
3. Increase financing and investment in ES
4. Enforce regulation policy and standards
5. Establish performance accountability mechanism at all levels

Program Components

Drinking-water supply, Sanitation (e.g excreta, sewage and septage management), Zero Open Defecation Program (ZODP), Food Sanitation, Air Pollution (indoor and ambient), Chemical Safety, WASH in Emergency situations, Climate Change for Health and Health Impact Assessment (HIA)

Home About Contact us Services Login

NAKAHAHAWANG SAKIT



Infectious Diseases

1. Acute Respiratory Infection
2. Influenza A (H1N1)
3. Bird Flu (Avian Influenza)
4. Chickenpox
5. Cholera
6. Dengue
7. Diarrhea
8. Diphtheria
9. Ebola
10. Hand, Foot, and Mouth Disease
11. Hepatitis A
12. Hepatitis B
13. Hepatitis C
14. HIV/AIDS
15. Influenza
16. Leprosy
17. Malaria

Home About Contact us Services Login

PAGBABAKUNA AT KALUSUGAN NG MGA BATA

1. Pabakunahan natin ang ating mga anak!

Ang natural na panlaban sa sakit (resistensiya o immunity) na ibinibigay ng nanay sa kaniyang anak ay naglalaho nang natural sa ika-3 buwan mula nang ipanganak para sa Pertussis, at ika-12 buwan naman mula nang isilang para sa Tigdas. Makalipas ang panahong nabanggit ay dapat magkaroon ang mga sanggol ng resistensiya o bumuo ng immunity upang maiwasan ang pagkakaroon ng sakit. Ang pagbabakuna ay isang paraan upang makabuo ng immunity.

Habang papalaki ang bata, magiging madalas ang kanilang pagkakataon na makisalamuha sa ibang tao, kaya't mas malaki ang pagkakataong sila ay mahawa sa mga sakit. Layunin naming ipagbigay-alam ang kahalagahan ng pagbabakuna upang maiwasan ang mga sakit na ito.

- Impeksiyon

Ang impeksiyon ay isang uri ng sakit na dala ng pathogen tulad ng virus at bacteria na pumapasok sa loob ng katawan ng tao at bumubuo ng iba't ibang uri ng sintomas kapag ito ay dumami sa loob ng katawan. Depende sa uri ng pathogen, nakakaranas ng iba't ibang sintomas at ang pangkaraniwang palatandaan nito ay pagkakaroon ng lagnat, ubo at sakit ng ulo.

2. Ano ang pagbabakuna?

Ang pagbabakuna ay ang paglalagat o pag-inject sa ating katawan ng bakuna na mula sa pinahinang uri ng virus o bacteria tulad ng Tigdas, Pertussis athp. Ito ay upang bumuo ang ating katawan ng sariling panlaban o immunity para sa uri ng sakit na ito. Ang uri ng gamot na ginagamit sa pagbabakuna ay tinatawag na bakuna o vaccine.

Hindi lahat ng uri ng impeksiyon ay maaaring buuin bilang isang bakuna. May mga impeksiyon o sakit na maaaring gawing bakuna, depende sa uri ng virus o bacteria.

3. Bisa (effectiveness) ng pagbabakuna

Ang layunin ng pagbabakuna ay upang maiwasan ang pagkakasakit o maibsan ang paglala ng kalagayan kahit na magkasakit, subalit hindi basta nagkakaroon ng resistensiya (immunity) ang taong nabibigyan nito laban sa sakit. Ito'y depende sa pangangatawan ng mga bata o sa pisikal na kondisyon noong araw na siya'y binakunahan. Kung nais nimyong malaman kung kayo ay nakagawa ng resistensiya (immunity), maari mo rin masukat ang inyong antibody sa dugo sa pagsusuri sa pamamagitan ng pagsa-sample ng dugo (blood sampling).

Home About Contact us Services Login

PAGPAPLANO NG PAMILYA

Bawat taon, kalahating milyong babae ang namamatay mula sa mga problema sa pagbubuntis, panganganak at hindi ligtas na pagpapalaglag. Maaaring mapigilan ng pagpaplano ng pamilya ang karamihan sa pagkamatay na ito. Halimbawa, makakapigil ang pagpaplano ng pamilya sa panganib mula sa pagbubuntis na:

- Masyadong Maaga
- Masyadong Huli
- Masyadong magkakalapit
- Masyadong Madami

Bukod sa pagliligtas ng buhay, may iba pang mga benepisyo ang pagpaplano ng pamilya

- Mas lulusog ang mga nanay at sanggol, dahil maiiwasan ang delikadong pagbubuntis.
- Mas kaunti ang anak, maaaring mas marami ang pagkain para sa bawat isa.
- Kapag naghintay bago magkaanak, maaaring magkapanahon ka para magtapos ng pag-aaral ang mga kabataang babae't lalaki.
- Kapag mas kaunti ang anak, maaaring mas may panahon para sa sarili at mga anak.
- Makakatulong din ang pagpaplano ng pamilya para mas masiyahan kayong magpartner sa pagtatalik, dahil hindi nangangamba sa di-gustong pagbubuntis. May iba pang benepisyo sa kalusugan ang ibang paraan. Halimbawa, makakatulong

Home About Contact us Services Login

PANGANGALAGA SA BUNTIS

1. Magtungo agad sa health center kapag nalaman mo na ikaw ay buntis.
2. Gawing regular ang iyong check up (DAPAT hindi mababa sa 4 na beses sa loob ng 9 na buwan) upang masigurado ang kaligtasan ninyo ni baby.
3. Mag-paplano tayo ukol sa iyong panganganak (birth planning) dahil bawal na manganak sa bahay.
4. Muling magpa-check up 24 oras matapos manganak (post natal).

MEDICAL & MATERNITY CLINICS IN SAN MATEO RIZAL

*Padre Pio Maternity and Medical Hospital Inc.

*My Little Love Maternity Clinic

*Sta. Clara Medical & Maternity Clinic

*St. Mattheus Hospital

*Chiombion-Panganiban Medical & Maternity Clinic

*San Mateo Medical Center

*Super Health Center San Mateo

Home About Contact us Services Login

DI-NAKAHAHAWANG SAKIT

Non-Communicable Diseases

1. Alzheimer's Disease
2. Cancer
3. Epilepsy
4. Osteoarthritis
5. Osteoporosis
6. Cerebrovascular Disease (Stroke)
7. Chronic Obstructive Pulmonary Disease (COPD)
8. Coronary Artery Disease
9. Heat Stroke
10. High Blood Pressure or Hypertension
11. Obesity and Overweight
12. Diabetes
13. Depressive Disorders
14. Substance Abuse: Alcohol
15. Substance Abuse: Ecstasy

Appendix I

The Proponent

Appendix J

Other Documents